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**Report prepared for the WHO annual consultation on
the composition of influenza vaccines for the
Southern Hemisphere 2023**

19th - 22nd September 2022

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Influenza Epidemiology in the WHO European Region

The epidemiological section of this report was prepared using data reported to The European Surveillance System (TESSy) and presented in Flu News Europe (<http://www.flunewseurope.org>) the joint ECDC-WHO Europe weekly influenza update, with the assistance of members of the Influenza and Other Respiratory Pathogens Group at the WHO European Office and the ECDC Influenza Surveillance Group:

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WHO Vaccine Recommendation Meeting for the 2023 Southern Hemisphere influenza season

WHO CC, London Final Report for the meeting: 19-22 September 2022

Epidemiology

Summary of Influenza activity in the WHO European Region, October 2021 – August 2022

Note

The novel coronavirus disease 2019 (**COVID-19**) pandemic, caused by severe acute respiratory syndrome coronavirus 2 (**SARS-CoV-2**), continues to adversely affect healthcare presentations and testing capacities of some countries, territories and areas in the Region. Consequently, the analysis presented here, based on data reported to The European Surveillance System (TESSy), should be interpreted with care.

Distribution of viruses

Between weeks 40/2021 and 34/2022, a total of 146,890 influenza detections from sentinel and non-sentinel primary care sources were reported to TESSy (**Table E1-A**). This represents 294-fold and 1.1-fold increases compared to 499 influenza detections for the same period in 2020-2021 and 131,104 in 2019-2020, respectively.

This increase has occurred in the context of increased numbers of samples from patients fulfilling Influenza-Like Illness (ILI) and/or Acute Respiratory Infection (ARI) criteria being tested; there were 3,119,226 specimens tested in 2021-2022 which represents 3.5-fold and 4.4-fold increases compared to 897,825 in the same period in 2020-2021 and 717,777 in 2019-2020, respectively (**Table E1-A**). Combined sentinel and non-sentinel data for type A viruses in the Region showed that most viruses detected (98%) were type A and 2% were type B (a ratio of ~ 51.8:1 and representing an increase from ~2.3:1 in 2019-2020) (**Table E1-A**). This distribution was generally consistent across the Region with all subregions detecting at least 96% type A, except for Central Asia where there was a higher proportion of type B viruses (10%) amongst 283 influenza detections (**Table E2-A**). Combined sentinel and non-sentinel data for type A viruses showed a majority (92%) to be A(H3N2) viruses, with 8% being A(H1N1)pdm09 viruses (a ratio of ~ 11.3:1, increased from ~1.5:1 in 2020-2021) (**Table E1-A**). These ratios were consistent across the Region except for western Europe where the detections ratio was 66% A(H3N2) and 34% A(H1N1)pdm09 viruses (**Table E2-A**). Of 142 type B viruses ascribed to a lineage, 140 (99%) were B/Victoria and only two (1%) were reported (unconfirmed, but likely to be LAIV-associated based on timing, patient age and limited HA sequence data) as B/Yamagata viruses.

Table E1-B shows a summary of influenza virus testing reported to TESSy for weeks 21-35/2022, a period when influenza activity is normally at 'inter-seasonal' levels, compared to similar periods in the 2020-2021 and 2019-2020 seasons. Compared to

the 2019-20 season (representative of influenza seasons before the COVID-19 pandemic), there were approximately 6- and 13-fold increases in the numbers of sentinel specimens tested for influenza in the consecutive seasons, which translated into 8- and 299-fold increases in the numbers of influenza detections respectively. Similar trends were seen for non-sentinel testing, and for combined sentinel and non-sentinel data there were 6- and 7-fold increases in the numbers of specimens tested with 8- and 289-fold increases in the numbers of influenza detections. Overall, during this late period of the 2021-2022 season in the Region, numbers of specimens tested and influenza detections made accounted for 17% and 10%, respectively, of the season's totals.

When the analysis was restricted to sentinel source data only for weeks 40/2021 to 34/2022, the regional distribution of influenza by type was similar: of the viruses detected (n=7,900), the majority were type A (99%). Of type A viruses, the majority were A(H3N2) subtype (94%) and of the type B viruses, all were B/Victoria-lineage (**Table E2-B**). These distributions were consistent across the Region with all subregions reporting a higher proportion of A(H3N2) viruses, although the proportion (82%) was lower in western Europe than elsewhere. Western Europe was the only subregion reporting at least 10 type B viruses, all were B/Victoria-lineage. Of the total detections, 2,537 (32%) were from south-west Europe and 2,086 (32%) were from western Europe (**Table E2-B**).

A dominant virus (sub)type (comprising at least 60% of influenza virus detections) could be assigned by 41 countries and areas within the WHO European Region (11 having insufficient numbers [ten or fewer] of detections). All countries and areas reported predominance of influenza type A virus (**Figure E1**). Five reported dominance of type A without specifying a subtype, and 36 were able to ascribe dominance (60% or greater) or co-dominance (between 41% and 59%) of influenza A virus subtypes for the season: all reported A(H3N2) dominance. No countries or areas reported dominance of type B, and very few B/Victoria-lineage viruses were detected.

Between weeks 20 and 34/2022, a total of 5,570 influenza detections from sentinel and non-sentinel primary care sources were reported to TESSy, of which, 5,270 (95%) were type A viruses (90% were A(H3N2) and 10% were A(H1N1)pdm09). All Influenza type B viruses assigned to a lineage (n=300) were B/Victoria.

Influenza virus characterisation

Of 146,890 influenza virus detections between weeks 40/2021 and 34/2022, genetic characterisation of 5,394 viruses was reported to TESSy.

Of the 4,904 influenza A(H3N2) viruses characterised, 4,855 were attributed to subclade 3C.2a1b.2a.2 represented by A/Bangladesh/4005/2020, 20 were attributed to subclade 3C.2a1b.1a represented by A/Denmark/3264/2019 and three were attributed to subclade 3C.2a1b.2a.1 represented by A/Cambodia/e0826360/2020. Twenty-six were not attributed to a clade/subclade.

Of the 396 A(H1N1)pdm09 viruses characterised, 364 were attributed to subclade 6B.1A.5a.1 represented by A/Guangdong-Maonan/SWL1536/2019 and 32 were attributed to subclade 6B.1A.5a.2 represented by A/Victoria/2570/2019.

Of the 85 B/Victoria-lineage viruses characterised, 55 belonged to subgroup V1A.3a.2, represented by B/Austria/1359417/2021 and 29 belonged to subclade V1A.3, represented by B/Washington/02/2019. One virus was not attributed to a clade/group.

Nine viruses were reported as B/Yamagata-lineage viruses with six being clade Y3, represented by B/Phuket/3073/2013, while three viruses were not attributed to a clade. However, the possibility that these viruses were derived from administration of live attenuated influenza vaccine (LAIV) could not be excluded.

Susceptibility to Neuraminidase (NA) and Polymerase Acidic protein (PA) inhibitors

Between weeks 40/2021 to 32/2022, 3,329 viruses from the 2021-2022 season were assessed genotypically and/or phenotypically for susceptibility to oseltamivir (2,897 A(H3), 350 A(H1)pdm09 and 82 B viruses) and 3,257 viruses to zanamivir (2,826 A(H3), 349 A(H1)pdm09 and 82 B viruses). Genotypically, one A(H1)pdm09 virus was found to carry NA H275Y amino acid substitution that has been associated with resistance to oseltamivir and four A(H3) viruses carried S331R substitution that has been reported to be associated with reduced susceptibility to both oseltamivir and zanamivir.

During the same period, 2,215 viruses were assessed genotypically for susceptibility to baloxavir marboxil (1,904 A(H3), 267 A(H1)pdm09 and 44 B viruses). Two A(H3) viruses carried PA E23G amino acid substitution that has been associated with reduced susceptibility to baloxavir marboxil. For one of these viruses reduced susceptibility has been confirmed by phenotypic testing.

Timing and spread

To indicate the start of the influenza season in the Region a threshold of 10% of ILI or ARI specimens from sentinel primary care sites testing positive for influenza viruses within a calendar week has been chosen.

There was effectively no period of increased influenza activity in 2020-2021 during the COVID-19 pandemic. During the 2021-2022 season, with weeks 40/2021 to 20/2022 representing the period in most seasons when influenza detections can be at or above the 10% threshold, the initial increase in influenza activity in the WHO European Region occurred in week 49/2021 (**Figures E2-A/B**). This was a relatively late start of increased activity compared to a range of week 46 to 49 for the 2016-2017 to 2019-2020 seasons.

The 2021-2022 season was characterised by two waves of activity at or above the 10% threshold (**Figure E2-A**). The first, weeks 49/2021 to 01/2022, was mainly driven by non-EU countries in the eastern part of the Region. The second, weeks 09/2022 to 20/2022, was largely driven by reports from countries within the EU and others in the western part of the Region. The proportion of specimens testing positive for influenza on a weekly basis reached a peak for the season of 27% in week 12/2022, which was lower than the peaks of 53-60% in the 2016-17 to 2019-20 seasons, with positivity in 2020-2021 never exceeding 2% (**Figures E2-A/B**). Activity in which $\geq 10\%$ of sentinel specimens tested positive for influenza lasted for 16 non-consecutive weeks, compared to an average of 20.8 consecutive weeks (range: 18-25 weeks) for the previous seasons (2016-2017 to 2019-2020). Similar patterns of activity and proportions of influenza type and subtype distributions were observed with non-sentinel data (**Figure E2-C**).

Based on weekly reporting on the intensity of influenza activity by each of the countries and areas sustained activity first increased as early as week 40/2021, mainly in the

east of the WHO European Region, and included Armenia, Georgia, Kyrgyzstan, Turkmenistan and Kosovo (in accordance with Security Council resolution 1244 (1999)) (**Figure E3**). Some countries in the west of the Region only experienced the second wave of increased influenza activity.

Over the season, a total of 42 countries or areas reported activity above their calculated baselines, with 25 having reported medium activity or higher in some weeks. Only two reported very high intensity (Luxembourg and Finland) but it is acknowledged that this was influenced by COVID-19 related activity. (*Note: this indicator is typically based on ILI and/or ARI activity but at least one virus detection per week is required to meet the definition for an intensity level above baseline. Not all countries, territories or areas reporting intensity above baseline have reported detections*) (**Figure E3**).

Hospitalised cases

Only six countries, territories and areas (the Czech Republic, France, Ireland, Sweden, Ukraine and the United Kingdom (England)) reported case-based hospital data in 2021-2022. Between weeks 40/2021 and 34/2022, a total of 1,486 hospitalised laboratory-confirmed influenza cases were reported to TESSy. Of these, 859 cases were admitted to intensive care units (ICUs) in France (n=426, 50%), United Kingdom (England) (n=288, 34%), Sweden (n=113, 13%), Ireland (n=19, 2%) and the Czech Republic (n=13, 2%). The remaining 627 cases were reported from other hospital wards in Ireland (n=576, 92%), Ukraine (n=34, 5%) and the Czech Republic (n=17, 3%). Among the 859 cases admitted to ICUs, 174 were infected with type A viruses (104 A(H3N2) and 70 A(H1N1)pdm09) and 25 with a type B virus (not ascribed to a lineage). A total of 565 cases had a recorded age, falling within different age groups: 251 were 15-64 years (44%), 203 were 65 years or older (36%), 58 were 00-04 years (10%) and 53 were 05-14 years (9%) (**Figure E4**).

A total of 25 countries, territories and areas (Albania, Armenia, Belarus, Belgium, Bosnia and Herzegovina, Croatia, Georgia, Germany, Ireland, Kazakhstan, Kyrgyzstan, Lithuania, Malta, Montenegro, North Macedonia, Republic of Moldova, Romania, Russian Federation, Serbia, Spain, Türkiye, Turkmenistan, Ukraine, Uzbekistan and Kosovo (in accordance with Security Council resolution 1244 (1999))) with surveillance systems for severe acute respiratory infection (SARI) reported a total of 178,633 cases, between weeks 40/2021 and 34/2022, together with the ages of 118,797 of the patients. Overall, children aged 00-04 years accounted for the largest proportion of cases (41%), followed by those aged 30-65 years (24%), those aged 65 years or older (18%) and those aged 05-30 years (18%). SARI cases were predominantly among children aged less than 05 years in Armenia, Belgium, Georgia, Kazakhstan, Kyrgyzstan, Spain, Türkiye, Turkmenistan and Uzbekistan but predominantly among those aged 65 years or older in Bosnia and Herzegovina, Montenegro and Serbia (**Table E3**). Of 28,956 SARI cases tested for influenza virus over the period, 5% (n=1,486) were positive. The majority (n=1,252, 84%) were infected with type A viruses and of these, 1,208 (96%) were A(H3N2) and 44 (4%) were A(H1N1)pdm09. Only Kyrgyzstan (n=33), Russian Federation (n=13), Serbia (n=12), Uzbekistan (n=11), Kazakhstan (n=8), Türkiye (n=3), Georgia (n=1), Montenegro (n=1) and Ukraine (n=1) reported infection by influenza type B viruses (12 were ascribed to a lineage, all were B/Victoria); all other patient samples were positive for influenza type A.

Mortality data

Pooled analyses of all-cause excess and influenza-attributable mortality from 29 European countries or sub-national regions participating in the EuroMOMO network (Austria, Belgium, Cyprus, Berlin (Germany), Denmark, England (UK), Estonia, Finland, France, Germany, Greece, Hesse (Germany), Hungary, Ireland, Israel, Italy, Luxembourg, Malta, the Netherlands, Northern Ireland (UK), Norway, Portugal, Scotland (UK), Slovenia, Spain, Sweden, Switzerland, Ukraine and Wales (UK)) showed a substantial increased excess all-cause mortality overall for the participating European countries during the 2021-2022 season. This excess mortality was driven by a substantial excess in a few countries. The significantly increased excess all-causes mortality was particularly prevalent among persons aged 45 years and older and coincided with periods of SARS-CoV-2 circulation.

Table E1. 2021-22 Influenza Season in the WHO European Region

A. Weeks 40/2021-34/2022 of three consecutive seasons

Virus type/ subtype/ lineage	Detections weeks 40/2021-34/2022					Detections weeks 40/2020-34/2021					Detections weeks 40/2019-34/2020				
	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios
Influenza A	7,784	136,324	144,108	98.1	51.8:1	26	254	280	56.1	1.3:1	9,241	81,559	90,800	69.3	2.3:1
A(H1N1)pdm09	417	2,738	3,155	8.2		11	20	31	40.3		5,293	17,915	23,208	64.3	
A(H3N2)	6,176	29,326	35,502	91.8	11.3:1	7	39	46	59.7	1.5:1	2,998	9,882	12,880	35.7	0.6:1
A not subtyped	1,191	104,260	105,451			8	195	203			950	53,762	54,712		
Influenza B	116	2,666	2,782	1.9		7	212	219	43.9		5,512	34,792	40,304	30.7	
Victoria lineage	23	117	140	98.6	70:1	2	8	10	90.9	10:1	2,146	1,757	3,903	98.9	92.9:1
Yamagata lineage	0	2	2	1.4		0	1	1	9.1		17	25	42	1.1	
Lineage not ascribed	93	2,547	2,640			5	203	208			3,349	33,010	36,359		
Total detections	7,900	138,990	146,890			33	466	499			14,753	116,351	131,104		
Total tested	79,398	3,039,828	3,119,226			31,300	866,525	897,825			38,528	679,249	717,777		

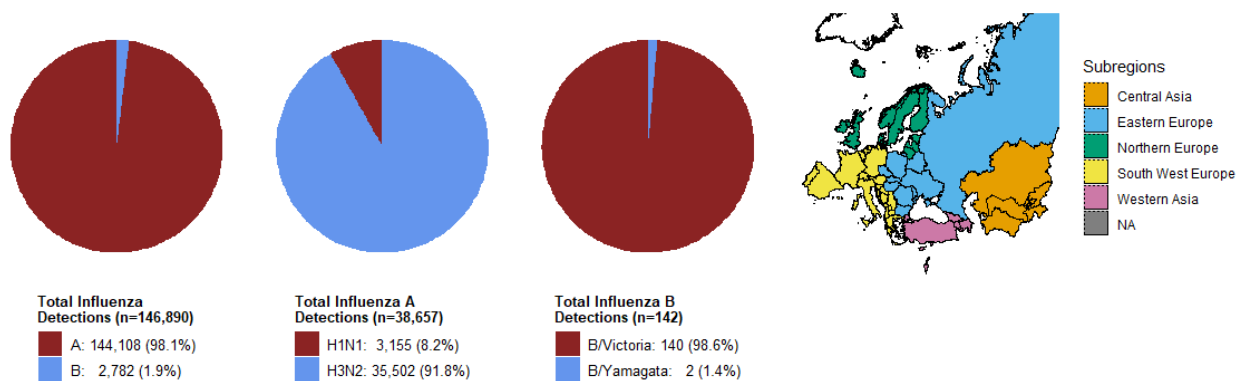
*Percentages are shown for total detections (types A & B [in bold type], and for viruses ascribed to influenza A subtype and influenza B lineage). Ratios are given for type A:B [in bold type], A(H3N2):A(H1N1)pdm09 and Victoria:Yamagata lineages.

B. Terminal weeks of three consecutive seasons

Virus type/subtype/lineage	Cumulative number of detections for weeks 21-35/2022					Cumulative number of detections for weeks 21-39/2021					Cumulative number of detections for weeks 21-39/2020				
	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios	Sentinel sources	Non-sentinel sources	Totals	%	Ratios
Influenza A	882	12788	13670	96.5	27:1	24	272	296	79.4	3.8:1	1	28	29	59.2	1.5:1
A(H1N1)pdm09	40	288	328	10.0		0	5	5	3.3		0	3	3	33.3	
A(H3N2)	775	2163	2938	90.0	9:1	4	185	189	96.7	38:1	1	6	7	66.7	2:1
A not subtyped	67	10337	10404			20	82	102			0	19	19		
Influenza B	15	483	498	3.5		0	77	77	20.6		2	18	20	40.8	
Victoria lineage	8	35	43	100.0		0	2	2	100.0		1	0	1		
Yamagata lineage	0	0	0			0	0	0			0	0	0		
Lineage not ascribed	7	448	455			0	75	75			1	18	19		
Total detections (total tested)	897 (20 375)	13 267 (>511 565)	14 168 (>531 940)			24 (9 313)	349 (>394 871)	373 (>404 184)			3 (1 523)	46 (>70 981)	49 (>72 504)		

Table E2. Distribution of Influenza Detections in the WHO European Region^a, weeks 40/2021-34/2022

Laboratory confirmed influenza from sentinel and non-sentinel sources by type and subtype/lineage



^a WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: (<http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM>)

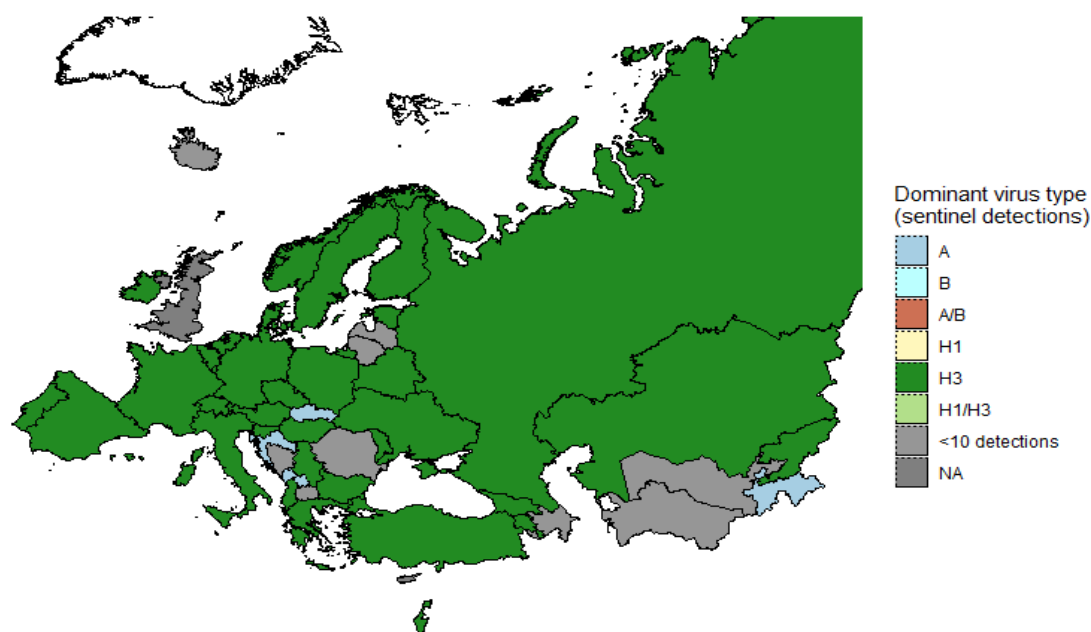
A. Laboratory confirmed influenza from sentinel and non-sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	26,314 (99%)	47,327 (99%)	12,029 (96%)	56,711 (97%)	1,473 (100%)	254 (90%)	144,108 (98%)
Influenza A subtyped	10,263	7,410	9,226	10,053	1,472	233	38,657
A(H1N1)pdm09	186 (2%)	2,545 (34%)	42 (0%)	381 (4%)	1 (0%)	0 (0%)	3,155 (8%)
A(H3N2)	10,077 (98%)	4,865 (66%)	9,184 (100%)	9,672 (96%)	1,471 (100%)	233 (100%)	35,502 (92%)
Influenza B	219 (1%)	454 (1%)	452 (4%)	1,622 (3%)	6 (0%)	29 (10%)	2,782 (2%)
Influenza B lineage determined	10	82	0	48	1	1	142
B/Yamagata-lineage	0 (0%)	2 (2%)	0	0 (0%)	0	0	2 (1%)
B/Victoria-lineage	10 (100%)	80 (98%)	0	48 (100%)	1	1	140 (99%)
Total	26,533	47,781	12,481	58,333	1,479	283	146,890

B. Laboratory confirmed influenza from sentinel sources by geographic sub-region and influenza type and subtype/lineage

Influenza type	South-West Europe	Western Europe	Eastern Europe	Northern Europe	West Asia	Central Asia	WHO European Region
Influenza A	2,516 (99%)	2,051 (98%)	824 (98%)	1,314 (98%)	878 (100%)	201 (98%)	7,784 (99%)
Influenza A subtyped	1,795	1,852	627	1,261	878	180	6,593
A(H1N1)pdm09	44 (2%)	337 (18%)	11 (2%)	24 (2%)	1 (0%)	0 (0%)	417 (6%)
A(H3N2)	1,751 (98%)	1,515 (82%)	616 (98%)	1,237 (98%)	877 (100%)	180 (100%)	6,176 (94%)
Influenza B	21 (1%)	35 (2%)	21 (2%)	31 (2%)	4 (0%)	4 (2%)	116 (1%)
Influenza B lineage determined	3	10	0	8	1	1	23
B/Yamagata-lineage	0	0 (0%)	0	0	0	0	0 (0%)
B/Victoria-lineage	3	10 (100%)	0	8	1	1	23 (100%)
Total	2,537	2,086	845	1,345	882	205	7,900

Figure E1. Influenza circulation^b in the WHO European Region by (sub)type and Country, territory or area, 2021-2022 season



^b Dominance of (sub)type is based on at least 60% of detections and co-dominance is based on between 40% and 60% of detections. Calculations are only considered valid if at least 10 viruses were detected

Figure E2. Seasonal influenza activity in the WHO European Region: weeks 40/2021-34/2022

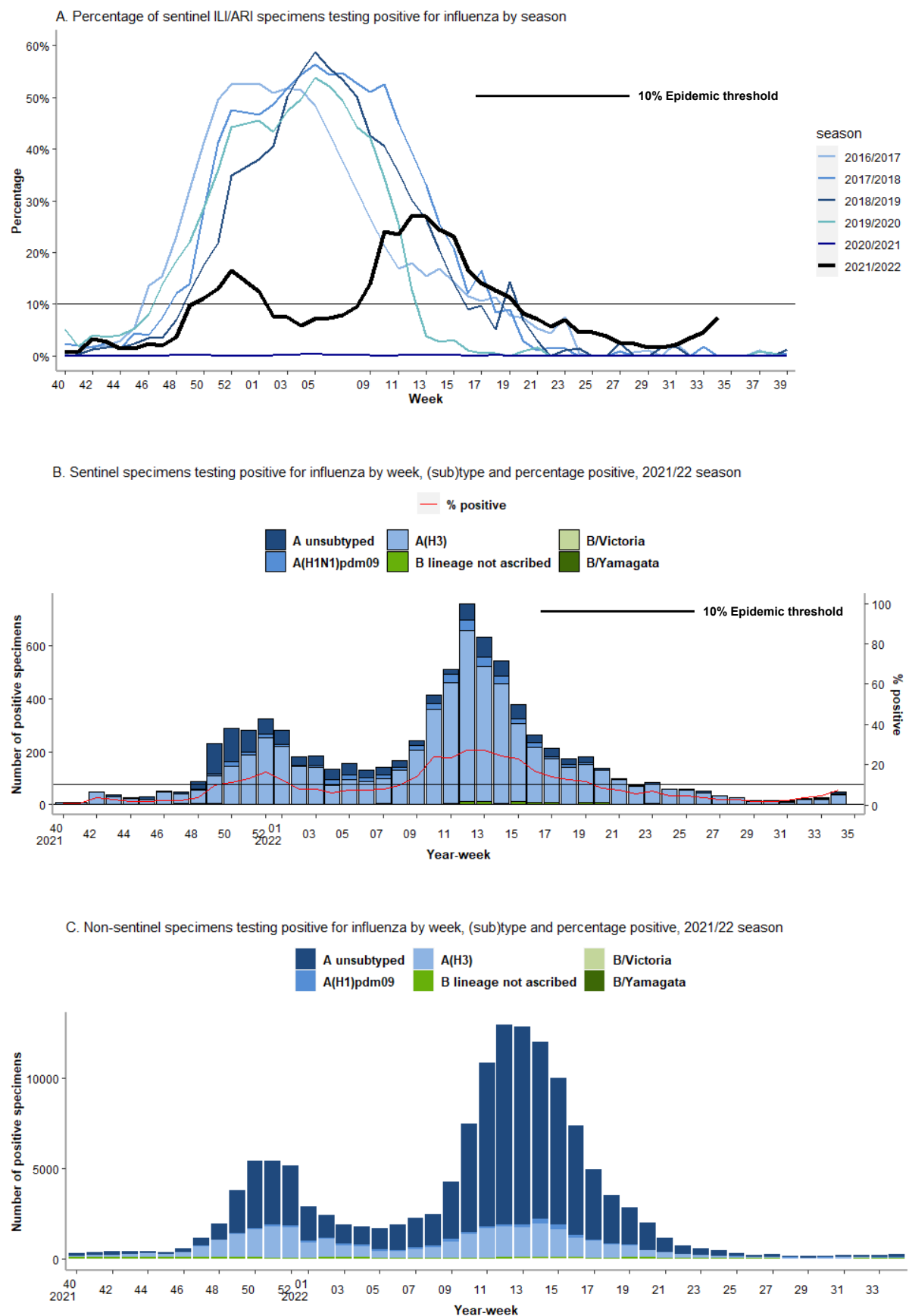
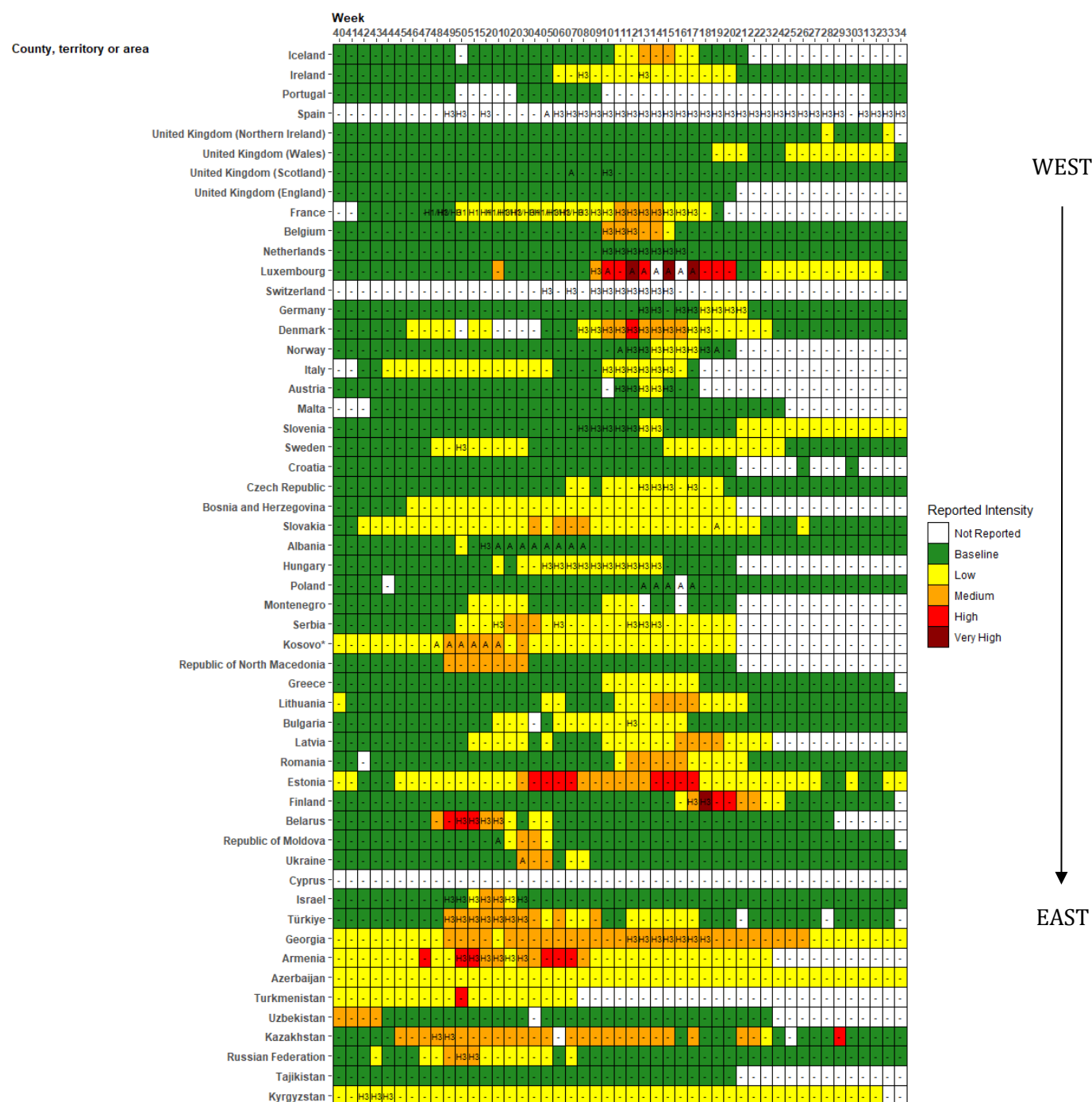


Figure E3. Intensity of influenza activity and dominant virus (sub)type in the WHO European Region by week and country, territory or area 202-/2022 season



*In accordance with United Nations Security Council Resolution 1244 (1999).

Baseline intensity influenza activity is the level at which clinical influenza activity remains throughout the summer.

Figure E4. Distribution of virus (sub)type in laboratory-confirmed hospitalised cases in ICUs by age group in the WHO European Region, 2021-2022 season

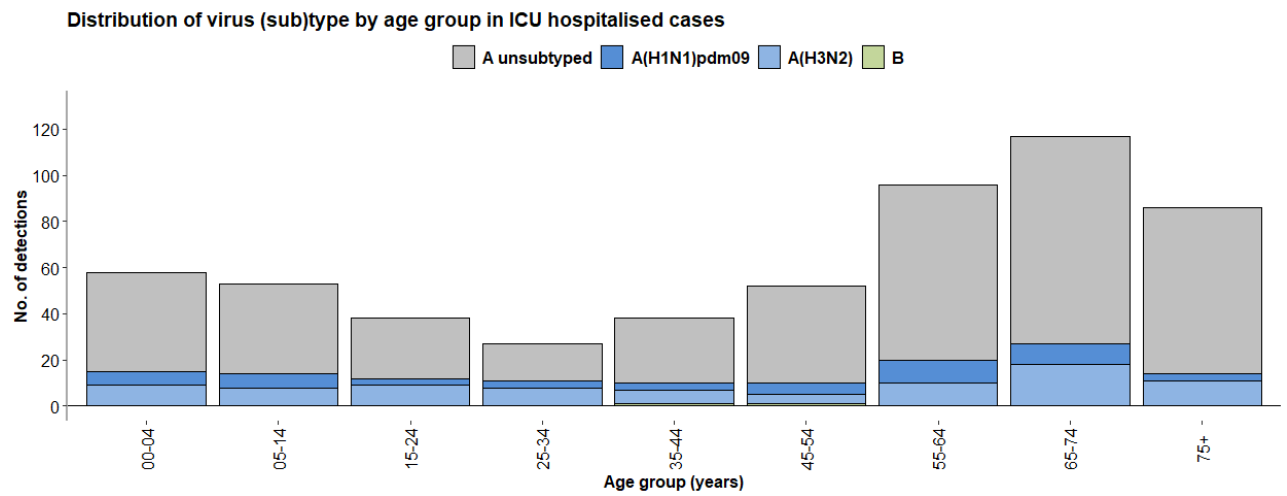


Table E3. Percentage distribution of severe acute respiratory infection (SARI) cases by age group and country, territory or area, and influenza positives, 2021-2022 season

Country, territory or area	Total SARI cases	SARI cases with age data	Percentage (%) of total cases					SARI specimens	Positive detections (%)
			0<5 years	5<15 years	15<30 years	30<65 years	>=65 years		
Albania	7,524	3,802	24	8	7	61	0	4,496	79 (2%)
Armenia	5,332	5,332	64	13	4	11	8	731	143 (20%)
Belarus	11,922	11,797	2	1	30	35	32	602	16 (3%)
Belgium	663	632	54	8	6	32	0	1,007	81 (8%)
Bosnia and Herzegovina	1,123	1,098	9	5	6	23	56	132	7 (5%)
Croatia	1,536	445	23	8	5	64	0	1,536	16 (1%)
Georgia	8,121	7,870	55	17	7	13	9	3,552	210 (6%)
Germany	27,932	12,161	44	5	5	46	0	0	0 (-%)
Ireland	203	203	0	0	11	89	0	538	17 (3%)
Kazakhstan	19,298	19,298	62	16	8	9	5	2,253	196 (9%)
Kyrgyzstan	7,889	7,889	77	10	3	7	3	1,833	137 (7%)
Lithuania	338	9	22	78	0	0	0	322	114 (35%)
Malta	1,221	362	0	0	12	88	0	603	29 (5%)
Montenegro	1,730	1,728	5	1	3	36	55	111	7 (6%)
North Macedonia	2,326	2,326	21	8	6	38	27	166	13 (8%)
Republic of Moldova	6,067	6,067	38	6	2	28	26	274	25 (9%)
Romania	61	61	33	8	13	46	0	113	18 (16%)
Russian Federation	2,984	2,983	29	15	17	18	21	2,309	147 (6%)
Serbia	5,693	5,693	1	1	4	41	52	194	69 (36%)
Spain	35,404	4,735	84	16	0	0	0	0	0 (-%)
Türkiye	9,727	6,474	53	14	0	0	34	2,443	95 (4%)
Turkmenistan	4,366	3,451	70	17	0	0	13	4,323	0 (0%)
Ukraine	16,723	13,975	11	5	7	36	40	1,095	26 (2%)
Uzbekistan	403	403	74	13	7	5	1	323	41 (13%)
Kosovo*	47	3	0	33	33	33	0	0	0 (-%)
Total	178,633	118,797	48,464 (41%)	11,205 (9%)	9,174 (8%)	28,308 (24%)	21,646 (18%)	28,956	1,486 (5%)

*In accordance with United Nations Security Council Resolution 1244 (1999).

Influenza specimens received at the Francis Crick Institute.

Seasonal influenza virus samples with collection dates after 2022-01-31, totalling 1839, have been received from 45 countries in five WHO Regions: African (n = 6; Ghana, Madagascar, Mauritius, Senegal, South Africa and Togo, Eastern Mediterranean (n = 5; Egypt, Morocco, Qatar, Somalia and Tunisia), European (n = 32), The Americas (n = 1; Argentina) and Western Pacific (n = 1; Hong Kong-SAR China) (**Figure 1 and Table 1**). These samples comprised 190 clinical specimen/virus pairs, 1,033 clinical specimens, 530 viruses and 86 clinical specimens in lysis mix. An additional 160 samples, with collection dates prior to 2022-02-01, were received in shipments from some countries. Overall, of the virus samples with collection dates after January 2022, influenza A viruses represented 91% of which A(H3N2) viruses outnumbered A(H1N1)pdm09 viruses at a ratio of ~6:1, while influenza B viruses represented 9% of the samples received. All influenza B viruses ascribed to a lineage were B/Victoria (**Table 1** and Pie-chart below).

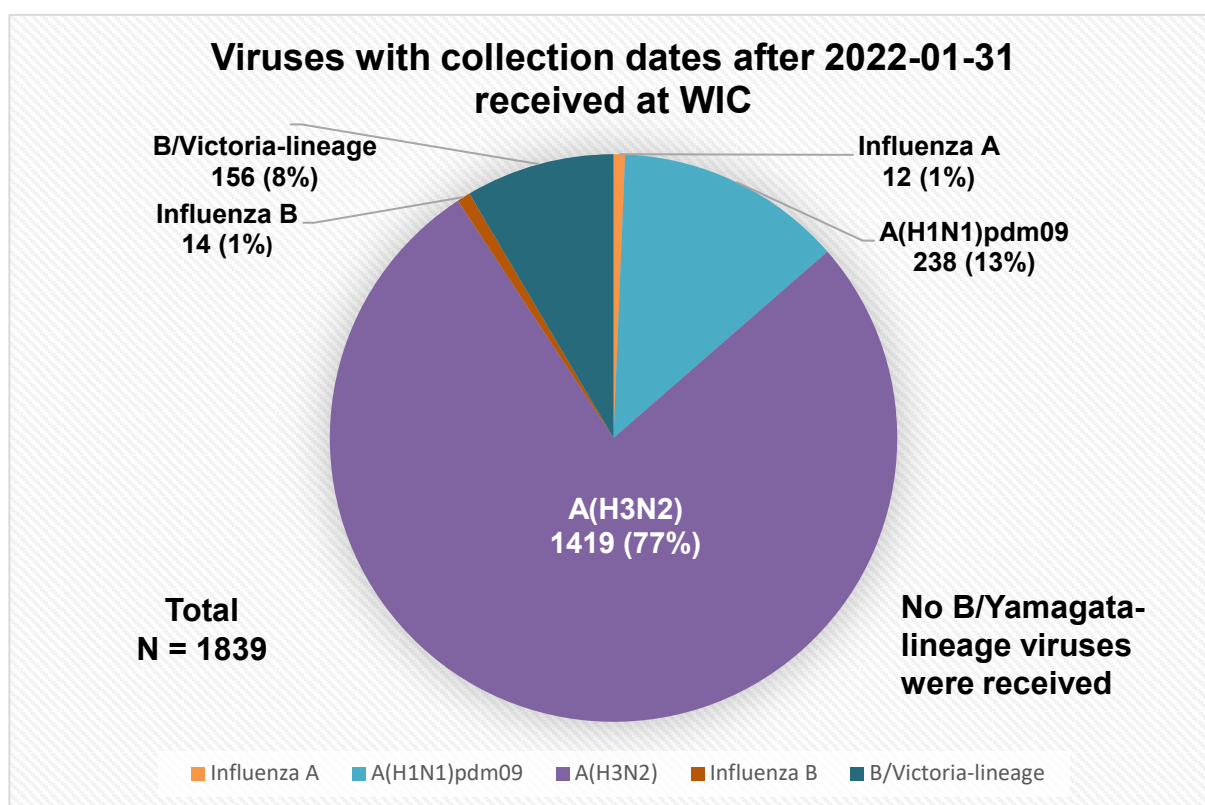


Table 2 shows the proportion of clinical samples that have been successfully characterised by next generation sequencing as of 2022-09-12. For those with collection dates pre-February 2022 approximately 50% yielded full-length HA gene sequences, while for those with collection dates post-January 2022 approximately 61% yielded full length HA gene sequences.

Table 3 shows the recovery/isolation rates from the samples with collection dates after 2022-08-31. Recoveries for samples sent as clinical specimens that were taken for virus culture were 68% for A(H1N1)pdm09, 80% for A(H3N2) and 51% for influenza B viruses, respectively. Recoveries for samples sent as virus isolates were 95% for A(H1N1)pdm09, 97% for A(H3N2) and 98% for influenza B viruses, respectively. All bar three A(H3N2)

viruses showed sufficient agglutination of guinea pig red blood cells to allow assessment by HI assay in the presence of 20nM oseltamivir.

Table 4 summarises the reactivity of A(H1N1)pdm09, A(H3N2) and B/Victoria-lineage viruses collected after 2022-01-31 with post-infection ferret antisera raised against either vaccine viruses or reference viruses in HI assays and in plaque reduction neutralisation assays (PRNA) for A(H3N2) viruses. For all assays, numbers and percentages of viruses showing ≤ 2 -, 4- and ≥ 8 -fold reductions in titres relative to the titres of the antisera with their respective homologous reference/vaccine viruses are shown.

For A(H1N1)pdm09 test viruses the numbers displaying specified folds-drop compared to homologous HI titres with antisera raised against egg-propagated A/Guangdong-Maonan/SWL1536/2019 and IVR-215 (A/Victoria/2570/2019) vaccine viruses, together with the folds-drop for antisera raised against cell culture-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 and A/Denmark/3280/2019 (A/Victoria/2570/2019-like) are shown.

For A(H3N2) test viruses in the HI assay the numbers displaying specified folds-drop compared to homologous HI titres with antisera raised against egg-propagated A/Cambodia/e0826360/2020 and A/Darwin/9/2021 vaccine viruses and cell culture-propagated A/Cambodia/925256/2020 and A/Stockholm/5/2021 (A/Darwin/6/2021-like) are summarised.

For the A(H3N2) test viruses assessed by PRNA the numbers displaying specified folds-drop compared to homologous 50% neutralising titres are summarised for antisera raised against cell culture-propagated A/Denmark/3264/2019, A/Cambodia/925256/2020, A/Bangladesh/4005/2020, A/Stockholm/5/2021, A/Slovenia/8720/2022 and an antiserum raised against egg-propagated A/Darwin/9/2021 vaccine virus.

For test viruses of the Influenza B/Victoria-lineage the numbers displaying specified folds-drop compared to homologous HI titres are given for antisera raised against the egg-propagated cultivar of the northern hemisphere 2021-2022 vaccine virus B/Washington/02/2019 (clade V1A.3) and the cell culture-propagated cultivar of B/Austria/1359417/2021 (subclade V1A.3a.2 with additional HA1 substitutions A127T, P144L and K203R). Results for antisera raised against two distinct egg-propagated cultivars of B/Austria/1359417/2021 – one with an HA sequence identical to the cell culture-propagated cultivar and one with the HA1 substitution G141R are also summarised in **Table 4**.

Figure 1. Countries Sharing Influenza-positive Samples Collected after 2022-01-31 with FCI-WIC.

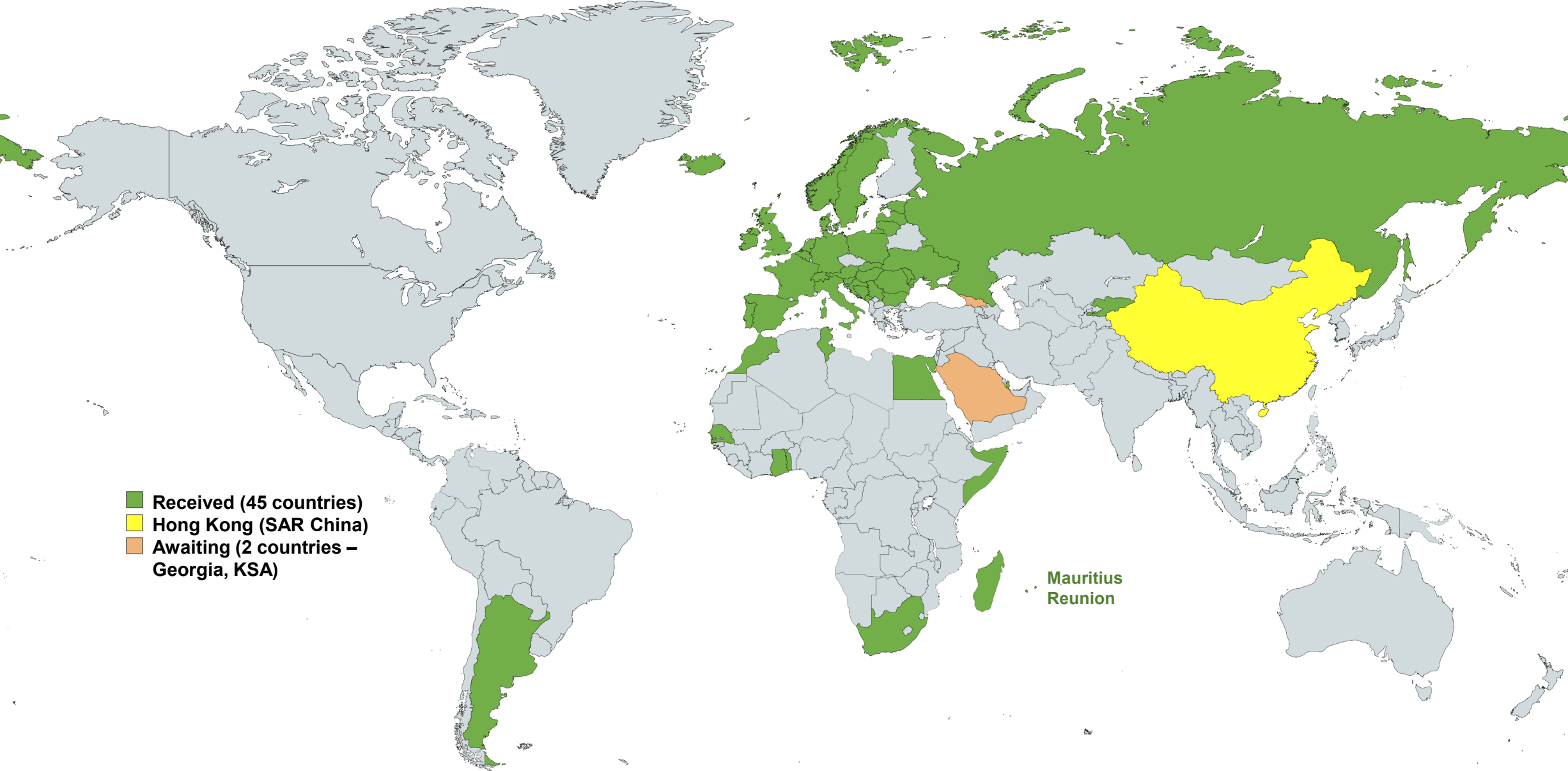


Table 1. Summary for influenza samples received (with collection dates after 2022-01-31), since the February 2022 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
2022													
FEBRUARY													
Austria	3			1	1	2	2						
Belgium	15			5	5	10	1	0					
Bosnia and Herzegovina	1	1	0										
Bulgaria	4					4	4						
Croatia	2					2	0	0					
Denmark	1					1	1						
Egypt	2					2	0	0					
Estonia	4					3	2	0		1	1		
France	8			4	2	4	4						
Germany	12			1	1	11	11						
Hungary	3					3	3						
Italy	6			1	1	5	5						
Moldova	1					1	1						
Morocco	4					4	0	0					
Netherlands	4			3	3					1	1		
Norway	2					2	2						
Poland	1					1	1						
Portugal	32			1	0	31	23	0					
Senegal	1					1	1						
Slovakia	1					1	0	0					
Slovenia	12					12	12						
Spain	12			2	1	10	10						
Sweden	2									2	2		
Switzerland	4			4	3								
Togo	2					1	0	0		1	0		
Tunisia	3					3	1	0					
Ukraine	1					1	1						
MARCH													
Austria	27			4	4	23	21	0					
Belgium	91			16	14	75	14	0					
Bosnia and Herzegovina	2					2	0	0					
Bulgaria	16					16	16						
Croatia	12					12	1	0					
Denmark	12			1	1	11	11						
Egypt	3								1	0	2	in process	
France	15			8	7	7	5	1					
Germany	8			1	1	7	7						
Hungary	4					4	4						
Iceland	4					4	0	0					
Ireland	39					39	4	0					
Italy	12			2	2	10	10						
Kyrgyzstan	6									6	6		
Lithuania	1					1	0	0					
Moldova	7			2	2	5	5						
Madagascar	6					6	2	0					
Montenegro	19					19	4	0					
Netherlands	12			4	4	4	4			4	4		
Norway	20			2	1	16	13	0		2	2		
Poland	27	2	0			25	in process						
Portugal	10					10	8	0					
Qatar	1									1	1		
Romania	1			1	0								
Russia	3									3	in process		
Serbia	13			4	1	9	7	0					
Slovakia	2					2	0	0					
Slovenia	40					40	39	0					
South Africa	5			4	4	1	1						
Spain	31			5	3	26	26						
Sweden	7					4	4			3	3		
Switzerland	20			6	6	14	13	0					
Togo	52			5	0	34	in process			13	0		
Ukraine	3					2	1	0		1	0		
UK (N. Ireland)	3					3							
APRIL													
Argentina	2					2	in process						
Austria	23			1	1	22	13	0					
Belgium	19			4	4	15	14						
Bosnia and Herzegovina	5					5	3						
Bulgaria	3					3	2	0					
Croatia	37					37	6	1					
Denmark	8			6	6	2	2						
Estonia	26					26	24	0					
France	9			3	in process	5	5			1	1		
Germany	8			2	2	6	6						
Hungary	2					2	2						
Iceland	27			2	0	25	1	0					
Ireland	11			1	1	10	3	0					
Italy	14					13	12	0		1	1		
Kyrgyzstan	6									6	6		
Latvia	15			2	2	12	12			1	0		
Lithuania	46					46	7	0					
Madagascar	20			1	0	18	2						
Moldova	7			1	1	6	6						
Montenegro	6					6	1	0					
Netherlands	27			1	1	6	5	0		20	16		
Norway	38			10	10	25	21	0		3	1		
Poland	64	5	0			45	in process		3	0	11	5	
Portugal	10					10	10						
Qatar	34			7	0	16	3	0		11	8		
Romania	28			4	4	24	24						
Russia	8									8	in process		
Serbia	11			5	2	4	4			2	2		
Slovakia	7					7	0	0					

Table 1. Summary for influenza samples received (with collection dates after 2022-01-31), since the February 2022 VCM, by country*

MONTH	TOTAL RECEIVED Seasonal	A		H1N1pdm09		H3N2		B		B Victoria lineage		B Yamagata lineage	
Country		Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ²	Number received	Number propagated ¹	Number received	Number propagated ¹	Number received	Number propagated ¹
Slovenia	11					11	11						
Somalia	17			17	0								
South Africa	11			11	10								
Spain	60			5	4	54	9	0		1	1		
Sweden	9			1	1	7	7			1	1		
Switzerland	14			2	0	12	6	0					
Togo	40	1	0			35	0	0		4	0		
Tunisia	1					1	0	0					
Ukraine	3					2	2			1	1		
UK (N. Ireland)	24					24							
MAY													
Argentina	19					19	in process						
Austria	2					2	2						
Belgium	2					1	1			1	1		
Bosnia and Herzegovina	1					1	0	0					
Croatia	3					3	2	0					
Denmark	3			2	2					1	1		
France	6			1	1	4	4			1	1		
Germany	11			1	1	8	8			2	2		
Ghana	21					20	in process			1	0		
Iceland	18					18	1	0					
Italy	1					1	1						
Lithuania	2					2	2						
Madagascar	14					9	1	0	1	0	4	0	
Mauritius	13			1	0	12							
Netherlands	2									2	0		
Norway	3									3	3		
Poland	9					7	in process	1	0	1	1		
Portugal	7			3	3	4	4						
Qatar	34			12	3	17	3	0	2	3	2		
Romania	9			1	1	5	4	0		3	2		
Russia	4									4	4		
Senegal	10					10	7	0					
Serbia	5							1	0	4	3		
Somalia	2			2	0								
South Africa	24			22	20	1	0	0		1	1		
Spain	50	2	0			48	29	0					
Sweden	1					1	1						
Togo	17					16	in process			1	0		
Tunisia	1							1	0				
Ukraine	1									1	1		
UK (N. Ireland)	3					3							
JUNE													
Argentina	14					14	in process						
Croatia	1			1	1					1	1		
Denmark	1												
France	2					2	2						
Ghana	14					14	in process						
Hong Kong	6					4	3	1		2	2		
Madagascar	11			1	0	6	2	0	2	2	1		
Mauritius	26			8		18							
Norway	5			3	in process	2	2						
Qatar	1					1	1						
Romania	2					2	2						
Russia	1									1	1		
Senegal	12					12	6	0					
Spain	5			1	0	4	2	0					
UK (N. Ireland)	6			1		4		1					
JULY													
Croatia	2			1	1	1	0	0					
France	11			1	1	10	10						
Ghana	5					2	in process			3	in process		
Hong Kong	1					1	0	0					
Madagascar	1							1	0				
Mauritius	11			1		10							
Norway	12			1	in process	10	in process			1	in process		
AUGUST													
Norway	3			1	in process					2	in process		
45 Countries	1839	12	0	238	150	1419	628	3	14	0	156	90	0
		0.65%		12.94%		77.16%			0.76%		8.48%		0.00%
				90.8%							9.2%		

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in the presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process)

Numbers in red indicate viruses recovered but with insufficient HA titre to permit HI assay

Samples received were in lysis buffer - only genetic characterisation possible

Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-COV-2 positive

* As of 2022-09-13

Table 2. Genetic characterisation¹ of samples supplied as clinical specimens only, since the February 2022 VCM*

Period	TOTAL RECEIVED Seasonal	SPECIMENS SUPPLIED IN						
		VTM ²				LM ³		
Country		Total	Good	Fail	ND	Total	Good	Fail
JANUARY 2021-22								
Bosnia and Herzegovina	8	8	7	1				
Egypt	5	5	0	5				
France	3	3	0	0	3			
Georgia	15	15	6	9				
Ghana	17	17	7	10				
Kazakhstan	30	30	7	23				
Morocco	26	26	5	21				
Norway	1	1	0	0	1			
Saudi Arabia	18	18	11	7				
Senegal	26	26	24	2				
Serbia	1	1	0	1				
Somalia	11	11	9	2				
Spain	16	16	9	7				
Switzerland	1	1	0	1				
Tunisia	18	18	9	9				
Ukraine	7	7	7	0				
FEBRUARY-AUGUST 2022								
Argentina	35	35	25	10				
Austria	20	20	14	6				
Bosnia and Herzegovina	9	9	4	5				
Bulgaria	1	1	0	1				
Croatia	57	57	50	7				
Egypt	5	5	2	3				
France	4	4	1	0	3			
Ghana	36	36	15	21				
Hong Kong	1	1	0	0	1			
Iceland	49	49	44	5				
Ireland	50	50	31	19				
Lithuania	49	49	12	37				
Madagascar	52	52	33	19				
Mauritius	43	43	14	29				
Moldova	15	15	13	2				
Montenegro	25	25	15	10				
Morocco	4	4	1	3				
Netherlands	6	6	0	2	4			
Norway	30	30	15	5	10			
Poland	101	101	60	41				
Portugal	6	6	3	3				
Qatar	70	70	51	19				
Romania	10	10	8	2				
Senegal	17	17	11	6				
Serbia	15	15	9	6				
Slovakia	10	10	2	8				
Somalia	19	19	13	6				
Spain	104	104	68	36				
Switzerland	38	38	28	10				
Togo	111	111	58	53				
Tunisia	5	5	2	3				
Ukraine	8	8	8	0				
UK (N. Ireland)	36					36	8	28
Pre-February total	203	203	101 (49.8%)	98 (48.3%)	4 (1.9%)	0	0	0
Post-January total	1041	1005	610 (60.7%)	377 (37.5%)	18 (1.8%)	36	8 (22.2%)	28 (77.8%)
Total	1244	1208	711	475	22	36	0	0

1. NGS yielded full-length HA gene sequence as a minimum

2. Influenza-positive specimens supplied in Virus Transport Medium

3. Influenza-positive specimens supplied in Lysis Mix

* As of 2022-09-12

Table 3. Isolation rates for specimens collected after 2022-01-31

Seasonal Virus	Clinical sample			Virus isolate		
	Number processed	Number propagated¹	Percent recovered	Number processed	Number propagated¹	Percent recovered
H1	62	42	68%	115	109	95%
H3	304	241²	79%	458	441²	96%
		2³	80% (99%)⁴ 0.7%		1³	97% (100%)⁴ 0.2%
B Victoria	70	36	51%	64	63	98%
B Yamagata						

1. Propagated to sufficient HA titre to perform HI assay

2. Propagated to sufficient HA titre in the presence of 20nM oseltamivir to allow HI assay

3. Viruses recovered (NA activity by MUNANA-based assay) but with insufficient HA titre to allow HI assay

4. The percentages for viruses recovered that could be assayed by HI are shown in parentheses

Table 4. Summary of isolates with collection dates after 2022-01-31 showing reduction in reactivity with antisera raised against reference viruses in HI[§] & plaque-reduction neutralisation assays (PRNA)[#]

Virus type/subtype	Number tested	≤2-fold ^a	4-fold ^a	≥8-fold ^a
H1 ^{1a}	151	100 (66%)	5 (3%)	46 (30%)
H1 ^{1b}	151	90 (60%)	9 (6%)	52 (34%)
H1 ^{1c}	151	42 (28%)	4 (2.5%)	105 (69.5%)
H1 ^{1d}	151	45 (30%)	1 (0.5%)	105 (69.5%)
H3 ^{2a}	682	313 (46%)	258 (37.5%)	111 (16.5%)
H3 ^{2b}	682	587 (86%)	88 (13%)	7 (1%)
H3 ^{2c}	682	24 (3.5%)	87 (13%)	571 (83.5%)
H3 ^{2d}	682	9 (1.5%)	15 (2%)	658 (96.5%)
H3 ^{3a}	133	0	0	133 (100%)
H3 ^{3b}	133	26 (19.6%)	68 (51.1%)	39 (29.3%)
H3 ^{3c}	133	60 (45.1%)	67 (50.4%)	6 (4.5%)
H3 ^{3d}	103	89 (86.4%)	10 (9.7%)	4 (3.9%)
H3 ^{3e}	133	4 (3.0%)	12 (9.0%)	117 (88.0%)*
H3 ^{3f}	133	1 (1.5%)	9 (6.0%)	123 (92.5%)
B/Victoria ^{4a}	99	86 (87%)	2 (2%)	11 (11%)
B/Victoria ^{4b}	99	86 (87%)	2 (2%)	11 (11%)
B/Victoria ^{4c}	99	0	12 (12%)	87 (88%)
B/Victoria ^{4d}	99	0	7 (7%)	92 (93%)

* Only 8 viruses in this category yielded titres of <160

[§] Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

[#] Reduction in titre, compared to the homologous titre based on 'rounded' values from PRNA:

1a. Compared to homologous titres for ferret antiserum raised against cell culture-propagated vaccine virus A/Guangdong-Maonan/SWL1536/2019 (genetic subgroup 6B.1A.5a.1)

1b. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus A/Guangdong-Maonan/SWL1536/2019 (genetic subgroup 6B.1A.5a.1)

1c. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Denmark/3280/2019 (genetic subgroup 6B.1A.5a.2)

1d. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus IVR-215 (A/Victoria/2570/2019) (genetic subgroup 6B.1A.5a.2)

2a. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Darwin/9/2021 (genetic subclade 3C.2a1b.2a.2)

2b. Compared to homologous titres for ferret antiserum raised against cell culture-propagated vaccine virus A/Stockholm/5/2021 (genetic subgroup 3C.2a1b.2a.2)

2c. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Cambodia/e0826360/2020 (genetic subgroup 3C.2a1b.2a.1)

2d. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Cambodia/925256/2020 (genetic subgroup 3C.2a1b.2a.1)

3a. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Cambodia/925256/2020 (genetic subgroup 3C.2a1b.2a.1)

3b. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 (genetic subgroup 3C.2a1b.2a.2)

3c. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Stockholm/5/2021 (genetic subgroup 3C.2a1b.2a.2)

3d. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Slovenia/8720/2022 (genetic subgroup 3C.2a1b.2a.2)

3e. Compared to homologous titres for ferret antiserum raised against egg-propagated A/Darwin/09/2021 (genetic subgroup 3C.2a1b.2a.2)

3f. Compared to homologous titres for ferret antiserum raised against cell culture-propagated A/Denmark/3264/2019 (genetic subgroup 3C.2a1b.1a)

4a. Compared to homologous titres for ferret antiserum raised against cell culture-propagated B/Austria/1359417/2021 - Clade 1A(Δ3)B N150K gp + A127T, P144L, K203R

4b. Compared to homologous titres for ferret antiserum raised against egg-propagated B/Austria/1359417/2021 - Clade 1A(Δ3)B N150K gp + A127T, P144L, K203R (G141)

4c. Compared to homologous titres for ferret antiserum raised against egg-propagated B/Austria/1359417/2021 - Clade 1A(Δ3)B N150K gp + A127T, P144L, K203R (G141R)

4d. Compared to homologous titres for ferret antiserum raised against egg-propagated vaccine virus B/Washington/02/2019 - Clade 1A(Δ3)B

H1N1pdm09

Influenza A(H1N1)pdm09 viruses with collection dates after 2022-01-31 were received from NICs in 26 countries in three WHO Regions: African (n = 4; Madagascar, Mauritius, South Africa and Togo) Eastern Mediterranean (n = 2; Qatar and Somalia) and European (n = 20) (**Table 1**). **Table 5** summarises antigenic (HI) results for the recognition of the viruses, propagated at the time of preparing this report, by antisera raised against the egg-propagated and cell culture-propagated cultivars of A/Guangdong-Maonan/SWL1536/2019 (the recommended vaccine virus for the Northern Hemisphere 2020-2021 influenza season), IVR-215 with its HA derived from egg-propagated A/Victoria/2570/2019 (the vaccine virus for the Southern Hemisphere 2021 and 2022, and the Northern Hemisphere 2021-2022 and 2022-2023 seasons) and against A/Denmark/3280/2019, a cell culture-propagated virus similar to A/Victoria/2570/2019.

HI analysis

Tables 7-1 to 7-10 show the results of HI analysis of samples analysed after the February 2022 VCM. Viruses below the red line in **Tables 7-5 to 7-10** had collection dates after 2022-01-31. A summary of the HI results on samples collected after 2022-01-31 is shown in **Table 7-11**.

Approximately 70% of viruses had HA genes falling into subclade 6B.1A.5a.1, 30% in subclade 6B.1A.5a.2.

Antisera raised against the vaccine virus IVR-215 (A/Victoria/2570/2019) or against cell culture-propagated A/Denmark/3280/2019 and cell culture-propagated A/Stockholm/10/2021, recognised all 6B.1A.5a.2 test viruses at titres within 4-fold of the homologous titres, and 98%, 91% and 73%, respectively, at titres within 2-fold of the homologous titres. An antiserum raised against an additional virus in subclade 6B.1A.5a.2, egg-propagated A/Sydney/5/2021, was also used and failed to recognise two test viruses in subclade 6B.1A.5a.2 at titres within 4-fold of the homologous titre, but still recognised 95% of the subclade 6B.1A.5a.2 viruses at titres within 4-fold of the homologous titre, 70% within 2-fold. These four antisera recognised none of the test viruses in subclade 6B.1A.5a.1 at titres within 4-fold of the homologous titres of the antisera (**Table 7-11**).

Antisera raised against the egg-propagated cultivar of A/Guangdong-Maonan/SWL1536/2019 (subclade 6B.1A.5a.1) and other related reference viruses (egg-propagated A/Ireland/87733/2019, A/Ghana/1894/2021, and A/Lyon/820/2021) recognised 93% to 98% of the test viruses that fell in subclade 6B.1A.5a.1 at titres within 4-fold of the homologous titres of the antisera. Although the antiserum raised against egg-propagated A/Lyon/820/2021 recognised only 58% of viruses at titres within 2-fold of the homologous titres, the antisera raised against the other egg-propagated reference/vaccine 6B.1A.5a.1 viruses recognised between 84% and 94% of the 6B.1A.5a.1 test viruses at titres within 2-fold of the homologous titres. Moreover, the antiserum raised against the cell culture-propagated cultivar of A/Guangdong-Maonan/SWL1536/2019 recognised all the 6B.1A.5a.1 test viruses at titres within 4-fold of the homologous titre of the antiserum. None of the antisera raised against the vaccine reference viruses from subclade 6B.1A.5a.1 recognised test viruses in subclade 6B.1A.5a.2 at titres within 4-fold of the homologous titres (**Table 7-11**).

Genetic analysis

All test viruses, for which HA gene sequences were known, fell into subclade 6B.1A which is defined by the amino acid substitutions **S74R**, **S164T** (altering a glycosylation motif introduced at residues **162-164** in clade 6B.1 viruses) and **I295V** in **HA1**, compared with the clade 6B.1 prototype virus, A/Michigan/45/2019. The vast majority of A(H1N1)pdm09-positive clinical specimens and viruses analysed had HA genes in group 6B.1A.5a in subclades 6B.1A.5a.1 or 6B.1A.5a.2.

Figure 2a shows a phylogenetic tree for HA genes of A(H1N1)pdm09 viruses analysed at the WIC, most with collection dates from 2022-02-01, with relevant reference virus HA gene sequences. **Figure 2b** shows a phylogenetic tree for HA genes of representative viruses as deposited in GISAID at the time of preparing this report. **Table 6** shows the geographic distribution of the 1,129 A(H1N1)pdm09 viruses with collection dates after 2022-01-31 for which genetic information was deposited in GISAID at the time of preparing this report.

The HA genes of recent viruses fall within group 6B.1A.5a. All test viruses carried **N129D**, **T185I** and **N260D** amino acid substitutions in **HA1**, and the two major subclades in circulation recently are:

- 6B.1A.5a.1: the HA genes of viruses in this subgroup encode **D187A** and **Q189E** substitutions in **HA1**. Some viruses have the additional **HA1** substitutions **P137S** and **G155E**, some of which also have **S128T** in **HA1** (including the reference virus A/Lyon/820/2021), the predominance of such viruses seems to be declining; another group has a defining substitution **I183T** in **HA2**; another has **R113K** in **HA1** and **H72N** in **HA2**; a third grouping can be adduced with the substitution **S326P** at the C-terminus of **HA1**.
- 6B.1A.5a.2: the HA genes of most viruses in this subgroup encode **HAs** with **K130N**, **N156K**, **L161I** and **V250A** substitutions in **HA1** and **E179D** substitution in **HA2**. The HA genes of a subgroup of viruses also encode **K54Q**, **A186T**, **Q189E**, **E224A**, **R259K** and **K308R** in **HA1**; within this subgroup some viruses have HA genes that encode **T216A** in **HA1** and some of these also encode **D94N** in **HA1**; another group has the substitutions **K142R** and **D260E** in **HA1** and **HA2** substitutions **I91V** and **N124H**, many of these have HA1 substitutions **P137S** and **T277A** and **E29D** in **HA2**.

Figure 3 shows H1pdm09-HA structures with the positions of amino acid substitutions in **HA1** that define the two subclades, 6B.1A.5a.1 and 6B.1A.5a.2. Also marked are amino acid substitutions that are associated with emerging and declining genetic subgroups within these two subclades.

An amino acid sequence alignment of the HAs for representative A(H1N1)pdm09 viruses is shown in **Figure 4**; the vaccine viruses are shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted, the genetic group/subgroup is shown in green and the HA1/HA2 proteolytic-processing site is highlighted.

Figure 5a shows a phylogenetic tree for the NA genes of viruses sequenced at the Francis Crick Institute (*cf* **Figure 2a**) and **Figure 5b** shows a NA gene phylogenetic tree of the viruses shown in **Figure 2b**. The NA trees and HA trees were largely congruent, however evidence of gene reassortment between the subclades has been seen in examples of viruses from South Africa. Viruses with HA genes in both subclades 6B.1A.5a.1 and 6B.1A.5a.2 shared the NA substitutions **M19T**, **Y66F** and **N222K**. Most viruses with HA genes in subclade 6B.1A.5a.1 had NA genes encoding **T16A**, some with the additional substitutions **S366N** or **K469N**. Most of the recent viruses with HA genes in subclade 6B.1A.5a.2 had NA genes encoding the substitutions **V453M**, some also with **I29M**, **S200N**, **R220K**, **I246T** or **K469N**.

Figure 6 shows a NA amino acid alignment for the same representative set of A(H1N1)pdm09 viruses as used in **Figure 4**.

Antiviral analysis

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 311 A(H1N1)pdm09 viruses detected and received within the course of the 2021-2022 influenza season (**Table 15**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

All of the A(H1N1)pdm09 full-length PA gene sequences generated at the Francis Crick Institute, from **308** individual viruses detected in the course of the 2021-2022 influenza season, encoded isoleucine at position 38 (**I38**) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion

Viruses received at the Francis Crick Institute fell into two distinct antigenic and genetic groups. The majority fell into subclade 6B.1A.5a.1. These viruses were recognised generally well by antisera raised against viruses in the same subclade (e.g. A/Guangdong-Maonan/SWL1536/2019). Viruses in subclade 6B.1A.5a.2 were recognised well by antisera raised against reference viruses and vaccine viruses in the same subclade (e.g. A/Victoria/2570/2019).

All viruses showed normal inhibition by oseltamivir and zanamivir and none showed amino acid substitutions indicative of reduced susceptibility to baloxavir marboxil.

Table 5. Summary of influenza A(H1N1)pdm09 viruses collected after 2022-01-31, analysed by HI

MONTH	H1N1pdm09													
Country	Number received	Number propagated ^{1,2}	Post-infection ferret antisera raised against											
			Clade 6B.1A.5a.1 virus			Clade 6B.1A.5a.1 virus			Clade 6B.1A.5a.2 virus			Clade 6B.1A.5a.2 virus		
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2022														
FEBRUARY														
Austria	1	1	1			1					1			1
Belgium	5	5	5			5					5			5
France	4	2	2			2					2			2
Germany	1	1	1			1					1			1
Italy	1	1		1				1			1			1
Netherlands	3	3	3			2	1				3			3
Portugal	1	0												
Spain	2	1		1				1			1			1
Switzerland	4	3	3			3					3			3
MARCH														
Austria	4	4	2		2	2		2	2		2	2		2
Belgium	16	14	12	2		13	1				14			14
Denmark	1	1	1			1					1			1
France	8	7	7			6		1			7			7
Germany	1	1	1			1					1			1
Italy	2	2	1		1	1		1	1		1	1		1
Moldova	2	2	2			2					2			2
Netherlands	4	4	3		1	2	1	1	1		3	1		3
Norway	2	1			1			1	1			1		
Romania	1	0							1					
Serbia	4	1	1			1					1			1
South Africa	4	4	3		1	3		1	1		3	1		3
Spain	5	3	3			3					3			3
Switzerland	6	6	6			5		1			6			6
Togo	5	0												
APRIL														
Austria	1	1	1			1					1			1
Belgium	4	4	4			4					4			4
Denmark	6	6	5	1		2	3	1			6			6
France	3	1	1			1					1			1
Germany	2	2	2			2					2			2
Iceland	2	0												
Ireland	1	1	1			1					1			1
Latvia	2	2	2			2					2			2
Madagascar	1	0												
Moldova	1	1	1				1				1			1
Netherlands	1	1			1			1	1			1		
Norway	10	10	2		8	2		8	8		2	8		2
Qatar	7	0												
Romania	4	4	4			4					4			4
Serbia	5	2	2			2					2			2
Somalia	17	0												
South Africa	11	10	1		9	1		9	8	1	1	9		1
Spain	5	4	4			4					4			4
Sweden	1	1	1			1					1			1
Switzerland	2	0												
MAY														
Denmark	2	2	2			1		1			2			2
France	1	1	1			1					1			1
Germany	1	1	1			1					1			1
Mauritius	1	0												
Portugal	3	3	3			1	2				3			3
Qatar	12	3			3			3	3			3		
Romania	1	1	1			1					1			1
Somalia	2	0												
South Africa	22	20	3		17	3		17	14	3	3	16	1	3
JUNE														
Croatia	1	1			1			1	1			1		
Madagascar	1	0												
Mauritius	8	0												
Norway	3	0												
Spain	1	0												
UK (N. Ireland)	1	0												
JULY														
Croatia	1	1			1			1	1			1		
France	1	1	1			1					1			1
Mauritius	1	0												
Norway	1	0												
AUGUST														
Norway	1	0												
26 Countries	238	151	100	5	46	90	9	52	42	4	105	45	1	105
			66.2%	3.3%	30.5%	59.6%	6.0%	34.4%	27.8%	2.6%	69.5%	29.8%	0.7%	69.5%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Reduction in HI titre with post-infection ferret antiserum raised against A/Guangdong-Maonan/SWL1536/2019 (cell-propagated vaccine virus)

4 Reduction in HI titre with post-infection ferret antiserum raised against A/Guangdong-Maonan/SWL1536/2019 (egg-propagated vaccine virus)

5 Reduction in HI titre with post-infection ferret antiserum raised against A/Denmark/3280/2019 (cell-propagated virus)

6 Reduction in HI titre with post-infection ferret antiserum raised against IVR-215 A/Victoria/2570/2019 (egg-propagated vaccine virus)

Samples received were in lysis buffer - only genetic characterisation possible

Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-COV-2 positive

As of 2022-09-07

Table 6. Global count of A(H1N1)pdm09 HA sequences for viruses with collection dates after 2022-01-31

Clade	Group	Number	Country and (number of sequences)
6B.1A.5a.1	D187A	467	Australia (13), Belgium (23), Canada (2), Denmark (5), France (35), Germany (6), Iceland (2), Ireland (1), Italy (1), Latvia (2), Luxembourg (16), Mauritius (1), Moldova (3), Netherlands (280), Norway (2), Portugal (8), Romania (8), Serbia (7), South Africa (3), Spain (13), Sweden (3), Switzerland (14), UK (12), USA (7)
	& R113K	25	Australia (1), Netherlands (2), Norway (1), South Africa (16), Spain (2), Sweden (2), UK (1)
	& P137S, G155E	3	Italy (1), Spain (1), Switzerland (1)
	& K211R	7	France (3), Germany (1), Netherlands (2), South Africa (1)
	& D129N, H138Q, T184N	8	Netherlands (8)
	& N125K	2	Luxembourg (2)
	& A187T	4	Denmark (4)
	& A139D	4	Spain (4)
6B.1A.5a.2	K54Q, K130N, N156K, L161I, A186T, E224A, V250A, R259K, K308R	127	Australia (98), Canada (2), India (2), Pakistan (5), UAE (2), UK (18)
	& K142R, D260E	10	South Africa (10)
	& K142R, D260E, P137S, T277A	75	Canada (1), Croatia (2), Netherlands (10), Norway (2), South Africa (46), Spain (9), Sweden (3), UK (2)
	& T216A	132	Australia (13), Canada (4), India (1), Norway (11), Pakistan (13), Qatar (7), Somalia (13), Spain (1), UAE (3), UK (63), USA (3)
	& T216A, D129N	7	UK (7)
	& T216A, D94N	211	Australia (176), Bahrain (1), India (1), Netherlands (3), New Zealand (3), Pakistan (4), Qatar (9), Spain (1), Sweden (1), UK (12)
	& T216A, D94N, P137L	11	Australia (11)
	& T216A, D94N, R205G	25	Australia (20), Bahrain (4), Pakistan (1)
	& T216A, D94N, T186A	11	Australia (11)
	Total count	1129	

WHO Region:

African (n = 90; 3 countries)

Americas (n = 19; 2 countries)

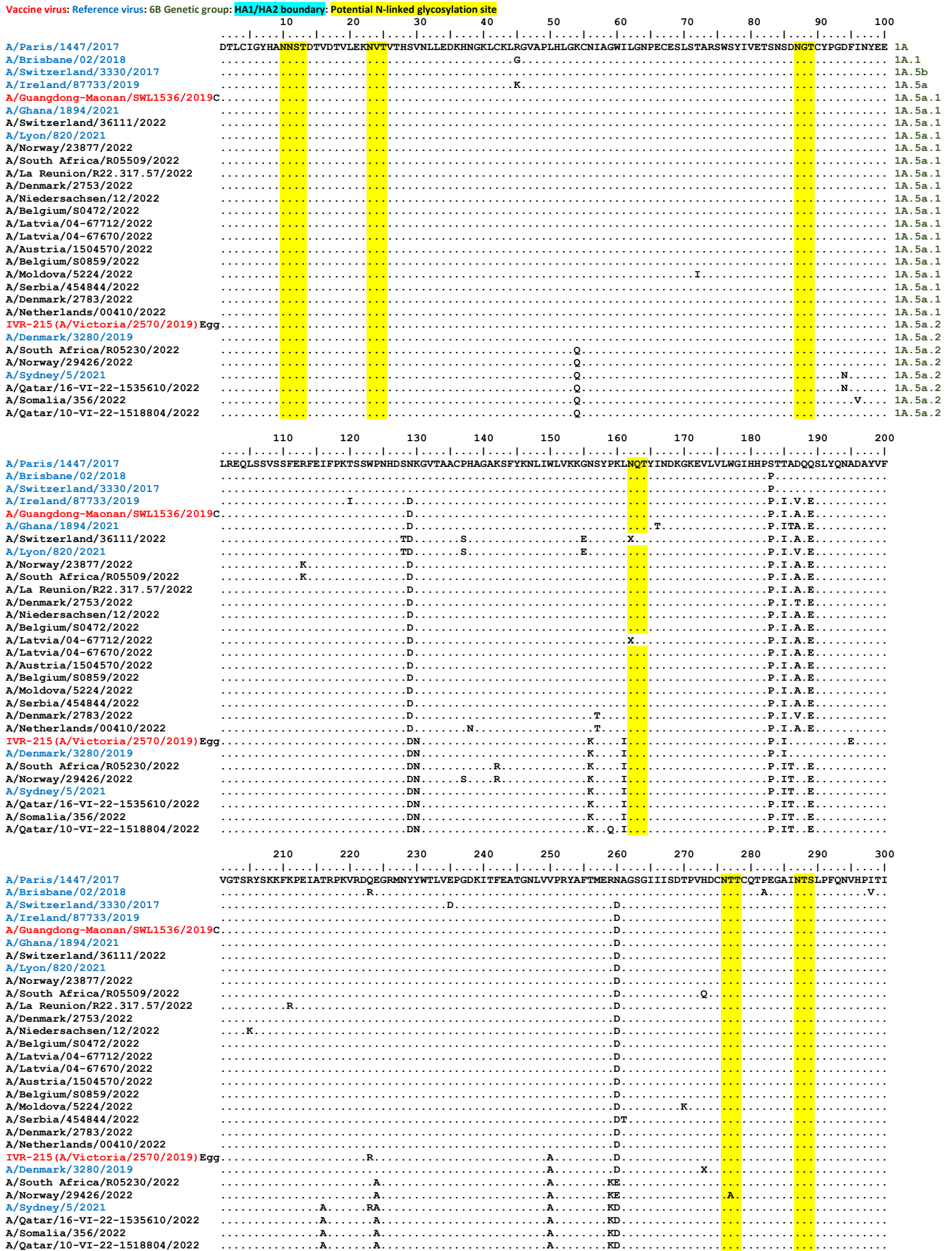
Eastern Mediterranean (n = 49; 4 countries)

European (n = 621; 20 countries)

South-East Asia (n = 4; 1 country)

Western Pacific (n = 346; 2 countries)

Full-length HA sequences downloaded from GISAID and de-duplicated on 2022-09-05



Vaccine virus: Reference virus: 6B Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 5a. Phylogenetic comparison of A(H1N1)pdm09 NA genes (WIC sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

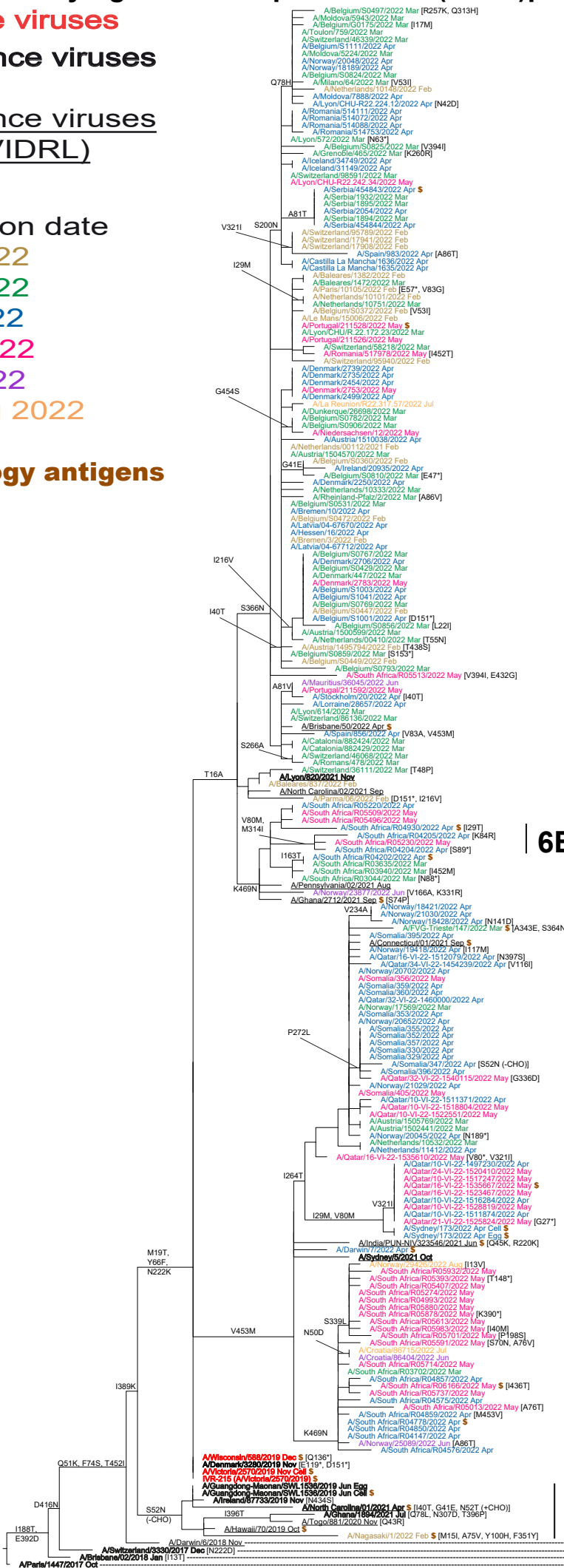
Apr 2022

May 2022

Jun 2022

Jul-Aug 2022

\$ serology antigens



6B.1A.5a.1

6B.1A.5a.2

6B.1A.5a.1

6B.1A.5a.2

6B.1A.5a.1

Figure 5b. Phylogenetic comparison of A(H1N1)pdm09 NA genes (non-WIC sequences)

Vaccine viruses

Reference viruses (WIC)

Reference viruses (CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

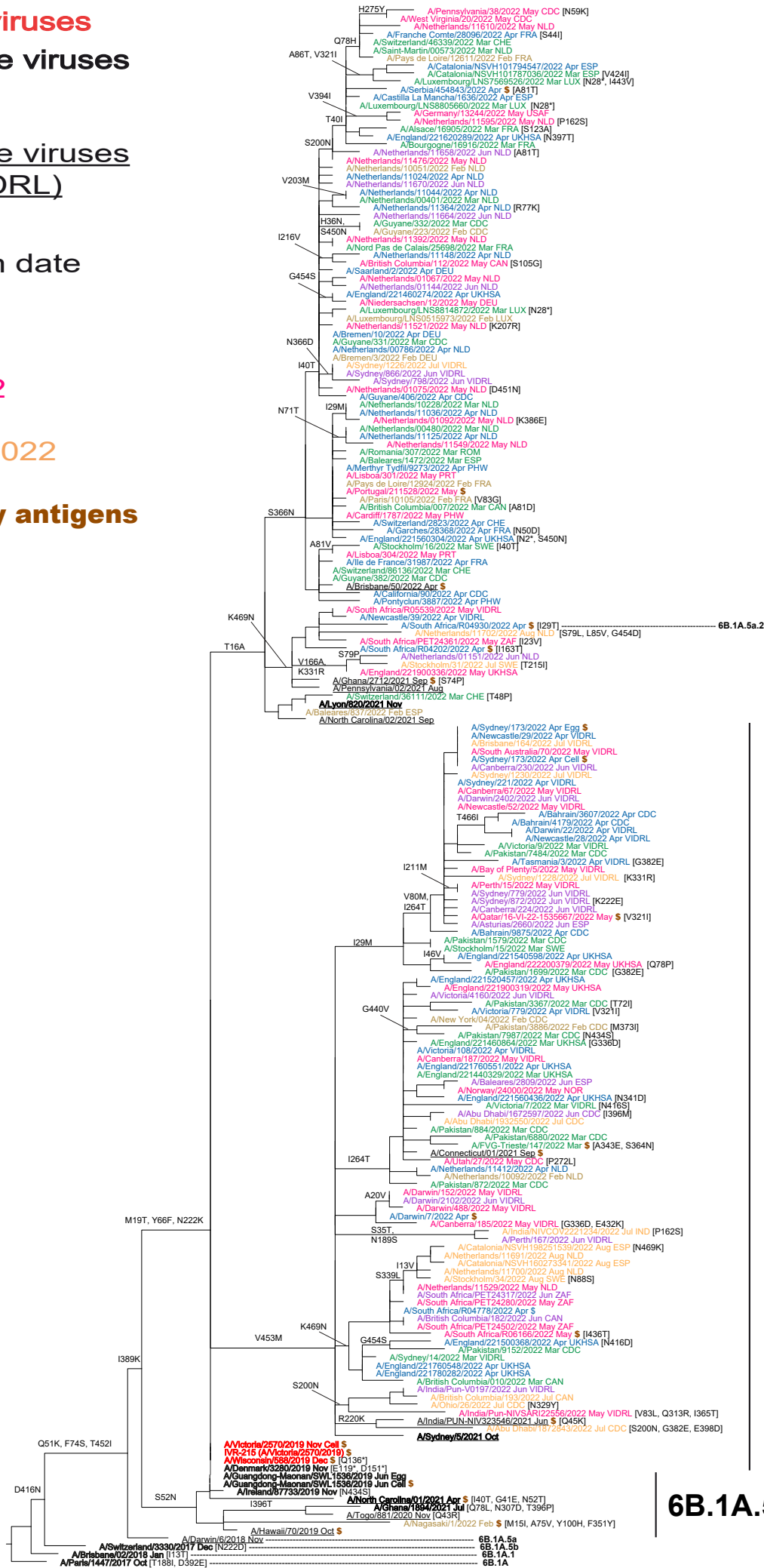
Apr 2022

May 2022

Jun 2022

Jul-Aug 2022

\$ serology antigens



6B.1A.5a.1

6B.1A.5a.2

6B.1A.5a.1

	310	320	330	340	350	360	370	380	390	400							
A/Paris/1447/2017	RPWVSFNQ	LEYGQMG	YICSGV	FGDNPR	PNDKTG	SCGPVSS	NGANGV	KGFSFK	YGNVW	IGRTKSI	SSRKG	FEMIWD	PNGWTG	TDNKFS	IKQEI	VGINEWS	1A
A/Brisbane/02/2018															D		1A.1
A/Switzerland/3330/2017															D		1A.5b
A/Ireland/87733/2019															K	D	1A.5a
A/Guangdong-Maonan/SWL1536/2019C															K	D	1A.5a.1
A/Ghana/1894/2021		D													K	D	1A.5a.1
A/Switzerland/36111/2022															K	D	1A.5a.1
A/Lyon/820/2021															K	D	1A.5a.1
A/Norway/23877/2022				R											K	D	1A.5a.1
A/South_Africa/R05509/2022															K	D	1A.5a.1
A/La_Reunion/R22.317.57/2022									N						K	D	1A.5a.1
A/Denmark/2753/2022									N						K	D	1A.5a.1
A/Niedersachsen/12/2022									N						K	D	1A.5a.1
A/Belgium/S0472/2022									N						K	D	1A.5a.1
A/Latvia/04-67712/2022									N						K	D	1A.5a.1
A/Latvia/04-67670/2022									N						K	D	1A.5a.1
A/Austria/1504570/2022									N						K	D	1A.5a.1
A/Belgium/S0859/2022									N						K	D	1A.5a.1
A/Moldova/5224/2022									N						K	D	1A.5a.1
A/Serbia/454844/2022									N						K	D	1A.5a.1
A/Denmark/2783/2022									N						K	D	1A.5a.1
A/Netherlands/00410/2022									N						K	D	1A.5a.1
IVR-215 (A/Victoria/2570/2019) Egg															K	D	1A.5a.2
A/Denmark/3280/2019															K	D	1A.5a.2
A/South_Africa/R05230/2022			I												K	D	1A.5a.2
A/Norway/29426/2022							L								K	D	1A.5a.2
A/Sydney/5/2021															K	D	1A.5a.2
A/Qatar/16-VI-22-1535610/2022															K	D	1A.5a.2
A/Somalia/356/2022															K	D	1A.5a.2
A/Qatar/10-VI-22-1518804/2022															K	D	1A.5a.2

	410	420	430	440	450	460						
A/Paris/1447/2017	GYSGSFVQ	HPHPELT	GLDCIR	PCFWV	ELIRGR	PEENTI	IWTSGSS	ISFCG	VSDSTV	GWSWPD	GAE	LPFTIDK
A/Brisbane/02/2018												
A/Switzerland/3330/2017												
A/Ireland/87733/2019						S						
A/Guangdong-Maonan/SWL1536/2019C												
A/Ghana/1894/2021												
A/Switzerland/36111/2022												
A/Lyon/820/2021												
A/Norway/23877/2022												N
A/South_Africa/R05509/2022												N
A/La_Reunion/R22.317.57/2022												
A/Denmark/2753/2022												
A/Niedersachsen/12/2022												
A/Belgium/S0472/2022												
A/Latvia/04-67712/2022												
A/Latvia/04-67670/2022												
A/Austria/1504570/2022												
A/Belgium/S0859/2022												
A/Moldova/5224/2022												
A/Serbia/454844/2022												
A/Denmark/2783/2022												
A/Netherlands/00410/2022												
IVR-215 (A/Victoria/2570/2019) Egg												
A/Denmark/3280/2019												
A/South_Africa/R05230/2022												N
A/Norway/29426/2022												N
A/Sydney/5/2021												
A/Qatar/16-VI-22-1535610/2022												
A/Somalia/356/2022												
A/Qatar/10-VI-22-1518804/2022												

Vaccine virus: Reference virus: 6B Genetic group: Potential N-linked glycosylation site

Table 7-1. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-02-08)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Swit 3330/17 Egg	A/Ire 87733/19 Egg	A/N Carolina 01/21 Cell	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
				F03/18 ²	F09/19 ¹	F23/18 ¹	St Jude's F18/20 ¹	CDC F2021-105	F09/20 ¹	F12/20 ¹	F02/22 ¹	F08/20 ¹	F37/21 ¹	F04/22 ¹
	Passage history			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	2560	1280	640	640	80	1280	2560	640	<	80	40
A/Brisbane/02/2018		2018-01-04	E3/E2	2560	1280	640	1280	80	2560	2560	1280	40	160	80
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	1280	640	1280	640	80	1280	1280	640	<	80	40
A/Ireland/87733/2019		2019-11-03	E4	2560	1280	640	640	160	2560	2560	640	<	160	40
A/North Carolina/01/2021			SIAT3/MDCK1	160	80	80	80	640	640	160	640	<	<	40
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	C2/MDCK1	1280	640	320	640	160	1280	2560	1280	<	80	80
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	E3/E2	2560	640	640	1280	160	2560	1280	640	<	160	40
A/Ghana/1894/2021		2021-07-21	E2	2560	1280	1280	1280	1280	>5120	>5120	2560	ND	ND	160
A/Denmark/3280/2019		2019-11-10	MDCK4/MDCK6	160	80	40	80	40	160	80	40	2560	2560	2560
IVR-215 (A/Victoria/2570/2019)		2018-11-22	E4/D7/E2	80	80	40	80	160	160	160	80	1280	2560	1280
A/Sydney/5/2021		2021-10-16	E3/E1	80	40	40	40	80	80	80	40	1280	2560	2560
TEST VIRUSES														
A/Belgium/S0063/2022		2021-12-07	MDCK1	1280	320	640	640	320	1280	1280	1280	<	80	80
A/Belgium/G0001/2022	A261T	2021-12-17	MDCK1	1280	320	320	640	640	1280	1280	1280	<	40	80
A/Belgium/H0007/2022		2022-01-11	MDCK1	160	80	80	160	640	640	160	640	<	40	40
A/Belgium/H0038/2022		2022-01-17	MDCK1	640	320	320	640	320	1280	1280	640	<	40	40
A/Belgium/H0042/2022		2022-01-20	MDCK1	1280	320	640	640	640	1280	1280	1280	<	80	80
A/India/Pun-NIV323546/2021		2021-06-22	X3/SIAT2/MDCK1	<	<	<	<	40	40	<	<	640	640	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2019-20
SH 2020

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-02-23)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Swit 3330/17 Egg	A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
				F03/18 ²	F09/19 ¹	F23/18 ¹	St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F08/20 ¹	F37/21 ¹	F04/22 ¹
	Passage history			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
	Ferret number												
	Genetic group												
REFERENCE VIRUSES													
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	1280	1280	1280	1280	2560	2560	1280	40	80	40
A/Brisbane/02/2018		2018-01-04	E3/E2	1280	1280	1280	640	1280	2560	1280	40	160	80
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	1280	640	1280	640	1280	1280	640	40	80	40
A/Ireland/87733/2019		2019-11-03	E4	1280	640	640	1280	2560	2560	1280	40	80	80
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	C2/MDCK1	1280	640	640	1280	2560	2560	1280	80	160	80
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	E3/E2	640	640	320	640	1280	1280	640	40	80	40
A/Ghana/1894/2021		2021-07-21	E3	1280	640	640	640	1280	1280	640	40	80	40
A/Denmark/3280/2019		2019-11-10	MDCK4/MDCK6	160	80	80	160	160	160	80	2560	>5120	2560
IVR-215 (A/Victoria/2570/2019)		2018-11-22	E4/D7/E2	80	40	40	80	160	160	80	1280	2560	1280
A/Sydney/5/2021		2021-10-16	E3/E1	80	40	80	40	80	80	80	1280	2560	2560
TEST VIRUSES													
A/Belgium/H0025/2022		2021-12-10	MDCK1/MDCK1	640	320	320	640	1280	1280	640	<	40	40
A/Belgium/H0018/2022		2021-12-20	MDCK1/MDCK1	1280	320	320	640	1280	1280	640	40	40	40
A/Belgium/S0060/2022		2021-12-26	MDCK1/MDCK1	640	320	320	320	1280	1280	1280	<	80	40
A/Belgium/S0048/2022		2021-12-26	MDCK1/MDCK1	1280	320	320	640	1280	1280	640	40	80	40
A/Belgium/H0017/2022		2022-01-05	MDCK1/MDCK1	320	160	160	160	1280	320	640	40	80	80
A/Belgium/H0032/2022		2022-01-18	MDCK1/MDCK1	640	320	320	320	1280	640	640	<	80	40
A/Belgium/H0041/2022		2022-01-19	MDCK1/MDCK1	640	320	320	640	1280	1280	640	<	80	80
A/Khasab/72126274/2021		2021-09-28	MDCK1	80	40	40	40	80	80	40	1280	2560	2560
A/Haima/72127498/2021		2021-10-26	MDCK1	<	<	<	<	40	40	40	1280	1280	1280
A/Ibri/52121896/2021		2021-11-09	MDCK1	<	<	<	<	40	<	<	320	320	640
A/Sur/72128529/2021		2021-11-11	MDCK1	<	<	<	<	<	<	<	320	640	640
A/AI Musana/72128566/2021		2021-11-14	MDCK1	<	<	<	<	40	<	<	640	640	640
A/Mahoot/72129184/2021		2021-11-18	MDCK1	<	<	<	<	40	40	<	640	640	640
A/Seeb/72130049/2021		2021-11-30	MDCK1	<	<	40	40	40	40	40	1280	1280	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-03-15)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Swit 3330/17 Egg	A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
				F03/18 ²	F09/19 ¹	F23/18 ¹	St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F08/20 ¹	F37/21 ¹	F04/22 ¹
	Passage history			6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
	Ferret number													
	Genetic group													
REFERENCE VIRUSES														
A/Paris/1447/2017		2017-10-20	MDCK1/MDCK3	2560	1280	1280	640	1280	2560	640	80	<	80	40
A/Brisbane/02/2018		2018-01-04	E3/E2	2560	2560	1280	1280	2560	2560	1280	160	80	160	80
A/Switzerland/3330/2017	clone 35	2017-12-20	E6/E2	1280	640	1280	640	1280	1280	640	160	<	80	40
A/Ireland/87733/2019		2019-11-03	E4	2560	1280	640	1280	2560	2560	1280	640	40	160	80
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	C2/MDCK1	2560	1280	640	640	2560	2560	1280	640	<	80	40
A/Guangdong-Maonan/SWL1536/2019		2019-06-17	E3/E2	1280	640	640	640	1280	1280	640	320	<	80	40
A/Ghana/1894/2021		2021-07-21	E3	1280	1280	640	640	2560	2560	1280	320	<	160	80
A/Lyon/820/2021		2021-11-16	E1/E2	320	160	80	160	640	320	320	640	<	80	80
A/Denmark/3280/2019		2019-11-10	MDCK4/MDCK6	80	40	<	40	80	80	<	80	1280	2560	2560
IVR-215 (A/Victoria/2570/2019)		2018-11-22	E4/D7/E2	80	80	<	40	160	80	40	80	640	2560	1280
A/Sydney/5/2021		2021-10-16	E3/E1	<	40	<	40	80	160	40	80	1280	2560	2560
TEST VIRUSES														
A/Cameroon/6425/2021		2021-09-08	MDCK1	2560	1280	640	1280	2560	2560	1280	160	<	80	<
A/Cameroon/7074/2021		2021-09-20	MDCK1	1280	640	640	640	2560	2560	1280	<	40	<	<
A/Cameroon/7073/2021		2021-09-20	MDCK2	80	160	<	80	320	160	320	80	160	320	80
A/Cameroon/7076/2021		2021-09-21	MDCK1	2560	1280	640	1280	2560	2560	1280	80	<	<	<
A/Cameroon/7079/2021		2021-09-22	MDCK1	1280	640	640	640	2560	1280	1280	<	40	<	<
A/Ghana/2712/2021		2021-09-22	MDCKx/MDCK1	2560	1280	1280	1280	2560	5120	2560	640	80	160	160
A/Ghana/433/2021		2021-11-27	MDCKx/MDCK1	2560	2560	1280	1280	2560	5120	2560	640	80	160	80
A/Albania/8093/2021		2021-12-10	MDCK1	<	40	<	40	80	40	80	40	1280	5120	2560
A/Albania/942/2021		2021-12-21	MDCK1	<	<	<	<	40	40	40	40	640	2560	1280
A/Albania/905/2021		2021-12-21	MDCK1	<	<	<	<	40	<	<	40	1280	2560	1280

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-04-12)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection fereret antisera										
				A/Paris 1447/17 MDCK	A/Bris 02/18 Egg	A/Swit 3330/17 Egg	A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
				F03/18 ¹²	F09/19 ¹	F23/18 ¹	St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F08/20 ¹	F37/21 ¹	F04/22 ¹
	Passage history	Genetic group		6B.1A	6B.1A.1	6B.1A.5b	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES														
A/Paris/1447/2017		6B.1A	2017-10-20	MDCK1/MDCK3	2560	1280	1280	640	2560	2560	1280	80	<	40
A/Brisbane/02/2018	clone 35	6B.1A.1	2018-01-04	E3/E2	2560	1280	1280	640	2560	2560	1280	160	40	160
A/Switzerland/3330/2017		6B.1A.5b	2017-12-20	E6/E2	1280	640	1280	640	1280	1280	640	160	<	80
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	1280	1280	640	640	2560	2560	1280	640	40	160
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	1280	1280	1280	2560	2560	1280	640	<	80
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	1280	640	640	640	2560	2560	1280	320	<	80
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E3	1280	1280	640	1280	2560	1280	1280	320	<	80
A/Lyon/820/2021		6B.1A.5a.1	2021-11-16	E1/E2	160	160	80	80	320	320	160	640	<	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK6	80	40	40	40	80	80	40	80	2560	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	80	80	40	80	320	160	80	160	1280	5120
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	80	40	40	40	80	80	80	80	1280	2560
TEST VIRUSES														
A/Dakar/41/2021		6B.1A.5a.1	2021-09-13	MDCK1	320	320	160	160	1280	320	1280	640	<	<
A/Dakar/21/2021		6B.1A.5a.1	2021-09-25	MDCK1	80	80	80	40	320	160	160	320	<	80
A/Dakar/10/2021		6B.1A.5a.1	2021-09-28	MDCK1	80	80	40	40	320	160	320	320	<	<
A/Dakar/11/2021		6B.1A.5a.1	2021-09-28	MDCK1	160	160	160	160	640	320	640	640	<	80
A/Dakar/16/2021		6B.1A.5a.1	2021-09-28	MDCK1	320	160	160	160	640	320	640	640	<	<
A/Dakar/08/2021		6B.1A.5a.1	2021-09-29	MDCK1	2560	1280	1280	1280	2560	2560	2560	640	40	80
A/Dakar/09/2021		6B.1A.5a.1	2021-09-29	MDCK1	160	160	80	80	640	320	320	640	<	<
A/Dakar/15/2021		6B.1A.5a.1	2021-09-29	MDCK1	640	320	320	640	1280	1280	640	320	<	<
A/Dakar/18/2021		6B.1A.5a.1	2021-09-30	MDCK2	160	160	160	160	640	320	640	640	<	<
A/Dakar/34/2021		6B.1A.5a.1	2021-10-01	MDCK1	320	320	160	320	1280	640	1280	1280	<	80
A/Dakar/36/2021		6B.1A.5a.1	2021-10-01	MDCK1	320	160	160	160	1280	320	640	640	<	<
A/Dakar/30/2021		6B.1A.5a.1	2021-10-02	MDCK1	160	160	80	80	640	320	640	640	<	<
A/Dakar/24/2021		6B.1A.5a.1	2021-10-03	MDCK1	160	160	80	80	640	320	640	320	<	<
A/Dakar/38/2021		6B.1A.5a.1	2021-10-04	MDCK1	320	160	80	160	1280	320	640	1280	<	<
A/Dakar/39/2021		6B.1A.5a.1	2021-10-04	MDCK1	320	160	160	160	1280	320	640	640	<	80
A/Dakar/40/2021		6B.1A.5a.1	2021-10-04	MDCK1	320	160	160	160	1280	320	640	640	<	<
A/Dakar/37/2021		6B.1A.5a.1	2021-10-05	MDCK3	<	80	40	40	320	80	80	320	<	<
A/Dakar/25/2021		6B.1A.5a.1	2021-10-06	MDCK1	2560	1280	640	1280	2560	2560	1280	640	40	80
A/Dakar/27/2021		6B.1A.5a.1	2021-10-06	MDCK1	320	160	160	160	640	320	640	640	<	80
A/Dakar/44/2021		6B.1A.5a.1	2021-10-13	MDCK1	160	160	80	80	640	160	640	640	<	<
A/Dakar/43/2021		6B.1A.5a.1	2021-10-13	MDCK1	320	160	160	160	640	320	640	640	<	<
A/Dakar/42/2021		6B.1A.5a.1	2021-10-20	MDCK2	160	80	80	80	320	160	160	320	<	<
A/Dakar/51/2021		6B.1A.5a.1	2021-10-26	MDCK1	160	160	80	160	640	320	640	640	<	<
A/Dakar/52/2021		6B.1A.5a.1	2021-11-04	MDCK1	320	160	160	160	1280	640	1280	1280	<	40
A/Dakar/49/2021		6B.1A.5a.1	2021-11-04	MDCK1	320	160	160	160	640	320	640	640	<	<
A/Dakar/53/2021		6B.1A.5a.1	2021-11-07	MDCK1	160	160	80	80	320	320	320	640	<	<
A/Dakar/55/2021		6B.1A.5a.1	2021-11-10	MDCK1	160	80	80	80	320	320	320	320	<	<
A/Dakar/50/2021		6B.1A.5a.1	2021-11-10	MDCK1	320	160	80	80	640	320	640	640	<	<
A/Dakar/48/2021		6B.1A.5a.1	2021-11-10	MDCK1	320	160	160	160	1280	640	1280	1280	<	<
A/Dakar/54/2021		6B.1A.5a.1	2021-11-15	MDCK2	160	80	80	80	320	160	320	320	<	<
A/Dakar/57/2021		6B.1A.5a.1	2021-11-19	MDCK1	160	80	80	80	640	320	320	320	<	<
A/Dakar/56/2021		6B.1A.5a.1	2021-11-23	MDCK1	320	160	160	160	640	320	640	640	<	<
A/Dakar/59/2021		6B.1A.5a.1	2021-11-26	MDCK1	160	80	80	80	640	160	320	320	<	<
A/Dakar/58/2021		6B.1A.5a.1	2021-11-26	MDCK1	320	160	160	160	1280	640	640	1280	40	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2019-20
SH 2020Vaccine
NH 2020-21Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-5. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-05-10)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection fereret antisera							
				A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/Lyon 820/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21
				Egg	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
				Ferret number F18/20 ^{*1}	F09/20 ^{*1}	F12/20 ^{*1}	F02/22 ^{*1}	F06/22 ^{*1}	F08/20 ^{*1}	F37/21 ^{*1}	F04/22 ^{*1}
Genetic group	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2			
REFERENCE VIRUSES											
A/Ireland/87733/2019	6B.1A.5a.1	2019-11-03	E4	640	2560	2560	1280	640	40	160	80
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	C2/MDCK1	1280	2560	2560	1280	640	<	80	80
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	E3/E2	640	2560	1280	1280	320	<	80	40
A/Ghana/1894/2021	6B.1A.5a.1	2021-07-21	E3	1280	1280	2560	2560	320	<	80	80
A/Lyon/820/2021	6B.1A.5a.1	2021-11-16	E1/E2	80	320	320	160	640	<	40	40
A/Denmark/3280/2019	6B.1A.5a.2	2019-11-10	MDCK4/MDCK6	40	80	80	40	80	2560	2560	2560
IVR-215 (A/Victoria/2570/2019)	6B.1A.5a.2	2018-11-22	E4/D7/E2	80	320	160	80	160	1280	2560	2560
A/Sydney/5/2021	6B.1A.5a.2	2021-10-16	E3/E1	40	80	80	80	80	1280	2560	2560
TEST VIRUSES											
A/Bremen/3/2022	6B.1A.5a.1	2022-02-23	C1/MDCK1	640	2560	2560	1280	320	<	80	40
A/Denmark/447/2022	6B.1A.5a.1	2022-03-06	MDCK4/MDCK1	320	1280	1280	640	160	<	80	40
A/Catalonia/882424/2022	6B.1A.5a.1	2022-03-17	MDCK1	640	1280	1280	1280	320	<	<	40
A/Catalonia/882429/2022	6B.1A.5a.1	2022-03-17	MDCK1	640	1280	2560	1280	320	<	80	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 <= <40; 2 <= <80; ND = Not Done

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-06-28)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre								
					Post-infection fereret antisera								
					NEW								
					A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	A/Stock 10/21 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
					St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F08/20 ¹	F22/22 ¹	F37/21 ¹	F04/22 ¹
Genetic group	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2				
REFERENCE VIRUSES													
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	320	1280	1280	320	160	<	<	40	<
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	640	1280	1280	640	320	<	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	640	1280	1280	640	160	<	<	40	<
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2/E1	640	1280	1280	640	160	<	<	80	40
A/Lyon/820/2021		6B.1A.5a.1	2021-11-16	E1/E2	80	320	320	160	320	<	<	40	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK6	<	40	80	40	80	640	640	1280	640
A/Stockholm/10/2021		6B.1A.5a.2	2021-09-13	SIAT1/MDCK1	40	40	40	<	<	1280	1280	2560	2560
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	40	80	40	40	80	640	640	1280	1280
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	<	80	40	40	40	640	640	1280	1280
TEST VIRUSES													
A/Switzerland/50178/2022		6B.1A.5a.1	2022-01-25	MDCK1	640	1280	1280	320	320	<	<	40	40
A/Switzerland/95789/2022		6B.1A.5a.1	2022-02-14	MDCK1	320	1280	1280	640	320	<	40	40	40
A/Switzerland/17908/2022		6B.1A.5a.1	2022-02-17	MDCK1	320	1280	1280	640	160	<	<	40	40
A/Switzerland/17941/2022		6B.1A.5a.1	2022-02-17	MDCK1	320	1280	1280	640	160	<	<	40	40
A/Austria/1495794/2022		6B.1A.5a.1	2022-02-24	Cx/MDCK1	640	1280	1280	640	160	<	<	40	40
A/Rheinland-Pfalz/2/2022		6B.1A.5a.1	2022-03-08	P1/MDCK1	640	1280	1280	640	320	<	<	40	40
A/Switzerland/36111/2022	S128T, P137S, G155E, A187V	6B.1A.5a.1	2022-03-10	MDCK1	80	640	160	320	320	<	<	40	40
A/Austria/1500599/2022		6B.1A.5a.1	2022-03-14	Cx/MDCK1	160	640	640	640	160	<	<	40	40
A/Switzerland/86136/2022		6B.1A.5a.1	2022-03-15	MDCK1	640	1280	2560	1280	320	<	40	80	40
A/Switzerland/98591/2022		6B.1A.5a.1	2022-03-16	MDCK1	640	1280	1280	1280	320	<	<	40	40
A/Switzerland/46339/2022		6B.1A.5a.1	2022-03-18	MDCK1	640	1280	1280	1280	320	<	40	40	40
A/Switzerland/46068/2022		6B.1A.5a.1	2022-03-22	MDCK1	640	2560	2560	1280	320	<	40	80	40
A/Switzerland/58218/2022		6B.1A.5a.1	2022-03-22	MDCK2	320	1280	1280	640	160	<	<	40	<
A/Austria/1504570/2022		6B.1A.5a.1	2022-03-28	Cx/MDCK1	640	1280	1280	640	160	<	<	40	40
A/Norway/18189/2022		6B.1A.5a.1	2022-04-02	MDCK1	640	1280	1280	640	160	<	<	40	40
A/Bremen/10/2022		6B.1A.5a.1	2022-04-04	P1/MDCK1	640	1280	1280	1280	160	<	<	40	40
A/Ireland/20935/2022		6B.1A.5a.1	2022-04-10	MDCK2	320	640	1280	640	160	<	<	40	<
A/Norway/20048/2022		6B.1A.5a.1	2022-04-11	MDCK2	320	640	1280	640	160	<	<	40	<
A/Hessen/16/2022		6B.1A.5a.1	2022-04-11	P2/MDCK1	640	1280	1280	640	320	<	<	40	40
A/Latvia/04-67712/2022		6B.1A.5a.1	2022-04-20	P1/MDCK1	160	1280	640	320	80	<	<	<	<
A/Latvia/04-67670/2022		6B.1A.5a.1	2022-04-20	P1/MDCK1	320	640	640	320	160	<	<	40	<
A/Niedersachsen/12/2022		6B.1A.5a.1	2022-05-04	P1/MDCK1	640	1280	1280	1280	160	<	<	40	40
A/Austria/1502441/2022		6B.1A.5a.2	2022-03-21	Cx/MDCK1	<	<	<	<	<	640	320	640	640
A/Norway/17569/2022		6B.1A.5a.2	2022-03-22	MDCK2	<	40	40	<	<	640	640	640	640
A/Austria/1505769/2022		6B.1A.5a.2	2022-03-31	Cx/MDCK1	<	<	40	<	<	640	640	1280	640
A/Norway/21029/2022		6B.1A.5a.2	2022-04-02	MDCK2	<	<	<	<	<	640	640	1280	640
A/Norway/18421/2022		6B.1A.5a.2	2022-04-03	MDCK1	<	<	40	<	<	640	1280	2560	1280
A/Norway/21030/2022		6B.1A.5a.2	2022-04-04	MDCK1	<	<	<	<	<	320	320	640	640
A/Norway/18428/2022		6B.1A.5a.2	2022-04-05	MDCK2	<	<	40	<	<	1280	1280	1280	1280
A/Norway/20702/2022		6B.1A.5a.2	2022-04-07	MDCK1	<	<	<	<	<	320	320	640	320
A/Norway/19418/2022		6B.1A.5a.2	2022-04-08	MDCK1	<	40	40	<	<	640	640	1280	640
A/Norway/20045/2022		6B.1A.5a.2	2022-04-11	MDCK1	<	<	<	<	<	640	640	1280	640
A/Norway/20652/2022		6B.1A.5a.2	2022-04-12	MDCK1	<	<	40	<	<	640	640	1280	1280

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-07-20)

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection fereret antisera							
					A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
					St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F28/20 ¹	F37/21 ¹	F04/22 ¹
	Genetic group				6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
REFERENCE VIRUSES												
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	320	1280	1280	640	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	640	1280	1280	1280	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	320	640	2560	640	160	40	80	40
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2/E1	640	2560	320	1280	320	80	160	80
A/Lyon/820/2021		6B.1A.5a.1	2021-11-16	E1/E2	80	320	80	160	640	40	80	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	40	40	80	40	80	1280	1280	1280
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	40	80	80	40	160	640	1280	640
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	40	40	40	40	80	640	1280	1280
TEST VIRUSES												
A/Murcia/10206/2021		6B.1A.5a.1	2021-12-10	SIAT2/MDCK1	640	1280	2560	1280	320	80	80	80
A/Netherlands/00023/2022		6B.1A.5a.1	2022-01-17	Hck1/MDCK1	320	640	1280	640	160	40	40	40
A/Netherlands/00028/2022		6B.1A.5a.1	2022-01-19	Hck1/MDCK1	320	640	1280	640	160	40	40	40
A/Netherlands/00031/2022		6B.1A.5a.1	2022-01-20	Hck1/MDCK1	320	1280	1280	640	160	40	40	40
A/Baleares/837/2022	S128T, P137S, G155E, T184S	6B.1A.5a.1	2022-02-05	SIAT2/MDCK1	80	320	160	320	320	40	40	40
A/Netherlands/10148/2022		6B.1A.5a.1	2022-02-15	MDCK-MIX2/MDCK1	640	1280	1280	1280	320	80	80	40
A/Netherlands/10101/2022		6B.1A.5a.1	2022-02-21	MDCK-MIX2/MDCK1	1280	2560	2560	1280	320	40	160	80
A/Parma/06/2022	A73T, P137S, G155E, N162(-cho)	6B.1A.5a.1	2022-02-23	MDCK4/MDCK1	80	320	80	320	320	<	40	40
A/Netherlands/00112/2021		6B.1A.5a.1	2022-02-24	Hck1/MDCK1	320	640	640	320	160	40	40	40
A/Netherlands/10333/2022		6B.1A.5a.1	2022-03-07	MDCK-MIX2/MDCK1	640	2560	2560	2560	640	80	160	80
A/Netherlands/00410/2022		6B.1A.5a.1	2022-03-20	Hck1/MDCK1	160	640	640	640	160	40	40	40
A/Moldova/5224/2022		6B.1A.5a.1	2022-03-21	MDCK1	640	1280	1280	640	160	40	80	40
A/Netherlands/10751/2022		6B.1A.5a.1	2022-03-23	MDCK-MIX2/MDCK1	640	1280	2560	1280	320	80	80	80
A/Serbia/1895/2022		6B.1A.5a.1	2022-03-28	MDCK-ATL1/MDCK1	320	640	1280	640	160	40	80	40
A/Milano/64/2022		6B.1A.5a.1	2022-03-28	MDCK3/MDCK1	320	1280	1280	640	160	40	40	40
A/Moldova/5943/2022		6B.1A.5a.1	2022-03-29	MDCK1	320	640	1280	640	160	40	80	40
A/Castilla La Mancha/1636/2022		6B.1A.5a.1	2022-04-02	SIAT1/MDCK1	1280	2560	2560	2560	640	80	160	80
A/Castilla La Mancha/1635/2022		6B.1A.5a.1	2022-04-02	SIAT1/MDCK1	1280	2560	2560	2560	640	80	160	80
A/Serbia/2054/2022		6B.1A.5a.1	2022-04-04	MDCK-ATL2/MDCK1	640	1280	2560	1280	320	80	80	40
A/Serbia/454843/2022		6B.1A.5a.1	2022-04-04	MDCK-ATL1/MDCK1	640	1280	1280	1280	320	80	80	40
A/Stockholm/20/2022		6B.1A.5a.1	2022-04-06	SIAT1/MDCK3	640	1280	1280	1280	320	80	80	40
A/Romania/514072/2022		6B.1A.5a.1	2022-04-09	SIAT1/MDCK1	640	2560	2560	1280	320	80	80	40
A/Romania/514111/2022		6B.1A.5a.1	2022-04-10	SIAT1/MDCK1	640	1280	1280	1280	320	40	80	40
A/Romania/514088/2022		6B.1A.5a.1	2022-04-11	SIAT1/MDCK1	640	1280	1280	640	160	40	40	40
A/Denmark/2250/2022		6B.1A.5a.1	2022-04-11	MDCK2/MDCK1	320	1280	1280	640	320	40	80	40
A/Romania/514753/2022		6B.1A.5a.1	2022-04-12	SIAT1/MDCK1	640	1280	1280	640	320	40	80	40
A/Moldova/7888/2022		6B.1A.5a.1	2022-04-18	MDCK1	320	640	640	640	160	40	40	40
A/Denmark/2454/2022	A187T	6B.1A.5a.1	2022-04-18	MDCK2/MDCK1	320	640	640	320	80	<	40	<
A/Denmark/2706/2022	A135*, E283K	6B.1A.5a.1	2022-04-20	MDCK3/MDCK1	160	320	320	320	80	<	40	<
A/Denmark/2499/2022		6B.1A.5a.1	2022-04-20	MDCK2/MDCK1	640	1280	2560	1280	320	40	80	40
A/Denmark/2739/2022	N162(-cho)	6B.1A.5a.1	2022-04-29	MDCK2/MDCK1	320	640	640	320	80	<	<	<
A/Denmark/2735/2022	A187T	6B.1A.5a.1	2022-04-29	MDCK2/MDCK1	320	640	640	320	80	<	40	<
A/Denmark/2753/2022		6B.1A.5a.1	2022-05-01	MDCK2/MDCK1	320	640	1280	320	160	<	40	<
A/Portugal/211526/2022		6B.1A.5a.1	2022-05-03	MDCK1	640	640	640	640	320	40	80	40
A/Denmark/2783/2022	A187V	6B.1A.5a.1	2022-05-03	MDCK4/MDCK1	80	160	320	160	80	<	40	<
A/Romania/517978/2022		6B.1A.5a.1	2022-05-04	MDCK0	320	640	1280	640	160	40	40	40
A/Portugal/211528/2022		6B.1A.5a.1	2022-05-05	MDCK1	640	1280	1280	1280	320	40	80	40
A/Portugal/211592/2022		6B.1A.5a.1	2022-05-22	MDCK1	320	640	640	640	160	40	40	40
A/Portugal/210197/2022		6B.1A.5a.1		SIAT1/MDCK1	640	2560	2560	1280	320	80	80	80
A/FVG-Trieste/147/2022		6B.1A.5a.2	2022-03-15	MDCK3/MDCK1	<	40	40	<	<	640	640	1280
A/Netherlands/10532/2022		6B.1A.5a.2	2022-03-18	MDCK-MIX2/MDCK1	40	40	40	<	40	1280	1280	1280
A/Netherlands/11412/2022		6B.1A.5a.2	2022-04-26	MDCK-MIX2/MDCK1	40	40	40	40	40	1280	2560	1280
A/South Africa/R05932/2022		6B.1A.5a.2	2022-05-20	MDCK1/MDCK1	<	40	40	<	40	1280	1280	1280
A/South Africa/R05880/2022		6B.1A.5a.2	2022-05-20	MDCK1/MDCK2	<	40	40	<	<	640	1280	640
A/South Africa/R05878/2022		6B.1A.5a.2	2022-05-20	MDCK1/MDCK1	<	40	40	<	40	1280	1280	1280
A/South Africa/R06166/2022		6B.1A.5a.2	2022-05-23	MDCK1/MDCK1	<	<	<	<	<	1280	1280	1280
A/South Africa/R05983/2022		6B.1A.5a.2	2022-05-23	MDCK1/MDCK2	<	40	40	<	<	1280	1280	1280

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2020-21

Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

□ < 4-fold □ 4-fold □ 8-fold □ > 8-fold □ not recognised by the antiserum □ ≥ 160 (when homologous titre ≥ 2560) □ ≥ 160 (no homologous titre)

Table 7-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-07-27)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Ire 87733/19 Egg	A/G-M SWL1536/19 MDCK	A/G-M SWL1536/19 Egg	A/Ghana 1894/21 Egg	A/Lyon 820/21 Egg	A/Denmark 3280/19 MDCK	IVR-215 A/Vic/2570/19 Egg	A/Sydney 5/21 Egg
				St Jude's F18/20 ¹	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F28/20 ¹	F37/21 ¹	F04/22 ¹
	Passage history			6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2
	Ferret number										
	Genetic group										
REFERENCE VIRUSES											
A/Ireland/87733/2019	6B.1A.5a.1	2019-11-03	E4	640	1280	1280	640	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	C2/MDCK1	640	1280	1280	1280	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019	6B.1A.5a.1	2019-06-17	E3/E2	320	640	1280	640	160	40	80	40
A/Ghana/1894/2021	6B.1A.5a.1	2021-07-21	E2/E1	640	1280	640	1280	160	80	80	40
A/Lyon/820/2021	6B.1A.5a.1	2021-11-16	E1/E2	80	320	320	160	640	40	40	40
A/Denmark/3280/2019	6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	<	<	40	<	40	1280	1280	1280
IVR-215 (A/Victoria/2570/2019)	6B.1A.5a.2	2018-11-22	E4/D7/E2	40	80	80	40	160	1280	1280	1280
A/Sydney/5/2021	6B.1A.5a.2	2021-10-16	E3/E1	40	80	80	40	80	1280	2560	2560
TEST VIRUSES											
A/South Africa/R00459/2022	6B.1A.5a.2	2022-01-11	MDCK1MDCK1	<	40	40	40	<	1280	2560	2560
A/South Africa/R03044/2022	6B.1A.5a.1	2022-03-10	MDCK1MDCK1	640	1280	2560	1280	320	<	40	40
A/South Africa/R03635/2022	6B.1A.5a.1	2022-03-18	MDCK1MDCK1	640	1280	2560	1280	320	40	80	40
A/South Africa/R03940/2022	6B.1A.5a.1	2022-03-28	MDCK1MDCK1	640	1280	1280	1280	320	40	80	40
A/Austria/1510038/2022	6B.1A.5a.1	2022-04-12	MDCK1	640	2560	2560	1280	320	40	80	40
A/Spain/983/2022	6B.1A.5a.1	2022-04-15	MDCK1	640	1280	2560	1280	320	40	80	40
A/South Africa/R05220/2022	6B.1A.5a.1	2022-04-29	MDCK1MDCK1	640	1280	2560	1280	320	<	80	40
A/South Africa/R05509/2022	6B.1A.5a.1	2022-05-09	MDCK1MDCK1	320	640	1280	640	160	<	40	<
A/South Africa/R05496/2022	6B.1A.5a.1	2022-05-11	MDCK1MDCK1	320	640	640	640	160	<	40	<
A/South Africa/R05513/2022	6B.1A.5a.1	2022-05-12	MDCK1MDCK1	320	640	640	640	160	<	40	40
A/South Africa/R03702/2022	6B.1A.5a.2	2022-03-15	MDCK1MDCK1	<	<	40	<	<	1280	1280	1280
A/South Africa/R04204/2022	6B.1A.5a.2	2022-04-01	MDCK1MDCK1	<	<	<	<	<	1280	1280	640
A/South Africa/R04205/2022	6B.1A.5a.2	2022-04-06	MDCK1MDCK1	<	<	40	<	<	640	1280	640
A/South Africa/R04147/2022	6B.1A.5a.2	2022-04-06	MDCK1MDCK1	<	<	<	<	<	640	1280	640
A/South Africa/R04575/2022	6B.1A.5a.2	2022-04-11	MDCK1MDCK1	<	<	40	<	<	1280	1280	1280
A/South Africa/R04576/2022	6B.1A.5a.2	2022-04-13	MDCK1MDCK1	<	<	<	<	<	320	640	640
A/South Africa/R04859/2022	6B.1A.5a.2	2022-04-20	MDCK1MDCK1	<	<	40	<	<	1280	1280	1280
A/South Africa/R04850/2022	6B.1A.5a.2	2022-04-20	MDCK1MDCK1	<	<	<	<	<	640	1280	1280
A/South Africa/R04930/2022	6B.1A.5a.2	2022-04-25	MDCK1MDCK1	<	<	40	<	<	640	1280	640
A/South Africa/R04857/2022	6B.1A.5a.2	2022-04-25	MDCK1MDCK1	<	<	40	<	<	640	1280	1280
A/South Africa/R05230/2022	6B.1A.5a.2	2022-05-03	MDCK1MDCK1	<	<	40	<	<	1280	1280	640
A/South Africa/R05013/2022	6B.1A.5a.2	2022-05-03	MDCK1MDCK1	<	<	40	<	<	640	640	1280
A/South Africa/R04993/2022	6B.1A.5a.2	2022-05-03	MDCK1MDCK1	<	<	40	<	<	1280	2560	1280
A/South Africa/R05274/2022	6B.1A.5a.2	2022-05-05	MDCK1MDCK1	<	<	<	<	<	640	640	640
A/South Africa/R05591/2022	6B.1A.5a.2	2022-05-09	MDCK1MDCK1	<	<	40	<	<	640	1280	1280
A/South Africa/R05393/2022	6B.1A.5a.2	2022-05-10	MDCK1MDCK1	<	<	40	<	<	640	1280	1280
A/South Africa/R05407/2022	6B.1A.5a.2	2022-05-11	MDCK1MDCK1	<	<	<	<	<	320	320	320
A/South Africa/R05737/2022	6B.1A.5a.2	2022-05-16	MDCK1MDCK1	<	<	<	<	<	640	1280	1280
A/South Africa/R05613/2022	6B.1A.5a.2	2022-05-16	MDCK1MDCK1	<	<	<	<	<	320	640	640
A/South Africa/R05714/2022	6B.1A.5a.2	2022-05-17	MDCK1MDCK1	<	<	40	<	80	1280	1280	1280
A/South Africa/R05701/2022	6B.1A.5a.2	2022-05-17	MDCK1MDCK1	<	<	<	<	<	640	640	640

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2020-21Vaccine
SH 2021
NH 2021-22
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 7-9. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-08-05)

Haemagglutination inhibition titre												
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection fereret antisera							
					A/Ire	A/G-M	A/G-M	A/Ghana	A/Lyon	A/Denmark	IVR-215	A/Sydney
					87733/19	SWL1536/19	SWL1536/19	1894/21	820/21	3280/19	A/Vic/2570/19	5/21
					Egg	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg
					St Jude's	F09/20 ¹	F12/20 ¹	F02/22 ¹	F06/22 ¹	F28/20 ¹	F37/21 ¹	F04/22 ¹
F18/20 ¹												
Genetic group	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2				
REFERENCE VIRUSES												
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	640	1280	1280	640	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	640	1280	1280	1280	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	320	1280	1280	640	160	40	80	40
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2/E1	640	1280	640	1280	160	40	80	40
A/Lyon/820/2021		6B.1A.5a.1	2021-11-16	E1/E2	80	320	320	160	640	40	40	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	<	<	40	<	40	1280	1280	1280
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	40	80	80	80	160	1280	1280	1280
A/Sydney/5/2021		6B.1A.5a.2	2021-10-16	E3/E1	40	40	80	40	40	1280	2560	2560
TEST VIRUSES												
A/Murcia/10207/2021		6B.1A.5a.1	2021-12-05	MDCK1	320	640	1280	640	160	40	80	40
A/Canarias/558/2022		6B.1A.5a.1	2022-01-01	MDCK1	640	1280	2560	1280	320	40	80	40
A/Belgium/S0141/2022		6B.1A.5a.1	2022-01-06	MDCK1	640	2560	640	320	160	<	40	<
A/Galicia/1460/2022		6B.1A.5a.1	2022-01-09	MDCK1	320	1280	1280	640	320	40	40	40
A/Galicia/1459/2022		6B.1A.5a.1	2022-01-09	MDCK1	640	1280	2560	1280	320	40	80	40
A/Belgium/S0120/2022		6B.1A.5a.1	2022-01-17	MDCK1	640	2560	1280	640	320	40	40	40
A/Belgium/S0197/2022		6B.1A.5a.1	2022-01-30	MDCK1	320	2560	2560	640	160	40	40	40
A/Belgium/S0360/2022		6B.1A.5a.1	2022-02-10	MDCK1	320	1280	1280	640	160	40	40	40
A/Belgium/S0372/2022		6B.1A.5a.1	2022-02-16	MDCK1	320	1280	1280	640	160	40	40	40
A/Belgium/S0449/2022		6B.1A.5a.1	2022-02-24	MDCK1	320	1280	1280	640	160	40	40	40
A/Belgium/S0447/2022	A135 ¹ , A187 ¹ , V295 ¹	6B.1A.5a.1	2022-02-24	MDCK1	160	640	640	160	80	<	40	<
A/Belgium/S0472/2022		6B.1A.5a.1	2022-02-25	MDCK1	320	1280	640	320	160	40	40	40
A/Belgium/S0810/2022		6B.1A.5a.1	2022-03-03	MDCK1	320	1280	1280	640	160	40	40	40
A/Belgium/S0429/2022	A187 ¹ , N320 ¹	6B.1A.5a.1	2022-03-03	MDCK1	160	320	640	160	160	<	40	40
A/Belgium/S0782/2022		6B.1A.5a.1	2022-03-04	MDCK1	640	2560	2560	1280	320	40	80	40
A/Belgium/S0531/2022		6B.1A.5a.1	2022-03-07	MDCK1	640	2560	2560	1280	320	40	80	40
A/Belgium/S0906/2022		6B.1A.5a.1	2022-03-18	MDCK1/MDCK1	320	2560	1280	640	160	40	40	40
A/Belgium/S0859/2022		6B.1A.5a.1	2022-03-18	MDCK1	640	2560	2560	1280	320	40	80	40
A/Belgium/S0824/2022		6B.1A.5a.1	2022-03-20	MDCK1/MDCK1	320	2560	1280	1280	160	40	40	40
A/Belgium/S0825/2022		6B.1A.5a.1	2022-03-21	MDCK1/MDCK1	640	2560	1280	1280	320	40	80	40
A/Belgium/S0856/2022		6B.1A.5a.1	2022-03-22	MDCK1/MDCK1	320	1280	640	640	160	40	80	40
A/Baleares/1472/2022		6B.1A.5a.1	2022-03-23	MDCK1	320	1280	640	640	160	40	40	40
A/Belgium/S0793/2022		6B.1A.5a.1	2022-03-27	MDCK1/MDCK1	640	2560	1280	1280	320	40	80	40
A/Belgium/G0175/2022		6B.1A.5a.1	2022-03-28	MDCK1/MDCK1	640	2560	1280	1280	320	40	80	40
A/Belgium/S0767/2022		6B.1A.5a.1	2022-03-30	MDCK1/MDCK1	160	1280	640	320	160	40	80	40
A/Belgium/S0769/2022	A187 ¹ , P282S	6B.1A.5a.1	2022-03-31	MDCK1/MDCK1	80	320	320	160	80	<	40	<
A/Belgium/S1111/2022		6B.1A.5a.1	2022-04-05	MDCK1/MDCK1	640	1280	2560	1280	320	40	80	40
A/Belgium/S1001/2022		6B.1A.5a.1	2022-04-10	MDCK1/MDCK1	320	640	640	640	160	40	80	40
A/Belgium/S1041/2022		6B.1A.5a.1	2022-04-12	MDCK1/MDCK1	320	640	640	640	160	40	80	40
A/Belgium/S1003/2022		6B.1A.5a.1	2022-04-13	MDCK1/MDCK1	320	640	640	640	160	40	80	40
A/Spain/856/2022		6B.1A.5a.1	2022-04-15	MDCK2	1280	1280	1280	1280	320	40	80	40
A/Belgium/S0418/2022		6B.1A.5a.1		MDCK1	160	1280	640	320	160	40	40	<
A/Qatar/10-VI-22-1518804/2022		6B.1A.5a.2	2022-05-01	MDCK1	<	40	40	<	40	1280	640	1280
A/Qatar/10-VI-22-1522551/2022		6B.1A.5a.2	2022-05-02	MDCK1	<	40	40	<	40	640	640	640
A/South Africa/R05114/2022		6B.1A.5a.2	2022-05-04	MDCK1/MDCK3	<	<	<	<	<	320	640	320
A/Qatar/16-VI-22-1535667/2022		6B.1A.5a.2	2022-05-07	MDCK1	<	40	40	<	40	1280	1280	1280
A/Croatia/86404/2022		6B.1A.5a.2	2022-06-30	MDCK1	<	<	<	<	<	640	1280	640
A/Croatia/86715/2022		6B.1A.5a.2	2022-07-01	MDCK1	<	40	40	<	<	1280	1280	1280
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)					Vaccine NH 2020-21				Vaccine SH 2021 NH 2021-22 SH 2022 NH 2022-23			
1 < = <40; 2 < = <80; ND = Not Done												

Table 7-10. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2022-08-24)

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection fereret antisera								
				A/Ire 87733/19	A/G-M SWL1536/19	A/G-M SWL1536/19	A/Ghana 1894/21	A/Lyon 820/21	A/Denmark 3280/19	IVR-215 A/Vic/2570/19	A/Sydney 5/21	
				Egg	MDCK	Egg	Egg	Egg	MDCK	Egg	Egg	
				St Jude's F18/20 ^{*1}	F09/20 ^{*1}	F12/20 ^{*1}	F02/22 ^{*1}	F06/22 ^{*1}	F28/20 ^{*1}	F37/21 ^{*1}	F04/22 ^{*1}	
	Passage history			6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.1	6B.1A.5a.2	6B.1A.5a.2	6B.1A.5a.2	
	Ferret number											
	Genetic group											
REFERENCE VIRUSES												
A/Ireland/87733/2019		6B.1A.5a.1	2019-11-03	E4	640	1280	1280	640	320	40	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	C2/MDCK1	640	1280	1280	1280	320	80	80	40
A/Guangdong-Maonan/SWL1536/2019		6B.1A.5a.1	2019-06-17	E3/E2	320	1280	1280	640	160	40	80	40
A/Ghana/1894/2021		6B.1A.5a.1	2021-07-21	E2/E1	640	2560	2560	1280	320	80	160	80
A/Lyon/820/2021		6B.1A.5a.1	2021-11-16	E1/E2	80	320	320	160	640	40	40	40
A/Denmark/3280/2019		6B.1A.5a.2	2019-11-10	MDCK4/MDCK5	40	40	80	<	80	1280	1280	1280
IVR-215 (A/Victoria/2570/2019)		6B.1A.5a.2	2018-11-22	E4/D7/E2	40	80	80	80	80	640	1280	1280
A/Sydney/5/2021		6B.1A.5a.2		E3/E1	40	80	40	40	40	1280	1280	1280
TEST VIRUSES												
A/Tours/37554/2021		6B.1A.5a.1	2021-11-02	MDCK1	640	1280	1280	640	320	40	80	40
A/Dijon/48658/2021		6B.1A.5a.1	2021-12-08	MDCK2	640	1280	1280	640	160	40	80	80
A/Alsace/48917/2021		6B.1A.5a.1	2021-12-15	MDCK1	640	1280	1280	1280	320	40	80	80
A/Ile de France/52132/2021		6B.1A.5a.1	2021-12-27	MDCK1	640	1280	1280	640	160	40	80	80
A/Ile de France/54452/2021		6B.1A.5a.1	2022-01-03	MDCK1	640	1280	1280	640	320	40	80	40
A/Lyon/215/2022		6B.1A.5a.1	2022-01-24	MDCKx/MDCK1	80	320	160	320	80	160	320	160
A/Clermont-Ferrand/184/2022		6B.1A.5a.1	2022-01-25	MDCKx/MDCK1	320	1280	1280	640	160	40	40	40
A/Paris/10105/2022		6B.1A.5a.1	2022-02-03	MDCK1	320	640	1280	640	160	40	80	40
A/Le Mans/15006/2022		6B.1A.5a.1	2022-02-24	MDCK1	640	1280	1280	1280	320	80	80	80
A/Grenoble/465/2022		6B.1A.5a.1	2022-03-02	MDCKx/MDCK1	640	1280	1280	1280	320	40	80	80
A/Belgium/S0497/2022		6B.1A.5a.1	2022-03-07	MDCK1/MDCK2	640	1280	2560	1280	320	80	80	40
A/Romans/478/2022		6B.1A.5a.1	2022-03-14	MDCKx/MDCK1	640	1280	1280	640	160	40	80	40
A/Lyon/614/2022		6B.1A.5a.1	2022-03-21	MDCKx/MDCK1	640	1280	1280	640	160	40	80	80
A/Lyon/572/2022		6B.1A.5a.1	2022-03-21	MDCKx/MDCK1	320	1280	1280	640	160	40	80	40
A/Lyon/CHU-R.22.172.23/2022	N156S, N162* (-cho)	6B.1A.5a.1	2022-03-27	MDCK2/MDCK1	80	640	160	320	160	40	80	80
A/Toulon/759/2022		6B.1A.5a.1	2022-03-27	MDCKx/MDCK1	640	1280	1280	640	160	40	80	80
A/Dunkerque/26698/2022		6B.1A.5a.1	2022-03-31	MDCK2	640	1280	2560	1280	320	80	80	40
A/Lyon/CHU-R22.224.12/2022		6B.1A.5a.1	2022-04-25	MDCKx/MDCK1	320	1280	1280	640	160	40	80	80
A/Lyon/CHU-R22.242.34/2022		6B.1A.5a.1	2022-05-09	MDCKx/MDCK1	640	1280	1280	640	320	40	80	80
A/La Reunion/R22.317.57/2022		6B.1A.5a.1	2022-07-09	MDCKx/MDCK1	640	2560	2560	1280	320	40	80	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40; 2 < = <80; ND = Not Done

Vaccine
NH 2020-21Vaccine
SH 2021
NH 2021-22

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 7-11. Summary of HI results for A(H1N1)pdm09 viruses with collection dates after 2022-01-31 by genetic group

Genetic group	Antiserum raised against Fold reduction	A/Ire 87733/19 Egg St Jude's F18/20 ^{**} 6B.1A.5a.1	A/G-M SWL1536/19 MDCK F09/20 ^{**} 6B.1A.5a.1	A/G-M SWL1536/19 Egg F12/20 ^{**} 6B.1A.5a.1	A/Ghana 1894/21 Egg F02/22 ^{**} 6B.1A.5a.1	A/Lyon 820/21 Egg F06/22 ^{**} 6B.1A.5a.1	A/Denmark 3280/19 MDCK F08/20 ^{**} 6B.1A.5a.2	A/Stock 10/21 MDCK F22/22 ^{**} 6B.1A.5a.2	IVR-215 A/Vic/2570/19 Egg F37/21 ^{**} 6B.1A.5a.2	A/Sydney 5/21 Egg F04/22 ^{**} 6B.1A.5a.2
6B.1A.5a.1	No tested	107	107	107	107	107	107	21	107	107
	≤2	97	101	92	90	62	0	0	0	0
	%	90.7	94.4	86.0	84.1	57.9	0	0	0	0
	4	8	6	9	13	38	0	0	0	0
	%	7.5	5.6	8.4	12.1	35.5	107	21	107	107
	≥8	2	0	6	4	7	107	21	107	107
	%	1.8		5.6	3.8	6.6	100	100	100	100
6B.1A.5a.2	No tested	46	46	46	46	46	46	11	46	46
	≤2	0	0	0	0	0	42	8	45	32
	%						91.3	72.7	97.8	69.6
	4	0	0	0	0	0	4	3	1	12
	%						8.7	27.3	2.2	26.1
	≥8	46	46	46	46	46	0	0	0	2
	%	100	100	100	100	100				4.3

All fold-reductions are based on homologous titres determined on the day of individual assays.

Influenza A(H3N2) virus analyses.

Influenza A(H3N2) viruses collected since 2022-01-31 were received from NICs in 42 countries or administrative regions in five WHO Regions: African (n = 4; Ghana, Madagascar, South Africa and Togo), American (n = 1; Argentina), Eastern Mediterranean (n = 3; Egypt, Qatar and Tunisia), Western Pacific (n = 1; Hong Kong-SAR China) and European (33 countries) (**Table 1**).

Influenza A(H3N2) viruses had been difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells (RBCs) from guinea pigs, turkeys and humans or the loss of the ability of viruses to agglutinate any of these RBCs. However, since the February VCM HI analysis was able to be done on 682 cell culture-propagated viruses that agglutinated guinea pig RBCs in the presence of oseltamivir (**Tables 8 and 10-1 to 10-26**). Virus neutralisation by plaque reduction neutralisation assay (PRNA) was done on 133 test viruses (**Tables 11-1 to 11-11**).

HI analyses.

A total of 682 viruses were analysed by HI assay and summaries of the HI results are shown in **Tables 8 and 10-27**. Individual HI results are shown in **Tables 10-1 to 10-26**. Test viruses in each table are sorted by date of collection and genetic clade/subclade/group/subgroup for those viruses that were sequenced, with those below the red horizontal lines in the tables being viruses with collection dates after 2022-01-31 (**Tables 10-10 to 10-26**). The summary shown in **Table 8** is of the HI results for viruses collected since 2022-01-31 with antisera raised against egg-propagated vaccine viruses A/Cambodia/e0826360/2020 (subclade 3C.2a1b.2a.1) and A/Darwin/9/2021 (subclade 3C.2a1b.2a.2), and cell culture-propagated A/Cambodia/925256/2020 (subclade 3C.2a1b.2a.1) and A/Stockholm/5/2021 (A/Darwin/9/2021-like, subclade 3C.2a1b.2a.2). This summary indicates that the tested viruses were recognised well by the antiserum raised against A/Stockholm/5/2021 with all but seven viruses (99%) being recognised at titres within 4-fold of the homologous titre, 86% within 2-fold, while the antiserum raised against egg-propagated A/Darwin/9/2021 recognised 84% of the analysed viruses at titres within 4-fold of the homologous titre, 46% within 2-fold. The antisera raised against cell culture-propagated A/Cambodia/925256/2020 and egg-propagated A/Cambodia/e0826360/2020 recognised a minority of viruses at titres within 4-fold of the homologous titres (16% and 2.5%, respectively).

The viruses sequenced and analysed by HI fell predominantly into clade 3C.2a subclades 3C.2a1b.2a.2 (99% of the sequenced viruses) and 3C.2a1b.1a (eight viruses, from France, Madagascar, Qatar and Sweden).

Subclade 3C.2a1b.2a.2 viruses (n = 672), characterised by **HA1** amino acid substitutions **Y159N**, **T160I** (resulting in loss of a glycosylation site), **L164Q**, **G186D** and **D190N**, have various additional substitutions. In **Table 10-27** the results of the HI assays have been summarised according to four genetic groups: those retaining **H156** in **HA1** (n=53), those

with **H156S** without additional substitutions (n=115), those with **H156S** and **D53G** in **HA1** (n=66) and those with **D53G**, **H156S** and **D104G** and **K276R** in **HA1** (n=438).

The results can be summarised:

- an antiserum raised against egg-propagated A/Darwin/9/2021 recognised well viruses in each of the groups of 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, the genetic group with **H156** was recognised somewhat less well with 56% being recognised at titres within 4-fold of the homologous titres, the antiserum recognising 80% of the other genetic groups at titres within 4-fold of the homologous titre;
- an antiserum raised against cell culture-propagated A/Stockholm/5/2021 however recognised 99% of viruses in each of the 3C.2a1b.2a.2 genetic groups at titres within 4-fold of the homologous titre, and over 75% at titres within 2-fold;
- an antiserum raised against cell culture-propagated A/England/21491723/2021 (with **D53N**) also recognised the large majority of viruses in each of the genetic groups of 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, although 10 test viruses of 438 (2.2%) were recognised less well;
- an antiserum raised against cell culture-propagated A/Slovenia/8720/2022 (with **D53G**, **H156S**, **D104G** and **K276R**) was used with 43 test viruses, 42 in the homologous group and it recognised 97% of these viruses at titres within 2-fold of the homologous titre and;
- an antiserum raised against an egg-propagated cultivar of A/Slovenia/8720/2022 also recognised all of the viruses against which it was used at titres within 4-fold of the homologous titres.

Other antisera raised against 3C.2a1b.2a.2 viruses propagated in hens' eggs have been produced and show varied recognition of the test viruses - most have been used with viruses in the 3C.2a1b.2a.2 subgroup with **D53G**, **H156S**, **D104G** and **K276R**. Those antisera raised against A/Norway/29511/2022, A/Slovenia/9216/2022 and A/Slovenia/9318/2022, recognised the majority of test viruses at titres within 4-fold of the homologous titres, with the latter two being in the same genetic subgroup as the test viruses and recognising most within 2-fold, while the antiserum raised against A/Norway/29511/2022 failed to recognise any of the test viruses at titres within 2-fold of the homologous titre. Antisera raised against egg-propagated A/Florida/02/2021 and A/Wyoming/01/2021 recognised very few viruses tested at titres within 2-fold of the homologous titres and recognised 17 of 38 (45%) and 27 of 38 (71%), respectively, of the test viruses at titres over 4-fold lower than the homologous titres.

An antiserum raised against cell culture-propagated A/Cambodia/925256/2020 (subclade 3C.2a1b.2a.1) recognised the majority of viruses in each genetic subgroup of subclade 3C.2a1b.2a.2 at titres over 4-fold lower than the homologous titre. Similarly, an antiserum raised against egg-propagated A/Cambodia/e0826360/2020 recognised the majority of viruses in the genetic subgroups of subclade 3C.2a1b.2a.2 at titres over 4-fold lower than the homologous titre, although this antiserum recognised just over half of the viruses in the **D53G**, **H156S** group at titres within 4-fold of the homologous titre.

An antiserum raised against cell culture-propagated A/Hong Kong/2671/2019 (subgroup 3C.2a1b.1b) recognised only 4 (<1%) of the subgroup 3C.2a1b.2a.2 test viruses at titres within 4-fold of the homologous titre.

An antiserum raised against cell culture-propagated A/Denmark/3264/2019 (subgroup 3C.2a1b.1a) recognised 53% of the subgroup 3C.2a1b.2a.2 test viruses at titres within 4-fold of the homologous titre, 14% within 2-fold.

The eight 3C.2a1b.1a viruses from France, Madagascar, Qatar and Sweden did not show a consistent pattern of recognition by the reference antisera, for example the antisera raised against cell culture-propagated A/Stockholm/5/2021 and egg-propagated A/Darwin/9/2021 recognised some of these test viruses at titres within 4-fold of the homologous titres, both recognising six of the eight tested viruses at titres within 4-fold of the homologous titres, but the other two less well. Similarly, the antisera raised against cell culture-propagated A/Cambodia/925256/2020 and egg-propagated A/Cambodia/e0826360/2020 recognised some of these test viruses at titres within 4-fold of the homologous titres, four of eight and three of eight respectively, but the others less well.

Virus Neutralisation results

Results of virus neutralisation, assessed by PRNA, are shown in **Tables 11-1 to 11-11**. A total of 133 test viruses from NICs were analysed. All viruses analysed were in subgroup 3C.2a1b.2a.2. A summary of the results is shown in **Table 4** and a phylogenetic tree of the HA genes of viruses analysed by PRNA is shown in **Figure 12**.

For viruses in subclade 3C.2a1b.2a.2 (n = 133):

- an antiserum raised against cell culture-propagated A/Stockholm/5/2021 recognised over 85% of the 3C.2a1b.2a.2 viruses in each of the genetic groups described above for the HI assay at titres within 4-fold of the homologous titres, 91% of those with **H156**, 87% of those with **H156S** only, 92% with **H156S** and **D53G** only and 97% with **D53G**, **H156S**, **D104G** and **K276R**; however only 18%, 73%, 84% and 37%, respectively, were recognised within 2-fold of the homologous titres;
- an antiserum raised against cell culture-propagated A/Bangladesh/4005/2020 recognised 20% of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, but only 3% of viruses with the HA1 substitutions **D53G**, **H156S**, **D104G** and **K276R**;
- an antiserum raised against cell culture-propagated A/Slovenia/8270/2022 (with the HA1 substitutions **D53G**, **H156S**, **D104G** and **K276R**) was used in a subset of the assays: notably this antiserum recognised all of the test viruses with the **HA1** substitutions **D53G** with **H156S**, and **D53G** with **H156S**, **D104G** and **K276R** at titres within 4-fold of the homologous titres, and 86% and 63% within 2-fold. The test viruses in the other genetic groups in subclade 3C.2a1b.2a.2 were recognised less well;
- an antiserum raised against egg-propagated A/Darwin/9/2021 (an antiserum raised against NIB-126A was used in **Table 10-1**) recognised only 5% of the 3C.2a1b.2a.2 viruses at titres within 4-fold of the homologous titre, none within 2-fold.

Antisera raised against cell culture-propagated A/Norway/3275/2018 (subclade 3C.2a1b.2), A/Cambodia/925256/2020 (3C.2a1b.2a.1) and A/Denmark/3264/2019 (3C.2a1b.1a) recognised very few 3C.2a1b.2a.2 viruses at titres within 4-fold of the respective homologous titres.

There are two sets of heat maps shown adjacent to the phylogenetic tree in **Figure 12**. These summarise the VN results and the HI results for the same viruses, where available, and a selection from the panel of reference viruses used to generate the antisera used in the HI assay. The heat map for the VN results illustrates the better recognition of test viruses by antisera raised against viruses within the same subclade.

Genetic analyses.

Figure 7a shows a phylogenetic tree for HA genes of A(H3N2) viruses analysed at the WIC most with collection dates from 2022-02-01, with relevant reference virus HA gene sequences. **Figure 7b** shows a phylogenetic tree for HA genes for representative viruses as deposited in GISAID at the time of preparing this report. **Table 9** shows the geographic distribution of the 2,359 A(H3N2) viruses with collection dates after 2022-04-30 for which genetic information was deposited in GISAID at the time of preparing this report.

The HA genes of the vast majority fell within subclade 3C.2a1b.2a.2.

Major HA clades/subclades

- The HA genes of recently detected clade **3C.2a** viruses share a series of amino acid substitutions **E62G**, **N91S**, **N121K**, **K144S**, **S159Y**, **K160T** (+CHO), **Q311H** and **R326K** in **HA1** and **I77V**, **M149I** and **G155E** in **HA2** that define subclade **3C.2a1b**.
- Subclade **3C.2a1b.2** is defined by substitutions **A128T** (+CHO), **T131K** and **S138A** in **HA1**, and **V200I** in **HA2**, with recently circulating viruses also carrying **Y83E**, **Y94N**, **Y195F** in **HA1** and **I193M** in **HA2**.
 - Subclade **3C.2a1b.2a.1** viruses (e.g. A/Cambodia/e0826360/2020) have HA glycoproteins with the additional substitutions **K171N**, **G186S** and **S198P** in **HA1**.
 - Subclade **3C.2a1b.2a.2** viruses have HA glycoproteins with the substitutions **Y159N**, **T160I** (resulting in the loss of a glycosylation site), **L164Q**, **G186D** and **D190N** in **HA1**. The majority of viruses within subclade **3C.2a1b.2a.2** now also have **HA1 H156S** substitution (e.g. A/Stockholm/5/2021 and A/Darwin/9/2021), many with **D53N** or **D53G** in **HA1**. Some of the subgroup with **D53G** (e.g. A/Darwin/6/2021) also have HA glycoproteins with **D104G** and **K276R** substitutions in **HA1**. A subgroup that has retained **HA1 H156** has additional **HA1** substitutions of **E50K**, **F79V**, **I140K** and **S262N**.

Figure 8 show the locations on a 2005 H3 HA structure of amino acid substitutions that define subclade 3C.2a1b viruses, and subclades (3C.2a1b.2a.1 and 2a.2) and the different subgroups within subclade 3C.2a1b.2a.2.

Figure 9 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogenies. The vaccine viruses A/Cambodia/e0826360/2020 (subclade 3C.2a1b.2a.1) and A/Darwin/9/20121 and A/Darwin/9/2021 (subclade 3C.2a1b.2a.2) are shown in red, glycosylation motifs are highlighted, reference viruses are coloured blue and the HA1/HA2 proteolytic cleavage site is highlighted. The clade/subclade/group/subgroup to which each of the HA sequences belongs is also shown.

Figure 10a shows a phylogenetic tree for NA genes of A(H3N2) viruses analysed at the WIC most with collection dates from 2022-02-01, with relevant reference virus HA gene sequences. **Figure 10b** shows a phylogenetic tree for NA genes for representative viruses as deposited in GISAID at the time of preparing this report.

Figure 11 shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, the vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the clade/subclade/group/subgroup to which the corresponding HA gene sequences belong is shown.

The NA gene sequences of viruses with HA genes within subclade **3C.2a1b** show generally similar subclade/group clustering. Viruses with HA genes in subclade **3C.2a1b.2** have the substitutions **S315R** and **G346D** in **NA**. Viruses with HA genes in subclade **3C.2a1b.2a.1** and subclade **3C.2a1b.2a.2** share the substitutions **D463N** and **N465S** in **NA**, creating a potential glycosylation site. The NA substitution **S329N** introduces/restores a glycosylation site and is associated with HA genes encoding the substitutions **D53N**, **N96S** and **I192F** in **HA1** and **N49S** in **HA2**. A reversion in **NA** (**D346G**) is associated with many viruses with HA genes encoding the substitutions **D53G**, **D104G**, **H156S** and **K276R** in **HA1**.

Antiviral resistance

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 1077 A(H3N2) viruses detected and received within the course of the 2021-2022 influenza season (**Table 15**). All but one virus showed normal inhibition (NI: no reduced susceptibility) by both antivirals. A/Tripoli/5646/2021, reported on in the February VCM report, showed reduced inhibition by zanamivir, associated with **NA A246V** substitution, within a glycosylation site (NAT → NVT).

All A(H3N2) full-length PA gene sequences generated at the Crick, from **1,497** individual viruses detected during the 2021-2022 influenza season, encoded isoleucine at position 38 (**I38**). Amino acid substitutions at this position have been associated with reduced susceptibility to baloxavir marboxil. However, three PA sequences from three viruses showed potential evidence of reduced susceptibility involving substitutions at other positions: A/Netherlands/10253/2022, with **PA E23G**, and A/Estonia/169374/2022 and A/Madrid/1445/2022 both encoded **PA L28P**.

Conclusion.

Received A(H3N2) samples with collection dates after 2022-01-31 have HAs in subclade **3C.2a1b** with the vast majority in subclade **3C.2a1b.2a.2**. The viruses falling into this subclade were recognised well in HI and PRNA by antisera raised against cell culture-propagated A/Stockholm/5/2021 and slightly less well by antiserum raised against egg-propagated vaccine virus A/Darwin/9/2021 (subclade **3C.2a1b.2a.2** viruses with the

additional substitution **H156S** in **HA1**). Some viruses with **HA1** substitutions **D53G**, **H156S**, **D104G** and **K276R** may be better recognised by antisera from a reference virus in the homologous group.

All but three A(H3N2) viruses characterised genetically showed no markers of reduced susceptibility to baloxavir marboxil and all but one of those analysed phenotypically showed normal inhibition by both oseltamivir and zanamivir.

Table 8. Summary of influenza A(H3N2) viruses collected after 2022-01-31, analysed by HI

MONTH	H3N2															
	Number received ¹	Number propagated ²	3C.2a1b.2a.2: Egg-propagated A/Darwin/9/2021			3C.2a1b.2a.2: Cell-propagated A/Stockholm/5/2021			3C.2a1b.2a.1: Egg-propagated A/Cambodia/e0826360/2020			3C.2a1b.2a.1: Cell-propagated A/Cambodia/925256/2020				
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶		
2022																
FEBRUARY																
Austria	2	2		2		2					2			2		
Belgium	10	1		1		1					1			1		
Bulgaria	4	4	4			4					4			4		
Croatia	2	0														
Denmark	1	1	1			1					1			1		
Egypt	2	0														
Estonia	3	2		1	1	2				1	1			2		
France	4	4		3	1	1	2	1			4	1	2	1		
Germany	11	11	11			11			1	3	7			11		
Hungary	3	3	3			3				1	2			3		
Italy	5	5	5			4	1				5			5		
Moldova	1	1	1			1					1			1		
Morocco	4	0														
Norway	2	2	2			2				1	1			2		
Poland	1	1		1		1					1			1		
Portugal	31	23	14	8	1	23					23		2	21		
Senegal	1	1		1		1					1			1		
Slovakia	1	0														
Slovenia	12	12	4	8		10	2				12			12		
Spain	10	10	9	1		8	2				10			10		
Togo	1	0														
Tunisia	3	1		1			1				1			1		
Ukraine	1	1		1		1					1			1		
MARCH																
Austria	23	21	7	5	9	21			2	3	16			21		
Belgium	75	14	2		12	9	5			1	13			14		
Bosnia and Herz	2	0														
Bulgaria	16	16	16			16					16		1	15		
Croatia	12	1	1			1					1			1		
Denmark	11	11	11			11					11			11		
France	7	5	1	4		5					5			5		
Germany	7	7	5	2		7			2	1	4		2	5		
Hungary	4	4	4			4					4			4		
Iceland	4	0														
Ireland	39	4		2	2	4					4			4		
Italy	10	10	6	4		7	2	1	1	2	7			10		
Lithuania	1	0														
Moldova	5	5	2	3		3	2			2	3			5		
Madagascar	6	2	1		1	1		1			2			2		
Montenegro	19	4	2	2		4					4		1	3		
Netherlands	4	4	1	3		1	3				4			4		
Norway	16	13	13			13					13			13		
Poland	25	14	2	10	2	14				2	12			14		
Portugal	10	8	1	3	4	8					8			8		
Serbia	9	7	4	1	2	7					7			7		
Slovakia	2	0														
Slovenia	40	39	10	27	2	36	3			5	34			39		
South Africa	1	1		1		1					1			1		
Spain	26	26	2	8	16	26			5	8	13			26		
Sweden	4	4	3	1		4				1	3			4		
Switzerland	14	13	4	8	1	11	2		2	2	9	1	1	11		
Togo	34	0														
Ukraine	2	1			1	1				1				1		
UK (N. Ireland)	3	0														
APRIL																
Argentina	2	0														
Austria	22	13	3	8	2	11	2			3	10			13		
Belgium	15	14		8	6	11	3				14			14		
Bosnia and Herz	5	3	3			3					3			3		
Bulgaria	3	2	2			2					2			2		
Croatia	37	6	4	2		3	2	1			6			6		
Denmark	2	2	2			2					2			2		
Estonia	26	24	8	14	2	24				1	23			24		
France	5	5	1	4		5				1	4			5		
Germany	6	6	2	4		6					6			6		
Hungary	2	2	1	1		2					2			2		
Iceland	25	1			1	1								1		
Ireland	10	3		2	1	2	1			1	3			3		
Italy	13	12	8	4		7	5				12			12		
Latvia	12	12		10	2	12					12			12		
Lithuania	46	7	3	2	2	5	2			1	6			7		
Madagascar	18	2	1	1		2					2			2		
Moldova	6	6	4	2		6				4	2			6		
Montenegro	6	1	1			1					1			1		
Netherlands	6	5	2	1	2	3	2			2	3			5		
Norway	25	21	12	7	2	21				3	18			21		
Poland	45	17	6	11		14	3			1	16			17		
Portugal	10	10	5	2	3	9	1			2	8			10		
Qatar	16	3	1	2		2	1			1	2		1	2		
Romania	24	24	23	1		8	16		1	1	22			24		
Serbia	4	4	4			4					4			4		
Slovakia	7	0														
Slovenia	11	11	9	2		11			1	4	6		3	8		
Spain	54	9	2	7		8	1		1	4	4			9		

Table 8. Summary of influenza A(H3N2) viruses collected after 2022-01-31, analysed by HI

MONTH	H3N2													
	Number received ¹	Number propagated ²	3C.2a1b.2a.2: Egg-propagated A/Darwin/9/2021			3C.2a1b.2a.2: Cell-propagated A/Stockholm/5/2021			3C.2a1b.2a.1: Egg-propagated A/Cambodia/e0826360/2020			3C.2a1b.2a.1: Cell-propagated A/Cambodia/925256/2020		
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
Sweden	7	7	3	3	1	4	3		2	3	2	1		6
Switzerland	12	6	1		5	6			1	5			1	5
Togo	35	0												
Tunisia	1	0												
Ukraine	2	2		1	1	2				2				2
UK (N. Ireland)	24	0												
MAY														
Argentina	19	0												
Austria	2	2		1	1	2					2			2
Belgium	1	1			1	1					1			1
Bosnia and Herz	1	0												
Croatia	3	2	2			2					2			2
France	4	4	1	3		4					4			4
Germany	8	8	4	4		7	1				8			8
Ghana	20	5		1	4	3	2				5			5
Iceland	18	1			1	1			1					1
Italy	1	1	1			1					1			1
Lithuania	2	2	2			2				1	1			2
Madagascar	9	1		1		1					1			1
Mauritius	12	0												
Poland	7	6	6			3	3				6			6
Portugal	4	4	3	1		4					4			4
Qatar	17	3	1	1	1	3				1	2	1		2
Romania	5	4	2	2			4				4			4
Senegal	10	7	6	1		7				1	6			7
South Africa	1	0												
Spain	48	29	5	11	13	29			1	8	20			29
Sweden	1	1			1				1			1		
Togo	16	8	7	1		3	5				8			8
UK (N. Ireland)	3	0												
JUNE														
Argentina	14	0												
France	2	2	2			2					2			2
Ghana	14	0												
Hong Kong	4	3		3				3			3	2	1	
Madagascar	6	2		2		2					2			2
Mauritius	18	0												
Norway	2	2	2			2			2			2		
Qatar	1	1		1		1					1			1
Romania	2	2	2			1	1				2			2
Senegal	12	6	3	3		4	2				6			6
Spain	4	2			2	1	1			2				2
UK (N. Ireland)	4	0												
JULY														
Croatia	1	0												
France	10	10		8	2	10				1	9			10
Ghana	2	0												
Hong Kong	1	0												
Mauritius	10	0												
Norway	10	4	1	3		3	1				4			4
	1419	682	313	258	111	587	88	7	24	87	571	9	15	658
42 Countries			45.9%	37.8%	16.3%	86.1%	12.9%	1.0%	3.5%	12.8%	83.7%	1.3%	2.2%	96.5%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir

3. Compared to homologous titres for ferret antisera raised against egg-propagated A/Darwin/9/2021 (3C.2a1b.2a.2)

4. Compared to homologous titres for ferret antisera raised cell-propagated A/Stockholm/5/2021 (3C.2a1b.2a.2)

5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Cambodia/e0826360/2020 (3C.2a1b.2a.1)

6. Compared to homologous titres for ferret antisera raised against cell-propagated A/Cambodia/925256/2020 (3C.2a1b.2a.1)

Samples received were in lysis buffer - only genetic characterisation possible

Numbers highlighted in yellow show the number of viruses subjected to HI assay for 'completed' sample sets. Under a 'sequence first' virus characterisation scheme: where multiple viruses shared the same HA sequence only a selection were propagated to allow assay by HI

As of 2022-09-11

Figure 7a. Phylogenetic comparison of A(H3N2) HA genes (WIC sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

Apr 2022

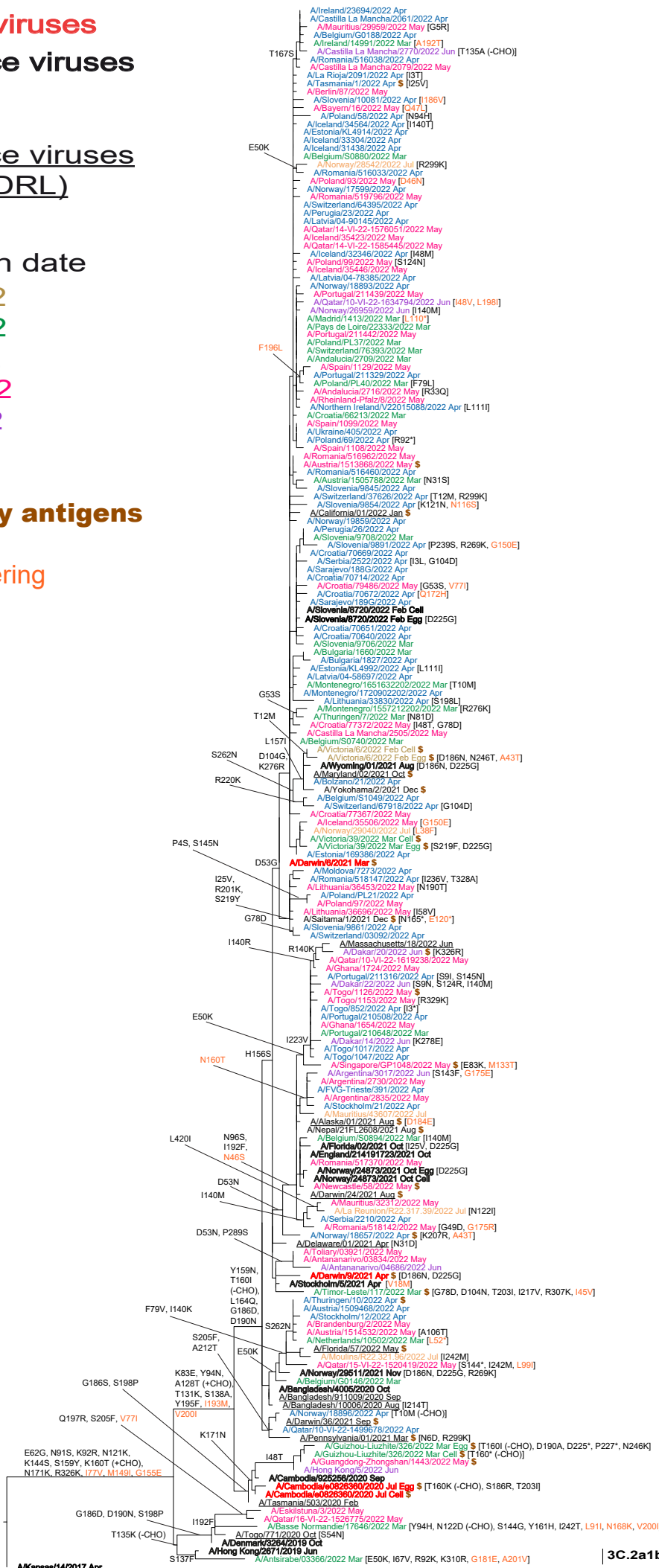
May 2022

Jun 2022

Jul 2022

\$ serology antigens

HA2 numbering



3C.2a1b.2a.1

3C.2a1b.1a

3C.2a1b.1b

3C.3a.1

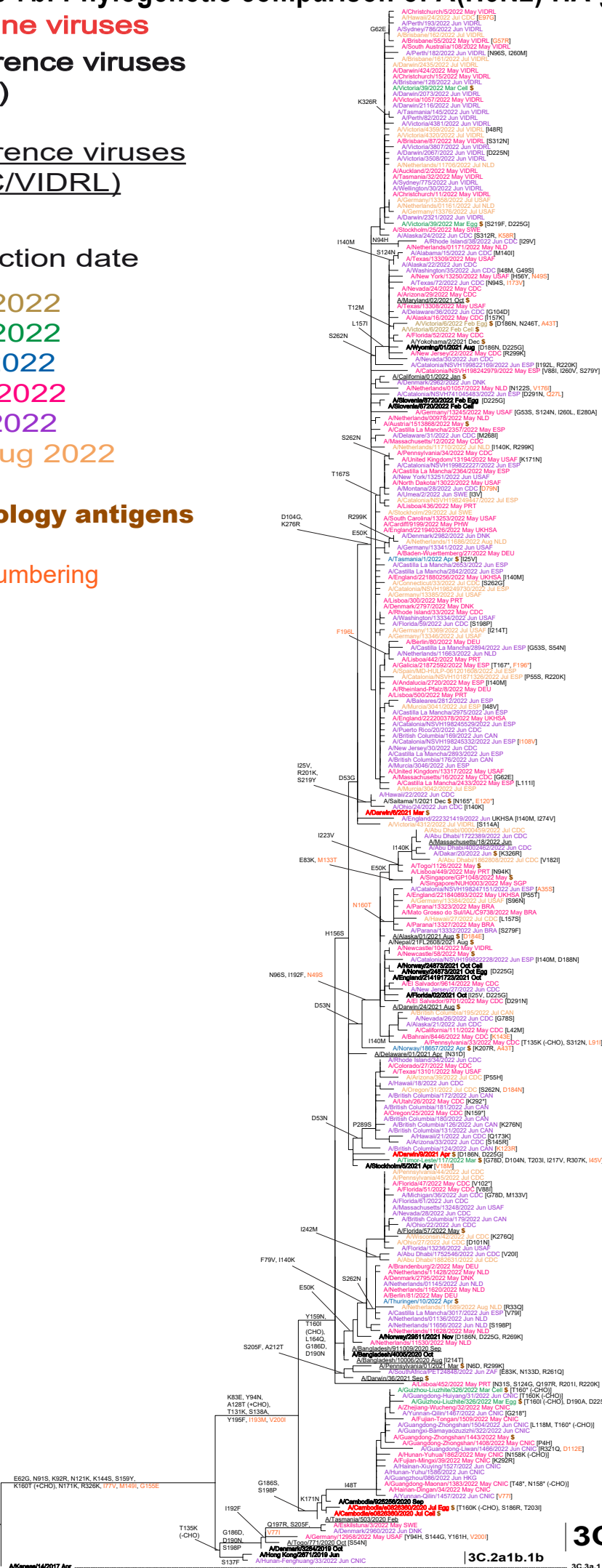


Table 6. Global count of A(H1N1)pdm09 HA sequences for viruses with collection dates after 2022-01-31

Clade	Group	Number	Country and (number of sequences)
6B.1A.5a.1	D187A	467	Australia (13), Belgium (23), Canada (2), Denmark (5), France (35), Germany (6), Iceland (2), Ireland (1), Italy (1), Latvia (2), Luxembourg (16), Mauritius (1), Moldova (3), Netherlands (280), Norway (2), Portugal (8), Romania (8), Serbia (7), South Africa (3), Spain (13), Sweden (3), Switzerland (14), UK (12), USA (7)
	& R113K	25	Australia (1), Netherlands (2), Norway (1), South Africa (16), Spain (2), Sweden (2), UK (1)
	& P137S, G155E	3	Italy (1), Spain (1), Switzerland (1)
	& K211R	7	France (3), Germany (1), Netherlands (2), South Africa (1)
	& D129N, H138Q, T184N	8	Netherlands (8)
	& N125K	2	Luxembourg (2)
	& A187T	4	Denmark (4)
6B.1A.5a.2	& A139D	4	Spain (4)
	K54Q, K130N, N156K, L161I, A186T, Q189E, E224A, V250A, R259K, K308R	127	Australia (98), Canada (2), India (2), Pakistan (5), UAE (2), UK (18)
	& K142R, D260E	10	South Africa (10)
	& K142R, D260E, P137S, T277A	75	Canada (1), Croatia (2), Netherlands (10), Norway (2), South Africa (46), Spain (9), Sweden (3), UK (2)
	& T216A	132	Australia (13), Canada (4), India (1), Norway (11), Pakistan (13), Qatar (7), Somalia (13), Spain (1), UAE (3), UK (63), USA (3)
	& T216A, D129N	7	UK (7)
	& T216A, D94N	211	Australia (176), Bahrain (1), India (1), Netherlands (3), New Zealand (3), Pakistan (4), Qatar (9), Spain (1), Sweden (1), UK (12)
	& T216A, D94N, P137L	11	Australia (11)
	& T216A, D94N, R205G	25	Australia (20), Bahrain (4), Pakistan (1)
	& T216A, D94N, T186A	11	Australia (11)
	Total count	1129	

WHO Region:

African (n = 90; 3 countries)

Americas (n = 19; 2 countries)

Eastern Mediterranean (n = 49; 4 countries)

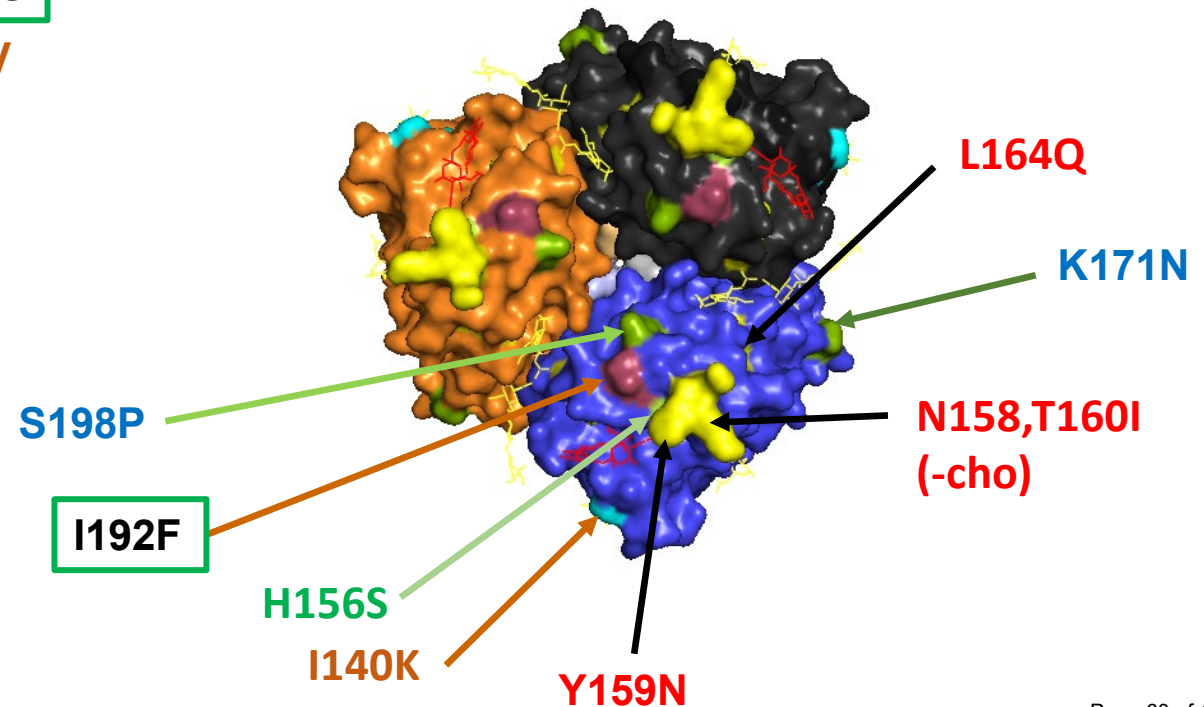
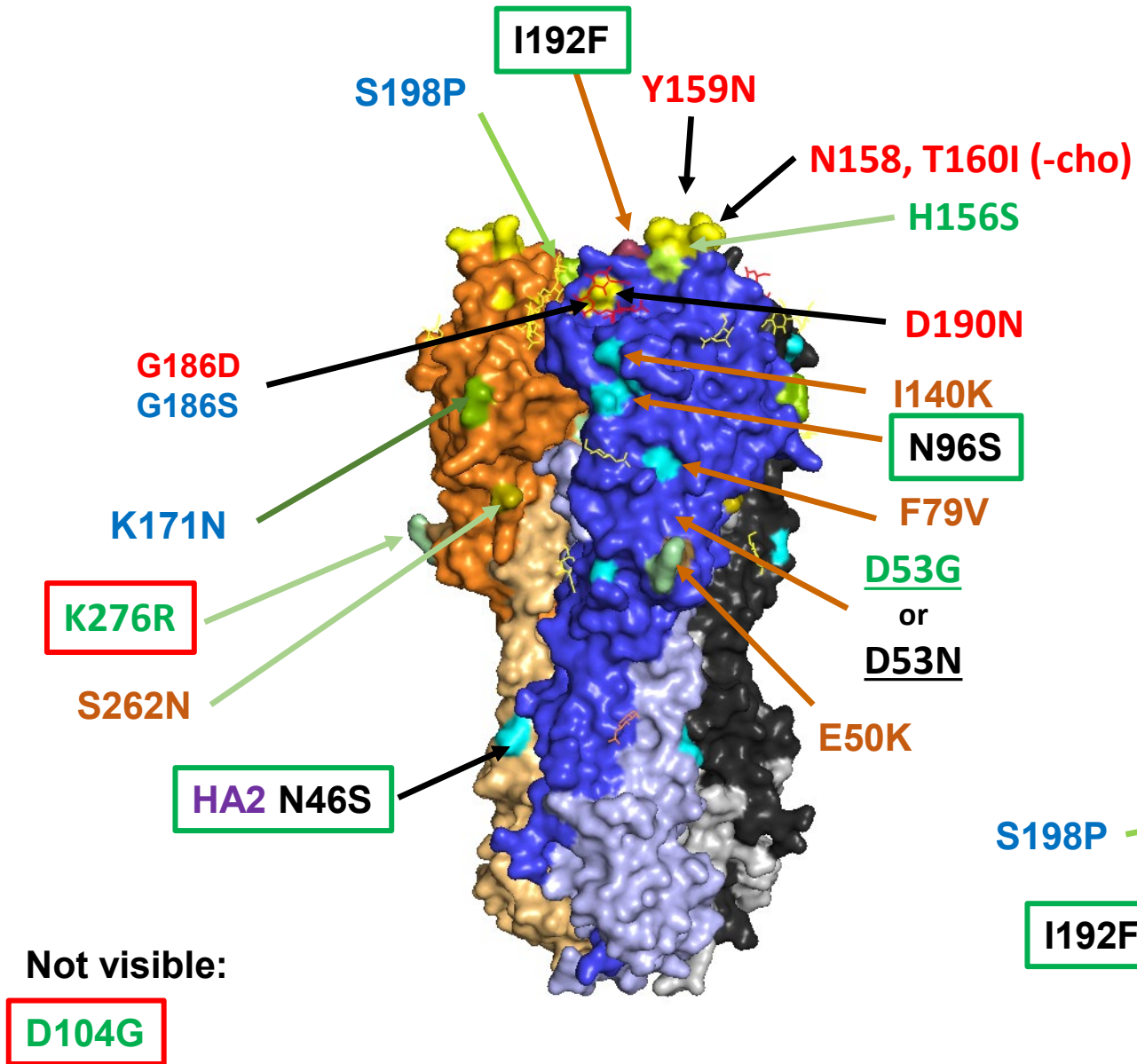
European (n = 621; 20 countries)

South-East Asia (n = 4; 1 country)

Western Pacific (n = 346; 2 countries)

Full-length HA sequences downloaded from GISAID and de-duplicated on 2022-09-05

Figure 8. HA amino acid substitutions defining H3N2 3C.2a1b virus groups



Vaccine virus: Reference virus: 3C.2a1b Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

	10	20	30	40	50	60	70	80	90	100		
A/Hong Kong/2671/2019	QKIPGNDNSTATLCLGHAVP	NGTIVKTI	TNDRIEVT	NATELVQ	NSSIGEICDS	SPHQILDGG	NCTLIDALLGDP	QCDGFQ	NKKWDLF	VERSRAYS	NCYPY	1b
A/Antsirabe/03366/2022						K.		V.			K.	1b
A/Denmark/3264/2019												1a
A/Eskilstuna/3/2022												1a
A/Cambodia/e0826360/2020 Cell									E.		N.	2a.1
A/Hong Kong/5/2022						T.			E.		N.	2a.1
A/Norway/18896/2022		M.							E.		N.	2a.2
A/Bangladesh/4005/2020									E.		N.	2a.2
A/Austria/1514532/2022						K.		V.	E.		N.	2a.2
A/Darwin/9/2021									E.		N.	2a.2
A/Toliary/03921/2022						N.			E.		N.	2a.2
A/Romania/518142/2022						D.	N.		E.		N.S	2a.2
A/La Reunion/R22.317.39/2022						N.			E.		N.S	2a.2
A/Norway/24873/2021 Cell						N.			E.		N.S	2a.2
A/Argentina/2835/2022						N.			E.		N.S	2a.2
A/Dakar/14/2022						K.	N.		E.		N.S	2a.2
A/Ghana/1724/2022						K.	N.		E.		N.S	2a.2
A/Switzerland/03092/2022			V.			G.			D.	E.	N.	2a.2
A/Lithuania/36696/2022			V.			G.	V.		E.		N.	2a.2
A/Darwin/6/2021						G.			E.		N.	2a.2
A/Iceland/35506/2022						G.			E.		N.	2a.2
A/Belgium/S1049/2022						G.			E.		N.	2a.2
A/Bolzano/21/2022						G.			E.		N.	2a.2
A/Thuringen/7/2022						S.			D.	E.	N.	2a.2
A/Slovenia/8720/2022						G.			E.		N.	2a.2
A/Croatia/79486/2022						S.			E.		N.	2a.2
A/Slovenia/9854/2022						G.			E.		N.	2a.2
A/Spain/1129/2022						G.			E.		N.	2a.2
A/Norway/28542/2022						K.	G.		E.		N.	2a.2
A/Castilla La Mancha/2770/2022						G.			E.		N.	2a.2

	110	120	130	140	150	160	170	180	190	200			
A/Hong Kong/2671/2019	DVPDYASLRSLVASSG	TLEFKN	ESFNWAG	VTQNGKSF	SCIRGSSSS	FFSRSLNWL	THLNYTYP	ALNV	TM	PNKEQ	FDKLYIWGV	HHPTGDKDQISLYAQSSG	
A/Antsirabe/03366/2022													
A/Denmark/3264/2019					S.					D.	N.		P.
A/Eskilstuna/3/2022					S.					D.	N.	F.	RP.
A/Cambodia/e0826360/2020 Cell			T.	K.	T.	SA						F.	P.
A/Hong Kong/5/2022			T.	K.	T.	SA				N.			F.
A/Norway/18896/2022			T.	K.	T.	SA				N.			F.
A/Bangladesh/4005/2020			T.	K.	T.	SA				NI.	Q.		F.
A/Austria/1514532/2022			T.	K.	T.	SA	K.			NI.	Q.		F.
A/Darwin/9/2021			T.	K.	T.	SA				S.	NI.	Q.	F.
A/Toliary/03921/2022			T.	K.	T.	SA				S.	NI.	Q.	F.
A/Romania/518142/2022		X.		T.	K.	T.	SA	M.		S.	NI.	Q.	F.
A/La Reunion/R22.317.39/2022			I.		T.	K.	T.	SA	M.		S.	NI.	Q.
A/Norway/24873/2021 Cell			T.	K.	T.	SA				S.	NI.	Q.	F.
A/Argentina/2835/2022			T.	K.	T.	SA				S.	NI.	Q.	F.
A/Dakar/14/2022			X.		T.	K.	T.	SA		S.	NI.	Q.	F.
A/Ghana/1724/2022			T.	K.	T.	SA	R.			S.			

	310	320	330	340	350	360	370	380	390	400	
A/Hong Kong/2671/2019	TYGACPRYVKQSTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS										1b
A/Antsirabe/03366/2022	R.....										1b
A/Denmark/3264/2019											1a
A/Eskilstuna/3/2022											1a
A/Cambodia/e0826360/2020 Cell											2a.1
A/Hong Kong/5/2022											2a.1
A/Norway/18896/2022											2a.2
A/Bangladesh/4005/2020											2a.2
A/Austria/1514532/2022											2a.2
A/Darwin/9/2021											2a.2
A/Toliary/03921/2022											2a.2
A/Romania/518142/2022								S			2a.2
A/La Reunion/R22.317.39/2022								S			2a.2
A/Norway/24873/2021 Cell								S			2a.2
A/Argentina/2835/2022								S			2a.2
A/Dakar/14/2022						X		S			2a.2
A/Ghana/1724/2022								S			2a.2
A/Switzerland/03092/2022											2a.2
A/Lithuania/36696/2022											2a.2
A/Darwin/6/2021											2a.2
A/Iceland/35506/2022											2a.2
A/Belgium/S1049/2022											2a.2
A/Bolzano/21/2022											2a.2
A/Thuringen/7/2022											2a.2
A/Slovenia/8720/2022											2a.2
A/Croatia/79486/2022											2a.2
A/Slovenia/9854/2022											2a.2
A/Spain/1129/2022											2a.2
A/Norway/28542/2022											2a.2
A/Castilla La Mancha/2770/2022											2a.2

	410	420	430	440	450	460	470	480	490	500	
A/Hong Kong/2671/2019	EVEGRVQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSSEMNKLFERTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNETYDHNHYRDEALNNRF										
A/Antsirabe/03366/2022											
A/Denmark/3264/2019											
A/Eskilstuna/3/2022	I.....										
A/Cambodia/e0826360/2020 Cell											
A/Hong Kong/5/2022											
A/Norway/18896/2022											
A/Bangladesh/4005/2020											
A/Austria/1514532/2022											
A/Darwin/9/2021											
A/Toliary/03921/2022									X		
A/Romania/518142/2022											
A/La Reunion/R22.317.39/2022	I.....										
A/Norway/24873/2021 Cell											
A/Argentina/2835/2022									T		
A/Dakar/14/2022											
A/Ghana/1724/2022											
A/Switzerland/03092/2022											
A/Lithuania/36696/2022											
A/Darwin/6/2021											
A/Iceland/35506/2022								E			
A/Belgium/S1049/2022											
A/Bolzano/21/2022											
A/Thuringen/7/2022											
A/Slovenia/8720/2022											
A/Croatia/79486/2022	I.....										
A/Slovenia/9854/2022					S						
A/Spain/1129/2022											
A/Norway/28542/2022											
A/Castilla La Mancha/2770/2022											

	510	520	530	540	550	
A/Hong Kong/2671/2019	QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIWACQKGNIRCNICI					
A/Antsirabe/03366/2022	E.....			V		
A/Denmark/3264/2019						
A/Eskilstuna/3/2022						
A/Cambodia/e0826360/2020 Cell		M	I			
A/Hong Kong/5/2022		M	I			
A/Norway/18896/2022		M	I			
A/Bangladesh/4005/2020		M	I			
A/Austria/1514532/2022		M	I			
A/Darwin/9/2021		M	I			
A/Toliary/03921/2022		M	I			
A/Romania/518142/2022	R.....	M	I			
A/La Reunion/R22.317.39/2022		M	I			
A/Norway/24873/2021 Cell		M	I			
A/Argentina/2835/2022		M	I			
A/Dakar/14/2022		M	I			
A/Ghana/1724/2022		M	I			
A/Switzerland/03092/2022		M	I			
A/Lithuania/36696/2022		M	I			
A/Darwin/6/2021		M	I			
A/Iceland/35506/2022		M	I			
A/Belgium/S1049/2022		M	I			
A/Bolzano/21/2022		M	I			
A/Thuringen/7/2022		M	I			
A/Slovenia/8720/2022		M	I			
A/Croatia/79486/2022		M	I			
A/Slovenia/9854/2022		M	I			
A/Spain/1129/2022		M	L	I		
A/Norway/28542/2022		M	I			
A/Castilla La Mancha/2770/2022		M	I			

Vaccine virus: Reference virus: 3C.2a1b Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 10a. Phylogenetic comparison of A(H3N2) NA genes (WIC sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

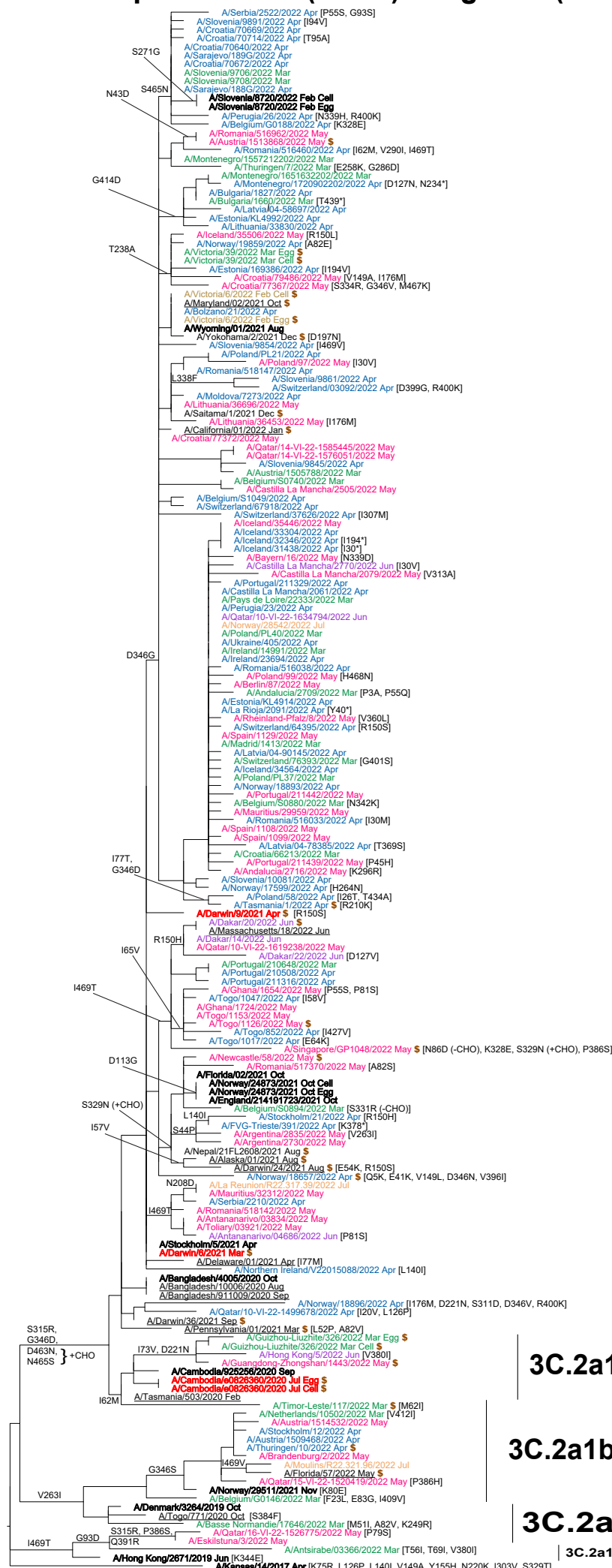
Apr 2022

May 2022

Jun 2022

Jul 2022

\$ serology antigens



3C.2a1b.2a.1

3C.2a1b.2a.2

3C.2a1b.1a

3C.2a1b.1b

3C.3a.1

Figure 10b. Phylogenetic comparison of A(H3N2) NA genes (non-WIC sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

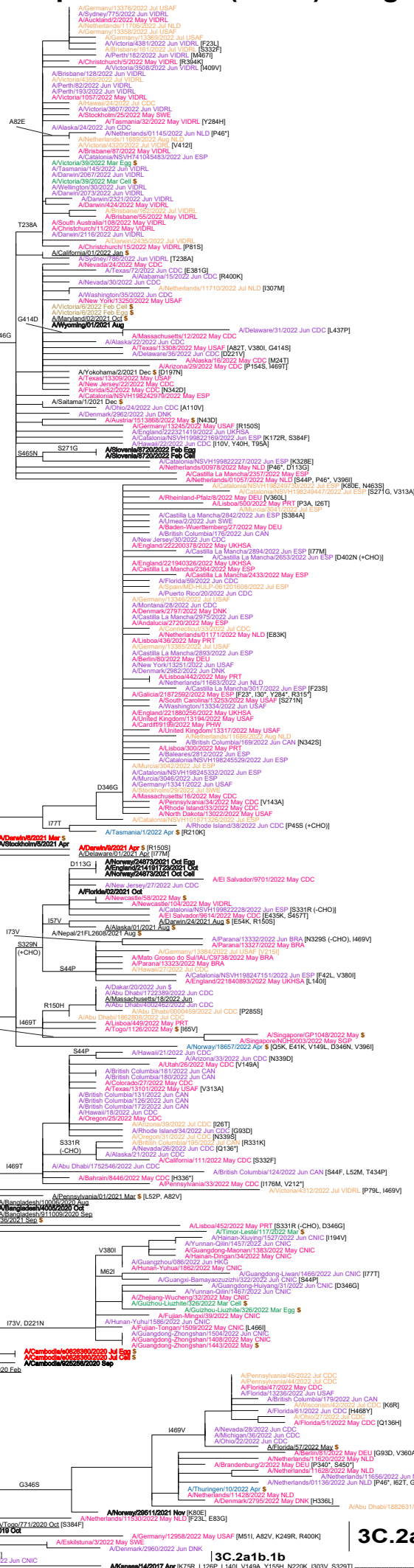
Apr 2022

May 2022

Jun 2022

Jul-Aug 2022

\$ serology antigens



3C.2a1b.2a.2

3C.2a1b.2a.1

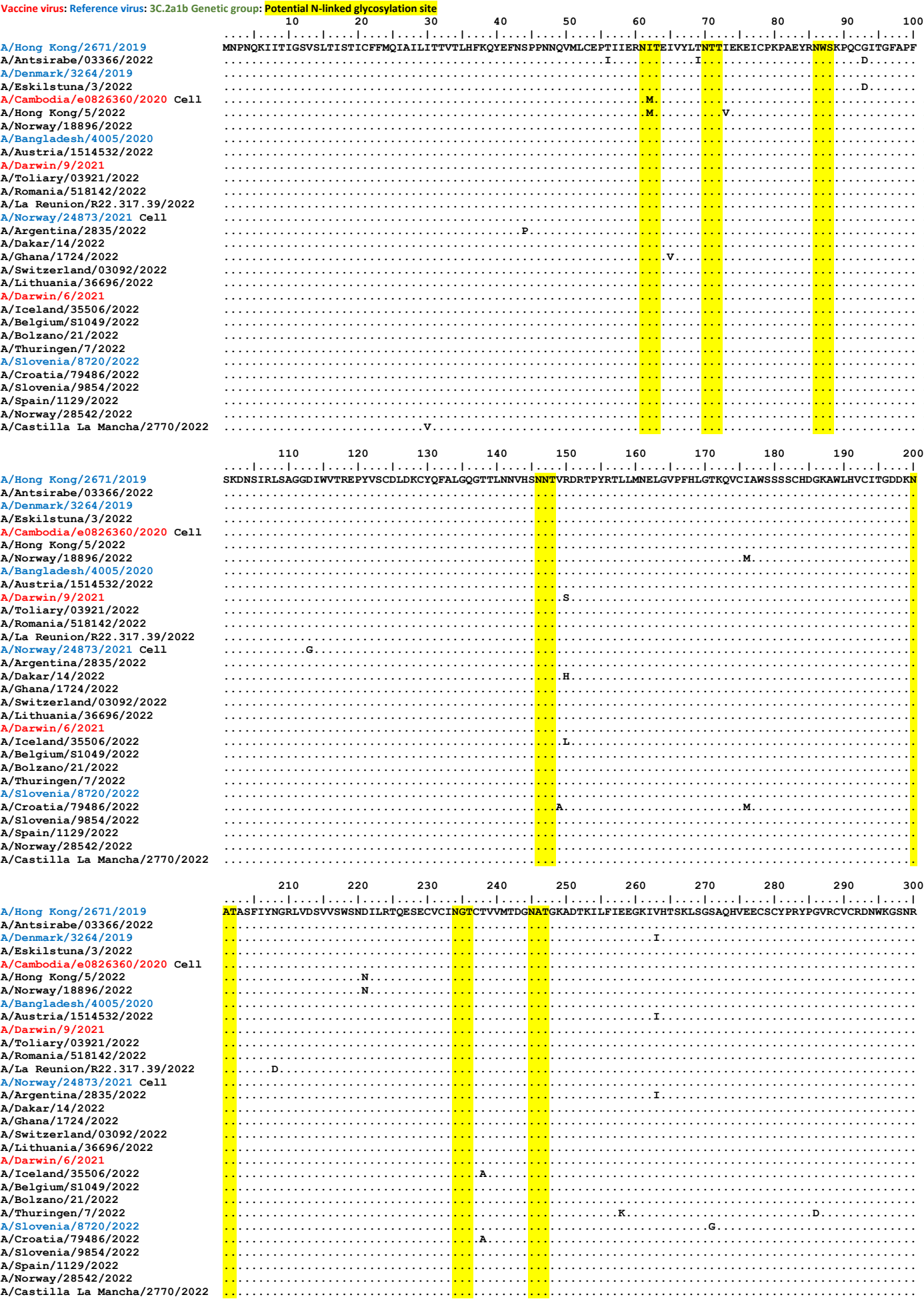
3C.2a1b.2a.2

3C.2a1b.1a

3C.2a1b.1b

3C.3a.1

Figure 11. NA protein alignment for A(H3N2) viruses



	310	320	330	340	350	360	370	380	390	400	
A/Hong Kong/2671/2019	PIIDINIKDHSIVSSIVCSGLVGDTPRKSDSSSSSHCLNPNNEEGGHGVKGWAFDDGNDVWMGR	TINET	SR	LG	YET	FKV	VEGWSNPKSKLQINRQIVDR				1b
A/Antsirabe/03366/2022				K.				I.			1b
A/Denmark/3264/2019				K.							1a
A/Eskilstuna/3/2022		R.		K.				S.	R.		1a
A/Cambodia/e0826360/2020 Cell		R.		K.D							2a.1
A/Hong Kong/5/2022				K.D				I.			2a.1
A/Norway/18896/2022	D.	R.		K.V						K	2a.2
A/Bangladesh/4005/2020		R.		K.D							2a.2
A/Austria/1514532/2022				K.S							2a.2
A/Darwin/9/2021		R.		K.D							2a.2
A/Toliary/03921/2022		R.		K.D							2a.2
A/Romania/518142/2022				K.D							2a.2
A/La Reunion/R22.317.39/2022		R.		K.D							2a.2
A/Norway/24873/2021 Cell		R.	N.	K.D							2a.2
A/Argentina/2835/2022		R.	N.	K.D							2a.2
A/Dakar/14/2022		R.		K.D							2a.2
A/Ghana/1724/2022				K.D							2a.2
A/Switzerland/03092/2022			F.	K.						GK	2a.2
A/Lithuania/36696/2022		R.		K.							2a.2
A/Darwin/6/2021		R.		K.D							2a.2
A/Iceland/35506/2022		R.		K.							2a.2
A/Belgium/S1049/2022		R.		K.							2a.2
A/Bolzano/21/2022		R.		K.							2a.2
A/Thuringen/7/2022		R.		K.							2a.2
A/Slovenia/8720/2022		R.		K.							2a.2
A/Croatia/79486/2022		R.		K.							2a.2
A/Slovenia/9854/2022		R.		K.							2a.2
A/Spain/1129/2022		R.		K.							2a.2
A/Norway/28542/2022		R.		K.							2a.2
A/Castilla La Mancha/2770/2022		R.		K.							2a.2

	410	420	430	440	450	460	
A/Hong Kong/2671/2019	GDRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTS	GTGTG	SWPDGADLNL	MHT			
A/Antsirabe/03366/2022							I
A/Denmark/3264/2019							
A/Eskilstuna/3/2022							
A/Cambodia/e0826360/2020 Cell					N.S.		I
A/Hong Kong/5/2022					N.S.		I
A/Norway/18896/2022					N.S.		I
A/Bangladesh/4005/2020					N.S.		I
A/Austria/1514532/2022							I
A/Darwin/9/2021					N.S.		I
A/Toliary/03921/2022					N.S.		
A/Romania/518142/2022					N.S.		
A/La Reunion/R22.317.39/2022					N.S.		
A/Norway/24873/2021 Cell					N.S.		I
A/Argentina/2835/2022					N.S.		I
A/Dakar/14/2022					N.S.		
A/Ghana/1724/2022					N.S.		
A/Switzerland/03092/2022					N.S.		I
A/Lithuania/36696/2022					N.S.		I
A/Darwin/6/2021					N.S.		I
A/Iceland/35506/2022					N.S.		I
A/Belgium/S1049/2022					N.S.		I
A/Bolzano/21/2022					N.S.		I
A/Thuringen/7/2022					N.S.		I
A/Slovenia/8720/2022					N.		I
A/Croatia/79486/2022					N.S.		I
A/Slovenia/9854/2022					N.S.		V
A/Spain/1129/2022					N.S.		I
A/Norway/28542/2022					N.S.		I
A/Castilla La Mancha/2770/2022					N.S.		I

Vaccine virus: Reference virus: 3C.2a1b Genetic group: Potential N-linked glycosylation site

Table 10-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-02-04

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre							
					Post-infection ferret antisera							
					A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb 925256/20 SIAT	A/Camb e0826360/20 Egg	A/Bang 4005/20 SIAT	A/Stock 5/21 SIAT	A/Darwin 9/21 Egg	A/Kansas 14/17 SIAT
					F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F38/21 ¹	F17/19 ¹
					3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT5	320	640	1280	320	320	320	640	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	1280	320	160	160	640	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	160	160	160	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	160	2560	160	160	320	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	320	640	640	1280	320
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	40	160	160	320	1280	2560	80
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	160	640	320	640	2560	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	80	80	40	80	160	640
TEST VIRUSES												
A/Wisconsin/02/2021		3C.2a1b.2a.2	2021-01-23	SIAT3/SIAT1	160	80	160	320	640	1280	2560	160
A/Delaware/01/2021		3C.2a1b.2a.2	2021-04-16	SIAT2/SIAT1	80	40	80	160	320	640	1280	80
A/Nepal/21FL2608/2021		3C.2a1b.2a.2	2021-08-10	Hck3/SIAT1	40	<	40	80	160	320	640	40
A/Nepal/21FL2699/2021		3C.2a1b.2a.2	2021-08-11	Hck3/SIAT1	80	<	80	80	160	320	1280	40
A/Alaska/01/2021		3C.2a1b.2a.2	2021-08-20	SIAT1/SIAT1	40	40	40	80	160	160	640	40
A/Belgium/H0007/2021		3C.2a1b.2a.2	2021-08-06	P1/SIAT1	80	40	40	80	160	640	1280	40
A/Belgium/H0006/2021		3C.2a1b.2a.2	2021-09-04	P1/SIAT1	40	40	40	40	80	160	640	40
A/Yaroslavl/6/2021		3C.2a1b.2a.2	2021-10-25	SIAT2/SIAT1	320	40	160	320	320	640	2560	80
A/Vladivostok/1/2021		3C.2a1b.2a.2	2021-10-31	MDCK1/SIAT1/SIAT1	160	80	320	320	320	640	1280	160
A/Vladimir/9/2021		3C.2a1b.2a.2	2021-11-01	SIAT3/SIAT1	160	40	160	320	320	640	1280	160
A/Vladivostok/2/2021		3C.2a1b.2a.2	2021-11-01	MDCK1/SIAT1/SIAT1	320	40	320	320	320	640	2560	160
A/Yaroslavl/24/2021		3C.2a1b.2a.2	2021-11-08	SIAT3/SIAT1	80	<	40	160	160	160	640	80
A/Yaroslavl/19/2021		3C.2a1b.2a.2	2021-11-08	SIAT2/SIAT1	160	40	160	320	320	640	1280	160
A/Uglich/22/2021		3C.2a1b.2a.2	2021-11-12	SIAT2/SIAT1	160	80	160	320	320	640	2560	160
A/Belgium/H0010/2021		3C.2a1b.2a.2	2021-11-15	P1/SIAT1	80	40	80	160	160	640	1280	80
A/Vladivostok/11/2021		3C.2a1b.2a.2	2021-11-16	MDCK1/SIAT1/SIAT1	160	40	80	320	320	320	640	160
A/Yaroslavl/32/2021		3C.2a1b.2a.2	2021-11-18	SIAT2/SIAT1	160	80	640	320	320	640	1280	320
A/Belgium/G0127/2021		3C.2a1b.2a.2	2021-11-22	P1/SIAT1	40	<	80	80	160	160	640	40
A/Vladivostok/12/2021		3C.2a1b.2a.2	2021-11-23	MDCK1/SIAT1/SIAT1	160	40	80	320	320	320	640	80
A/Velikiy Novgorod/46/2021		3C.2a1b.2a.2	2021-11-24	SIAT2/SIAT1	160	80	160	320	320	640	1280	80
A/Orenburg/44/2021		3C.2a1b.2a.2	2021-11-24	SIAT3/SIAT1	160	80	160	320	320	640	2560	160
A/Vladivostok/14/2021		3C.2a1b.2a.2	2021-11-26	MDCK1/SIAT1/SIAT1	160	40	80	320	320	320	640	80
A/Vladimir/40/2021		3C.2a1b.2a.2	2021-11-29	SIAT2/SIAT1	160	80	160	320	320	640	1280	160
A/Orenburg/41/2021		3C.2a1b.2a.2	2021-11-30	SIAT2/SIAT1	160	40	80	320	320	320	640	160
A/Moscow/38/2021		3C.2a1b.2a.2	2021-11-30	SIAT2/SIAT1	160	80	160	320	320	640	1280	160
A/Orenburg/45/2021		3C.2a1b.2a.2	2021-12-01	SIAT3/SIAT1	160	80	160	320	320	640	1280	160
A/Moscow/43/2021		3C.2a1b.2a.2	2021-12-06	SIAT2/SIAT1	160	80	160	320	320	640	1280	160
A/Velikiy Novgorod/49/2021		3C.2a1b.2a.2	2021-12-08	SIAT2/SIAT1	160	80	160	320	320	640	1280	80
A/Mahajanga/13255/2021		3C.2a1b.2a.2	2021-12-09	SIAT2	320	80	160	640	320	1280	2560	160
A/Spain/89/2021		3C.2a1b.2a.2	2021-12-22	SIAT1	80	40	80	80	160	320	1280	40
A/Spain/87/2021		3C.2a1b.2a.2	2021-12-23	SIAT1	40	40	80	80	160	320	640	40
A/Spain/84/2021		3C.2a1b.2a.2	2021-12-25	SIAT1	40	<	80	80	160	320	640	40
A/Spain/83/2021		3C.2a1b.2a.2	2021-12-26	SIAT1	40	<	40	80	160	320	640	40
A/Spain/82/2021		3C.2a1b.2a.2	2021-12-27	SIAT1	<	<	80	80	160	320	1280	40
A/Spain/81/2021		3C.2a1b.2a.2	2021-12-28	SIAT1	<	40	80	80	80	160	640	40
A/Lebanon/8515/2021		3C.2a1b.2a.2	2021-12-29	SIAT1	80	40	40	80	160	320	1280	40
A/Lebanon/8510/2021		3C.2a1b.2a.2	2021-12-29	SIAT1	80	40	40	80	160	640	1280	40
A/Lebanon/8505/2021		3C.2a1b.2a.2	2021-12-29	SIAT1	80	40	40	80	160	640	1280	40
A/Lebanon/8500/2021		3C.2a1b.2a.2	2021-12-29	SIAT1	80	40	80	80	160	640	1280	80
A/Lebanon/8497/2021		3C.2a1b.2a.2	2021-12-29	SIAT1	80	40	40	80	160	320	1280	40
A/Estonia/168139/2022		3C.2a1b.2a.2	2022-01-03	SIAT1/SIAT1	160	40	80	160	320	640	1280	80
A/Estonia/168142/2022		3C.2a1b.2a.2	2022-01-05	SIAT1/SIAT1	80	40	40	80	160	320	1280	40
A/Estonia/168141/2022		3C.2a1b.2a.2	2022-01-05	SIAT1/SIAT1	80	40	40	80	160	320	640	40
A/Estonia/168131/2022		3C.2a1b.2a.2	2022-01-06	MDCKx/SIAT1	80	40	80	160	160	320	640	80
A/Armenia/19/2022		3C.2a1b.1b	2022-01-05	SIAT2	320	640	640	80	80	80	160	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-02-11

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb 925256/20 SIAT	A/Camb e0826360/20 Egg	A/Bang 4005/20 SIAT	A/Stock 5/21 SIAT	A/Darwin 9/21 Egg	A/Kansas 14/17 SIAT
				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F38/21 ¹	F17/19 ¹
	Passage history			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
	Ferret number										
	Genetic group										
REFERENCE VIRUSES											
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	640	1280	320	320	640	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	160	320
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	160	160	160	320
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	160	2560	320	640	160
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	320	640	1280	320
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	40	160	160	320	1280	80
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	160	320	640	5120	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	160	80	80	160	640
TEST VIRUSES											
A/Ireland/92285/2021		3C.2a1b.2a.2	2021-10-21	SIAT1	160	40	160	640	640	1280	320
A/Ireland/2707/2021		3C.2a1b.2a.2	2021-12-05	SIAT1	40	<	80	80	160	640	40
A/Berlin/4/2021		3C.2a1b.2a.2	2021-12-16	P2/SIAT1	160	<	80	80	320	640	40
A/Ireland/6979/2021		3C.2a1b.2a.2	2021-12-20	SIAT1	80	<	80	80	160	640	40
A/Nordrhein-Westfalen/1/2021		3C.2a1b.2a.2	2021-12-20	P1/SIAT1	1280	320	320	1280	640	2560	320
A/Rheinland-Pfalz/2/2021		3C.2a1b.2a.2	2021-12-22	P1/SIAT1	160	<	80	80	320	640	40
A/Nordrhein-Westfalen/2/2021		3C.2a1b.2a.2	2021-12-22	P1/SIAT1	80	<	40	80	160	160	40
A/Berlin/5/2021		3C.2a1b.2a.2	2021-12-23	P1/SIAT1	80	<	40	80	160	320	40
A/Baden-Wuerttemberg/2/2021		3C.2a1b.2a.2	2021-12-27	P1/SIAT1	160	80	160	160	320	640	160
A/Nordrhein-Westfalen/3/2021		3C.2a1b.2a.2	2021-12-28	P1/SIAT1	80	<	80	80	160	640	40
A/Berlin/6/2021		3C.2a1b.2a.2	2021-12-29	P1/SIAT1	80	<	80	80	160	640	40
A/Berlin/7/2021		3C.2a1b.2a.2	2021-12-31	P1/SIAT1	160	40	80	80	320	640	40
A/Saarland/1/2022		3C.2a1b.2a.2	2022-01-03	P1/SIAT1	80	<	80	80	160	640	40
A/Thuringen/1/2022		3C.2a1b.2a.2	2022-01-03	P1/SIAT1	160	40	80	80	320	640	40
A/Kyiv/41/2022		3C.2a1b.2a.2	2022-01-04	SIATx/SIAT1	160	80	160	320	640	2560	160
A/Norway/409/2022		3C.2a1b.2a.2	2022-01-04	SIAT1	40	<	40	40	80	320	<
A/Hessen/1/2022		3C.2a1b.2a.2	2022-01-04	P1/SIAT1	80	<	80	80	160	640	40
A/Kyiv/7/2022		3C.2a1b.2a.2	2022-01-05	SIATx/SIAT1	160	40	80	160	320	640	40
A/Kyiv/5/2022		3C.2a1b.2a.2	2022-01-05	SIATx/SIAT1	160	40	80	160	320	640	40
A/Kyiv/4/2022		3C.2a1b.2a.2	2022-01-05	SIATx/SIAT1	80	<	80	80	320	640	40
A/Kyiv/3/2022		3C.2a1b.2a.2	2022-01-05	SIATx/SIAT1	160	<	80	160	320	640	40
A/Kyiv/1/2022		3C.2a1b.2a.2	2022-01-05	SIATx/SIAT1	160	<	80	160	320	640	40
A/Norway/530/2022		3C.2a1b.2a.2	2022-01-06	SIAT1	80	<	40	80	160	320	<
A/Norway/676/2022		3C.2a1b.2a.2	2022-01-11	SIAT1	320	160	160	320	640	320	320
A/Serbia/206/2022		3C.2a1b.2a.2	2022-01-13	SIAT1	160	40	80	160	320	640	40
A/Serbia/416478/2022		3C.2a1b.2a.2	2022-01-17	SIAT1	160	40	80	160	320	640	40
A/Serbia/416476/2022		3C.2a1b.2a.2	2022-01-17	SIAT1	160	40	80	160	320	640	40
A/Serbia/416473/2022		3C.2a1b.2a.2	2022-01-17	SIAT1	80	<	80	160	320	640	40
A/Serbia/336/2022		3C.2a1b.2a.2	2022-01-17	SIAT1	160	40	80	80	320	640	40
A/South Africa/R14962/2021		3C.2a1b.1a	2021-10-12	MDCK3/MDCK2/SIAT1	320	160	160	640	320	320	160

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

 < 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-02-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre							
				Post-infection ferret antisera							
				A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb 925256/20 SIAT	A/Camb e0826360/20 Egg	A/Bang 4005/20 SIAT	A/Stock 5/21 SIAT	A/Darwin 9/21 Egg	A/Kansas 14/17 SIAT
				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F38/21 ¹	F17/19 ¹
	Passage history										
	Ferret number										
	Genetic group										
REFERENCE VIRUSES											
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	160	320	160	320	160
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	640	160	320	160	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	320	320	320	640	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	40	160	1280	160	320	640	80
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	320	80	160	320	640	640	1280	320
A/Stockholm/5/2021	H156S	2021-04-16	S0/S3	80	<	80	160	320	640	2560	80
A/Darwin/9/2021	H156S	2021-04-17	E3/E2	160	<	160	320	320	640	2560	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	160	320
TEST VIRUSES											
A/Akkar.LBN/6574/2021	3C.2a1b.2a.2	2021-10-21	SIAT1	40	<	40	40	160	320	640	<
A/Zghorta.LBN/6633/2021	3C.2a1b.2a.2	2021-10-26	SIAT1	40	<	40	80	160	320	640	40
A/Akkar.LBN/6728/2021	3C.2a1b.2a.2	2021-10-28	SIAT1	80	<	40	80	320	640	1280	<
A/Minieh Dinieh.LBN/6926/2021	3C.2a1b.2a.2	2021-11-09	SIAT1	80	<	40	80	320	640	1280	<
A/Zahleh.LBN/6895/2021	3C.2a1b.2a.2	2021-11-09	SIAT1	80	<	80	80	320	640	1280	40
A/Seeb/72128338/2021	3C.2a1b.2a.2	2021-11-09	SIAT2	40	<	40	40	320	640	1280	<
A/Bazoureyeh.LBN/7037/2021	3C.2a1b.2a.2	2021-11-11	SIAT1	80	<	40	80	160	320	1280	<
A/Zahleh.LBN/7063/2021	3C.2a1b.2a.2	2021-11-16	SIAT1	80	<	40	40	160	640	1280	<
A/Zahleh.LBN/7060/2021	3C.2a1b.2a.2	2021-11-16	SIAT1	80	<	40	40	160	640	1280	<
A/Zahleh.LBN/7292/2021	3C.2a1b.2a.2	2021-11-23	SIAT1	40	<	40	40	160	320	640	<
A/Ireland/4594/2021	3C.2a1b.2a.2	2021-12-12	SIAT2	40	<	40	40	160	320	640	<
A/BuenosAires/R1/2022	3C.2a1b.2a.2	2021-12-13	MDCKx/SIAT1	40	<	40	80	160	320	640	<
A/Latvia/12-045953/2021	3C.2a1b.2a.2	2021-12-14	P2/SIAT1	40	<	40	80	160	320	640	<
A/Switzerland/98693/2021	3C.2a1b.2a.2	2021-12-21	SIAT1/SIAT1	80	<	40	80	320	640	1280	40
A/Latvia/12-091326/2021	3C.2a1b.2a.2	2021-12-24	P2/SIAT1	160	<	80	160	320	640	1280	40
A/Latvia/12-091325/2021	3C.2a1b.2a.2	2021-12-24	P2/SIAT1	320	40	160	640	640	1280	2560	160
A/Latvia/12-091317/2021	3C.2a1b.2a.2	2021-12-26	P2/SIAT1	160	<	40	160	320	320	1280	40
A/Latvia/12-097744/2021	3C.2a1b.2a.2	2021-12-28	P2/SIAT1	40	<	40	40	80	160	640	<
A/BuenosAires/R12/2022	3C.2a1b.2a.2	2022-01-04	MDCKx/SIAT1	40	<	40	40	160	320	640	<
A/BuenosAires/R11/2022	3C.2a1b.2a.2	2022-01-05	MDCKx/SIAT1	40	<	80	80	160	320	640	<
A/Latvia/01-036169/2022	3C.2a1b.2a.2	2022-01-08	P2/SIAT1	160	<	160	320	320	640	1280	80
A/BuenosAires/R15/2022	3C.2a1b.2a.2	2022-01-10	MDCKx/SIAT1	40	<	40	80	160	320	640	<
A/Norway/675/2022	3C.2a1b.2a.2	2022-01-11	SIAT2	160	<	80	320	320	640	1280	80
A/BuenosAires/R2/2022	3C.2a1b.2a.2	2022-01-13	MDCKx/SIAT1	40	<	40	40	160	320	640	<
A/Latvia/01-089648/2022	3C.2a1b.2a.2		P2/SIAT1	40	<	40	40	160	320	640	<
A/Norway/31262/2021	3C.2a1b.1a	2021-12-02	SIAT2	160	80	320	80	80	40	80	40

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
 4-fold
 8-fold
 > 8-fold
 not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-03-18

				Haemagglutination inhibition titre									
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
Passage history													
Ferret number				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
Genetic group				3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	640	640	640	160	160	320	40	320	80
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	640	160	320	160	40	320	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	160	160	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	320	40	320	2560	320	320	640	1280	160
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	320	80	320	320	640	640	640	2560	320
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	160	40	160	80	320	640	640	2560	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	80	40	160	320	640	1280	<
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	320	40	160	320	320	640	640	5120	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	80	80	80	160	80	160	640
TEST VIRUSES													
A/Qatar/15-VI-21-4138232/2021		3C.2a1b.2a.2	2021-08-17	SIAT1	160	<	80	320	320	320	320	640	80
A/Qatar/10-VI-21-4396149/2021		3C.2a1b.2a.2	2021-08-26	SIAT1	160	<	80	160	320	640	1280	2560	40
A/Qatar/47-VI-21-4486313/2021		3C.2a1b.2a.2	2021-08-29	SIAT1	160	<	80	160	320	640	1280	2560	40
A/Qatar/32-VI-21-4519262/2021		3C.2a1b.2a.2	2021-08-30	SIAT1	160	<	80	320	320	320	320	640	80
A/Andalucia/9561/2021		3C.2a1b.2a.2	2021-10-08	SIAT1/SIAT1	320	40	160	320	640	320	640	1280	160
A/Norway/23748/2021		3C.2a1b.2a.2	2021-10-14	SIAT1	160	<	80	320	320	320	320	640	80
A/Norway/23662/2021		3C.2a1b.2a.2	2021-10-14	SIAT1	160	<	80	320	320	320	320	640	80
A/Norway/24121/2021		3C.2a1b.2a.2	2021-10-15	SIAT1	160	<	80	320	320	320	320	640	80
A/Cameroon/7554/2021		3C.2a1b.2a.2	2021-10-15	SIAT2	160	40	160	320	640	320	320	640	160
A/Norway/24931/2021		3C.2a1b.2a.2	2021-10-24	SIAT1	40	<	40	<	80	160	320	320	<
A/Tajikistan/2/2021		3C.2a1b.2a.2	2021-10-26	SIAT1	160	<	80	160	320	640	320	1280	40
A/Nabatieh.LBN/6814/2021		3C.2a1b.2a.2	2021-11-03	SIAT1	40	<	40	40	160	320	320	1280	<
A/Cameroon/8639/2021		3C.2a1b.2a.2	2021-11-04	SIAT1	320	40	160	640	640	640	640	1280	320
A/Andalucia/9929/2021		3C.2a1b.2a.2	2021-11-11	SIAT1/SIAT1	40	<	80	80	160	320	640	640	<
A/Tripoli.LBN/7180/2021		3C.2a1b.2a.2	2021-11-16	SIAT1	80	<	40	40	160	640	320	1280	<
A/Castilla La Mancha/9755/2021		3C.2a1b.2a.2	2021-11-16	SIAT1	40	<	40	40	80	160	320	640	<
A/Ksara.LBN/7293/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	80	40	160	320	320	1280	<
A/Baalbeck.LBN/7288/2021		3C.2a1b.2a.2	2021-11-23	SIAT1	80	<	40	40	160	320	320	1280	<
A/Andalucia/9928/2021		3C.2a1b.2a.2	2021-11-23	SIAT1/SIAT1	40	<	40	40	160	320	640	640	<
A/Navarra/9875/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	40	<	40	40	80	320	640	640	<
A/Navarra/9843/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	80	<	80	80	160	640	640	2560	40
A/Mahajanga/12570/2021		3C.2a1b.2a.2	2021-11-24	SIAT1	320	160	640	640	640	640	640	1280	320
A/Norway/29512/2021		3C.2a1b.2a.2	2021-11-25	SIAT1	80	<	80	40	160	320	640	1280	<
A/Mahajanga/12604/2021		3C.2a1b.2a.2	2021-11-25	SIAT1	320	<	80	320	640	320	640	640	80
A/Moscow/53/2021		3C.2a1b.2a.2	2021-12-06	SIAT2/SIAT2	160	40	160	320	320	640	640	2560	80
A/Switzerland/02353/2021		3C.2a1b.2a.2	2021-12-07	SIAT1	160	<	80	320	320	320	640	640	80
A/Spain/58/2021		3C.2a1b.2a.2	2021-12-14	SIAT1	40	<	40	40	160	160	640	640	<
A/Belgium/H0024/2022		3C.2a1b.2a.2	2021-12-14	SIAT1/SIAT1	160	<	80	320	320	320	160	640	80
A/Portugal/58/2021		3C.2a1b.2a.2	2021-12-15	SIAT1	40	<	40	40	160	160	640	640	<
A/Portugal/57/2021		3C.2a1b.2a.2	2021-12-15	SIAT1	160	40	80	320	320	640	640	1280	80
A/Spain/90/2021		3C.2a1b.2a.2	2021-12-21	SIAT1	80	<	40	40	160	160	160	640	<
A/Spain/88/2021		3C.2a1b.2a.2	2021-12-22	SIAT1	40	<	40	40	80	160	320	640	<
A/Armenia/869/2021		3C.2a1b.2a.2	2021-12-23	SIAT1	40	40	80	40	160	160	640	640	<
A/Switzerland/79079/2021		3C.2a1b.2a.2	2021-12-27	SIAT1/SIAT1	40	<	40	40	160	320	640	640	<
A/Switzerland/87473/2021		3C.2a1b.2a.2	2021-12-28	SIAT1/SIAT1	80	<	40	40	160	640	320	1280	<
A/Switzerland/12229/2022		3C.2a1b.2a.2	2022-01-02	SIAT1	80	<	40	80	160	160	320	640	<
A/Switzerland/42321/2021		3C.2a1b.2a.2	2022-01-03	MDCK1/SIAT1	320	40	160	320	320	1280	640	2560	80
A/Switzerland/28536/2022		3C.2a1b.2a.2	2022-01-03	SIAT1	160	40	80	320	320	320	320	640	80
A/Switzerland/44070/2022		3C.2a1b.2a.2	2022-01-04	SIAT1	40	<	40	40	160	160	640	640	<
A/Belgium/H0026/2022		3C.2a1b.1a	2021-12-10	SIAT1/SIAT1	40	40	160	40	80	40	<	80	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-04-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
Ferret number	F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹				
Genetic group	3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1				
REFERENCE VIRUSES													
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	160	160	160	40	320	80	
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	160	160	40	320	160	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT6	160	160	640	160	160	160	<	320	160	
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	320	1280	320	320	320	640	160	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	160	320	320	640	640	1280	320	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	160	<	80	160	320	640	320	80	
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	80	<	80	80	160	320	640	40	
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	80	640	320	640	640	80	
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	80	80	40	80	80	160	640	
TEST VIRUSES													
A/Estonia/167744/2021	3C.2a1b.2a.2	2021-12-08	SIAT1	160	<	80	320	160	640	640	1280	80	
A/Lebanon/8137/2021	3C.2a1b.2a.2	2021-12-14	SIAT1	40	<	40	80	160	320	320	640	40	
A/Lebanon/8074/2021	3C.2a1b.2a.2	2021-12-14	SIAT1	80	<	40	80	160	320	320	640	40	
A/Israel/R-9278/2021	3C.2a1b.2a.2	2021-12-15	SIAT1	80	<	40	80	160	320	320	1280	40	
A/Israel/R-9337/2021	3C.2a1b.2a.2	2021-12-17	SIAT1	40	<	40	80	160	320	320	640	40	
A/Israel/R-9526/2021	3C.2a1b.2a.2	2021-12-21	SIAT1	320	40	160	320	320	640	320	1280	160	
A/Lebanon/8276/2021	3C.2a1b.2a.2	2021-12-21	SIAT1	80	<	40	80	160	320	320	1280	40	
A/Israel/R-9603/2021	3C.2a1b.2a.2	2021-12-22	SIAT1	80	<	40	80	160	320	320	1280	40	
A/Lebanon/8348/2021	3C.2a1b.2a.2	2021-12-22	SIAT1	80	<	40	80	160	320	320	1280	40	
A/Lebanon/8325/2021	3C.2a1b.2a.2	2021-12-22	SIAT1	80	<	40	80	160	320	320	1280	40	
A/Montenegro/5968412102/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	80	<	40	160	160	640	320	1280	40	
A/Israel/R-9997/2021	3C.2a1b.2a.2	2021-12-28	SIAT1	40	<	40	80	160	160	640	640	40	
A/Lebanon/8484/2021	3C.2a1b.2a.2	2021-12-29	SIAT1	40	<	40	80	160	320	320	640	40	
A/Montenegro/1082622202/2022	3C.2a1b.2a.2	2022-01-10	SIAT1	80	<	40	160	160	640	320	1280	80	
A/Belgium/H0013/2022	3C.2a1b.2a.2	2022-01-17	SIAT1	<	<	40	80	160	320	640	640	40	
A/Belgium/H0008/2022	3C.2a1b.2a.2	2022-01-17	SIAT1	80	<	40	80	80	160	160	320	40	
A/Rouen/41245/2021	3C.2a1b.1a	2021-11-15	SIAT1	40	80	80	<	40	<	<	40	40	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-04-08

Haemagglutination inhibition titre												
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera								
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT
	Passage history			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹
	Ferret number			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	640	640	640	320	160	320	80	640	320
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	160	160	40	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	640	160	160	160	40	160	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	80	1280	80	320	160	320	80
A/Bangladesh/4005/2020	H156 3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	160	320	640	320	1280	160
A/Stockholm/5/2021	H156S 3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	160	640	320	1280	40
A/England/214191723/2021	H156S 3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	40	440	80	320	320	640	<
A/Darwin/9/2021	H156S 3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	80	320	160	640	640	2560	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	40	40	40	80	40	160	320
TEST VIRUSES												
A/Netherlands/10018/2021	3C.2a1b.2a.2		MDCK-MIX2/SIAT1	40	<	40	80	80	160	640	640	40
A/Netherlands/10011/2021	3C.2a1b.2a.2	2021-09-20	MDCK-MIX2/SIAT1	40	<	40	40	80	160	320	640	<
A/Netherlands/10012/2021	3C.2a1b.2a.2	2021-09-30	MDCK-MIX1/SIAT1	40	<	40	80	80	320	640	1280	<
A/Bremen/1/2021	3C.2a1b.2a.2	2021-10-11	P1/SIAT2	40	160	80	80	80	320	640	640	40
A/Netherlands/10062/2021	3C.2a1b.2a.2	2021-10-21	MDCK-MIX2/SIAT1	160	<	80	160	160	320	320	640	80
A/Netherlands/10057/2021	3C.2a1b.2a.2	2021-10-22	MDCK-MIX2/SIAT2	40	<	40	80	80	320	640	640	<
A/Netherlands/10063/2021	3C.2a1b.2a.2	2021-10-23	MDCK-MIX2/SIAT1	40	<	40	40	40	160	320	320	<
A/Berlin/2/2021	3C.2a1b.2a.2	2021-10-25	P1/SIAT1	80	160	80	160	160	320	640	1280	80
A/Schleswig-Holstein/1/2021	3C.2a1b.2a.2	2021-11-15	P1/SIAT1	160	160	80	320	160	320	160	640	80
A/Berlin/3/2021	3C.2a1b.2a.2	2021-11-16	P1/SIAT1	40	160	40	80	80	320	640	640	<
A/Hessen/1/2021	3C.2a1b.2a.2	2021-11-22	P1/SIAT1	40	160	<	40	80	320	320	1280	<
A/Baden-Wuerttemberg/1/2021	3C.2a1b.2a.2	2021-11-26	P1/SIAT1	40	160	<	40	80	160	160	640	<
A/Slovenia/19583/2021	3C.2a1b.2a.2	2021-11-27	MDCKx/SIAT1	160	<	80	160	320	640	640	1280	80
A/Kyiv/393/2021	3C.2a1b.2a.2	2021-12-13	SIAT1	40	160	40	80	160	320	640	640	40
A/Rheinland-Pfalz/1/2021	3C.2a1b.2a.2	2021-12-16	P1/SIAT1	40	160	<	40	80	320	320	640	40
A/Slovenia/679/2022	3C.2a1b.2a.2	2021-12-30	MDCKx/SIAT1	160	<	80	160	160	640	640	1280	80
A/Slovenia/1054/2022	3C.2a1b.2a.2	2022-01-05	MDCKx/SIAT1	160	<	80	320	320	640	640	1280	80
A/Eph Blida/97875/2021	3C.2a1b.1a	2021-12-21	SIAT1	80	160	80	80	80	160	<	320	80
A/Slovenia/8821/2022	D104G 3C.2a1b.2a.2	2022-02-02	SIAT1/SIAT1	<	<	40	80	160	320	320	1280	<
A/Slovenia/8830/2022	D104G 3C.2a1b.2a.2	2022-02-03	MDCK1/SIAT1	<	<	40	80	160	320	320	640	40
A/Slovenia/8828/2022	D104G 3C.2a1b.2a.2	2022-02-03	SIAT1/SIAT1	<	<	40	80	80	160	320	640	<
A/Slovenia/8720/2022	D104G 3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT1	<	<	<	40	80	320	320	640	<
A/Slovenia/8723/2022	D104G 3C.2a1b.2a.2	2022-02-11	MDCK1/SIAT1	<	<	<	40	80	320	320	640	<
A/Slovenia/8799/2022	D104G 3C.2a1b.2a.2	2022-02-15	SIAT1/SIAT1	<	<	40	40	160	320	320	1280	<
A/Slovenia/9091/2022	D104G 3C.2a1b.2a.2	2022-02-22	MDCKx/SIAT1	<	<	<	<	80	160	160	640	<
A/Slovenia/9090/2022	D104G 3C.2a1b.2a.2	2022-02-22	MDCKx/SIAT1	<	<	<	40	160	320	320	1280	<
A/Slovenia/9146/2022	D104G 3C.2a1b.2a.2	2022-02-23	SIAT1/SIAT1	<	<	<	40	160	320	320	640	<
A/Slovenia/9216/2022	D104G 3C.2a1b.2a.2	2022-02-24	MDCKx/SIAT1	<	<	40	80	160	320	320	640	<
A/Slovenia/9318/2022	D104G 3C.2a1b.2a.2	2022-03-02	SIATx/SIAT1	<	<	40	40	160	320	320	1280	<
A/Slovenia/9356/2022	D104G 3C.2a1b.2a.2	2022-03-03	SIATx/SIAT1	40	160	40	40	80	320	320	1280	<
A/Slovenia/8965/2022	D53G 3C.2a1b.2a.2	2022-02-21	MDCK1/SIAT3	80	<	40	160	160	320	320	640	<
A/Slovenia/9302/2022	D53G 3C.2a1b.2a.2	2022-02-28	MDCKx/SIAT1	<	<	40	160	320	640	640	1280	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-04-13

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT
	Ferret number			F19/20 ¹	St Judes F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	160	160	160	40	320	80
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	160	160	40	320	160
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT6	160	160	320	160	160	160	<	320	160
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	320	1280	320	320	320	640	160
A/Bangladesh/4005/2020	H156 3C.2a1b.2a.2	2020-10-04	SIAT3	160	160	320	320	640	640	640	1280	320
A/Stockholm/5/2021	H156S 3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	160	<	80	160	320	640	320	1280	80
A/England/214191723/2021	H156S 3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	80	<	80	80	160	320	640	1280	40
A/Darwin/9/2021	H156S 3C.2a1b.2a.2	2021-04-17	E3/E2	160	40	80	640	320	640	640	2560	80
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	80	80	40	80	80	160	640
TEST VIRUSES												
A/Serbia/5/2021	3C.2a1b.2a.2	2021-12-31	SIAT1	80	<	40	160	160	320	320	1280	<
A/Serbia/91/2022	3C.2a1b.2a.2	2022-01-06	SIAT1	40	<	40	40	160	320	320	1280	<
A/Serbia/112/2022	3C.2a1b.2a.2	2022-01-06	SIAT1	80	<	40	80	160	320	320	1280	<
A/Serbia/199/2022	3C.2a1b.2a.2	2022-01-10	SIAT1	80	<	40	80	160	320	320	1280	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-05-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
	Ferret number			F19/20 ¹	St Jude's	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	320	160	160	<	320	80	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	640	160	160	<	160	160	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	80	160	640	160	160	<	320	160	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	160	1280	320	160	320	80	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	80	320	320	640	640	640	1280	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	1280	40
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	80	<	80	80	160	320	640	640	<
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	160	640	640	640	2560	80	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	80	80	80	80	160	320	
TEST VIRUSES													
A/Netherlands/10066/2021		3C.2a1b.2a.2	2021-10-20	MDCK-MIX2/SIAT1	<	<	40	40	80	80	320	320	<
A/Netherlands/10067/2021		3C.2a1b.2a.2	2021-10-21	MDCK-MIX2/SIAT2	40	<	80	80	160	320	640	640	<
A/Netherlands/10122/2021		3C.2a1b.2a.2	2021-10-26	MDCK-MIX2/SIAT2	80	<	40	160	160	160	640	640	40
A/Netherlands/10065/2021		3C.2a1b.2a.2	2021-10-29	MDCK-MIX2/SIAT1	40	<	80	80	160	160	640	640	<
A/Netherlands/10121/2021		3C.2a1b.2a.2	2021-10-31	MDCK-MIX2/SIAT2	40	<	40	80	160	160	640	640	<
A/Netherlands/10109/2021		3C.2a1b.2a.2	2021-10-31	MDCK-MIX2/SIAT1	<	<	40	80	160	160	320	320	<
A/Netherlands/10108/2021		3C.2a1b.2a.2	2021-11-07	MDCK-MIX2/SIAT2	40	<	80	80	160	320	640	640	<
A/Netherlands/10128/2021		3C.2a1b.2a.2	2021-11-08	MDCK-MIX2/SIAT1	<	<	40	80	80	160	640	640	<
A/Netherlands/10119/2021		3C.2a1b.2a.2	2021-11-09	MDCK-MIX2/SIAT1	<	<	40	40	80	160	320	640	<
A/Netherlands/10114/2021		3C.2a1b.2a.2	2021-11-09	MDCK-MIX2/SIAT1	40	<	40	80	160	320	640	1280	<
A/Netherlands/10129/2021		3C.2a1b.2a.2	2021-11-10	MDCK-MIX2/SIAT1	40	<	80	160	160	320	640	1280	40
A/Netherlands/10138/2021		3C.2a1b.2a.2	2021-11-15	MDCK-MIX2/SIAT1	80	<	40	80	80	160	640	640	<
A/Netherlands/10137/2021		3C.2a1b.2a.2	2021-11-16	MDCK-MIX2/SIAT1	40	<	40	80	160	160	640	640	<
A/Netherlands/10132/2021		3C.2a1b.2a.2	2021-11-16	MDCK-MIX2/SIAT1	40	<	40	80	160	160	640	640	<
A/Netherlands/10145/2021		3C.2a1b.2a.2	2021-11-17	MDCK-MIX2/SIAT1	40	<	80	80	160	160	640	640	40
A/Netherlands/10144/2021		3C.2a1b.2a.2	2021-11-17	MDCK-MIX2/SIAT1	40	<	80	80	160	160	640	640	<
A/Netherlands/10140/2021		3C.2a1b.2a.2	2021-11-17	MDCK-MIX2/SIAT1	40	<	<	80	160	160	640	640	<
A/Netherlands/10148/2021		3C.2a1b.2a.2	2021-11-18	MDCK-MIX2/SIAT1	40	<	80	80	160	320	640	640	<
A/Netherlands/10147/2021		3C.2a1b.2a.2	2021-11-20	MDCK-MIX2/SIAT2	40	<	80	160	320	320	1280	1280	40
A/Netherlands/10150/2021		3C.2a1b.2a.2	2021-11-23	MDCK-MIX2/SIAT1	40	<	80	80	160	160	640	640	<
A/Netherlands/10151/2021		3C.2a1b.2a.2	2021-11-24	MDCK-MIX2/SIAT1	40	40	80	80	160	160	640	640	<
A/Netherlands/10154/2021		3C.2a1b.2a.2	2021-11-25	MDCK-MIX2/SIAT1	<	<	40	80	160	160	640	640	<
A/Kosova/778/2021		3C.2a1b.2a.2	2021-11-29	SIAT1	160	<	80	320	320	640	640	1280	80
A/Netherlands/10180/2021		3C.2a1b.2a.2	2021-12-02	MDCK-MIX2/SIAT1	<	<	<	40	160	160	640	640	<
A/Netherlands/10182/2021		3C.2a1b.2a.2	2021-12-03	MDCK-MIX2/SIAT1	80	<	80	320	160	320	320	640	40
A/Kosova/114/2021		3C.2a1b.2a.2	2021-12-03	SIAT1	80	<	40	160	160	320	320	640	40
A/Netherlands/10202/2021		3C.2a1b.2a.2	2021-12-06	MDCK-MIX2/SIAT1	<	<	40	80	160	160	320	640	<
A/Netherlands/10184/2021		3C.2a1b.2a.2	2021-12-08	MDCK-MIX2/SIAT1	80	<	80	160	160	640	320	640	40
A/Albania/8441/2021		3C.2a1b.2a.2	2021-12-10	SIAT1	<	<	40	80	80	160	320	640	<
A/Albania/8322/2021		3C.2a1b.2a.2	2021-12-10	SIAT1	80	<	80	320	320	640	640	1280	40
A/Netherlands/10203/2021		3C.2a1b.2a.2	2021-12-11	MDCK-MIX2/SIAT1	80	<	40	80	160	320	320	1280	<
A/Albania/8614/2021		3C.2a1b.2a.2	2021-12-11	SIAT1	80	<	80	320	320	640	640	1280	40
A/Kosova/368/2021		3C.2a1b.2a.2	2021-12-14	SIAT1	40	<	40	40	160	320	320	640	<
A/Netherlands/10193/2021		3C.2a1b.2a.2	2021-12-16	MDCK-MIX2/SIAT1	160	<	80	320	320	320	320	640	80
A/Albania/9833/2021		3C.2a1b.2a.2	2021-12-16	SIAT1	40	<	40	40	160	320	320	640	<
A/Albania/9725/2021		3C.2a1b.2a.2	2021-12-16	SIAT1	160	<	80	320	320	640	640	1280	80
A/Albania/78/2021		3C.2a1b.2a.2	2021-12-17	SIAT1	80	<	160	160	320	640	640	1280	40
A/Albania/31/2021		3C.2a1b.2a.2	2021-12-17	SIAT1	160	<	160	320	320	640	640	2560	80
A/Netherlands/10204/2021		3C.2a1b.2a.2	2021-12-18	MDCK-MIX2/SIAT1	40	<	40	40	160	160	320	640	<
A/Netherlands/10205/2021		3C.2a1b.2a.2	2021-12-19	MDCK-MIX2/SIAT1	160	<	80	160	640	320	320	640	80
A/Netherlands/10194/2021		3C.2a1b.2a.2	2021-12-19	MDCK-MIX2/SIAT1	40	<	40	40	80	160	1280	640	<
A/Netherlands/10197/2021		3C.2a1b.2a.2	2021-12-20	MDCK-MIX2/SIAT1	40	<	<	80	160	320	320	640	<
A/Netherlands/10195/2021		3C.2a1b.2a.2	2021-12-20	MDCK-MIX2/SIAT2	160	<	80	160	640	640	640	1280	80
A/Albania/622/2021		3C.2a1b.2a.2	2021-12-20	SIAT1	160	<	80	320	320	640	640	1280	80
A/Kosova/587/2021		3C.2a1b.2a.2	2021-12-21	SIAT1	80	<	80	320	320	640	640	1280	40
A/Albania/890/2021		3C.2a1b.2a.2	2021-12-21	SIAT1	40	<	40	40	160	320	320	640	<
A/Albania/862/2021		3C.2a1b.2a.2	2021-12-21	SIAT1	40	<	40	80	160	320	640	640	<
A/Kosova/607/2021		3C.2a1b.2a.2	2021-12-22	SIAT1	80	<	40	160	160	320	640	1280	40
A/Kosova/618/2021		3C.2a1b.2a.2	2021-12-22	SIAT1	<	<	40	80	80	160	640	640	<
A/Netherlands/10215/2021		3C.2a1b.2a.2	2021-12-23	MDCK-MIX1/SIAT1	80	<	80	320	320	320	320	640	80
A/Netherlands/10210/2021		3C.2a1b.2a.2	2021-12-23	MDCK-MIX2/SIAT1	160	<	40	320	320	640	640	640	40
A/Netherlands/10208/2021		3C.2a1b.2a.2	2021-12-26	MDCK-MIX2/SIAT1	40	<	40	40	80	160	320	320	<
A/Kosova/701/2021		3C.2a1b.2a.2	2021-12-27	SIAT1	80	<	40	160	160	320	320	1280	40
A/Kosova/710/2021		3C.2a1b.2a.2	2021-12-27	SIAT1	80	<	40	320	320	320	640	1280	40
A/Netherlands/10231/2021		3C.2a1b.2a.2	2021-12-28	MDCK-MIX1/SIAT1	160	<	80	320	320	320	160	640	80
A/Netherlands/10230/2021		3C.2a1b.2a.2	2021-12-28	MDCK-MIX1/SIAT1	160	<	80	320	320	320	160	320	80
A/Argentina/R21/2022		3C.2a1b.2a.2	2022-01-09	SIAT2	<	<	<	40	80	160	320	320	<
A/Netherlands/10206/2021		3C.2a1b.1a	2021-12-15	MDCK-MIX2/SIAT1	160	160	320	320	320	320	40	640	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-05-13

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT
	Ferret number			F19/20 [†]	St Jude's F21/20 [†]	F03/21 [†]	F10/21 [†]	F07/21 [†]	F35/21 [†]	F07/22 [†]	F38/21 [†]	F17/19 [†]
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019		2019-10-25	SIAT3/SIAT4	320	160	640	160	320	160	40	320	160
A/Hong Kong/2671/2019		2019-06-17	MDCK1/SIAT4	160	160	640	160	160	80	40	320	80
A/Cambodia/925256/2020		2020-09-25	SIAT4	160	160	640	160	320	160	40	320	160
A/Cambodia/e0826360/2020		2020-07-16	E5/E2	160	<	160	1280	320	160	320	640	80
A/Bangladesh/4005/2020	H156	2020-10-04	SIAT3	160	40	320	320	640	640	640	1280	160
A/Stockholm/5/2021	H156S	2021-04-16	SIAT0/SIAT3	160	<	80	160	320	640	640	1280	40
A/England/214191723/2021	H156S	2021-10-12	MDCK1/SIAT2	40	<	80	80	160	320	640	640	40
A/Darwin/9/2021	H156S	2021-04-17	E3/E2	160	<	80	640	320	640	640	2560	80
A/Kansas/14/2017		2017-12-14	SIAT3/SIAT2	80	40	80	80	80	160	80	160	640
TEST VIRUSES												
A/Dakar/60/2021		2021-12-06	SIAT1	40	<	80	80	160	320	1280	1280	40
A/Dakar/63/2021		2021-12-14	SIAT1	40	<	40	80	160	320	1280	640	40
A/Dakar/62/2021		2021-12-14	SIAT1	160	<	40	160	320	640	1280	1280	40
A/Georgia/2048/2021		2021-12-16	SIAT1	160	<	160	320	320	640	640	1280	80
A/Georgia/2052/2021		2021-12-17	SIAT1	160	40	80	320	320	640	640	1280	80
A/Georgia/2060/2021		2021-12-18	SIAT2	<	<	40	40	80	160	640	640	<
A/Georgia/2063/2021		2021-12-19	SIAT1	320	40	80	640	320	320	640	1280	80
A/Ghana/195/2022		2022-01-12	SIAT1	80	<	80	80	320	640	1280	1280	80
A/Georgia/87/2022		2022-01-12	SIAT1	160	<	160	320	320	640	640	2560	80
A/Georgia/132/2022		2022-01-18	SIAT1	80	<	80	80	160	640	640	1280	40
A/Dakar/61/2021		2021-12-08	SIAT2	80	80	80	<	80	40	<	160	<
A/Dakar/64/2021		2021-12-13	SIAT1	160	80	160	160	160	160	<	160	40

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 10-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-05-20

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre										
				Post-infection ferret antisera										
				A/Denmark 3264/19	A/HK 2671/19	A/Camb 925256/20	A/Camb e0826360/20	A/Bang 4005/20	A/Stock 5/21	A/Eng 214191723/21	A/Darwin 9/21	A/Kansas 14/17		
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT		
	Passage history			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹		
	Ferret number			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1		
	Genetic group													
REFERENCE VIRUSES														
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	640	320	640	320	320	320	80	320	320	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	1280	160	160	160	80	320	160	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT4	160	160	1280	320	320	160	40	320	160	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	80	1280	160	160	160	160	40	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT5	320	80	320	320	640	640	640	1280	320	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	160	<	80	160	320	640	640	1280	40	
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	80	<	80	80	320	640	1280	640	40	
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	160	640	320	640	640	1280	80	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	80	40	80	80	80	80	320	
TEST VIRUSES														
A/Catalonia/800505/2021		3C.2a1b.2a.2	2021-10-25	C0/SIAT1	80	<	80	80	160	320	640	640	40	
A/Catalonia/3551005/2021		3C.2a1b.2a.2	2021-12-07	C0/SIAT1	80	<	80	80	320	640	640	640	<	
A/Thuringen/2/2022		3C.2a1b.2a.2	2022-01-04	C2/SIAT1	80	<	40	40	160	320	320	640	<	
A/Catalonia/3556026/2022		3C.2a1b.2a.2	2022-01-06	C0/SIAT1	40	<	40	80	160	320	640	640	<	
A/Berlin/5/2022		3C.2a1b.2a.2	2022-01-10	C1/SIAT1	40	<	40	80	160	320	320	640	<	
A/Catalonia/13522/2022		3C.2a1b.2a.2	2022-01-17	C0/SIAT1	40	<	80	80	160	320	640	640	40	
A/Catalonia/13520/2022		3C.2a1b.2a.2	2022-01-17	C0/SIAT1	80	<	80	80	160	320	1280	640	40	
A/Baden-Wuerttemberg/8/2022		3C.2a1b.2a.2	2022-01-24	C1/SIAT1	160	<	80	320	320	640	640	1280	80	
A/Sachsen/1/2022		3C.2a1b.2a.2	2022-01-24	C1/SIAT1	40	<	40	40	160	320	320	640	<	
A/Hessen/4/2022		3C.2a1b.2a.2	2022-01-24	C1/SIAT1	80	<	40	80	160	320	320	640	<	
A/Berlin/10/2022		3C.2a1b.2a.2	2022-01-31	C1/SIAT1	80	<	80	40	160	320	320	1280	40	
A/Bremen/1/2022		3C.2a1b.2a.2	2022-01-31	C1/SIAT1	160	40	80	160	320	1280	640	1280	40	
A/Hessen/6/2022		3C.2a1b.2a.2	2022-01-31	C1/SIAT1	80	<	80	80	160	640	640	1280	40	
A/Bayern/4/2022	D104G	3C.2a1b.2a.2	2022-02-02	C1/SIAT1	80	<	40	40	160	320	320	640	<	
A/Niedersachsen/8/2022	D104G	3C.2a1b.2a.2	2022-02-11	C1/SIAT1	80	<	40	80	160	320	320	640	<	
A/Thuringen/4/2022	D104G	3C.2a1b.2a.2	2022-02-14	C1/SIAT1	40	<	40	40	160	320	320	640	<	
A/Catalonia/13756/2022	D104G	3C.2a1b.2a.2	2022-02-16	C0/SIAT1	80	<	80	80	160	640	320	640	40	
A/Catalonia/13755/2022	D104G	3C.2a1b.2a.2	2022-02-16	C0/SIAT1	80	<	80	80	160	320	320	640	<	
A/Bayern/7/2022	D104G	3C.2a1b.2a.2	2022-02-24	C1/SIAT1	40	<	40	40	160	320	160	640	<	
A/Rheinland-Pfalz/1/2022	D104G	3C.2a1b.2a.2	2022-02-25	C1/SIAT1	80	80	80	80	320	640	640	1280	40	
A/Thuringen/6/2022	D104G	3C.2a1b.2a.2	2022-02-28	C1/SIAT1	80	<	40	80	160	320	320	640	<	
A/Denmark/267/2022	D104G	3C.2a1b.2a.2	2022-02-28	SIAT2/SIAT1	80	<	80	80	160	320	640	1280	<	
A/Catalonia/2081558/2022	D104G	3C.2a1b.2a.2	2022-03-01	C0/SIAT1	80	<	40	80	160	320	640	1280	40	
A/Denmark/294/2022	D104G	3C.2a1b.2a.2	2022-03-01	SIAT2/SIAT1	80	<	80	80	160	640	640	1280	<	
A/Denmark/333/2022	D104G	3C.2a1b.2a.2	2022-03-02	SIAT2/SIAT1	80	<	40	80	160	320	320	640	<	
A/Denmark/320/2022	D104G	3C.2a1b.2a.2	2022-03-02	SIAT2/SIAT2	80	<	80	80	160	640	320	1280	<	
A/Denmark/335/2022	D104G	3C.2a1b.2a.2	2022-03-03	SIAT2/SIAT1	80	<	40	80	160	320	320	1280	<	
A/Thuringen/7/2022	D104G	3C.2a1b.2a.2	2022-03-06	C1/SIAT1	160	40	160	160	320	640	1280	1280	40	
A/Bremen/4/2022	D104G	3C.2a1b.2a.2	2022-03-07	C2/SIAT2	80	<	40	40	160	320	320	640	<	
A/Denmark/499/2022	D104G	3C.2a1b.2a.2	2022-03-07	SIAT2/SIAT1	80	<	40	40	160	320	320	640	<	
A/Catalonia/356654/2022	D104G	3C.2a1b.2a.2	2022-03-08	C0/SIAT1	80	<	40	80	160	640	640	1280	40	
A/Denmark/558/2022	D104G	3C.2a1b.2a.2	2022-03-09	SIAT2/SIAT1	80	<	80	80	160	320	640	1280	40	
A/Nordrhein-Westfalen/2/2022	D104G	3C.2a1b.2a.2	2022-03-10	C1/SIAT1	320	80	320	640	640	1280	1280	5120	80	
A/Denmark/1098/2022	D104G	3C.2a1b.2a.2	2022-03-10	SIAT2/SIAT1	80	<	80	80	160	320	320	640	<	
A/Denmark/602/2022	D104G	3C.2a1b.2a.2	2022-03-10	SIAT2/SIAT1	80	<	40	40	160	320	320	640	40	
A/Denmark/612/2022	D104G	3C.2a1b.2a.2	2022-03-11	SIAT2/SIAT1	80	<	40	80	320	320	320	1280	<	
A/Denmark/922/2022	D104G	3C.2a1b.2a.2	2022-03-16	SIAT2/SIAT1	80	<	40	40	160	320	320	640	<	
A/Denmark/1253/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT2/SIAT1	80	<	40	80	160	320	320	640	<	
A/Denmark/2031/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT2/SIAT1	80	<	40	80	320	320	320	1280	40	
A/Denmark/1836/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT2/SIAT1	80	<	40	80	160	320	320	1280	<	
A/Baden-Wuerttemberg/5/2022	D53G	3C.2a1b.2a.2	2022-02-04	C1/SIAT1	160	<	160	320	320	1280	640	1280	80	
A/Hessen/8/2022	D53G	3C.2a1b.2a.2	2022-02-07	C1/SIAT1	160	40	80	320	320	640	640	1280	80	
A/Bremen/2/2022	D53G	3C.2a1b.2a.2	2022-02-18	C1/SIAT1	160	<	80	320	320	640	640	1280	80	
A/Sachsen/2/2022	D53G	3C.2a1b.2a.2	2022-03-14	C1/SIAT1	160	<	80	320	320	640	640	640	80	
A/Baden-Wuerttemberg/6/2022	H156S	3C.2a1b.2a.2	2022-02-08	C1/SIAT1	40	<	40	40	160	320	1280	640	<	
A/Catalonia/13731/2022	H156S	3C.2a1b.2a.2	2022-02-09	C0/SIAT1	40	<	80	80	160	320	640	640	40	
A/Catalonia/13738/2022	H156S	3C.2a1b.2a.2	2022-02-10	C0/SIAT1	80	<	80	160	320	640	1280	640	80	
A/Catalonia/13737/2022	H156S	3C.2a1b.2a.2	2022-02-10	C0/SIAT1	40	<	40	<	40	160	640	640	40	
A/Catalonia/13758/2022	H156S	3C.2a1b.2a.2	2022-02-16	C0/SIAT1	<	<	40	40	80	160	320	320	<	
A/Catalonia/2076976/2022	H156S	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	40	<	80	80	160	320	640	640	<	
A/Catalonia/13775/2022	H156S	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	40	<	80	80	160	320	640	640	40	
A/Catalonia/13766/2022	H156S	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	80	<	80	160	320	640	1280	640	80	
A/Thuringen/3/2022	H156	3C.2a1b.2a.2	2022-02-01	C1/SIAT1	320	80	160	640	1280	1280	1280	1280	160	
A/Catalonia/13692/2022	no sequence	no sequence	2022-02-02	C0/SIAT1	40	<	40	80	160	320	640	640	40	
1 Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)				Vaccine NH 2021-22				Vaccine SH 2022 NH 2022-23						
1 < = <40, ND = Not Done														

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-05-27

					Haemagglutination inhibition titre											
					Post-infection ferret antisera											
Viruses	Other information	Collection date	Passage history	NEW												
				NEW												
				A/Denmark 3264/19	A/HK 2671/19	A/Camb 925256/20	A/Camb e0826360/20	A/Bang 4005/20	A/Stock 5/21	A/Eng 214191723/21	A/Darwin 9/21	A/Flor 02/21	A/Wyom 01/21	A/Kansas 14/17		
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	Egg	Egg	SIAT		
Passage history																
Ferret number																
Genetic group																
REFERENCE VIRUSES																
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	160	320	320	40	320	80	<	160	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	320	160	160	160	40	160	80	<	80	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	80	640	160	160	160	40	320	40	<	80	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	160	1280	320	160	160	320	80	40	80	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	160	640	640	160	640	160	40	160	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	80	320	640	160	1280	320	160	40	
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	40	80	160	320	160	640	320	80	<	
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	80	640	320	640	160	2560	320	640	80	
A/Florida/02/2021	H156S	3C.2a1b.2a.2	2021-10-20	E4 (Am2AL2)	80	<	80	320	320	640	320	1280	1280	320	80	
A/Wyoming/01/2021	D104G	3C.2a1b.2a.2	2021-08-27	E4 (Am2AL2)	160	<	80	320	320	640	160	1280	320	2560	80	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	40	80	80	40	80	40	<	320	
TEST VIRUSES																
A/Poland/PL39/2022	D104G	3C.2a1b.2a.2	2022-02-18	SIAT2	40	<	40	40	160	320	160	640	160	320	<	
A/Poland/PL16/2022	D104G	3C.2a1b.2a.2	2022-03-13	SIAT1	80	<	40	40	160	640	320	1280	320	640	<	
A/Poland/PL40/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT2	40	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL15/2022	D104G	3C.2a1b.2a.2	2022-03-22	SIAT1	80	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL42/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT2	80	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL38/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1	80	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL19/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1	40	<	40	40	160	320	160	320	160	320	<	
A/Poland/PL04/2022	D104G	3C.2a1b.2a.2	2022-04-03	SIAT1	80	<	40	80	160	320	320	640	160	320	<	
A/Poland/PL25/2022	D104G	3C.2a1b.2a.2	2022-04-07	SIAT1	80	<	40	40	160	320	160	640	160	320	<	
A/Poland/PL27/2022	D104G	3C.2a1b.2a.2	2022-04-09	SIAT1	40	<	40	40	160	320	160	640	160	320	<	
A/Poland/PL44/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1	80	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL49/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1	80	<	40	40	160	320	320	640	160	320	<	
A/Poland/PL14/2022	D53G	3C.2a1b.2a.2	2022-03-26	SIAT1	40	<	40	80	160	320	320	640	160	320	40	
A/Poland/PL13/2022	D53G	3C.2a1b.2a.2	2022-03-26	SIAT1	40	<	40	80	160	640	320	640	160	160	40	
A/Poland/PL12/2022	D53G	3C.2a1b.2a.2	2022-03-28	SIAT1	160	<	80	320	320	640	640	640	320	160	40	
A/Poland/PL10/2022	D53G	3C.2a1b.2a.2	2022-03-29	SIAT1	40	<	40	80	160	320	320	640	160	160	40	
A/Poland/PL09/2022	D53G	3C.2a1b.2a.2	2022-03-29	SIAT1	80	<	80	160	320	640	640	1280	320	320	40	
A/Poland/PL11/2022	D53G	3C.2a1b.2a.2	2022-03-29	SIAT1	160	<	80	160	160	640	320	640	160	160	40	
A/Poland/PL07/2022	D53G	3C.2a1b.2a.2	2022-03-31	SIAT1	80	<	40	80	320	640	640	640	320	320	40	
A/Poland/PL06/2022	D53G	3C.2a1b.2a.2	2022-04-02	SIAT1	40	<	40	160	160	640	320	1280	320	320	40	
A/Poland/PL03/2022	D53G	3C.2a1b.2a.2	2022-04-03	SIAT1	40	<	<	80	160	640	320	640	160	160	40	
A/Poland/PL21/2022	D53G	3C.2a1b.2a.2	2022-04-04	SIAT1	40	<	40	80	320	640	320	640	320	160	40	
A/Poland/PL24/2022	D53G	3C.2a1b.2a.2	2022-04-05	SIAT1	40	<	80	80	320	640	320	640	320	160	40	
A/Poland/PL23/2022	D53G	3C.2a1b.2a.2	2022-04-05	SIAT1	80	<	40	160	160	640	320	640	320	160	40	
A/Poland/PL22/2022	D53G	3C.2a1b.2a.2	2022-04-05	SIAT1	80	<	40	160	320	640	640	640	320	320	40	
A/Poland/PL28/2022	D53G	3C.2a1b.2a.2	2022-04-11	SIAT1	80	<	40	160	320	640	320	640	320	320	40	
A/Poland/PL45/2022	D53G	3C.2a1b.2a.2	2022-04-14	SIAT1	160	<	80	320	320	640	640	1280	320	320	80	
A/Poland/PL08/2022	H156	3C.2a1b.2a.2	2022-03-30	SIAT1	160	<	80	320	320	320	160	320	80	<	80	

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-06-17

Haemagglutination inhibition titre																	
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera													
				NEW													
				NEW													
				NEW													
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Slov	A/Norway	A/Norway	A/Darwin	A/Norway	A/Kansas	
Passage history	Passage history	Passage history	Passage history	3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	8720/2022	24873/21	24873/21	9/21	29511/21	14/17	
Ferret number	Ferret number	Ferret number	Ferret number	F19/20 ¹	St Judes F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F24/22 ¹	F10/22 ¹	F11/22 ¹	F38/21 ¹	F12/22 ¹	F17/19 ¹	
Genetic group	Genetic group	Genetic group	Genetic group	3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES																	
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	320	160	160	<	ND	ND	ND	320	ND	80	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	1280	320	320	40	<	160	80	320	40	320	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	80	160	640	160	160	<	ND	ND	ND	320	ND	160	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	160	2560	320	160	320	80	160	320	320	160	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	320	160	320	640	1280	640	1280	320	640	2560	640	160	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	160	40	160	320	640	1280	1280	160	320	1280	320	80	
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	80	<	80	160	320	640	1280	640	320	640	80	40	
A/Slovenia/8720/2022	D104G	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2	160	<	80	160	320	1280	640	1280	160	320	1280	160	
A/Norway/24873/2021	H156S	3C.2a1b.2a.2	2021-10-24	SIAT2	80	<	80	160	320	640	1280	640	320	640	1280	40	
A/Norway/24873/2021	H156S	3C.2a1b.2a.2	2021-10-24	E4 (Am2AI2)	160	<	160	640	640	1280	1280	320	1280	2560	320	160	
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	160	640	640	1280	1280	160	320	2560	320	80	
A/Norway/29511/2021	H156	3C.2a1b.2a.2	2021-11-25	E4 (Am2AI2)	ND	ND	160	ND	640	320	640	160	160	1280	640	ND	
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	80	40	160	160	80	80	80	<	<	160	80	640	
TEST VIRUSES																	
A/Bulgaria/193/2022		3C.2a1b.2a.2	2022-01-16	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/142/2022		3C.2a1b.2a.2	2022-01-16	SIAT2/SIAT1	160	<	80	160	320	640	640	640	ND	ND	1280	ND	40
A/Bulgaria/104/2022		3C.2a1b.2a.2	2022-01-16	SIAT2/SIAT1	160	<	80	160	320	640	640	640	ND	ND	1280	ND	40
A/Bulgaria/75/2022		3C.2a1b.2a.2	2022-01-16	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/404/2022		3C.2a1b.2a.2	2022-01-18	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/648/2022		3C.2a1b.2a.2	2022-01-30	SIAT2/SIAT1	160	<	160	160	320	640	640	640	ND	ND	1280	ND	80
A/Bulgaria/646/2022		3C.2a1b.2a.2	2022-01-30	SIAT2/SIAT1	160	40	160	160	320	640	640	1280	ND	ND	1280	ND	80
A/Bulgaria/644/2022		3C.2a1b.2a.2	2022-01-30	SIAT2/SIAT1	160	<	160	320	320	640	640	640	ND	ND	1280	ND	80
A/Bulgaria/1057/2022	D104G	3C.2a1b.2a.2	2022-02-11	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/1127/2022	D104G	3C.2a1b.2a.2	2022-02-16	SIAT2/SIAT1	160	<	80	160	320	640	640	640	ND	ND	1280	ND	40
A/Bulgaria/1195/2022	D104G	3C.2a1b.2a.2	2022-02-21	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/1273/2022	D104G	3C.2a1b.2a.2	2022-02-25	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1473/2022	D104G	3C.2a1b.2a.2	2022-03-10	SIAT2/SIAT1	80	<	40	160	320	640	320	640	80	160	1280	80	40
A/Bulgaria/1616/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/1611/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	160	<	80	160	320	640	320	1280	ND	ND	1280	ND	40
A/Bulgaria/1537/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT2/SIAT1	160	40	160	160	320	640	640	1280	ND	ND	1280	ND	80
A/Bulgaria/1536/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	160	320	1280	160	40
A/Bulgaria/1582/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	ND	ND	1280	ND	40
A/Bulgaria/1581/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	160	<	80	320	320	640	640	1280	ND	ND	1280	ND	80
A/Bulgaria/1579/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	160	<	80	160	320	640	320	1280	ND	ND	1280	ND	40
A/Bulgaria/1588/2022	D104G	3C.2a1b.2a.2	2022-03-22	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1666/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1665/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	80	<	40	160	320	640	640	640	80	320	1280	160	40
A/Bulgaria/1660/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1659/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	80	<	80	160	320	640	640	640	80	320	1280	160	40
A/Bulgaria/1646/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1763/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1726/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT2/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1813/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT2/SIAT1	80	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Bulgaria/1827/2022	D104G	3C.2a1b.2a.2	2022-04-07	SIAT2/SIAT1	80	<	80	160	320	640	640	640	80	320	1280	160	40
A/Slovenia/9891/2022	D104G	3C.2a1b.2a.2	2022-04-11	MDCKx/SIAT1	80	<	40	160	320	640	1280	ND	ND	1280	ND	40	
A/Slovenia/9967/2022	D104G	3C.2a1b.2a.2	2022-04-20	MDCKx/SIAT1	160	<	80	160	320	640	640	1280	80	320	1280	160	40
A/Slovenia/10081/2022	D104G	3C.2a1b.2a.2	2022-04-21	MDCKx/SIAT1	160	<	80	160	320	640	640	1280	160	320	1280	160	40
A/Slovenia/9905/2022	D53G	3C.2a1b.2a.2	2022-04-12	MDCKx/SIAT1	640	40	160	640	640	1280	640	1280	ND	ND	1280	ND	160

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

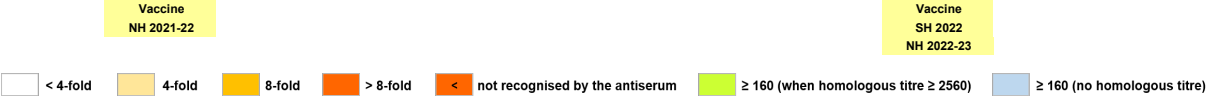


Table 10-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-06-24

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
	Ferret number			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F19/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	160	160	40	320	80	
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	160	40	160	160	
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	160	40	320	160	
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	160	1280	160	320	320	80	
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	320	320	640	640	640	320	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	160	<	160	160	320	640	320	1280	40
A/England/2141917/23/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	80	<	80	80	320	320	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	160	640	640	1280	640	2560	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	40	80	80	320	
TEST VIRUSES													
A/Kazakhstan/583/2021		3C.2a1b.2a.2	2021-12-06	SIAT1	160	40	160	320	640	320	640	160	
A/Kazakhstan/586/2021		3C.2a1b.2a.2	2021-12-09	SIAT1	160	<	160	320	320	320	640	160	
A/Switzerland/12981/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT1	40	<	40	80	160	320	320	640	<
A/Switzerland/73028/2022	D104G	3C.2a1b.2a.2	2022-03-16	SIAT1	40	<	40	80	160	160	320	640	<
A/Switzerland/34074/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT1	40	<	40	80	160	320	320	640	<
A/Switzerland/18016/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT1	80	<	40	160	160	320	320	640	40
A/Switzerland/19130/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT1	40	<	40	80	320	320	320	640	<
A/Mecklenburg-Vorpommern/48/2022	D104G	3C.2a1b.2a.2	2022-03-30	P1/SIAT1	40	<	40	80	160	320	320	640	<
A/Switzerland/76393/2022	D104G	3C.2a1b.2a.2	2022-03-31	SIAT1	80	<	40	80	160	320	320	640	<
A/Hessen/10/2022	D104G	3C.2a1b.2a.2	2022-03-31	P1/SIAT1	40	<	40	80	160	320	320	640	<
A/Berlin/40/2022	D104G	3C.2a1b.2a.2	2022-04-01	P2/SIAT1	80	<	40	160	320	640	320	1280	40
A/Niedersachsen/9/2022	D104G	3C.2a1b.2a.2	2022-04-04	P1/SIAT1	40	<	40	80	160	320	320	1280	<
A/Norway/18575/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1	40	<	40	80	160	320	320	640	<
A/Sachsen/4/2022	D104G	3C.2a1b.2a.2	2022-04-07	P1/SIAT1	40	<	40	80	160	320	320	640	<
A/Norway/18894/2022	D104G	3C.2a1b.2a.2	2022-04-08	SIAT2	40	<	80	160	320	640	640	1280	40
A/Norway/18893/2022	D104G	3C.2a1b.2a.2	2022-04-08	SIAT1	40	<	40	80	160	320	320	640	<
A/Norway/19252/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1	40	<	40	80	160	320	320	640	<
A/Norway/19053/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1	40	<	40	80	160	320	320	640	<
A/Norway/19865/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT1	40	<	40	40	160	320	320	640	<
A/Norway/19859/2022	D104G	3C.2a1b.2a.2	2022-04-15	SIAT1	40	<	40	80	160	320	320	640	<
A/Norway/19259/2022	D104G	3C.2a1b.2a.2	2022-04-15	SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-58897/2022	D104G	3C.2a1b.2a.2	2022-04-19	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Nordrhein-Westfalen/5/2022	D104G	3C.2a1b.2a.2	2022-04-19	P1/SIAT1	40	<	40	40	160	320	320	640	<
A/Latvia/04-78414/2022	D104G	3C.2a1b.2a.2	2022-04-22	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-78409/2022	D104G	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-78385/2022	D104G	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-78381/2022	D104G	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-82637/2022	D104G	3C.2a1b.2a.2	2022-04-24	P2/SIAT1	80	<	40	80	320	320	320	640	<
A/Latvia/04-82661/2022	D104G	3C.2a1b.2a.2	2022-04-25	P2/SIAT1	40	<	40	40	160	320	320	640	<
A/Hessen/16/2022	D104G	3C.2a1b.2a.2	2022-04-25	P1/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-86299/2022	D104G	3C.2a1b.2a.2	2022-04-26	P2/SIAT1	40	<	40	80	160	320	320	640	<
A/Latvia/04-86261/2022	D104G	3C.2a1b.2a.2	2022-04-26	P2/SIAT1	80	<	40	80	160	320	320	640	<
A/Latvia/04-90221/2022	D104G	3C.2a1b.2a.2	2022-04-28	P2/SIAT1	80	<	40	80	320	320	320	640	<
A/Latvia/04-90145/2022	D104G	3C.2a1b.2a.2	2022-04-28	P2/SIAT1	80	<	40	80	160	320	320	640	<
A/Mecklenburg-Vorpommern/69/2022	D104G	3C.2a1b.2a.2	2022-05-02	P1/SIAT1	80	<	40	40	160	640	320	1280	<
A/Bayern/16/2022	D104G	3C.2a1b.2a.2	2022-05-03	P1/SIAT1	80	<	40	80	160	640	320	1280	<
A/Baden-Württemberg/27/2022	D104G	3C.2a1b.2a.2	2022-05-04	P1/SIAT1	80	<	40	80	160	640	320	640	<
A/Berlin/87/2022	D104G	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	80	<	40	80	320	320	320	1280	<
A/Bremen/17/2022	D104G	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	80	<	40	80	160	320	320	1280	<
A/Rheinland-Pfalz/6/2022	D104G	3C.2a1b.2a.2	2022-05-09	P1/SIAT1	40	<	40	80	160	320	320	640	<
A/Switzerland/15559/2022	D50G	3C.2a1b.2a.2	2022-03-08	SIAT1	160	<	80	320	320	640	640	1280	80
A/Switzerland/10871/2022	D50G	3C.2a1b.2a.2	2022-03-28	SIAT1	80	<	80	160	320	640	640	640	40
A/Switzerland/09683/2022	D50G	3C.2a1b.2a.2	2022-03-30	MDCK/SIAT2	320	<	80	640	320	640	640	1280	160
A/Switzerland/54637/2022	D50G	3C.2a1b.2a.2	2022-03-30	SIAT1	320	40	320	640	640	1280	640	1280	160
A/Switzerland/89503/2022	D50G	3C.2a1b.2a.2	2022-03-31	SIAT1	160	<	160	320	320	1280	640	1280	80
A/Norway/19419/2022	D50G	3C.2a1b.2a.2	2022-04-09	SIAT1	160	<	80	320	320	640	640	1280	80
A/Slovenia/9904/2022	D50G	3C.2a1b.2a.2	2022-04-12	MDCK/SIAT2	160	<	80	320	640	640	640	640	80
A/Switzerland/60042/2022	H156S	3C.2a1b.2a.2	2022-03-10	SIAT1	40	<	40	80	160	160	640	640	<
A/Switzerland/45784/2022	H156S	3C.2a1b.2a.2	2022-03-21	SIAT1	40	<	40	80	160	320	640	640	<
A/Norway/20047/2022	H156S	3C.2a1b.2a.2	2022-04-17	SIAT1	160	<	80	320	320	320	320	640	80
A/Latvia/04-58976/2022	H156S	3C.2a1b.2a.2	2022-04-18	P2/SIAT1	40	<	40	80	160	320	640	320	<
A/Thuringen/10/2022	H156	3C.2a1b.2a.2	2022-04-01	P1/SIAT1	160	<	80	160	320	640	320	640	80
A/Slovenia/9960/2022	H156	3C.2a1b.2a.2	2022-04-06	MDCK1/SIAT2	160	40	160	320	640	320	640	640	160
A/Norway/18896/2022	H156	3C.2a1b.2a.2	2022-04-08	SIAT2	160	<	80	320	320	320	320	640	160
A/Brandenburg/2/2022	H156	3C.2a1b.2a.2	2022-05-05	P1/SIAT1	80	<	40	160	320	320	320	640	80
A/Sachsen/11/2022	H156	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	80	<	40	80	320	160	320	640	<

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
 1 < = <40, ND = Not Done

Vaccine
 NH 2021-22

Vaccine
 SH 2022
 NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-01

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
	Passage history			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
	Ferret number			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
	Genetic group												
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	160	320	160	80	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	640	160	160	80	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	160	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	80	640	160	80	160	160	40
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	160	320	640	320	640	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	80	160	320	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	320	640	640	1280	80	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	640	640
TEST VIRUSES													
A/Linkoping/3/2022	D104G	3C.2a1b.2a.2	2022-03-12	SIAT1/SIAT1	80	<	40	80	320	640	320	640	40
A/Stockholm/9/2022	D104G	3C.2a1b.2a.2	2022-03-15	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Stockholm/10/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT1/SIAT1	80	<	40	80	320	640	320	640	40
A/Austria/1504938/2022	D104G	3C.2a1b.2a.2	2022-03-29	Cx/SIAT1	80	<	40	160	160	640	320	640	40
A/Slovenia/9767/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Austria/1505788/2022	D104G	3C.2a1b.2a.2	2022-03-31	Cx/SIAT1	80	<	40	80	160	320	320	640	40
A/Austria/1505762/2022	D104G	3C.2a1b.2a.2	2022-03-31	Cx/SIAT1	80	<	40	80	160	320	320	640	40
A/Slovenia/9768/2022	D104G	3C.2a1b.2a.2	2022-04-01	MDCK1/SIAT1	80	<	40	160	320	320	320	640	40
A/Slovenia/9854/2022	D104G	3C.2a1b.2a.2	2022-04-01	MDCK1/SIAT1	80	<	40	80	320	640	320	640	40
A/Austria/1506181/2022	D104G	3C.2a1b.2a.2	2022-04-01	Cx/SIAT1	40	<	40	80	160	320	320	640	40
A/Slovenia/9806/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Austria/1507395/2022	D104G	3C.2a1b.2a.2	2022-04-04	Cx/SIAT1	40	<	40	80	160	320	320	640	40
A/Slovenia/9845/2022	D104G	3C.2a1b.2a.2	2022-04-05	MDCK1/SIAT1	40	<	40	80	160	320	320	640	40
A/Stockholm/19/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	80	<	40	160	160	320	320	640	40
A/Sweden/3/2022	D104G	3C.2a1b.2a.2	2022-04-11	SIAT1/SIAT1	40	<	<	80	80	160	320	320	40
A/Esikilistuna/1/2022	D104G	3C.2a1b.2a.2	2022-04-16	SIAT1/SIAT1	80	<	40	80	160	160	320	320	40
A/Portugal/211329/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Linkoping/5/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	80	<	40	160	160	320	320	640	40
A/Portugal/211317/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT1/SIAT1	80	<	40	160	160	320	320	640	40
A/Portugal/211412/2022	D104G	3C.2a1b.2a.2	2022-04-28	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Portugal/211439/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Portugal/211442/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT2/SIAT1	40	<	40	80	160	320	320	640	40
A/Portugal/211459/2022	D104G	3C.2a1b.2a.2	2022-05-05	SIAT2/SIAT1	80	<	40	80	160	640	320	640	40
A/Portugal/211462/2022	D104G	3C.2a1b.2a.2	2022-05-05	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Portugal/211456/2022	D104G	3C.2a1b.2a.2		SIAT2/SIAT1	80	<	40	160	320	640	320	640	80
A/Portugal/211458/2022	D104G	3C.2a1b.2a.2		SIAT2/SIAT1	40	<	40	80	160	320	320	640	40
A/Bayern/9/2022	D53G	3C.2a1b.2a.2	2022-03-29	PI/SIAT1	160	<	160	320	320	640	640	640	80
A/Austria/1505342/2022	D53G	3C.2a1b.2a.2	2022-03-29	Cx/SIAT1	160	<	80	640	640	1280	640	1280	160
A/Austria/1504947/2022	D53G	3C.2a1b.2a.2	2022-03-29	Cx/SIAT1	160	<	80	640	320	640	640	640	160
A/Slovenia/9861/2022	D53G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	160	<	160	320	640	640	640	640	160
A/Portugal/211330/2022	H156S	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	80	<	40	80	320	320	640	320	40
A/Portugal/211338/2022	H156S	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	160	<	80	160	320	640	1280	640	80
A/Stockholm/21/2022	H156S	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	40	<	40	160	160	320	640	320	40
A/Portugal/211316/2022	H156S	3C.2a1b.2a.2	2022-04-23	SIAT1/SIAT1	<	<	<	80	320	320	640	640	40
A/Austria/1504943/2022	H156	3C.2a1b.2a.2	2022-03-29	Cx/SIAT1	160	<	80	160	640	320	320	640	160
A/Halmstad/3/2022	H156	3C.2a1b.2a.2	2022-03-29	SIAT1/SIAT1	80	<	40	160	320	320	320	320	80
A/Stockholm/12/2022	H156	3C.2a1b.2a.2	2022-04-03	SIAT1/SIAT1	160	<	40	320	320	320	320	640	80
A/Austria/1507013/2022	H156	3C.2a1b.2a.2	2022-04-04	Cx/SIAT1	80	<	40	80	320	160	320	320	80
A/Austria/1506583/2022	H156	3C.2a1b.2a.2	2022-04-04	Cx/SIAT1	80	<	40	160	320	160	320	320	80
A/Austria/1506576/2022	H156	3C.2a1b.2a.2	2022-04-04	Cx/SIAT1	160	<	40	160	320	320	320	320	80
A/Halmstad/4/2022		3C.2a1b.1a	2022-04-11	SIAT1/SIAT1	160	160	320	320	320	160	40	160	80
A/Esikilistuna/3/2022		3C.2a1b.1a	2022-05-01	SIAT1/SIAT1	160	160	320	320	320	160	40	160	80

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
¹ < = <40, ND = Not Done

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 10-15. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-08

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
	Passage history			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
	Ferret number												
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	160	320	160	80	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	1280	160	320	160	40	320	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	160	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	160	2560	320	160	320	320	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	320	40	320	320	640	640	640	1280	320
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	160	<	160	160	320	640	320	1280	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	80	<	80	160	320	640	1280	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	160	640	640	640	1280	80	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	160	80	80	640	640
TEST VIRUSES													
A/Serbia/416495/2022		3C.2a1b.2a.2	2022-01-17	P1/SIAT1	80	<	160	80	320	640	640	1280	40
A/Serbia/491/2022		3C.2a1b.2a.2	2022-01-19	P2/SIAT1	160	<	160	80	320	640	640	1280	40
A/Serbia/422406/2022		3C.2a1b.2a.2	2022-01-25	P1/SIAT1	80	<	160	80	320	320	1280	1280	80
A/Hungary/39/2022	D104G	3C.2a1b.2a.2	2022-02-01	MDCK1/SIAT1	80	<	40	160	160	320	320	640	40
A/Hungary/42/2022	D104G	3C.2a1b.2a.2	2022-02-07	MDCK1/SIAT1	80	<	80	160	160	320	320	640	40
A/Norway/11716/2022	D104G	3C.2a1b.2a.2	2022-02-26	SIAT1	80	<	40	80	320	320	320	640	<
A/Norway/12664/2022	D104G	3C.2a1b.2a.2	2022-03-04	SIAT1	80	<	40	80	160	320	320	640	<
A/Norway/12392/2022	D104G	3C.2a1b.2a.2	2022-03-06	SIAT1	80	<	40	80	160	320	320	640	<
A/Norway/13415/2022	D104G	3C.2a1b.2a.2	2022-03-08	SIAT1	80	<	80	160	320	640	640	1280	160
A/Norway/13992/2022	D104G	3C.2a1b.2a.2	2022-03-13	SIAT1	160	<	80	160	320	640	640	1280	80
A/Serbia/1172/2022	D104G	3C.2a1b.2a.2	2022-03-14	P1/SIAT1	80	<	160	80	320	640	320	640	40
A/Norway/15331/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT1	80	<	80	80	160	320	640	640	<
A/Hungary/60/2022	D104G	3C.2a1b.2a.2	2022-03-18	MDCK1/SIAT1	80	<	40	80	160	320	640	1280	40
A/Norway/16898/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIAT1	80	<	40	80	320	320	320	640	<
A/Serbia/1549/2022	D104G	3C.2a1b.2a.2	2022-03-24	P1/SIAT1	80	<	80	40	160	320	320	640	<
A/Norway/16929/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT1	80	<	40	80	160	320	320	640	<
A/Serbia/1732/2022	D104G	3C.2a1b.2a.2	2022-03-28	P2/SIAT1	80	<	80	40	320	640	320	640	<
A/Hungary/51/2022	D104G	3C.2a1b.2a.2	2022-03-28	MDCK1/SIAT1	80	<	40	80	160	320	320	640	<
A/Serbia/452190/2022	D104G	3C.2a1b.2a.2	2022-03-29	P1/SIAT1	80	<	80	40	160	320	320	640	40
A/Hungary/52/2022	D104G	3C.2a1b.2a.2	2022-03-29	MDCK1/SIAT1	80	<	40	80	160	320	320	640	40
A/Norway/16622/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1	80	<	40	80	320	320	320	640	<
A/Norway/19137/2022	D104G	3C.2a1b.2a.2	2022-03-31	SIAT1	80	<	80	80	160	320	320	640	<
A/Hungary/55/2022	D104G	3C.2a1b.2a.2	2022-03-31	MDCK1/SIAT1	80	<	80	40	160	320	320	640	<
A/Norway/18088/2022	D104G	3C.2a1b.2a.2	2022-04-01	SIAT1	80	<	80	80	160	320	320	640	40
A/Norway/18089/2022	D104G	3C.2a1b.2a.2	2022-04-02	SIAT1	80	<	40	80	320	320	320	640	<
A/Norway/18090/2022	D104G	3C.2a1b.2a.2	2022-04-03	SIAT1	80	<	80	80	320	320	320	1280	40
A/Norway/18656/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT1	80	<	80	80	320	640	320	640	40
A/Norway/17599/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT1	80	<	80	80	320	640	640	1280	40
A/Norway/17522/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT1	80	<	80	80	320	320	320	640	40
A/Norway/18641/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT1	80	<	80	80	320	320	320	640	40
A/Norway/18562/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1	80	<	80	80	160	320	320	640	40
A/La Rioja/2092/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	80	<	40	80	320	640	640	1280	40
A/Hungary/63/2022	D104G	3C.2a1b.2a.2	2022-04-07	MDCK1/SIAT1	80	<	80	40	160	320	320	640	40
A/Hungary/64/2022	D104G	3C.2a1b.2a.2	2022-04-08	MDCK1/SIAT1	40	<	80	40	160	320	320	320	<
A/Serbia/2230/2022	D104G	3C.2a1b.2a.2	2022-04-10	P1/SIAT1	80	<	40	80	320	320	320	640	40
A/Estonia/169422/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	80	<	40	80	160	320	640	640	<
A/Estonia/169416/2022	D104G	3C.2a1b.2a.2	2022-04-22	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Estonia/169371/2022	D104G	3C.2a1b.2a.2	2022-04-22	MDCKx/SIAT1	80	<	40	80	160	320	320	640	40
A/Estonia/169440/2022	D104G	3C.2a1b.2a.2	2022-04-23	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Estonia/169485/2022	D104G	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	80	<	40	80	160	320	320	640	<
A/Estonia/169484/2022	D104G	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	80	<	40	80	160	640	320	640	40
A/Serbia/2489/2022	D104G	3C.2a1b.2a.2	2022-04-26	P1/SIAT1	80	<	40	80	320	320	160	640	<
A/Andalucia/2715/2022	D104G	3C.2a1b.2a.2	2022-04-27	Px/SIAT1	80	<	40	80	320	640	640	640	<
A/Serbia/2522/2022	D104G	3C.2a1b.2a.2	2022-04-30	P1/SIAT1	80	<	40	80	320	320	320	640	<
A/Andalucia/2717/2022	D104G	3C.2a1b.2a.2	2022-05-06	Px/SIAT1	80	<	40	80	320	640	640	1280	40
A/Andalucia/2716/2022	D104G	3C.2a1b.2a.2	2022-05-10	Px/SIAT1	160	<	40	80	320	640	640	640	40
A/Andalucia/2723/2022	D104G	3C.2a1b.2a.2	2022-05-12	Px/SIAT1	320	40	160	160	640	1280	1280	1280	80
A/Andalucia/2720/2022	D104G	3C.2a1b.2a.2	2022-05-14	Px/SIAT1	160	<	80	160	320	640	1280	1280	80
A/Andalucia/2718/2022	D104G	3C.2a1b.2a.2	2022-05-15	Px/SIAT1	160	<	80	160	320	640	640	1280	40
A/Hungary/41/2022	D53G	3C.2a1b.2a.2	2022-02-02	MDCK1/SIAT1	320	<	80	640	320	640	1280	1280	160
A/Norway/12650/2022	D53G	3C.2a1b.2a.2	2022-03-03	SIAT1	160	40	80	320	320	640	640	640	80
A/Serbia/447250/2022	H156S	3C.2a1b.2a.2	2022-03-15	P1/SIAT1	40	<	160	40	160	320	640	320	40
A/Norway/15385/2022	H156S	3C.2a1b.2a.2	2022-03-18	SIAT1	80	<	80	80	320	320	640	640	40
A/Norway/18657/2022	H156S	3C.2a1b.2a.2	2022-04-04	SIAT1	40	<	80	80	160	320	640	640	<
A/Norway/18658/2022	H156S	3C.2a1b.2a.2	2022-04-05	SIAT1	40	<	80	80	160	320	640	640	40
A/Serbia/2210/2022	H156S	3C.2a1b.2a.2	2022-04-08	P1/SIAT1	40	<	80	80	160	320	640	640	40
A/Norway/12245/2022	H156	3C.2a1b.2a.2	2022-03-02	SIAT1	160	<	80	320	640	320	320	640	160
A/Norway/13224/2022	H156	3C.2a1b.2a.2	2022-03-06	SIAT1	160	<	80	160	640	320	640	640	160
A/South Africa/R03557/2022	H156	3C.2a1b.2a.2	2022-03-22	MDCK1/SIAT1	160	<	80	320	320	320	320	320	160

¹Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-16. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-12

				Haemagglutination inhibition titre									
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
Genetic group				3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	320	320	320	80	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	160	320	640	160	160	160	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	80	<	160	1280	320	160	320	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	320	320	640	640	640	640	320
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	160	320	640	320	640	40
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	320	640	640	1280	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	160	80	80	640
TEST VIRUSES													
A/Netherlands/00092/2021		3C.2a1b.2a.2	2021-11-30	Hck1/SIAT1	<	<	40	40	80	160	320	320	<
A/Netherlands/00103/2021		3C.2a1b.2a.2	2021-12-13	Hck1/SIAT1	40	<	40	80	160	160	320	320	<
A/Netherlands/00001/2022		3C.2a1b.2a.2	2022-01-02	Hck1/SIAT1	160	40	160	160	640	640	320	640	160
A/Netherlands/00009/2022		3C.2a1b.2a.2	2022-01-05	Hck1/SIAT1	80	<	80	320	320	320	160	320	80
A/Netherlands/00110/2021		3C.2a1b.1a	2021-12-22	Hck1/SIAT1	320	160	320	180	160	160	<	80	80
A/Moldova/20837/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIAT1	40	<	<	80	160	320	320	640	<
A/Lithuania/33828/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT1	80	<	40	80	160	320	320	640	<
A/Norway/12347/2022	D53G	3C.2a1b.2a.2	2022-02-25	SIAT3	80	<	40	320	320	640	320	640	80
A/Moldova/18402/2022	D53G	3C.2a1b.2a.2	2022-02-28	SIAT1	40	<	40	80	320	640	320	640	40
A/Moldova/6662/2022	D53G	3C.2a1b.2a.2	2022-04-04	SIAT1	160	<	80	320	320	640	640	640	80
A/Moldova/6649/2022	D53G	3C.2a1b.2a.2	2022-04-05	SIAT1	160	<	80	320	320	640	640	640	80
A/Moldova/7267/2022	D53G	3C.2a1b.2a.2	2022-04-11	SIAT1	160	<	80	320	320	640	640	640	80
A/Moldova/7273/2022	D53G	3C.2a1b.2a.2	2022-04-13	SIAT1	160	<	80	320	320	640	640	640	80
A/Lithuania/33816/2022	D53G	3C.2a1b.2a.2	2022-04-14	SIAT1	40	<	40	80	160	320	320	640	<
A/Lithuania/36119/2022	D53G	3C.2a1b.2a.2	2022-04-28	SIAT1	160	<	80	320	320	640	640	640	80
A/Lithuania/36453/2022	D53G	3C.2a1b.2a.2	2022-05-02	SIAT1	40	<	40	80	160	320	320	640	<
A/Lithuania/36696/2022	D53G	3C.2a1b.2a.2	2022-05-05	SIAT1	160	<	80	320	320	640	640	1280	80
A/Dakar/03/2022	H156S	3C.2a1b.2a.2	2022-02-03	P1/SIAT1	40	<	40	80	160	320	640	320	40
A/Moldova/20046/2022	H156S	3C.2a1b.2a.2	2022-03-10	SIAT1	40	<	40	80	160	320	640	640	40
A/Moldova/20838/2022	H156S	3C.2a1b.2a.2	2022-03-17	SIAT1	40	<	40	80	160	160	640	320	<
A/Moldova/7877/2022	H156S	3C.2a1b.2a.2	2022-04-19	SIAT1	40	<	40	80	160	320	640	320	<
A/Moldova/7896/2022	H156S	3C.2a1b.2a.2	2022-04-21	SIAT1	40	<	40	80	160	320	640	320	<
A/Dakar/10/2022	H156S	3C.2a1b.2a.2	2022-05-23	P1/SIAT1	80	<	40	80	160	320	640	320	40
A/Dakar/11/2022	H156S	3C.2a1b.2a.2	2022-05-25	P1/SIAT1	160	<	40	160	320	1280	1280	1280	80
A/Dakar/20/2022	H156S	3C.2a1b.2a.2	2022-06-11	P1/SIAT1	80	<	40	80	160	160	640	320	40
A/Moldova/20035/2022	H156	3C.2a1b.2a.2	2022-03-07	SIAT1	80	<	40	320	320	160	160	320	80
A/Moldova/20813/2022	H156	3C.2a1b.2a.2	2022-03-15	SIAT1	160	<	80	320	320	320	160	320	80
A/Netherlands/10502/2022	H156	3C.2a1b.2a.2	2022-03-24	MDCK-MIX2/SIAT1	80	<	40	80	320	320	320	320	80
A/Netherlands/10884/2022	H156	3C.2a1b.2a.2	2022-04-05	MDCK-MIX2/SIAT1	160	<	80	320	320	320	320	320	160
A/Netherlands/10866/2022	H156	3C.2a1b.2a.2	2022-04-07	MDCK-MIX2/SIAT1	<	<	<	<	160	160	80	160	<
A/Netherlands/11119/2022	H156	3C.2a1b.2a.2	2022-04-13	MDCK-MIX2/SIAT1	160	<	80	160	320	320	320	640	160
A/Netherlands/11201/2022	H156	3C.2a1b.2a.2	2022-04-15	MDCK-MIX2/SIAT1	160	<	80	320	320	320	320	640	160
A/Netherlands/00832/2022	H156	3C.2a1b.2a.2	2022-04-19	Hck1/SIAT1	80	<	40	80	160	160	160	160	80

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-17. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-15

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
Genetic group				3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	160	640	160	320	80	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	1280	320	160	160	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	640	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	160	320	640	320	640	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	640	1280	640	1280	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	80	640
TEST VIRUSES													
A/Orenburg/5/2022		3C.2a1b.2a.2	2022-01-14	MDCK2/SIAT1/SIAT1	160	<	80	320	320	320	160	320	80
A/Dakar/02/2022		3C.2a1b.2a.2	2022-01-19	SIAT1	80	<	80	160	320	320	640	640	40
A/Birobidjan/12/2022		3C.2a1b.2a.2	2022-01-20	SIAT3/SIAT1	160	<	80	640	320	640	640	1280	160
A/FVG-Trieste/05/2022		3C.2a1b.2a.2	2022-01-25	SIAT2/SIAT1	80	<	40	80	160	320	320	640	40
A/Parma/2/2022	D104G	3C.2a1b.2a.2	2022-02-02	SIAT4/SIAT1	40	<	40	160	320	640	320	640	40
A/FVG-Trieste/13/2022	D104G	3C.2a1b.2a.2	2022-02-15	SIAT2/SIAT1	40	<	40	80	160	320	320	640	40
A/FVG-Trieste/16/2022	D104G	3C.2a1b.2a.2	2022-02-17	SIAT2/SIAT1	80	<	40	80	160	640	320	640	40
A/Milano/30/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT4/SIAT1	80	<	40	80	320	640	640	1280	40
A/Sassari/2/2022	D104G	3C.2a1b.2a.2	2022-03-16	SIAT1/SIAT2	80	<	40	80	320	640	640	640	40
A/Sassari/4/2022	D104G	3C.2a1b.2a.2	2022-03-29	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Sassari/7/2022	D104G	3C.2a1b.2a.2	2022-04-01	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Sassari/9/2022	D104G	3C.2a1b.2a.2	2022-04-08	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Perugia/23/2022	D104G	3C.2a1b.2a.2	2022-04-12	SIAT3/SIAT1	40	<	40	80	160	320	320	640	40
A/Bolzano/21/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT2/SIAT1	80	<	40	160	320	320	320	640	40
A/FVG-Trieste/366/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT3/SIAT1	40	<	40	80	160	320	160	320	40
A/Milano/112/2022	D104G	3C.2a1b.2a.2	2022-05-03	SIAT3/SIAT1	80	<	40	160	320	320	320	640	40
A/Milano/37/2022	D53G	3C.2a1b.2a.2	2022-03-14	SIAT3/SIAT1	160	<	80	320	320	640	640	640	80
A/Parma/51/2022	H156S	3C.2a1b.2a.2	2022-03-08	SIAT3/SIAT1	160	<	80	640	320	320	320	320	160
A/Bolzano/16/2022	H156S	3C.2a1b.2a.2	2022-03-14	SIAT2/SIAT1	80	<	40	160	320	320	320	320	160
A/Milano/32/2022	H156S	3C.2a1b.2a.2	2022-03-16	SIAT3/SIAT1	40	<	40	160	160	320	640	640	40
A/FVG-Trieste/385/2022	H156S	3C.2a1b.2a.2	2022-04-22	SIAT3/SIAT1	40	<	40	80	160	160	640	320	40
A/FVG-Trieste/381/2022	H156S	3C.2a1b.2a.2	2022-04-22	SIAT3/SIAT1	40	<	80	80	160	320	640	640	40
A/Bolzano/24/2022	H156S	3C.2a1b.2a.2	2022-04-29	SIAT2/SIAT1	80	<	40	160	320	320	320	320	80
A/Dakar/05/2022	H156S	3C.2a1b.2a.2	2022-05-09	SIAT1	80	<	40	80	320	320	1280	640	40
A/Dakar/23/2022	H156S	3C.2a1b.2a.2	2022-05-10	SIAT1	80	<	80	160	320	640	1280	640	80
A/Dakar/15/2022	H156S	3C.2a1b.2a.2	2022-05-17	SIAT1	80	<	40	160	320	320	1280	640	80
A/Dakar/06/2022	H156S	3C.2a1b.2a.2	2022-05-23	SIAT1	80	<	80	320	320	640	640	1280	80
A/Dakar/08/2022	H156S	3C.2a1b.2a.2	2022-05-27	SIAT1	40	<	40	160	320	320	640	640	80
A/Dakar/21/2022	H156S	3C.2a1b.2a.2	2022-06-09	SIAT1	80	<	80	160	320	320	1280	640	40
A/Dakar/25/2022	H156S	3C.2a1b.2a.2	2022-06-10	SIAT1	80	<	80	160	320	640	1280	640	80
A/Dakar/24/2022	H156S	3C.2a1b.2a.2	2022-06-13	SIAT1	80	<	80	160	320	320	1280	320	80
A/Dakar/22/2022	H156S	3C.2a1b.2a.2	2022-06-16	SIAT1	80	<	40	160	320	320	640	320	80

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-18. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-22

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antisera								
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17
	Passage history			SIAT	Call	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT
	Ferret number			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES												
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	160	40	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	160	640	160	320	80	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	ES/E2	160	<	80	1280	320	160	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	160	320	1280	320	640
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	1280	640
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	1280	640	1280	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	640
TEST VIRUSES												
A/Netherlands/10222/2021		3C.2a1b.2a.2	2021-12-30	MDCK-MIX2/SIAT1	80	<	40	80	160	320	160	320
A/Netherlands/10025/2022		3C.2a1b.2a.2	2022-01-17	MDCK-MIX2/SIAT1	80	<	80	320	320	640	640	1280
A/Dakar/01/2022		3C.2a1b.1a	2022-01-03	SIAT2	80	<	80	40	160	80	<	80
A/Bolzano/3/2022	D104G	3C.2a1b.2a.2	2022-02-24	SIAT3/SIAT2	80	<	40	80	160	320	160	640
A/Romania/516033/2022	D104G	3C.2a1b.2a.2	2022-04-07	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/516038/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT0/SIAT1	160	<	80	320	640	1280	1280	80
A/Romania/515116/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT1/SIAT1	80	<	40	80	160	640	320	640
A/Romania/516030/2022	D104G	3C.2a1b.2a.2	2022-04-15	SIAT0/SIAT1	80	<	40	80	160	320	320	640
A/Romania/515290/2022	D104G	3C.2a1b.2a.2	2022-04-15	SIAT1/SIAT1	80	<	40	80	160	320	320	640
A/Romania/515274/2022	D104G	3C.2a1b.2a.2	2022-04-15	SIAT1/SIAT1	80	<	40	80	160	320	320	640
A/Perugia/26/2022	D104G	3C.2a1b.2a.2	2022-04-17	SIAT3/SIAT2	80	<	40	80	160	320	320	640
A/Romania/515894/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1/SIAT1	80	<	40	80	160	640	320	640
A/Romania/516042/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/516041/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/516036/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	80	<	40	80	160	640	320	640
A/Romania/516032/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/518145/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	80	<	80	80	160	640	320	640
A/Bolzano/22/2022	D104G	3C.2a1b.2a.2	2022-04-22	SIAT3/SIAT2	40	<	40	80	160	320	320	640
A/Poland/76/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT2	80	<	40	80	160	320	320	640
A/Poland/75/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT1	80	<	40	80	160	640	320	640
A/Romania/516351/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/516348/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT1/SIAT1	80	<	40	80	160	320	320	640
A/Perugia/35/2022	D104G	3C.2a1b.2a.2	2022-04-24	SIAT3/SIAT2	80	<	40	80	160	320	320	640
A/Romania/516460/2022	D104G	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	80	<	40	80	320	640	320	640
A/Romania/516376/2022	D104G	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	80	<	40	80	160	320	320	640
A/Romania/516374/2022	D104G	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	80	<	40	160	160	640	640	640
A/Poland/78/2022	D104G	3C.2a1b.2a.2	2022-04-26	SIAT1	80	<	40	80	160	320	320	640
A/Poland/77/2022	D104G	3C.2a1b.2a.2	2022-04-26	SIAT1	40	<	40	80	160	320	320	640
A/Romania/516540/2022	D104G	3C.2a1b.2a.2	2022-04-26	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/518135/2022	D104G	3C.2a1b.2a.2	2022-04-27	SIAT1/SIAT1	40	<	40	80	160	320	320	640
A/Romania/516884/2022	D104G	3C.2a1b.2a.2	2022-04-29	SIAT1/SIAT2	80	<	80	80	320	320	320	1280
A/Romania/516962/2022	D104G	3C.2a1b.2a.2	2022-05-01	SIAT1/SIAT2	40	<	40	80	160	320	320	320
A/Romania/516961/2022	D104G	3C.2a1b.2a.2	2022-05-01	SIAT1/SIAT2	80	<	40	80	160	320	320	640
A/Poland/96/2022	D104G	3C.2a1b.2a.2	2022-05-08	SIAT1	40	<	40	80	160	320	320	640
A/Poland/98/2022	D104G	3C.2a1b.2a.2	2022-05-09	SIAT1	80	<	40	80	160	320	320	640
A/Poland/99/2022	D104G	3C.2a1b.2a.2	2022-05-10	SIAT1	40	<	40	80	160	320	320	640
A/Poland/101/2022	D104G	3C.2a1b.2a.2	2022-05-15	SIAT1	80	<	40	80	160	640	320	640
A/Romania/521767/2022	D104G	3C.2a1b.2a.2	2022-06-16	SIAT1	80	<	40	160	320	640	320	1280
A/Parma/5/2022	D53G	3C.2a1b.2a.2	2022-02-21	SIAT4/SIAT2	320	<	80	160	320	640	640	1280
A/Romania/518147/2022	D53G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	160	<	80	640	320	1280	640	1280
A/Poland/97/2022	D53G	3C.2a1b.2a.2	2022-05-09	SIAT1	40	<	<	80	160	640	320	640
A/Poland/100/2022	D53G	3C.2a1b.2a.2	2022-05-13	SIAT1	40	<	80	160	320	640	640	640
A/FVG-Trieste/28/2022	H156S	3C.2a1b.2a.2	2022-03-01	SIAT4/SIAT2	40	<	40	80	160	320	640	640
A/Bolzano/4/2022	H156S	3C.2a1b.2a.2	2022-03-04	SIAT3/SIAT2	<	<	40	80	160	160	640	320
A/Romania/515331/2022	H156S	3C.2a1b.2a.2	2022-04-13	SIAT1/SIAT1	40	<	40	80	160	320	640	320
A/Romania/515280/2022	H156S	3C.2a1b.2a.2	2022-04-15	SIAT1/SIAT1	80	<	40	160	160	320	640	640
A/FVG-Trieste/391/2022	H156S	3C.2a1b.2a.2	2022-04-28	SIAT3/SIAT2	40	<	40	80	160	320	320	320
A/Romania/516980/2022	H156S	3C.2a1b.2a.2	2022-04-30	SIAT1/SIAT2	40	<	40	80	160	320	640	640
A/Romania/517370/2022	H156S	3C.2a1b.2a.2	2022-05-03	SIAT1/SIAT2	80	<	40	80	160	320	640	320
A/Togo/1122/2022	H156S	3C.2a1b.2a.2	2022-05-04	SIAT1	80	<	40	160	320	320	640	640
A/Togo/1126/2022	H156S	3C.2a1b.2a.2	2022-05-05	SIAT1	80	<	80	160	320	640	1280	640
A/Togo/1132/2022	H156S	3C.2a1b.2a.2	2022-05-06	SIAT1	160	<	40	160	320	640	1280	320
A/Togo/1129/2022	H156S	3C.2a1b.2a.2	2022-05-06	SIAT1	40	<	40	80	160	320	640	640
A/Romania/518142/2022	H156S	3C.2a1b.2a.2	2022-05-09	SIAT1	40	<	40	160	160	320	640	640
A/Togo/1153/2022	H156S	3C.2a1b.2a.2	2022-05-11	SIAT1	160	<	80	160	320	640	1280	640
A/Togo/1165/2022	H156S	3C.2a1b.2a.2	2022-05-13	SIAT1	80	<	40	160	160	320	640	640
A/Togo/1184/2022	H156S	3C.2a1b.2a.2	2022-05-18	SIAT1	80	<	40	160	320	320	1280	640
A/Togo/1063/2022	H156S	3C.2a1b.2a.2	2022-05-25	SIAT1	80	<	40	160	320	320	640	640
A/Dakar/14/2022	H156S	3C.2a1b.2a.2	2022-06-08	SIAT2	80	<	80	80	160	320	320	640
A/Netherlands/10253/2022	H156	3C.2a1b.2a.2	2022-03-06	MDCK-MIX2/SIAT1	160	<	80	160	320	320	320	640
A/Netherlands/10182/2022	H156	3C.2a1b.2a.2	2022-03-10	MDCK-MIX2/SIAT1	80	<	40	160	320	320	320	320
A/Netherlands/10595/2022	H156	3C.2a1b.2a.2	2022-03-23	MDCK-MIX2/SIAT1	160	<	80	160	320	320	320	80
A/Milano/60/2022	H156	3C.2a1b.2a.2	2022-03-23	SIAT3/SIAT2	160	<	80	320	320	320	320	160
A/Romania/512422/2022	H156	3C.2a1b.2a.2	2022-06-18	SIAT1	80	<	80	160	640	320	320	640

¹Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-19. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-29A

Haemagglutination inhibition titre																
				Post-infection ferret antisera												
												NEW	NEW	NEW		
Viruses	Other information	Collection date	Passage history	A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Slo	A/Slo	A/Slo	A/Kansas	
	Passage history			3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	8720/22	9216/22	9318/22	14/17	
	Ferret number			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	Egg	Egg	Egg	SIAT	
	Genetic group			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/21 ¹	F39/21 ¹	F25/22 ¹	F26/22 ¹	F27/22 ¹	F17/19 ¹	
				3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES																
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	320	40	<	<	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	80	40	160	40	<	<	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	320	320	160	40	320	40	<	<	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	1280	320	160	160	160	80	80	80	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	320	160	160	160	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	640	320	160	320	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	640	640	320	160	160	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	640	1280	640	1280	320	320	640	160
A/Slovenia/87/20/2022	D104G	3C.2a1b.2a.2	2022-02-10	E3(Am1A12)	160	<	80	320	320	640	640	640	640	320	640	80
A/Slovenia/9216/2022	D104G	3C.2a1b.2a.2	2022-02-24	E3(Am1A12)	80	<	40	320	320	320	320	640	640	320	640	80
A/Slovenia/9318/2022	D104G	3C.2a1b.2a.2	2022-03-02	E3(Am1A12)	160	<	80	640	320	640	640	640	1280	640	640	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	<	<	80	80	640
TEST VIRUSES																
A/Estonia/168577/2022	D104G	3C.2a1b.2a.2	2022-02-01	SIAT1/SIAT1	80	<	40	80	320	640	320	320	ND	ND	ND	40
A/Austria/1491740/2022	D104G	3C.2a1b.2a.2	2022-02-11	Cx/SIAT1	40	<	40	80	160	320	160	320	ND	ND	ND	<
A/Austria/1494246/2022	D104G	3C.2a1b.2a.2	2022-02-18	Cx/SIAT1	80	<	40	80	160	640	320	320	ND	ND	ND	40
A/Austria/1498682/2022	D104G	3C.2a1b.2a.2	2022-03-02	Cx/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	<
A/Austria/1503342/2022	D104G	3C.2a1b.2a.2	2022-03-23	Cx/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	40
A/Austria/1504561/2022	D104G	3C.2a1b.2a.2	2022-03-28	Cx/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	40
A/Austria/1504529/2022	D104G	3C.2a1b.2a.2	2022-03-28	Cx/SIAT1	40	<	40	40	160	320	320	320	ND	ND	ND	<
A/Belgium/G0197/2022	D104G	3C.2a1b.2a.2	2022-04-08	SIAT1/SIAT1	80	<	40	80	160	320	320	320	ND	ND	ND	40
A/Belgium/S1049/2022	D104G	3C.2a1b.2a.2	2022-04-09	SIAT1/SIAT1	80	<	40	160	160	160	320	320	ND	ND	ND	40
A/Austria/KL4914/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	<
A/Estonia/KL4911/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1/SIAT1	40	<	<	80	160	320	320	320	ND	ND	ND	<
A/Estonia/KL4967/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT1/SIAT1	40	<	<	80	160	320	160	320	ND	ND	ND	<
A/Austria/1510167/2022	D104G	3C.2a1b.2a.2	2022-04-16	SIAT1	80	<	40	80	160	320	320	320	ND	ND	ND	40
A/Estonia/169447/2022	D104G	3C.2a1b.2a.2	2022-04-16	SIAT1/SIAT1	40	<	<	80	160	320	320	320	ND	ND	ND	<
A/Estonia/169303/2022	D104G	3C.2a1b.2a.2	2022-04-16	SIAT1/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	40
A/Estonia/169446/2022	D104G	3C.2a1b.2a.2	2022-04-17	SIAT1/SIAT1	80	<	40	80	160	320	320	640	ND	ND	ND	40
A/Belgium/S1030/2022	D104G	3C.2a1b.2a.2	2022-04-17	SIAT1/SIAT1	80	<	40	80	160	320	320	320	640	320	320	40
A/Belgium/S1027/2022	D104G	3C.2a1b.2a.2	2022-04-17	SIAT1/SIAT1	80	<	40	80	160	320	320	320	320	160	160	40
A/Estonia/KL4992/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1/SIAT1	40	<	40	80	160	320	320	320	ND	ND	ND	<
A/Estonia/169344/2022	D104G	3C.2a1b.2a.2	2022-04-18	MDCKCx/SIAT1	40	<	40	80	160	320	320	160	ND	ND	ND	<
A/Estonia/169343/2022	D104G	3C.2a1b.2a.2	2022-04-18	MDCKCx/SIAT1	80	<	40	80	320	640	320	320	ND	ND	ND	<
A/Austria/1510472/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1	80	<	40	160	320	640	320	640	ND	ND	ND	40
A/Estonia/169374/2022	D104G	3C.2a1b.2a.2	2022-04-19	MDCKCx/SIAT1	80	<	40	80	320	640	320	320	ND	ND	ND	40
A/Austria/1510800/2022	D104G	3C.2a1b.2a.2	2022-04-20	SIAT1	80	<	40	80	160	320	320	320	640	320	320	40
A/Estonia/KL4998/2022	D104G	3C.2a1b.2a.2	2022-04-20	SIAT1/SIAT1	40	<	40	80	160	320	320	160	ND	ND	ND	<
A/Estonia/169390/2022	D104G	3C.2a1b.2a.2	2022-04-20	SIAT1/SIAT1	40	<	40	80	160	320	160	320	ND	ND	ND	<
A/Estonia/169421/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	80	<	40	160	160	640	320	320	640	320	320	40
A/Estonia/169420/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	80	<	40	80	160	320	320	320	320	320	320	40
A/Estonia/169386/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	80	<	40	160	160	640	320	320	640	320	320	40
A/Estonia/169385/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	40	<	40	80	160	320	320	320	640	320	320	<
A/Estonia/169384/2022	D104G	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	40	<	40	80	160	320	160	320	ND	ND	ND	<
A/Austria/1512885/2022	D104G	3C.2a1b.2a.2	2022-04-27	SIAT1	80	<	40	80	320	640	320	320	640	640	640	40
A/Austria/1513777/2022	D104G	3C.2a1b.2a.2	2022-04-28	SIAT1	80	<	40	80	320	320	320	320	640	640	640	40
A/Austria/1513868/2022	D104G	3C.2a1b.2a.2	2022-05-01	SIAT1	80	<	40	80	160	320	320	320	160	80	80	40
A/Austria/1502410/2022	D53G	3C.2a1b.2a.2	2022-03-21	Cx/SIAT1	160	<	80	320	320	640	640	640	ND	ND	ND	80
A/Estonia/169301/2022	D53G	3C.2a1b.2a.2	2022-04-17	SIAT1/SIAT1	160	<	80	320	320	640	640	640	ND	ND	ND	80
A/Austria/1497981/2022	H156S	3C.2a1b.2a.2	2022-03-04	Cx/SIAT1	40	<	40	160	160	320	640	320	ND	ND	ND	40
A/Austria/1499572/2022	H156	3C.2a1b.2a.2	2022-03-10	Cx/SIAT1	80	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1501955/2022	H156	3C.2a1b.2a.2	2022-03-18	Cx/SIAT1	80	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1502449/2022	H156	3C.2a1b.2a.2	2022-03-20	Cx/SIAT1	160	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1503824/2022	H156	3C.2a1b.2a.2	2022-03-23	Cx/SIAT1	160	<	40	160	320	320	320	160	ND	ND	ND	160
A/Austria/1503368/2022	H156	3C.2a1b.2a.2	2022-03-23	Cx/SIAT1	80	<	40	160	320	320	160	160	ND	ND	ND	80
A/Austria/1503854/2022	H156	3C.2a1b.2a.2	2022-03-24	Cx/SIAT1	80	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1503823/2022	H156	3C.2a1b.2a.2	2022-03-24	Cx/SIAT1	160	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1504127/2022	H156	3C.2a1b.2a.2	2022-03-25	Cx/SIAT1	160	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1504562/2022	H156	3C.2a1b.2a.2	2022-03-28	Cx/SIAT1	80	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/15068277/2022	H156	3C.2a1b.2a.2	2022-04-08	SIAT1	160	<	40	160	320	320	320	160	ND	ND	ND	80
A/Austria/1509468/2022	H156	3C.2a1b.2a.2	2022-04-13	SIAT1	80	<	40	160	320	320	160	160	ND	ND	ND	80
A/Austria/1514532/2022	H156	3C.2a1b.2a.2	2022-05-02	SIAT1	80	<	40	160	320	320	320	160	ND	ND	ND	80
A/Belgium/S1074/2022	H156	3C.2a1b.2a.2	2022-05-03	SIAT1/SIAT1	80	<	40	160	320	320	160	160	80	40	40	80

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 ≤ = <40. ND = Not Done

Vaccine
NH 2021-22Vaccine
SH 2022

4-fold

> 8-fo

ised by t

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-20. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-07-29B

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
				F19/20 ¹	St Judes F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F38/21 ¹	F17/19 ¹	
Passage history													
Ferret number													
Genetic group				3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	80	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	1280	320	160	160	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	640	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	S0/S3	80	<	80	160	320	640	320	640	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	1280	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	640	1280	640	1280	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	80	640
TEST VIRUSES													
A/Saudi Arabia/22-28/2021		3C.2a1b.2a.2	2021-11-04	SIAT1	160	<	80	160	320	320	320	320	160
A/Saudi Arabia/22-35/2021		3C.2a1b.2a.2	2021-11-15	SIAT1	80	<	80	80	160	320	320	320	40
A/Saudi Arabia/6/2021		3C.2a1b.2a.2	2021-12-05	SIAT1	80	<	80	160	320	640	320	640	80
A/Saudi Arabia/22-21/2021		3C.2a1b.2a.2	2021-12-05	SIAT1	80	<	80	80	160	320	320	320	40
A/Saudi Arabia/5/2021		3C.2a1b.2a.2	2021-12-08	SIAT1	160	<	80	160	320	320	320	320	160
A/Saudi Arabia/22-23/2021		3C.2a1b.2a.2	2021-12-13	SIAT1	80	<	80	160	320	320	320	640	40
A/Saudi Arabia/11/2021		3C.2a1b.2a.2	2021-12-15	SIAT1	40	<	40	40	160	160	160	320	40
A/Saudi Arabia/18/2021		3C.2a1b.2a.2	2021-12-20	SIAT1	40	<	40	80	160	160	640	320	40
A/Saudi Arabia/17/2021		3C.2a1b.2a.2	2021-12-21	SIAT1	40	<	40	80	160	160	640	320	40
A/Slovenia/9449/2022	D104G	3C.2a1b.2a.2	2022-03-08	SIATx/SIAT1	80	<	40	80	160	320	320	320	<
A/Slovenia/9450/2022	D104G	3C.2a1b.2a.2	2022-03-08	MDCKx/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9454/2022	D104G	3C.2a1b.2a.2	2022-03-08	SIAT1/SIAT1	40	<	40	40	160	160	160	160	<
A/Slovenia/9456/2022	D104G	3C.2a1b.2a.2	2022-03-08	SIATx/SIAT2	80	<	40	80	160	320	320	320	40
A/Slovenia/9448/2022	D104G	3C.2a1b.2a.2	2022-03-09	MDCKx/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9558/2022	D104G	3C.2a1b.2a.2	2022-03-10	SIAT1/SIAT1	40	<	40	40	160	160	160	320	<
A/Slovenia/9565/2022	D104G	3C.2a1b.2a.2	2022-03-11	SIAT1/SIAT1	80	<	40	80	320	320	320	320	40
A/Slovenia/9560/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT1/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9599/2022	D104G	3C.2a1b.2a.2	2022-03-15	MDCK1/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9614/2022	D104G	3C.2a1b.2a.2	2022-03-17	MDCKx/SIAT1	80	<	40	40	160	320	320	320	<
A/Slovenia/9623/2022	D104G	3C.2a1b.2a.2	2022-03-17	MDCKx/SIAT1	40	<	40	40	160	320	320	320	<
A/Slovenia/9624/2022	D104G	3C.2a1b.2a.2	2022-03-17	MDCKx/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9626/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIATx/SIAT1	40	<	40	40	160	320	160	320	<
A/Slovenia/9633/2022	D104G	3C.2a1b.2a.2	2022-03-21	MDCK1/SIAT1	40	<	40	80	320	320	320	320	40
A/Slovenia/9644/2022	D104G	3C.2a1b.2a.2	2022-03-21	MDCKx/SIAT1	40	<	40	40	160	320	160	320	40
A/Slovenia/9641/2022	D104G	3C.2a1b.2a.2	2022-03-22	MDCKx/SIAT2	40	<	40	40	160	320	320	320	40
A/Slovenia/9670/2022	D104G	3C.2a1b.2a.2	2022-03-22	MDCKx/SIAT1	40	<	40	40	160	320	160	160	40
A/Slovenia/9659/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	80	<	40	80	160	320	320	640	40
A/Slovenia/9668/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIATx/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9706/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIATx/SIAT2	40	<	40	40	160	320	160	320	40
A/Slovenia/9707/2022	D104G	3C.2a1b.2a.2	2022-03-23	MDCKx/SIAT1	40	<	40	80	160	320	320	320	40
A/Slovenia/9671/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIATx/SIAT1	40	<	40	80	160	320	160	320	40
A/Slovenia/9676/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIATx/SIAT1	40	<	40	40	160	320	320	320	40
A/Slovenia/9705/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIATx/SIAT1	40	<	40	40	160	320	320	320	40
A/Slovenia/9708/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIATx/SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9726/2022	D104G	3C.2a1b.2a.2	2022-03-25	SIATx/SIAT2	40	<	40	80	160	320	320	320	40
A/Slovenia/9728/2022	D104G	3C.2a1b.2a.2	2022-03-25	MDCKx/SIAT1	40	<	40	80	160	320	320	320	40
A/Slovenia/9736/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIATx/SIAT2	40	<	40	80	160	320	320	320	40
A/Slovenia/9737/2022	D104G	3C.2a1b.2a.2	2022-03-28	MDCKx/SIAT1	40	<	40	40	160	160	320	320	40
A/Slovenia/9739/2022	D104G	3C.2a1b.2a.2	2022-03-29	SIAT1/SIAT1	80	<	40	80	160	320	320	320	40
A/Spain/1127/2022	D104G	3C.2a1b.2a.2	2022-05-03	SIAT1	40	<	40	40	160	320	320	320	40
A/Spain/1097/2022	D104G	3C.2a1b.2a.2	2022-05-03	SIAT1	40	<	40	40	160	320	320	320	40
A/Spain/1126/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	40	<	40	80	160	320	320	320	40
A/Spain/1105/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	80	<	40	80	160	320	320	320	40
A/Spain/1100/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	40	<	40	80	160	320	320	320	40
A/Spain/1099/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	80	<	40	80	160	320	320	320	40
A/Spain/1098/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	40	<	40	80	160	320	320	320	40
A/Spain/1091/2022	D104G	3C.2a1b.2a.2	2022-05-04	SIAT1	40	<	40	80	160	320	320	320	40
A/Spain/1109/2022	D104G	3C.2a1b.2a.2	2022-05-05	SIAT1	80	<	40	80	160	320	320	320	40
A/Spain/1108/2022	D104G	3C.2a1b.2a.2	2022-05-05	SIAT1	80	<	40	80	160	320	320	320	40
A/Slovenia/9362/2022	D53G	3C.2a1b.2a.2	2022-03-04	MDCK1/SIAT1	160	<	80	320	320	640	640	640	80
A/Slovenia/9528/2022	D53G	3C.2a1b.2a.2	2022-03-08	SIATx/SIAT1	160	<	80	320	320	640	640	640	80
A/Slovenia/9562/2022	D53G	3C.2a1b.2a.2	2022-03-14	SIAT1/SIAT1	160	<	80	320	320	640	640	640	80
A/Slovenia/9579/2022	D53G	3C.2a1b.2a.2	2022-03-15	SIAT1/SIAT1	160	<	80	320	320	640	640	640	80
A/Slovenia/9675/2022	D53G	3C.2a1b.2a.2	2022-03-24	SIATx/SIAT1	160	<	80	160	320	640	640	640	80
A/Slovenia/9740/2022	D53G	3C.2a1b.2a.2	2022-03-29	SIATx/SIAT1	160	<	80	320	320	640	640	640	80
A/Spain/1059/2022	H156S	3C.2a1b.2a.2	2022-04-22	SIAT1	40	<	40	40	80	160	320	160	40

¹Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 <= <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-21. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-08-05

Haemagglutination inhibition titre														
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera										
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Wyom	A/Kansas	
				3264/19	2671/19	926256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	01/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	Egg	SIAT	
Passage history														
Ferret number														
Genetic group														
REFERENCE VIRUSES														
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	320	<	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	80	40	160	<	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	<	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	2560	320	160	160	160	40	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	640	80	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	640	320	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	640	640	160	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	640	1280	640	2560	640	160
A/Wyoming/01/2021	D104G	3C.2a1b.2a.2	2021-08-27	E4 (Am2AL2)	160	<	80	320	320	640	160	1280	1280	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	80	<	640
TEST VIRUSES														
A/Ukraine/348/2022		3C.2a1b.2a.2	2022-01-24	SIAT1	80	40	160	320	320	1280	640	640	ND	80
A/Ukraine/360/2022		3C.2a1b.2a.2	2022-01-26	SIAT1	40	<	40	80	160	320	320	320	ND	40
A/Ukraine/358/2022		3C.2a1b.2a.2	2022-01-26	SIAT1	80	<	<	80	160	320	320	320	ND	40
A/Tunisia/7467/2022		3C.2a1b.2a.2	2022-01-28	SIAT2	80	<	40	80	160	320	320	320	320	40
A/Estonia/16877/2022	D104G	3C.2a1b.2a.2	2022-02-20	SIAT1/SIAT3	80	<	40	40	160	320	320	320	320	40
A/La Rioja/2087/2022	D104G	3C.2a1b.2a.2	2022-03-03	SIAT1/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/1374/2022	D104G	3C.2a1b.2a.2	2022-03-15	Px/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Ukraine/372/2022	D104G	3C.2a1b.2a.2	2022-03-15	SIAT1	40	<	40	80	160	320	320	320	ND	40
A/La Rioja/2099/2022	D104G	3C.2a1b.2a.2	2022-03-16	Px/SIAT1	80	<	40	80	320	640	320	640	ND	40
A/Castilla La Mancha/1376/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	80	<	40	80	<	320	320	640	ND	40
A/La Rioja/2097/2022	D104G	3C.2a1b.2a.2	2022-03-22	SIAT1/SIAT1	80	<	40	80	320	640	640	640	ND	40
A/Madrid/1436/2022	D104G	3C.2a1b.2a.2	2022-03-22	SIAT1/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Andalucia/2706/2022	D104G	3C.2a1b.2a.2	2022-03-23	Px/SIAT1	80	<	40	80	320	320	320	320	ND	40
A/Galicia/1451/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	80	<	40	80	320	320	320	640	ND	40
A/Madrid/1445/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	40	<	<	40	160	320	160	320	ND	40
A/Andalucia/2708/2022	D104G	3C.2a1b.2a.2	2022-03-24	Px/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Navarra/1477/2022	D104G	3C.2a1b.2a.2	2022-03-24	Px/SIAT1	80	<	40	80	320	640	320	320	ND	40
A/Andalucia/2709/2022	D104G	3C.2a1b.2a.2	2022-03-27	Px/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Andalucia/2714/2022	D104G	3C.2a1b.2a.2	2022-03-30	Px/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/La Rioja/2090/2022	D104G	3C.2a1b.2a.2	2022-04-01	SIAT2/SIAT1	80	<	40	80	320	320	320	320	ND	40
A/Switzerland/64395/2022	D104G	3C.2a1b.2a.2	2022-04-02	SIAT1	80	<	40	80	160	320	320	320	320	40
A/La Rioja/2088/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT2/SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Switzerland/63625/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT1	40	<	40	80	160	320	320	320	320	40
A/La Rioja/2094/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT1/SIAT1	40	<	40	80	160	320	320	320	ND	40
A/La Rioja/2093/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT1/SIAT1	40	<	40	80	160	320	320	320	ND	40
A/Switzerland/63466/2022	D104G	3C.2a1b.2a.2	2022-04-07	SIAT1	40	<	40	80	160	320	320	320	ND	40
A/Iceland/32842/2022	D104G	3C.2a1b.2a.2	2022-04-12	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Switzerland/64156/2022	D104G	3C.2a1b.2a.2	2022-04-12	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Ukraine/406/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1	80	<	40	80	160	320	320	320	320	40
A/Ukraine/405/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1	80	<	40	80	160	640	320	640	320	40
A/Switzerland/67859/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT1	40	<	40	40	160	320	320	320	ND	40
A/Austria/1513783/2022	D104G	3C.2a1b.2a.2	2022-04-29	SIAT2	80	<	40	80	160	640	320	640	640	40
A/Castilla La Mancha/2231/2022	D104G	3C.2a1b.2a.2	2022-05-02	SIAT1	40	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2513/2022	D104G	3C.2a1b.2a.2	2022-05-14	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2511/2022	D104G	3C.2a1b.2a.2	2022-05-14	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Iceland/37144/2022	D104G	3C.2a1b.2a.2	2022-05-15	SIAT1	40	<	40	40	320	320	160	160	320	40
A/Castilla La Mancha/2550/2022	D104G	3C.2a1b.2a.2	2022-05-17	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2508/2022	D104G	3C.2a1b.2a.2	2022-05-20	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2643/2022	D104G	3C.2a1b.2a.2	2022-05-21	SIAT1	160	<	40	320	320	640	320	320	ND	80
A/Castilla La Mancha/2505/2022	D104G	3C.2a1b.2a.2	2022-05-23	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2656/2022	D104G	3C.2a1b.2a.2	2022-05-27	SIAT1	80	<	40	80	160	320	320	320	ND	40
A/Castilla La Mancha/2637/2022	D104G	3C.2a1b.2a.2	2022-05-29	SIAT1	80	<	40	80	160	320	320	320	320	40
A/Castilla La Mancha/2697/2022	D104G	3C.2a1b.2a.2	2022-06-04	SIAT1	80	<	40	80	160	320	320	320	640	40
A/Castilla La Mancha/2770/2022	D104G	3C.2a1b.2a.2	2022-06-10	SIAT1	80	<	<	80	160	160	160	160	320	40
A/Switzerland/63092/2022	D53G	3C.2a1b.2a.2	2022-04-25	SIAT1	160	<	160	320	640	640	640	1280	ND	80
A/Andalucia/2703/2022	H156S	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	40	<	<	40	160	320	320	320	ND	<
A/La Rioja/2089/2022	H156S	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	40	<	80	160	160	320	320	320	ND	<

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-22. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-08-12

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
	Ferret number			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F39/21 ¹	F17/19 ¹	
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1	
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	640	160	640	320	320	160	40	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	640	160	320	80	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	2560	320	160	160	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	640	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	640	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	320	640	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	1280	640	2560	160	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	320	320
TEST VIRUSES													
A/Ukraine/354/2021		3C.2a1b.2a.2	2021-12-22	SIAT2	80	<	<	160	320	640	640	640	40
A/Tunisia/7468/2022		3C.2a1b.1a	2022-01-29	SIAT4	80	80	320	160	160	160	40	320	160
A/Tunisia/8259/2022	D104G	3C.2a1b.2a.2	2022-02-05	SIAT4	80	<	40	320	160	160	320	640	80
A/Andalucia/2702/2022	D104G	3C.2a1b.2a.2	2022-03-09	SIAT1/SIAT2	80	80	80	80	160	320	320	640	40
A/Andalucia/2701/2022	D104G	3C.2a1b.2a.2	2022-03-10	SIAT1/SIAT2	40	<	40	80	160	320	320	320	<
A/Ireland/14993/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT1	80	<	40	80	320	320	320	320	<
A/Ireland/15242/2022	D104G	3C.2a1b.2a.2	2022-03-14	SIAT1	40	<	40	80	320	320	320	320	<
A/Castilla La Mancha/1384/2022	D104G	3C.2a1b.2a.2	2022-03-15	Px/SIAT2	80	<	40	80	160	320	320	640	<
A/Castilla La Mancha/1383/2022	D104G	3C.2a1b.2a.2	2022-03-15	Px/SIAT2	40	<	40	80	160	320	320	320	<
A/La Rioja/2098/2022	D104G	3C.2a1b.2a.2	2022-03-16	Px/SIAT2	80	<	40	80	320	320	320	640	<
A/La Rioja/2096/2022	D104G	3C.2a1b.2a.2	2022-03-16	Px/SIAT2	40	<	40	80	160	320	320	320	<
A/La Rioja/2095/2022	D104G	3C.2a1b.2a.2	2022-03-18	SIAT1/SIAT2	80	<	40	80	160	320	320	320	<
A/Ireland/16348/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT1	80	<	80	320	640	640	640	640	<
A/Andalucia/2705/2022	D104G	3C.2a1b.2a.2	2022-03-22	SIAT1/SIAT2	80	<	40	80	160	320	320	640	<
A/Madrid/1413/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT2	80	<	40	80	160	320	320	320	<
A/Belgium/W0019/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	80	<	40	80	160	320	320	320	<
A/Belgium/S0814/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1/SIAT1	80	<	40	160	160	160	320	320	40
A/Belgium/S0740/2022	D104G	3C.2a1b.2a.2	2022-03-25	SIAT1/SIAT1	40	<	40	40	160	320	160	320	<
A/Belgium/S0834/2022	D104G	3C.2a1b.2a.2	2022-03-26	SIAT1/SIAT1	80	<	40	80	160	160	160	320	40
A/Belgium/S0162/2022	D104G	3C.2a1b.2a.2	2022-03-28	SIAT1/SIAT1	40	<	40	40	160	320	320	320	<
A/Serbia/2007/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1	80	<	40	80	160	320	320	320	<
A/Belgium/S0761/2022	D104G	3C.2a1b.2a.2	2022-03-30	SIAT1/SIAT1	40	<	40	80	160	320	320	320	<
A/Andalucia/2713/2022	D104G	3C.2a1b.2a.2	2022-03-31	Px/SIAT2	40	<	40	80	160	320	320	320	<
A/Belgium/S0747/2022	D104G	3C.2a1b.2a.2	2022-03-31	SIAT1/SIAT1	40	<	40	80	160	320	320	320	<
A/Belgium/S1045/2022	D104G	3C.2a1b.2a.2	2022-04-01	SIAT1/SIAT1	40	<	40	80	160	320	320	640	40
A/Belgium/S0166/2022	D104G	3C.2a1b.2a.2	2022-04-02	SIAT1/SIAT1	40	<	40	80	160	320	320	320	<
A/Belgium/S0188/2022	D104G	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	80	<	40	80	160	320	320	320	<
A/Portugal/211230/2022	D104G	3C.2a1b.2a.2	2022-04-05	SIAT2/SIAT1	40	<	40	80	160	320	320	320	<
A/La Rioja/2091/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT2	40	<	40	80	160	320	320	320	<
A/Belgium/S1065/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	80	<	40	80	160	320	320	320	<
A/Belgium/S0757/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	40	<	40	80	160	320	320	640	<
A/Belgium/S0201/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	80	<	40	80	160	160	160	320	<
A/Lithuania/33830/2022	D104G	3C.2a1b.2a.2	2022-04-14	SIAT1	40	<	40	80	160	160	320	320	<
A/Portugal/211242/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1/SIAT1	80	<	40	80	160	640	320	640	40
A/Portugal/211249/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1/SIAT1	40	<	40	80	160	320	320	320	<
A/Ireland/23694/2022	D104G	3C.2a1b.2a.2	2022-04-23	SIAT1	80	<	40	80	160	320	320	320	<
A/Ukraine/371/2022	D53G	3C.2a1b.2a.2	2022-02-22	SIAT2	80	<	40	160	320	640	320	640	40
A/Belgium/S0182/2022	D53G	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	160	<	40	320	320	640	640	640	80
A/Lithuania/30885/2022	D53G	3C.2a1b.2a.2	2022-04-06	SIAT1	160	<	80	320	320	640	640	640	80
A/Belgium/S0772/2022	D53G	3C.2a1b.2a.2	2022-04-07	SIAT1/SIAT1	80	<	40	160	320	640	320	640	40
A/Lithuania/33225/2022	D53G	3C.2a1b.2a.2	2022-04-13	SIAT2	160	<	80	320	320	320	640	640	80
A/Lithuania/33224/2022	D53G	3C.2a1b.2a.2	2022-04-13	SIAT1	40	<	40	40	160	160	320	320	<
A/Portugal/210648/2022	H156S	3C.2a1b.2a.2	2022-03-05	SIAT1/SIAT1	40	<	40	80	160	320	640	320	<
A/Serbia/1550/2022	H156S	3C.2a1b.2a.2	2022-03-24	SIAT1	40	<	40	160	160	320	640	320	40
A/Belgium/S0894/2022	H156S	3C.2a1b.2a.2	2022-03-26	SIAT1/SIAT1	40	<	40	80	80	160	320	320	<
A/Belgium/S0163/2022	H156S	3C.2a1b.2a.2	2022-03-28	SIAT1/SIAT1	<	<	40	40	80	160	320	320	<
A/Ireland/02000/2022	H156S	3C.2a1b.2a.2	2022-04-14	SIAT1	<	<	40	80	160	160	640	640	<
A/Portugal/210508/2022	H156S	3C.2a1b.2a.2	2022-04-18	SIAT2/SIAT1	40	<	40	40	160	160	640	320	<
A/Ghana/1115/2022	H156S	3C.2a1b.2a.2	2022-05-11	SIAT1	40	<	80	80	160	320	640	640	40
A/Ghana/1724/2022	H156S	3C.2a1b.2a.2	2022-05-18	SIAT1	80	<	40	160	320	320	640	320	40
A/Ghana/1654/2022	H156S	3C.2a1b.2a.2	2022-05-19	SIAT1	40	<	40	80	160	160	320	320	<
A/Ghana/1836/2022	H156S	3C.2a1b.2a.2	2022-05-20	SIAT1	<	<	<	40	80	160	160	320	<
A/Ghana/2123/2022	H156S	3C.2a1b.2a.2	2022-05-23	SIAT1	80	<	40	80	320	320	640	320	40
A/Belgium/S0159/2022	H156	3C.2a1b.2a.2	2022-03-25	SIAT1/SIAT1	160	<	80	160	320	320	320	320	80
A/Belgium/S0151/2022	H156	3C.2a1b.2a.2	2022-03-25	SIAT1/SIAT1	80	<	40	80	320	160	160	160	40
A/Belgium/S0164/2022	H156	3C.2a1b.2a.2	2022-03-28	SIAT1/SIAT1	80	<	40	80	320	320	160	160	80
A/Belgium/S0183/2022	H156	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	80	<	40	160	320	320	320	160	80
A/Belgium/S1128/2022	H156	3C.2a1b.2a.2	2022-04-05	SIAT1/SIAT1	80	<	40	320	320	160	160	80	80

Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)
1 <= 40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-23. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-08-17

				Haemagglutination inhibition titre									
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
				F19/20 [†]	St Jude's F21/20 [†]	F03/21 [†]	F10/21 [†]	F07/21 [†]	F35/21 [†]	F07/22 [†]	F39/21 [†]	F17/19 [†]	
Genetic group	3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1				
REFERENCE VIRUSES													
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	320	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	80	40	160	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	320	320	320	160	40	320	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	1280	320	160	160	160	80
A/Bangladesh/4005/2020	H156	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	640	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	640	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	80	160	160	320	640	640	40
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	640	1280	640	2560	160
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	80	320
TEST VIRUSES													
A/Portugal/210273/2022	D104G	3C.2a1b.2a.2	2022-02-21	SIAT1/SIAT2	40	<	40	40	160	320	320	320	<
A/Portugal/210274/2022	D104G	3C.2a1b.2a.2	2022-02-21	SIAT2/SIAT2	80	<	40	80	320	640	320	640	<
A/Portugal/210372/2022	D104G	3C.2a1b.2a.2	2022-03-08	SIAT2/SIAT2	80	<	40	80	160	640	320	640	<
A/Portugal/210588/2022	D104G	3C.2a1b.2a.2	2022-03-09	SIAT2/SIAT2	40	<	40	80	160	640	320	640	<
A/Portugal/210592/2022	D104G	3C.2a1b.2a.2	2022-03-15	SIAT1/SIAT2	80	<	40	80	160	640	320	640	<
A/Portugal/210603/2022	D104G	3C.2a1b.2a.2	2022-03-15	SIAT2/SIAT2	80	<	40	80	160	640	320	320	<
A/Portugal/210611/2022	D104G	3C.2a1b.2a.2	2022-03-15	SIAT2/SIAT2	80	<	40	80	160	640	320	320	<
A/Ireland/16347/2022	D104G	3C.2a1b.2a.2	2022-03-21	SIAT2	80	<	40	80	160	640	320	640	<
A/Ireland/24347/2022	D104G	3C.2a1b.2a.2	2022-04-27	SIAT2	80	<	40	80	320	640	320	640	<
A/Castilla La Mancha/2234/2022	D104G	3C.2a1b.2a.2	2022-05-03	SIAT1	80	<	40	80	160	640	320	640	<
A/Castilla La Mancha/2224/2022	D104G	3C.2a1b.2a.2	2022-05-08	SIAT1	80	<	40	80	160	320	320	320	<
A/Castilla La Mancha/2333/2022	D104G	3C.2a1b.2a.2	2022-05-16	SIAT1	40	<	40	40	160	320	320	320	<
A/Castilla La Mancha/2358/2022	D104G	3C.2a1b.2a.2	2022-05-17	SIAT1	40	<	40	40	160	320	320	320	<
A/Castilla La Mancha/2356/2022	D104G	3C.2a1b.2a.2	2022-05-19	SIAT1	80	<	40	40	160	320	320	320	<
A/Portugal/210269/2022	H156S	3C.2a1b.2a.2	2022-02-21	SIAT1/SIAT2	80	<	80	160	320	640	1280	640	<
A/Portugal/210307/2022	H156S	3C.2a1b.2a.2	2022-02-25	SIAT2/SIAT2	80	<	40	80	320	320	640	640	<
A/Portugal/210312/2022	H156S	3C.2a1b.2a.2	2022-02-26	SIAT2/SIAT2	80	<	80	80	320	640	1280	640	<
A/Portugal/210366/2022	H156S	3C.2a1b.2a.2	2022-03-04	SIAT2/SIAT2	40	<	40	80	160	320	640	320	<
* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)							Vaccine NH 2021-22		Vaccine SH 2022 NH 2022-23				
1 < = <40, ND = Not Done													

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-24. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-08-19

Haemagglutination inhibition titre													
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera									
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Darwin	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	9/21	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	Egg	SIAT	
				F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F35/21 ¹	F07/22 ¹	F39/21 ¹	F17/19 ¹	
Genetic group													
REFERENCE VIRUSES													
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	640	160	640	320	320	160	40	320	160	
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	640	640	160	320	80	40	160	160	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	640	320	320	160	40	320	160	
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	1280	320	160	160	160	80	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160	160	
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	320	640	80	
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	80	160	160	320	640	40	
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E4	160	<	80	640	1280	640	1280	160	
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	80	160	
TEST VIRUSES													
A/Saint-Denis/48407/2021	3C.2a1b.2a.2	2021-12-06	SIAT1	<	<	80	80	80	160	320	320	<	
A/Dunkerque/51979/2021	3C.2a1b.2a.2	2021-12-24	SIAT1	40	<	40	40	160	320	320	320	40	
A/Portugal/210259/2022	D104G	3C.2a1b.2a.2	2022-02-17	SIAT1/SIAT2	80	<	80	80	160	640	320	640	
A/Nord Pas de Calais/11659/2022	D104G	3C.2a1b.2a.2	2022-02-17	SIAT1	80	<	80	80	160	320	320	40	
A/Portugal/210266/2022	D104G	3C.2a1b.2a.2	2022-02-18	SIAT1/SIAT2	40	<	40	40	160	320	320	40	
A/Dunkerque/16908/2022	D104G	3C.2a1b.2a.2	2022-03-01	SIAT1	80	<	40	160	160	320	320	40	
A/Portugal/210597/2022	D104G	3C.2a1b.2a.2	2022-03-09	SIAT1/SIAT2	80	<	40	40	160	640	320	40	
A/Montenegro/1557212202/2022	D104G	3C.2a1b.2a.2	2022-03-09	SIAT1	80	<	80	80	160	320	320	40	
A/Montenegro/1614822202/2022	D104G	3C.2a1b.2a.2	2022-03-17	SIAT1	40	<	40	40	160	320	320	40	
A/Montenegro/1651632202/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1	40	<	40	40	160	320	320	40	
A/Montenegro/1651692202/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1	80	<	160	160	320	640	640	80	
A/Pays de Loire/22333/2022	D104G	3C.2a1b.2a.2	2022-03-24	SIAT1	80	<	40	80	160	640	320	40	
A/Montenegro/1714472202/2022	D104G	3C.2a1b.2a.2	2022-04-01	SIAT1	80	<	40	80	160	320	640	40	
A/Garches/28367/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1	80	<	40	80	160	320	320	40	
A/Alsace/28544/2022	D104G	3C.2a1b.2a.2	2022-04-08	SIAT1	80	<	80	80	160	640	320	40	
A/Picardie/29331/2022	D104G	3C.2a1b.2a.2	2022-04-13	SIAT1	40	<	40	40	80	320	320	40	
A/Sarajevo/192G/2022	D104G	3C.2a1b.2a.2	2022-04-19	SIAT2/SIAT2	80	<	40	80	160	320	640	40	
A/Qatar/10-VI-22-1634794/2022	D104G	3C.2a1b.2a.2	2022-06-01	SIAT1	40	<	40	40	160	320	320	40	
A/Portugal/210192/2022	H156S	3C.2a1b.2a.2	2022-02-08	SIAT2/SIAT2	80	<	40	80	160	320	640	40	
A/Portugal/210210/2022	H156S	3C.2a1b.2a.2	2022-02-09	SIAT2/SIAT2	80	<	80	160	160	320	640	40	
A/Portugal/210214/2022	H156S	3C.2a1b.2a.2	2022-02-10	SIAT1/SIAT2	80	<	80	160	320	320	640	40	
A/Portugal/210212/2022	H156S	3C.2a1b.2a.2	2022-02-11	SIAT2/SIAT2	80	<	80	160	320	640	1280	40	
A/Portugal/210220/2022	H156S	3C.2a1b.2a.2	2022-02-11	SIAT1/SIAT2	80	<	80	160	320	640	1280	40	
A/Portugal/210218/2022	H156S	3C.2a1b.2a.2	2022-02-14	SIAT1/SIAT2	80	<	80	80	320	320	640	40	
A/Portugal/210219/2022	H156S	3C.2a1b.2a.2	2022-02-14	SIAT2/SIAT2	80	<	40	40	160	320	640	40	
A/Portugal/210224/2022	H156S	3C.2a1b.2a.2	2022-02-14	SIAT2/SIAT2	80	<	40	80	320	320	640	40	
A/Portugal/210255/2022	H156S	3C.2a1b.2a.2	2022-02-16	SIAT1/SIAT2	80	<	80	160	320	640	1280	40	
A/Portugal/210256/2022	H156S	3C.2a1b.2a.2	2022-02-16	SIAT2/SIAT2	80	<	80	80	320	640	640	40	
A/Portugal/210257/2022	H156S	3C.2a1b.2a.2	2022-02-16	SIAT2/SIAT2	80	<	80	160	320	640	1280	80	
A/Portugal/210258/2022	H156S	3C.2a1b.2a.2	2022-02-16	SIAT2/SIAT2	80	<	80	80	320	640	1280	40	
A/Portugal/210270/2022	H156S	3C.2a1b.2a.2	2022-02-21	SIAT1/SIAT2	80	<	80	160	320	640	1280	40	
A/Portugal/210271/2022	H156S	3C.2a1b.2a.2	2022-02-22	SIAT1/SIAT2	80	<	80	80	160	320	640	40	
A/La Mans/16907/2022	H156S	3C.2a1b.2a.2	2022-03-02	SIAT1	40	<	40	80	160	320	640	40	
A/Qatar/15-VI-22-1528555/2022	H156S	3C.2a1b.2a.2	2022-05-04	SIAT1	40	<	80	80	160	320	640	40	
A/La Reunion/ R22.317.39/2022	H156S	3C.2a1b.2a.2	2022-07-03	MDCKx/SIAT1	<	<	40	40	80	160	320	<	
A/La Reunion/ R22.317.38/2022	H156S	3C.2a1b.2a.2	2022-07-04	MDCKx/SIAT1	40	<	40	80	160	320	640	40	
A/La Reunion/ R22.317.37/2022	H156S	3C.2a1b.2a.2	2022-07-04	MDCKx/SIAT1	40	<	40	80	160	320	640	40	
A/La Reunion/ R22.317.54/2022	H156S	3C.2a1b.2a.2	2022-07-07	MDCKx/SIAT1	<	<	40	40	80	160	320	40	
A/La Reunion/ R22.317.43/2022	H156S	3C.2a1b.2a.2	2022-07-07	MDCKx/SIAT1	40	<	80	80	160	320	320	40	
A/La Reunion/ R22.317.42/2022	H156S	3C.2a1b.2a.2	2022-07-07	MDCKx/SIAT1	40	<	40	80	160	320	640	40	
A/La Reunion/ R22.317.41/2022	H156S	3C.2a1b.2a.2	2022-07-07	MDCKx/SIAT1	40	<	80	80	160	320	640	40	
A/Qatar/10-VI-22-1499678/2022	H156	3C.2a1b.2a.2	2022-04-26	SIAT1	160	<	80	320	320	320	160	80	
A/Qatar/15-VI-22-1520419/2022	H156	3C.2a1b.2a.2	2022-05-02	SIAT1	160	<	80	80	320	320	320	80	
A/Moulin/R22.321.98/2022	H156	3C.2a1b.2a.2	2022-07-20	MDCKx/SIAT1	160	<	80	320	640	320	320	80	
A/Moulin/R22.321.96/2022	H156	3C.2a1b.2a.2	2022-07-20	MDCKx/SIAT1	160	<	80	160	320	320	320	80	
A/Lyon/CHU/R22.320.85/2022	H156	3C.2a1b.2a.2	2022-07-24	MDCKx/SIAT1	80	<	40	80	320	320	320	160	
A/Qatar/16-VI-22-1526775/2022		3C.2a1b.1a	2022-05-04	SIAT1	320	320	640	320	640	640	40	160	

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

† < = <40, ND = Not Done

Vaccine
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold 4-fold 8-fold > 8-fold < not recognised by the antiserum ≥ 160 (when homologous titre ≥ 2560) ≥ 160 (no homologous titre)

Table 10-25. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-08-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre												
				Post-infection ferret antisera												
				NEW						NEW						
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Bang	A/Stock	A/Eng	A/Eng	A/Slov	A/Darwin	A/Slo	A/Kansas
				3264/19	2671/19	925256/20	e0826360/20	4005/20	4005/20	5/21	214191723/21	214191723/21	8720/2022	9/21	8720/22	14/17
	Passage history			SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	SIAT	SIAT	SIAT	Egg	Egg	SIAT
	Ferret number			F19/20 ¹	St Jude's F21/20 ¹	F03/21 ¹	F10/21 ¹	F07/21 ¹	F31/22 ¹	F35/21 ¹	F07/22 ¹	F32/22 ¹	F24/22 ¹	F39/21 ¹	F25/22 ¹	F17/19 ¹
	Genetic group			3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1
REFERENCE VIRUSES																
A/Denmark/3264/2019		3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	160	640	320	320	160	40	<	ND	320	40	160
A/Hong Kong/2671/2019		3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	80	40	<	<	160	40	160
A/Cambodia/925256/2020		3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	320	640	160	40	<	ND	320	40	160
A/Cambodia/e0826360/2020		3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	80	2560	320	640	160	160	320	160	80	80
A/Bangladesh/4005/2020	H156S	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	160	320	640	640	640	160	1280	640	320	160
A/Stockholm/5/2021	H156S	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	160	1280	640	320	80
A/England/214191723/2021	H156S	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT2	40	<	80	160	160	320	640	320	1280	640	320	40
A/Slovenia/8720/2022	D104G	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2	160	<	160	80	320	ND	1280	1280	ND	320	640	ND
A/Darwin/9/2021	H156S	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	80	640	320	1280	640	320	640	640	1280	160
A/Slovenia/8720/2022	D104G	3C.2a1b.2a.2	2022-02-10	E3(Am1AI2)	160	<	80	320	ND	640	640	ND	320	640	640	80
A/Kansas/14/2017		3C.3a.1	2017-12-14	SIAT3/SIAT2	40	<	80	80	80	80	80	<	80	80	<	640
TEST VIRUSES																
A/Croatia/66213/2022	D104G	3C.2a1b.2a.2	2022-03-23	SIAT1	80	<	40	80	160	ND	320	320	ND	160	320	80
A/Croatia/70723/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1	40	<	40	80	160	ND	640	640	ND	640	1280	<
A/Croatia/70708/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT1	40	<	40	80	160	ND	640	320	ND	320	640	<
A/Croatia/70697/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT2	40	<	40	80	160	ND	640	320	ND	320	640	<
A/Croatia/70654/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT2	40	<	40	40	160	ND	80	160	ND	40	160	<
A/Croatia/70648/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT2	80	<	40	80	160	ND	160	160	ND	320	160	640
A/Croatia/70645/2022	D104G	3C.2a1b.2a.2	2022-04-06	SIAT2	80	<	40	160	160	ND	160	320	ND	160	320	<
A/Sarajevo/186G/2022	D104G	3C.2a1b.2a.2	2022-04-09	SIAT1	40	<	40	80	160	ND	640	320	ND	320	640	640
A/Qatar/10-VI-22-1463000/2022	D104G	3C.2a1b.2a.2	2022-04-17	SIAT2	40	<	<	80	160	ND	640	320	ND	320	640	40
A/Sarajevo/189G/2022	D104G	3C.2a1b.2a.2	2022-04-18	SIAT1	40	<	<	80	160	ND	320	320	ND	320	320	<
A/Croatia/77372/2022	D104G	3C.2a1b.2a.2	2022-05-07	SIAT1	80	<	40	80	160	ND	640	320	ND	320	320	640
A/Croatia/79486/2022	D104G	3C.2a1b.2a.2	2022-05-17	SIAT1	40	<	<	80	160	ND	320	320	ND	320	320	640
A/Portugal/210209/2022	H156S	3C.2a1b.2a.2	2022-02-09	SIAT1/SIAT3	40	<	40	80	160	ND	320	640	ND	80	320	40
A/Portugal/210207/2022	H156S	3C.2a1b.2a.2	2022-02-10	SIAT2/SIAT3	80	<	40	80	160	ND	320	1280	ND	320	640	40
A/Hong Kong/3/2022		3C.2a1b.2a.1	2022-06-13	MDCK1/SIAT3	80	<	40	80	160	ND	80	<	ND	<	160	80
A/Hong Kong/5/2022		3C.2a1b.2a.1	2022-06-20	MDCK1/SIAT3	40	<	40	80	160	ND	80	<	ND	<	160	80
A/Hong Kong/8/2022		3C.2a1b.2a.1	2022-06-21	MDCK1/SIAT3	40	<	40	80	160	ND	80	<	ND	<	160	80
A/Ile de France/08440/2022		3C.2a1b.1a	2022-02-01	SIAT3	40	<	40	80	160	ND	160	40	ND	<	160	80
A/Pays de Loire/13561/2022		3C.2a1b.1a	2022-02-24	SIAT2	80	<	40	80	320	ND	160	<	ND	<	160	80
A/Qatar/15-VI-22-1438736/2022		3C.2a1b.1a	2022-04-10	SIAT3	80	<	40	160	320	ND	160	<	ND	<	160	80

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-26. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2022-09-09

Haemagglutination inhibition titre															
Viruses	Other information	Collection date	Passage history	Post-infection ferret antisera											
				A/Denmark	A/HK	A/Camb	A/Camb	A/Bang	A/Stock	A/Eng	A/Slov	A/Darwin	A/Slo	A/Kansas	
				3264/19	2671/19	925256/20	e0826360/20	4005/20	5/21	214191723/21	8720/2022	9/21	8720/22	14/17	
				SIAT	Cell	SIAT	Egg	SIAT	SIAT	SIAT	SIAT	Egg	Egg	SIAT	
				F19/20 [†]	St Jude's F21/20 [†]	F03/21 [†]	F10/21 [†]	F07/21 [†]	F35/21 [†]	F07/22 [†]	F24/22 [†]	F39/21 [†]	F25/22 [†]	F17/19 [†]	
Genetic group	3C.2a1b.1a	3C.2a1b.1b	3C.2a1b.2a.1	3C.2a1b.2a.1	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.2a1b.2a.2	3C.3a.1				
REFERENCE VIRUSES															
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT4	320	320	640	320	320	160	40	ND	320	40	160	
A/Hong Kong/2671/2019	3C.2a1b.1b	2019-06-17	MDCK1/SIAT4	320	320	640	160	320	160	80	<	160	40	160	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT5	160	160	1280	320	320	160	80	ND	640	40	160	
A/Cambodia/e0826360/2020	3C.2a1b.2a.1	2020-07-16	E5/E2	160	<	160	1280	320	160	320	320	320	160	80	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT3	160	40	320	320	640	640	640	1280	640	320	160	
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3	80	<	80	160	320	640	320	1280	640	320	40	
A/England/214191723/2021	3C.2a1b.2a.2	2021-10-12	MDCK1/SIAT3	40	<	80	160	160	320	640	1280	640	320	<	
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2	80	<	80	80	160	640	640	640	640	1280	<	
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2	160	<	80	640	320	640	640	640	640	1280	80	
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	E3(Am1AI2)	160	<	80	320	320	640	640	320	640	1280	40	
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2	40	40	80	80	80	160	80	80	320	<	640	
TEST VIRUSES															
A/Belgium/S0446/2022	D104G	3C.2a1b.2a.2	2022-02-21	SIAT1/SIAT1	40	<	40	80	160	320	320	ND	320	640	<
A/Brest/932/2022	D104G	3C.2a1b.2a.2	2022-03-28	MDCKx/SIAT1	40	<	40	80	160	320	320	ND	320	640	<
A/Toulon/760/2022	D104G	3C.2a1b.2a.2	2022-03-28	MDCKx/SIAT1	80	<	80	160	320	320	320	ND	320	1280	<
A/Lyon/808/2022	D104G	3C.2a1b.2a.2	2022-04-15	MDCKx/SIAT1	80	<	40	80	160	320	320	ND	320	640	<
A/Lyon/854/2022	D104G	3C.2a1b.2a.2	2022-05-04	MDCKx/SIAT1	40	<	40	80	160	320	320	ND	320	640	<
A/Lyon/CHU-R22.242.82/2022	D104G	3C.2a1b.2a.2	2022-05-09	MDCKx/SIAT1	40	<	40	80	160	320	320	ND	320	640	<
A/Norway/25483/2022	D104G	3C.2a1b.2a.2	2022-06-13	SIAT1	80	<	40	80	160	320	320	ND	640	1280	<
A/Lyon/CHU-R22.285.92/2022	D104G	3C.2a1b.2a.2	2022-06-26	MDCKx/SIAT1	80	<	40	80	320	320	320	640	640	640	<
A/Norway/26959/2022	D104G	3C.2a1b.2a.2	2022-06-29	SIAT1	80	<	40	80	320	640	640	ND	640	1280	<
A/Norway/27942/2022	D104G	3C.2a1b.2a.2	2022-07-09	SIAT1	80	<	40	80	320	640	320	ND	640	640	<
A/Norway/27459/2022	D104G	3C.2a1b.2a.2	2022-07-11	SIAT1	40	<	<	40	160	320	320	320	320	640	<
A/Norway/28542/2022	D104G	3C.2a1b.2a.2	2022-07-12	SIAT1	40	<	40	80	160	320	320	320	320	640	<
A/Norway/27849/2022	D104G	3C.2a1b.2a.2	2022-07-13	SIAT1	40	<	<	40	160	160	160	320	320	640	<
A/Belgium/S0891/2022	D53G	3C.2a1b.2a.2	2022-03-20	SIAT1/SIAT1	160	<	160	320	640	1280	640	640	1280	1280	80
A/Belgium/G0116/2022	H156S	3C.2a1b.2a.2	2022-03-14	SIAT1/SIAT1	40	<	80	80	160	640	640	ND	640	320	40
A/Nosy Be/03427/2022	H156S	3C.2a1b.2a.2	2022-03-28	SIAT1	80	<	80	80	160	320	640	ND	640	320	<
A/Antananarivo/03601/2022	H156S	3C.2a1b.2a.2	2022-04-15	SIAT1	80	<	80	80	160	320	640	ND	640	320	40
A/Antsirabe/03799/2022	H156S	3C.2a1b.2a.2	2022-04-26	SIAT1	80	<	40	80	160	320	320	ND	320	320	<
A/Antananarivo/04020/2022	H156S	3C.2a1b.2a.2	2022-05-17	SIAT1	80	<	40	80	160	320	320	ND	320	320	<
A/La Reunion/R22.317.56/2022	H156S	3C.2a1b.2a.2	2022-05-17	MDCKx/SIAT1	40	<	80	80	160	320	640	ND	640	320	<
A/La Reunion/R22.317.55/2022	H156S	3C.2a1b.2a.2	2022-05-18	MDCKx/SIAT1	40	<	40	80	160	320	640	160	320	320	<
A/Lyon/CHU-R22.264.84/2022	H156S	3C.2a1b.2a.2	2022-06-02	MDCKx/SIAT1	80	<	80	160	320	320	640	160	640	320	40
A/Antananarivo/04650/2022	H156S	3C.2a1b.2a.2	2022-06-16	SIAT1	80	<	80	80	160	320	640	160	320	320	<
A/Antananarivo/05015/2022	H156S	3C.2a1b.2a.2	2022-06-30	SIAT1	40	<	80	80	160	320	640	160	320	320	<
A/Bourgoin/758/2022	H156	3C.2a1b.2a.2	2022-04-09	MDCKx/SIAT1	160	<	160	320	640	320	320	ND	320	160	80
A/Lyon/271/2022	H156	3C.2a1b.1b	2022-02-07	MDCKx/SIAT1	320	320	640	<	80	80	<	ND	80	<	40
A/Antsirabe/03366/2022	H156	3C.2a1b.1b	2022-03-31	SIAT1	80	80	160	<	40	<	<	ND	40	<	<

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used)

1 < = <40, ND = Not Done

Vaccine

NH 2021

Vaccine

NH 2022

NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 10-27. Summary of HI results for A(H3N2) 3C.2a1b.2a.2 viruses with collection dates after 2022-01-31 by genetic group

Genetic group defined by	Antiserum raised against	A/Denmark 3264/19 SIAT	A/HK 2671/19 Cell	A/Camb 925256/20 SIAT	A/Camb e0826360/20 Egg	A/Bang 4005/20 SIAT	A/Stock 5/21 214191723/21 SIAT	A/Eng 14191723/21 SIAT	A/Slov 8720/2022 SIAT	A/Norway 24873/21 SIAT	A/Norway 24873/21 Egg	A/Darwin 9/21 Egg	A/Flor 02/21 Egg	A/Wyom 01/21 Egg	A/Norway 29511/21 Egg	A/Slo 8720/22 Egg	A/Slo 9216/22 Egg	A/Slo 9318/22 Egg	A/Kansas 14/17 SIAT
		F19/20 ¹¹	St Jude's F21/20 ¹¹	F03/21 ¹¹	F10/21 ¹¹	F07/21 ¹¹	F35/21 ¹¹	F07/22 ¹¹	F24/22 ¹¹	F10/22 ¹¹	F11/22 ¹¹	F38/21 ¹¹	F19/22 ¹¹	F20/22 ¹¹	F12/22 ¹¹	F25/22 ¹¹	F26/22 ¹¹	F27/22 ¹¹	F17/19 ¹¹
		1a	1b	2a.1	2a.1	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	2a.2	3C.3a.1
D104G	No tested	438	438	438	438	438	438	438	41	15	15	438	12	22	15	35	10	10	438
	≤2	27	1	0	1	82	394	295	40	2	0	205	0	2	0	33	9	8	1
	%	6.2	0.2		0.2	18.7	90.0	67.4	97.6	13.3		46.8		9.1		94.3	90.0	80.0	0.2
	4	198	2	2	7	352	43	133	0	13	14	159	1	9	14	2	1	1	18
	%	45.2	0.5	0.5	1.6	80.4	9.8	30.4		86.7	93.3	36.3	8.3	40.9	93.3	5.7	10.0	10.0	4.1
	≥8	213	435	436	430	4	1	10	1	0	1	74	11	11	1	0	0	1	419
	%	48.6	99.3	99.5	98.2	0.9	0.2	2.2	2.4		6.7	16.9	91.7	50.0	6.7			10.0	95.7
D53G	No tested	66	66	66	66	66	66	66	2			66	15	15		1			66
	≤2	35	0	1	7	55	65	65	2			45	0	0		1			2
	%	53.0		1.5	10.6	83.3	98.5	98.5				68.2							3.0
	4	14	0	5	28	11	1	1	0			20	10	0		0			18
	%	21.2		7.6	42.4	16.7	1.5	1.5				30.3	66.7						27.3
	≥8	17	66	60	31	0	0	0	0			1	5	15		0			46
	%	25.8	100	90.9	47.0							1.5	33.3	100					69.7
H156S	No tested	115	115	115	115	115	115	115				115				12			115
	≤2	6	0	0	1	41	88	110				54				2			1
	%	5.2			0.8	35.7	76.5	95.7				47.0				16.7			0.8
	4	31	0	0	4	66	26	5				45				10			36
	%	27.00			3.5	57.4	22.7	4.3				39.1				83.3			31.4
	≥8	78	115	115	110	8	1	0				16				0			78
	%	67.8	100	100	95.7	6.9	0.8					13.9							67.8
H156	No tested	53	53	53	53	53	53	53				53	1	1		2	1	1	53
	≤2	24	0	0	2	51	42	35				9	0	0		0	0	0	8
	%	45.3			3.8	96.2	79.2	66.0				17.0							15.1
	4	23	1	1	15	2	11	17				21	0	0		0	0	0	16
	%	43.4	1.9	1.9	28.3	3.8	20.8	32.1				39.6							30.2
	≥8	6	52	52	36	0	0	1				23	1	1		2	1	1	29
	%	11.3	98.1	98.1	67.9			1.9				43.4							54.7

All fold-reductions are based on homologous titres determined on the day of individual assays.

80% - 100% of test viruses recognised within 4-fold of homologous titres

50% - <80% of test viruses recognised within 4-fold of homologous titres

Table 11-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-07-07

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus			
				A/Norway 3275/2018		A/Cambodia 925256/2020		A/Bangladesh 4005/2020		A/Stockholm 5/2021		A/Darwin 09/2021		A/Kansas 14/2017				A/Denmark 3264/19	
				SIAT F03/19 3C.2a1b.2	SIAT F03/21 3C.2a1b.2a.1	SIAT F07/21 3C.2a1b.2a.2	SIAT F35/21 3C.2a1b.2a.2	Egg F38/21 3C.2a1b.2a.2	SIAT F17/19 3C.3a.1	SIAT F19/20 3C.2a1b.1a									
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read				
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97		
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	80	71	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	<	<	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G	
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	<	<	<	<	<	<	<	<	<	160	142	<	<	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R	
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P	
TEST VIRUSES																			
A/Slovenia/8965/2022	3C.2a1b.2a.2	2022-02-21	MDCK1/SIAT1	<	<	<	<	160	184	640	610	640	922	<	<	40	46	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y	
A/Slovenia/9302/2022	3C.2a1b.2a.2	2022-02-28	MDCKx/SIAT1	<	<	<	<	320	407	1280	981	1280	1262	40	48	160	150	I25V, D53G, G73D, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y	
A/Slovenia/8723/2022	3C.2a1b.2a.2	2022-02-11	MDCK1/SIAT1	<	<	<	<	80	103	320	331	640	576	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Slovenia/9318/2022	3C.2a1b.2a.2	2022-03-02	SIATx/SIAT1	<	<	<	<	160	131	640	494	640	838	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Slovenia/9356/2022	3C.2a1b.2a.2	2022-03-03	SIATx/SIAT1	<	<	<	<	80	103	320	330	640	631	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Slovenia/9146/2022	3C.2a1b.2a.2	2022-02-23	SIAT1/SIAT1	<	<	<	<	160	167	640	491	640	750	<	<	<	<	N22K(-cho), D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Slovenia/9216/2022	3C.2a1b.2a.2	2022-02-24	MDCKx/SIAT1	<	<	<	<	160	128	320	249	320	447	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, N165X(-cho), G186D, D190N, F193S, Y195F, K276R	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-07-14

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				Post-infection ferret antisera														
				A/Norway 3275/2018		A/Cambodia 925256/2020		A/Bangladesh 4005/2020		A/Stockholm 5/2021		A/Darwin 09/2021		A/Kansas 14/2017		A/Denmark 3264/19		
				SIAT F03/19 3C.2a1b.2	SIAT F03/21 3C.2a1b.2a.1	SIAT F07/21 3C.2a1b.2a.2	SIAT F07/21 3C.2a1b.2a.2	SIAT F35/21 3C.2a1b.2a.2	SIAT F35/21 3C.2a1b.2a.2	Egg F38/21 3C.2a1b.2a.2	Egg F38/21 3C.2a1b.2a.2	SIAT F17/19 3C.3a.1	SIAT F17/19 3C.3a.1	SIAT F19/20 3C.2a1b.1a	SIAT F19/20 3C.2a1b.1a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	80	71	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	<	<	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	<	<	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	<	<	<	<	<	<	<	<	<	160	142	<	<	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
TEST VIRUSES																		
A/Catalonia/13756/2022	3C.2a1b.2a.2	2022-02-16	C0/SIAT1	<	<	<	<	80	76	320	250	320	448	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Catalonia/2081558/2022	3C.2a1b.2a.2	2022-03-01	C0/SIAT1	<	<	<	<	80	74	320	298	640	560	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Catalonia/3565654/2022	3C.2a1b.2a.2	2022-03-08	C0/SIAT1	<	<	<	<	80	78	320	377	640	531	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Catalonia/13766/2022	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	<	<	<	<	160	121	320	460	640	698	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190S, I192F, F193S, Y195F
A/Catalonia/2076976/2022	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	<	<	<	<	80	68	320	243	320	439	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, P239S
A/Catalonia/13775/2022	3C.2a1b.2a.2	2022-02-23	C0/SIAT1	<	<	<	<	80	73	320	300	320	475	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, P239S
A/Catalonia/13758/2022	3C.2a1b.2a.2	2022-02-16	C0/SIAT1	<	<	<	<	80	95	320	320	640	597	<	<	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, N122H(-cho), H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-07-21

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹																HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus	
				Post-infection ferret antisera																	
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Kansas 14/2017 SIAT F17/19 3C.3a.1		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a					
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read				
REFERENCE VIRUSES																					
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97				
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P			
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	80	71	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F			
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	<	<	<	<	< K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F			
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	<	<	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G			
A/Kansas/14/2017	3C.3a.1	2017-12-14	SIAT3/SIAT2 10 ⁻³	<	<	<	<	<	<	<	<	<	<	160	142	<	<	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R			
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P			
TEST VIRUSES																					
A/Poland/PL14/2022	3C.2a1b.2a.2	2022-03-26	SIAT1	<	<	<	<	160	189	640	527	640	630	<	<	<	<	P4S, I25V, D53G, K83E, Y94N, V106A, S145N, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y			
A/Poland/PL09/2022	3C.2a1b.2a.2	2022-03-29	SIAT1	<	<	<	<	160	156	320	411	640	584	<	<	<	<	P4S, I25V, D53G, K83E, Y94N, V106A, S145N, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y			
A/Poland/PL21/2022	3C.2a1b.2a.2	2022-04-04	SIAT1	<	<	<	<	160	146	320	347	640	512	<	<	<	<	P4S, I25V, D53G, K83E, Y94N, V106A, S145N, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y			
A/Nordrhein-Westfalen/2/2022	3C.2a1b.2a.2	2022-03-10	Cell1/SIAT1	<	<	<	<	80	80	320	320	640	580	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Poland/PL15/2022	3C.2a1b.2a.2	2022-03-22	SIAT1	<	<	<	<	80	92	320	271	640	480	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Poland/PL04/2022	3C.2a1b.2a.2	2022-04-03	SIAT1	<	<	<	<	80	65	160	189	320	362	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Denmark/1836/2022	3C.2a1b.2a.2	2022-04-05	SIAT2/SIAT1	<	<	<	<	80	61	160	232	320	320	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Denmark/499/2022	3C.2a1b.2a.2	2022-03-07	SIAT2/SIAT1	<	<	<	<	80	104	320	311	640	588	<	<	<	<	D53G, K83E, R92X, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Bayern/7/2022	3C.2a1b.2a.2	2022-02-24	Cell1/SIAT1	<	<	<	<	80	84	320	253	640	496	<	<	<	<	D53G, K83E, Y94N, D104G, V106A, N122I(-cho), H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, I236V, K276R			
A/Bremen/4/2022	3C.2a1b.2a.2	2022-03-07	Cell2/SIAT1	<	<	<	<	80	78	320	255	320	471	<	<	<	<	V43X, Q44X, D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Sachsen/2/2022	3C.2a1b.2a.2	2022-03-14	Cell1/SIAT1	<	<	<	<	320	253	640	640	640	873	160	147	40	50	D53G, G78D, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R01K, S219R, R220K			
A/Baden-Wuerttemberg/6/2022	3C.2a1b.2a.2	2022-02-08	Cell1/SIAT1	<	<	<	<	80	77	320	310	640	505	<	<	<	<	E50K, D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F			
A/Denmark/558/2022	3C.2a1b.2a.2	2022-03-09	SIAT2/SIAT1	<	<	<	<	80	75	320	330	640	539	<	<	<	<	D53S, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F			
A/Thuringen/4/2022	3C.2a1b.2a.2	2022-02-14	Cell1/SIAT1	<	<	<	<	80	74	320	315	640	562	<	<	<	<	D53S, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Thuringen/7/2022	3C.2a1b.2a.2	2022-03-06	Cell1/SIAT1	<	<	<	<	80	87	320	310	640	523	<	<	<	<	D53S, N81D, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R			
A/Thuringen/6/2022	3C.2a1b.2a.2	2022-02-28	Cell1/SIAT1	<	<	<	<	80	69	160	208	320	379	<	<	<	<	D53S, N81D, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R, D291E			

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-07-25

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				Post-infection ferret antisera														
				A/Norway 3275/2018		A/Cambodia 925256/2020		A/Bangladesh 4005/2020		A/Stockholm 5/2021		A/Darwin 09/2021		A/Slovenia 8720/2022		A/Denmark 3264/19		
				SIAT F03/19 3C.2a1b.2	SIAT F03/21 3C.2a1b.2a.1	SIAT F07/21 3C.2a1b.2a.2	SIAT F07/21 3C.2a1b.2a.2	SIAT F35/21 3C.2a1b.2a.2	SIAT F35/21 3C.2a1b.2a.2	Egg F38/21 3C.2a1b.2a.2	Egg F38/21 3C.2a1b.2a.2	SIAT F24/22 3C.2a1b.2a.2	SIAT F24/22 3C.2a1b.2a.2	SIAT F19/20 3C.2a1b.1a	SIAT F19/20 3C.2a1b.1a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, D225G
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	320	345	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
TEST VIRUSES																		
A/Hessen/8/2022	3C.2a1b.2a.2	2022-02-07	Cell1/SIAT1	<	<	<	<	160	149	320	342	640	503	320	391	80	69	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y
A/Bremen/2/2022	3C.2a1b.2a.2	2022-02-18	Cell1/SIAT1	<	<	<	<	160	128	320	240	320	297	320	313	80	61	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S205X, S219Y
A/Bulgaria/1195/2022	3C.2a1b.2a.2	2022-02-21	SIAT2/SIAT1	<	<	<	<	80	78	320	258	640	493	320	240	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1273/2022	3C.2a1b.2a.2	2022-02-25	SIAT2/SIAT1	<	<	<	<	80	73	160	169	320	299	320	286	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1616/2022	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	<	<	<	<	80	70	320	242	320	442	320	277	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1611/2022	3C.2a1b.2a.2	2022-03-17	SIAT2/SIAT1	<	<	<	<	80	65	160	206	320	328	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Denmark/1253/2022	3C.2a1b.2a.2	2022-03-18	SIAT2/SIAT1	<	<	<	<	40	59	160	139	320	240	160	211	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1582/2022	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	<	<	<	<	80	66	160	193	320	384	320	255	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1660/2022	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	<	<	<	<	80	78	320	310	640	516	320	279	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1646/2022	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	<	<	<	<	80	75	320	279	320	461	320	283	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1536/2022	3C.2a1b.2a.2	2022-03-18	SIAT2/SIAT1	<	<	<	<	80	88	320	305	640	539	160	152	<	<	D53X, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, I260L, K276R
A/Bulgaria/1581/2022	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	<	<	<	<	80	78	320	262	640	496	320	414	<	<	E50X, D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Niedersachsen/8/2022	3C.2a1b.2a.2	2022-02-11	Cell1/SIAT1	<	<	<	<	40	58	160	129	320	242	160	156	<	<	D53G, K83E, Y94N, D104G, V106A, H40K, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-5. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-08-04

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹												HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus		
				Post-infection ferret antisera														
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2			A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a	
2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	320	345	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
TEST VIRUSES																		
A/Switzerland/34074//2022	3C.2a1b.2a.2	2022-03-18	SIAT1	<	<	<	<	40	59	160	193	320	317	80	111	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Rheinland-Pfalz/8/2022	3C.2a1b.2a.2	2022-05-09	P1/SIAT1	<	<	<	<	40	54	160	138	320	274	80	102	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Berlin/87/2022	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	<	<	<	<	80	72	320	282	320	455	160	151	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bremen/17/2022	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	<	<	<	<	80	63	320	242	320	419	160	149	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Baden-Wuerttemberg/27/2022	3C.2a1b.2a.2	2022-05-04	P1/SIAT1	<	<	<	<	40	56	160	211	320	309	160	140	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bayern/16/2022	3C.2a1b.2a.2	2022-05-03	P1/SIAT1	<	<	<	<	40	52	160	136	320	259	160	124	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Mecklenburg-Vorpommern/69/2022	3C.2a1b.2a.2	2022-05-02	P1/SIAT1	<	<	<	<	80	61	160	122	320	255	160	139	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, N190S, F193S, Y195F, K276R
A/Hessen/18/2022	3C.2a1b.2a.2	2022-04-25	P1/SIAT1	<	<	<	<	80	72	160	187	320	326	160	152	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Nordrhein-Westfalen/5/2022	3C.2a1b.2a.2	2022-04-19	P1/SIAT1	<	<	<	<	80	62	160	153	320	294	160	146	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, V213A, K276R
A/Slovenia/9891/2022	3C.2a1b.2a.2	2022-04-11	MDCKx/SIAT1	<	<	<	<	40	57	160	173	320	332	160	137	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, P239S, R269K, K276R
A/Switzerland/12981/2022	3C.2a1b.2a.2	2022-03-14	SIAT1	<	<	<	<	80	64	160	192	320	344	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R, R299K
A/Norway/18896/2022	3C.2a1b.2a.2	2022-04-08	SIAT2	<	<	<	<	160	189	160	173	160	182	40	53	80	68	T10M(-cho), K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T
A/Brandenburg/2/2022	3C.2a1b.2a.2	2022-05-05	P1/SIAT1	<	<	<	<	80	112	160	157	80	112	<	<	<	<	E50K, F79V, K83E, Y94N, V106A, H40K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N
A/Thuringen/10/2022	3C.2a1b.2a.2	2022-04-01	P1/SIAT1	<	<	<	<	160	131	160	126	160	142	40	55	<	<	E50K, F79V, K83E, Y94N, V106A, H40K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-6. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-08-08

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	320	345	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
TEST VIRUSES																		
A/Latvia/04-58697/2022	3C.2a1b.2a.2	2022-04-19	P2/SIAT1	<	<	<	<	80	73	160	148	320	328	160	131	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-78409/2022	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	<	<	<	<	80	72	160	149	320	284	160	169	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-82661/2022	3C.2a1b.2a.2	2022-04-25	P2/SIAT1	<	<	<	<	80	78	160	196	320	391	160	178	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-90221/2022	3C.2a1b.2a.2	2022-04-28	P2/SIAT1	<	<	<	<	80	62	160	204	320	363	160	154	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-90145/2022	3C.2a1b.2a.2	2022-04-28	P2/SIAT1	<	<	<	<	80	60	160	141	320	262	80	115	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Portugal/211412/2022	3C.2a1b.2a.2	2022-04-28	SIAT1/SIAT1	<	<	<	<	80	69	320	276	320	410	160	157	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Portugal/211439/2022	3C.2a1b.2a.2	2022-05-04	SIAT1/SIAT1	<	<	<	<	80	64	160	206	320	379	80	85	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Portugal/211459/2022	3C.2a1b.2a.2	2022-05-05	SIAT2/SIAT1	<	<	<	<	80	62	160	150	320	260	160	120	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-86299/2022	3C.2a1b.2a.2	2022-04-26	P2/SIAT1	<	<	<	<	80	63	160	230	320	414	80	117	<	<	I25M, D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-78381/2022	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	<	<	<	<	40	57	160	142	320	264	160	134	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, P221X, K276R
A/Latvia/04-86261/2022	3C.2a1b.2a.2	2022-04-26	P2/SIAT1	<	<	<	<	40	52	160	172	320	300	80	91	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S198L, K276R
A/Bulgaria/1813/2022	3C.2a1b.2a.2	2022-04-04	SIAT2/SIAT1	<	<	<	<	40	52	160	140	320	271	80	78	<	<	V20I, D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, Q197R, K276R
A/Portugal/211316/2022	3C.2a1b.2a.2	2022-04-23	SIAT1/SIAT1	<	<	<	<	80	97	320	290	320	452	160	149	<	<	S9I, E50K, D53N, K83E, Y94N, N96S(+cho), V106A, S145N, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, I223V
A/Thuringen/3/2022	3C.2a1b.2a.2	2022-02-01	C1/SIAT1	<	<	<	<	160	218	160	171	160	144	80	65	80	75	K2Q, H56Y, K83E, Y94N, V106A, S124N(-cho), Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T, S270T, P273S, V309I

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-7. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-08-15

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				Post-infection ferret antisera														
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	320	345	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
TEST VIRUSES																		
A/Slovenia/9767/2022	3C.2a1b.2a.2	2022-03-30	SIAT1/SIAT1	<	<	<	<	80	65	320	247	320	406	160	141	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Austria/1506576/2022	3C.2a1b.2a.2	2022-04-04	Cx/SIAT1	<	<	<	<	160	148	160	186	160	130	40	44	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Slovenia/9806/2022	3C.2a1b.2a.2	2022-04-04	SIAT1/SIAT1	<	<	<	<	80	61	160	233	320	400	160	156	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Slovenia/9845/2022	3C.2a1b.2a.2	2022-04-05	MDCK1/SIAT1	<	<	<	<	80	66	160	229	320	378	160	166	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Sweden/3/2022	3C.2a1b.2a.2	2022-04-11	SIAT1/SIAT1	<	<	<	<	80	71	160	210	320	371	160	144	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Serbia/2489/2022	3C.2a1b.2a.2	2022-04-26	M-ATL1/SIAT1	<	<	<	<	80	61	160	180	160	223	160	130	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Slovenia/9768/2022	3C.2a1b.2a.2	2022-04-01	MDCK1/SIAT1	<	<	<	<	80	67	320	304	320	474	160	145	<	<	D53G, G78D, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Slovenia/9854/2022	3C.2a1b.2a.2	2022-04-01	MDCK1/SIAT1	<	<	<	<	80	73	160	235	320	404	160	196	<	<	D53G, K83E, Y94N, D104G, V106A, K121N, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Serbia/2522/2022	3C.2a1b.2a.2	2022-04-30	M-ATL1/SIAT1	<	<	<	<	40	50	160	154	160	226	160	135	<	<	I3L, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Slovenia/9861/2022	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	<	<	<	<	160	204	640	856	1280	1027	320	313	80	93	I25V, D53G, G78D, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y
A/Serbia/2210/2022	3C.2a1b.2a.2	2022-04-08	M-ATL1/SIAT1	<	<	<	<	80	67	320	252	320	367	80	108	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H40M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, H92F, F193S, Y195F
A/Poland/PL08/2022	3C.2a1b.2a.2	2022-03-30	SIAT2	<	<	<	<	40	40	<	<	<	<	40	47	<	<	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S205F, A212T
A/Austria/1506181/2022	3C.2a1b.2a.2	2022-04-01	Cx/SIAT1	<	<	<	<	80	69	160	223	320	368	160	154	<	<	E50K, F79V, K83E, Y94N, V106A, H40K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N, Q327X
A/Slovenia/9860/2022	3C.2a1b.2a.2	2022-04-06	MDCK1/SIAT1	<	<	<	<	160	158	160	157	160	180	80	69	80	68	D7N, K83E, Y94N, V106A, H40K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201I

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-8. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-08-22

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹																HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				Post-infection ferret antisera																
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F38/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a				
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read			
REFERENCE VIRUSES																				
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	80	69	<	<	80	97			
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	80	77	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P		
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	640	678	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F		
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	640	751	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F		
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	5120	4133	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G		
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	SIAT1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	320	345	160	210	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	40	50	<	<	320	261	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P		
TEST VIRUSES																				
A/La Rioja/2092/2022	3C.2a1b.2a.2	2022-04-06	SIAT1/SIAT1	<	<	<	<	80	60	160	208	320	333	80	115	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Hungary/64/2022	3C.2a1b.2a.2	2022-04-08	MDCK1/SIAT1	<	<	<	<	40	52	160	228	320	316	80	94	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Hungary/63/2022	3C.2a1b.2a.2	2022-04-07	MDCK1/SIAT1	<	<	<	<	40	59	160	238	320	359	160	140	<	<	D53G, F79L, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, T167S, G186D, D190N, F193S, Y195F, A272T, K276R		
A/Estonia/169422/2022	3C.2a1b.2a.2	2022-04-21	SIAT1/SIAT1	<	<	<	<	40	57	160	210	320	320	160	124	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Estonia/169371/2022	3C.2a1b.2a.2	2022-04-22	MDCKx/SIAT1	<	<	<	<	40	52	160	213	320	335	80	119	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Estonia/169440/2022	3C.2a1b.2a.2	2022-04-23	SIAT1/SIAT1	<	<	<	<	40	57	160	160	320	261	80	99	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Estonia/169485/2022	3C.2a1b.2a.2	2022-04-25	SIAT1/SIAT1	<	<	<	<	40	53	160	193	320	298	80	110	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Andalucia/2717/2022	3C.2a1b.2a.2	2022-05-06	Px/SIAT1	<	<	<	<	40	50	160	148	160	233	160	123	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Andalucia/2718/2022	3C.2a1b.2a.2	2022-05-15	Px/SIAT1	<	<	<	<	40	44	160	215	320	283	80	73	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Andalucia/2716/2022	3C.2a1b.2a.2	2022-05-10	Px/SIAT1	<	<	<	<	40	53	160	226	320	288	80	75	<	<	R33Q, D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Andalucia/2720/2022	3C.2a1b.2a.2	2022-05-14	Px/SIAT1	<	<	<	<	80	66	320	285	320	438	160	143	<	<	D53G, K83E, Y94N, D104G, V106A, H40M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R		
A/Hungary/55/2022	3C.2a1b.2a.2	2022-03-31	MDCK1/SIAT1	<	<	<	<	40	57	160	222	320	309	80	76	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, T167S, G186D, D190N, F193S, Y195F, K276R		
A/Andalucia/2723/2022	3C.2a1b.2a.2	2022-05-12	Px/SIAT1	<	<	<	<	40	51	320	243	320	320	80	102	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, T167S, G186D, D190N, F193S, Y195F, K276R		
A/Lithuania/33816/2022	3C.2a1b.2a.2	2022-04-14	SIAT1	<	<	<	<	40	41	80	78	160	123	40	54	<	<	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, Q178H, G186D, D190T, F193S, Y195F, R201K, S219Y		

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-9. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-08-25

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus	
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F39/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	<	<	<	<	80	97		
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	<	<	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	160	211	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	160	213	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	1280	1469	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G	
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	160	180	160	210	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P	
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	<	<	<	<	320	261	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R	
TEST VIRUSES																			
A/Denmark/267/2022	3C.2a1b.2a.2	2022-02-28	SIAT2/SIAT1	<	<	<	<	80	89	320	386	320	265	160	143	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Bulgaria/1579/2022	3C.2a1b.2a.2	2022-03-21	SIAT2/SIAT1	<	<	<	<	80	77	320	314	320	264	160	151	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Moldova/7273/2022	3C.2a1b.2a.2	2022-04-13	SIAT1	<	<	<	<	160	204	640	508	640	558	320	256	80	78	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y	
A/Moldova/6649/2022	3C.2a1b.2a.2	2022-04-05	SIAT1	<	<	<	<	160	184	320	458	640	517	320	284	80	67	I25V, I48M, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y	
A/Moldova/7896/2022	3C.2a1b.2a.2	2022-04-21	SIAT1	<	<	<	<	40	59	160	152	320	240	80	96	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F	
A/Moldova/7877/2022	3C.2a1b.2a.2	2022-04-19	SIAT1	<	<	<	<	80	66	160	189	320	271	80	93	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F	
A/Latvia/04-58975/2022	3C.2a1b.2a.2	2022-04-18	P2/SIAT1	<	<	<	<	80	68	320	262	320	290	80	88	<	<	D53N, K83E, Y94N, N96S(+cho), V106A, I140M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, S262N, S279F	
A/Sachsen/11/2022	3C.2a1b.2a.2	2022-05-06	P1/SIAT1	<	<	<	<	160	220	320	249	80	76	40	57	<	<	E50K, F79V, K83E, Y94N, V106A, I140K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-10. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-09-05

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus	
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F39/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a			
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read		
REFERENCE VIRUSES																			
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	<	<	<	<	80	97		
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	<	<	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P	
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	160	211	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	160	213	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F	
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	1280	1469	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G	
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	160	180	160	210	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P	
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	<	<	<	<	320	261	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R	
TEST VIRUSES																			
A/Slovenia/9967/2022	3C.2a1b.2a.2	2022-04-20	MDCKx/SIAT1	<	<	<	<	80	64	160	237	160	152	80	64	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Slovenia/10081/2022	3C.2a1b.2a.2	2022-04-21	MDCKx/SIAT1	<	<	<	<	80	70	320	320	320	268	80	68	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Portugal/211317/2022	3C.2a1b.2a.2	2022-04-24	SIAT1/SIAT1	<	<	<	<	40	47	160	182	160	179	80	63	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Bulgaria/1763/2022	3C.2a1b.2a.2	2022-03-28	SIAT2/SIAT1	<	<	<	<	40	52	160	143	160	163	80	65	<	<	D53S, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R	
A/Bulgaria/1666/2022	3C.2a1b.2a.2	2022-03-23	SIAT2/SIAT1	<	<	<	<	40	53	160	196	160	144	80	60	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S199T, K276R	
A/Bulgaria/1726/2022	3C.2a1b.2a.2	2022-03-28	SIAT2/SIAT1	<	<	<	<	80	60	160	195	160	197	80	72	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, L251X, K276R	
A/Slovenia/9905/2022	3C.2a1b.2a.2	2022-04-12	MDCKx/SIAT1	<	<	<	<	160	148	640	601	640	524	160	144	80	74	I25V, D53G, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R201K, S219Y	
A/Dakar/25/2022	3C.2a1b.2a.2	2022-06-10	SIAT1	<	<	<	<	160	134	320	300	320	436	80	73	<	<	E50K, D53N, K83E, Y94N, N96S(+cho), V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, I223V	
A/Dakar/10/2022	3C.2a1b.2a.2	2022-05-23	P1/SIAT1	<	<	<	<	80	101	320	453	320	476	80	62	<	<	E50K, D53N, K83E, Y94N, N96S(+cho), V106A, I140M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, I223V	
A/Netherlands/11119/2022	3C.2a1b.2a.2	2022-04-13	MDCK-MIX2/SIAT1	<	<	<	<	160	130	160	152	40	40	<	<	<	<	E50K, F79V, K83E, Y94N, V106A, T117X, I140K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S262N	
A/Netherlands/11201/2022	3C.2a1b.2a.2	2022-04-15	MDCK-MIX2/SIAT1	<	<	<	<	160	145	160	148	40	49	<	<	<	<	E50K, F79V, K83E, Y94N, V106A, I140K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, G218X, S262N	
A/Netherlands/00832/2022	3C.2a1b.2a.2	2022-04-19	hCK1/SIAT1	<	<	<	<	80	67	40	59	40	41	<	<	<	<	E50K, F79V, K83E, Y94N, V106A, I140K, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, R222K	

¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Table 11-11. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2022-09-08

Viruses	Passage history Ferret number Genetic group	Collection Date	Passage History	Neutralisation Titre ¹ Post-infection ferret antisera														HA1 substitutions for 3C.2a/3a viruses compared to A/Norway/3275/2018: Egg adaptation HA substitutions compared to the corresponding cell culture propagated virus
				A/Norway 3275/2018 SIAT F03/19 3C.2a1b.2		A/Cambodia 925256/2020 SIAT F03/21 3C.2a1b.2a.1		A/Bangladesh 4005/2020 SIAT F07/21 3C.2a1b.2a.2		A/Stockholm 5/2021 SIAT F35/21 3C.2a1b.2a.2		A/Darwin 09/2021 Egg F39/21 3C.2a1b.2a.2		A/Slovenia 8720/2022 SIAT F24/22 3C.2a1b.2a.2		A/Denmark 3264/19 SIAT F19/20 3C.2a1b.1a		
				2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	2-fold	Read	
REFERENCE VIRUSES																		
A/Norway/3275/2018	3C.2a1b.2	2018-10-04	SIAT2 10 ⁻³	640	516	80	80	<	<	<	<	<	<	<	<	80	97	
A/Cambodia/925256/2020	3C.2a1b.2a.1	2020-09-25	SIAT3 10 ⁻³	160	120	320	331	<	<	<	<	<	<	<	<	160	183	K83E, Y94N, V106A, K171N, G186S, F193S, Y195F, S198P
A/Bangladesh/4005/2020	3C.2a1b.2a.2	2020-10-04	SIAT2 10 ⁻³	<	<	<	<	320	462	640	720	160	211	160	136	40	41	K83E, Y94N, V106A, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Stockholm/5/2021	3C.2a1b.2a.2	2021-04-16	SIAT0/SIAT3 10 ⁻³	<	<	<	<	80	94	640	495	160	213	160	153	<	<	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F
A/Darwin/9/2021	3C.2a1b.2a.2	2021-04-17	E3/E2 10 ⁻³	<	<	80	78	640	597	1280	1307	1280	1469	640	594	80	110	K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186N, D190N, F193S, Y195F, D225G
A/Slovenia/8720/2022	3C.2a1b.2a.2	2022-02-10	1/MDCK1/SIAT2 10 ⁻³	<	<	<	<	80	65	320	253	160	180	160	210	<	<	V106A, T128A(-cho), K131T, T135K(-cho), A138S, G186D, D190N, F193S, S198P
A/Denmark/3264/2019	3C.2a1b.1a	2019-10-25	SIAT3/SIAT2 10 ⁻³	40	52	320	268	<	<	<	<	<	<	<	<	320	261	G62E, S91N, R92K, V106A, K121N, T128A(-cho), K131T, A138S, S144K, Y159S, T160K(-cho), K171N, F193S, K326R
TEST VIRUSES																		
A/Bulgaria/1827/2022	3C.2a1b.2a.2	2022-04-07	SIAT2/SIAT1	<	<	<	<	40	53	160	158	160	144	80	65	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Norway/18641/2022	3C.2a1b.2a.2	2022-04-05	SIAT1	<	<	<	<	40	44	80	119	160	128	80	66	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Norway/18656/2022	3C.2a1b.2a.2	2022-04-04	SIAT1	<	<	<	<	40	49	160	123	160	131	80	67	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Norway/17522/2022	3C.2a1b.2a.2	2022-04-04	SIAT1	<	<	<	<	40	51	160	134	160	145	80	86	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Perugia/23/2022	3C.2a1b.2a.2	2022-04-12	SIAT3/SIAT1	<	<	<	<	40	56	320	296	320	253	160	200	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Bulgaria/1588/2022	3C.2a1b.2a.2	2022-03-22	SIAT2/SIAT1	<	<	<	<	80	61	320	306	320	278	80	72	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-82637/2022	3C.2a1b.2a.2	2022-04-24	P2/SIAT1	<	<	<	<	40	52	160	192	160	142	80	61	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-78385/2022	3C.2a1b.2a.2	2022-04-23	P2/SIAT1	<	<	<	<	40	55	160	228	160	153	40	58	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Portugal/211462/2022	3C.2a1b.2a.2	2022-05-05	SIAT1/SIAT1	<	<	<	<	40	59	320	345	320	245	80	62	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Portugal/211442/2022	3C.2a1b.2a.2	2022-05-04	SIAT2/SIAT1	<	<	<	<	80	61	320	320	160	201	80	65	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K276R
A/Latvia/04-78414/2022	3C.2a1b.2a.2	2022-04-22	P2/SIAT1	<	<	<	<	40	45	160	180	160	131	40	59	<	<	D53G, K83E, Y94N, D104G, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, S198L, K276R
A/Norway/18658/2022	3C.2a1b.2a.2	2022-04-05	SIAT1	<	<	<	<	40	41	80	103	80	114	40	43	<	<	D53N, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K207R
A/Norway/18657/2022	3C.2a1b.2a.2	2022-04-04	SIAT1	<	<	<	<	<	<	80	96	80	108	40	50	<	<	D53N, K83E, Y94N, V106A, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, F193S, Y195F, K207R
A/Dakar/11/2022	3C.2a1b.2a.2	2022-05-25	P1/SIAT1	<	<	<	<	160	150	640	679	640	613	80	80	<	<	R33L, E50K, D53N, K82R, K83E, Y94N, N96S(+cho), V106A, I140M, H156S, Y159N, T160I(-cho), L164Q, G186D, D190N, I192F, F193S, Y195F, I223V, P273X

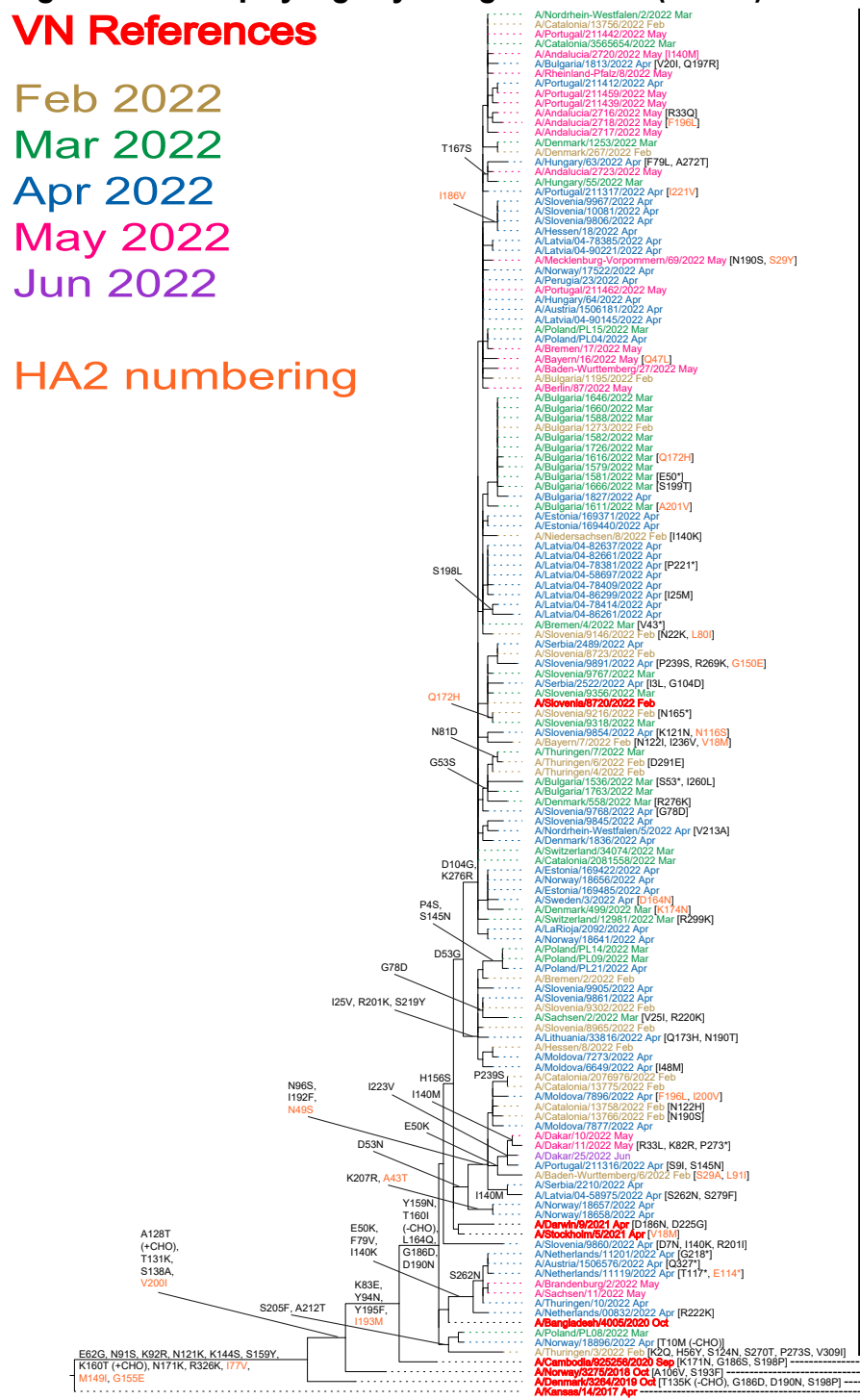
¹ Antiserum dilution value (2-fold), equivalent to HI reading, closest to the actual computer read value from a digitized image (Read) causing 50% reduction in plaque formation

Figure 12. HA phylogeny for genes of A(H3N2) viruses used in VN assay

VN References

Feb 2022
Mar 2022
Apr 2022
May 2022
Jun 2022

HA2 numbering



3C.2a1b.2a.2

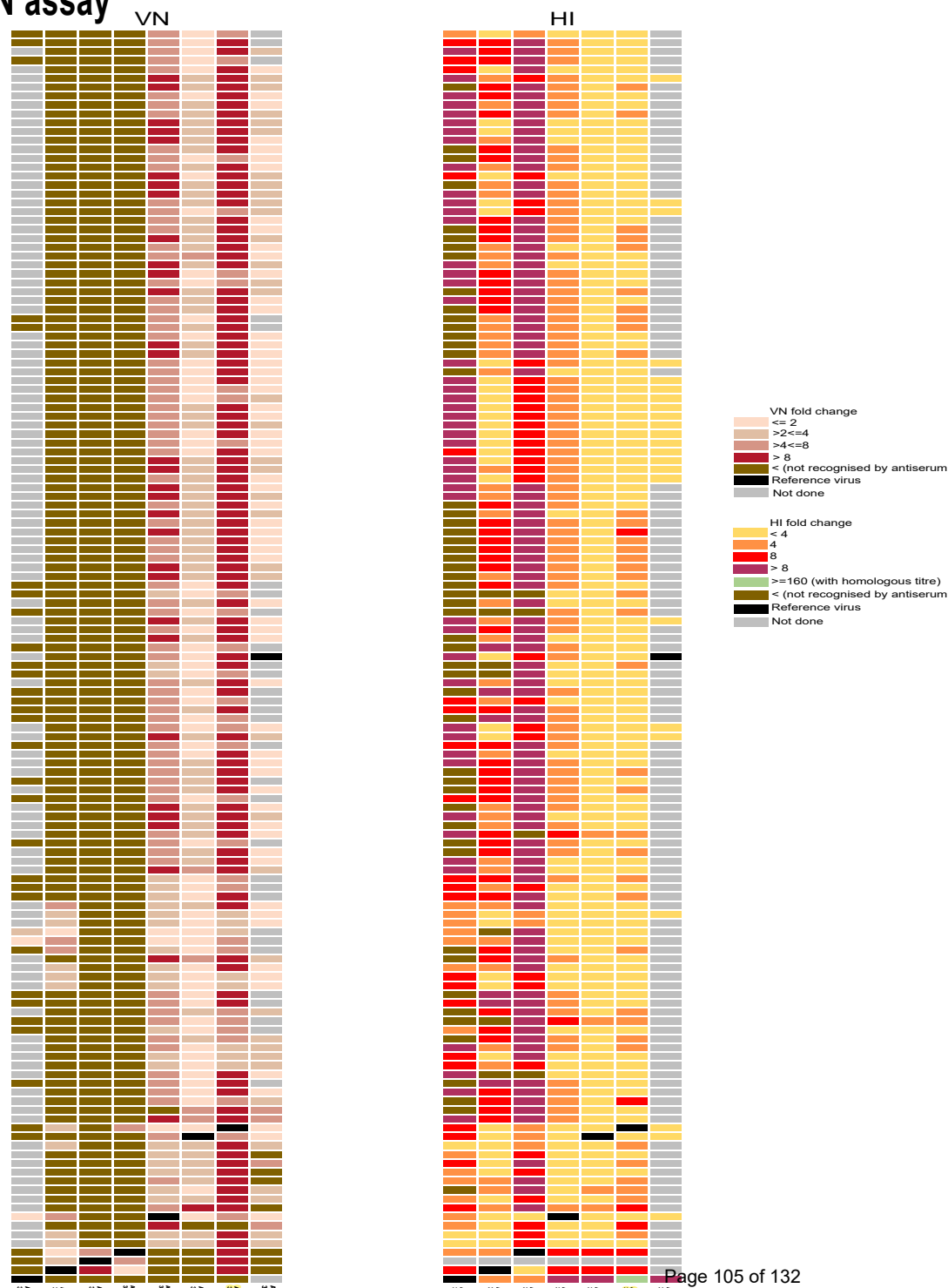
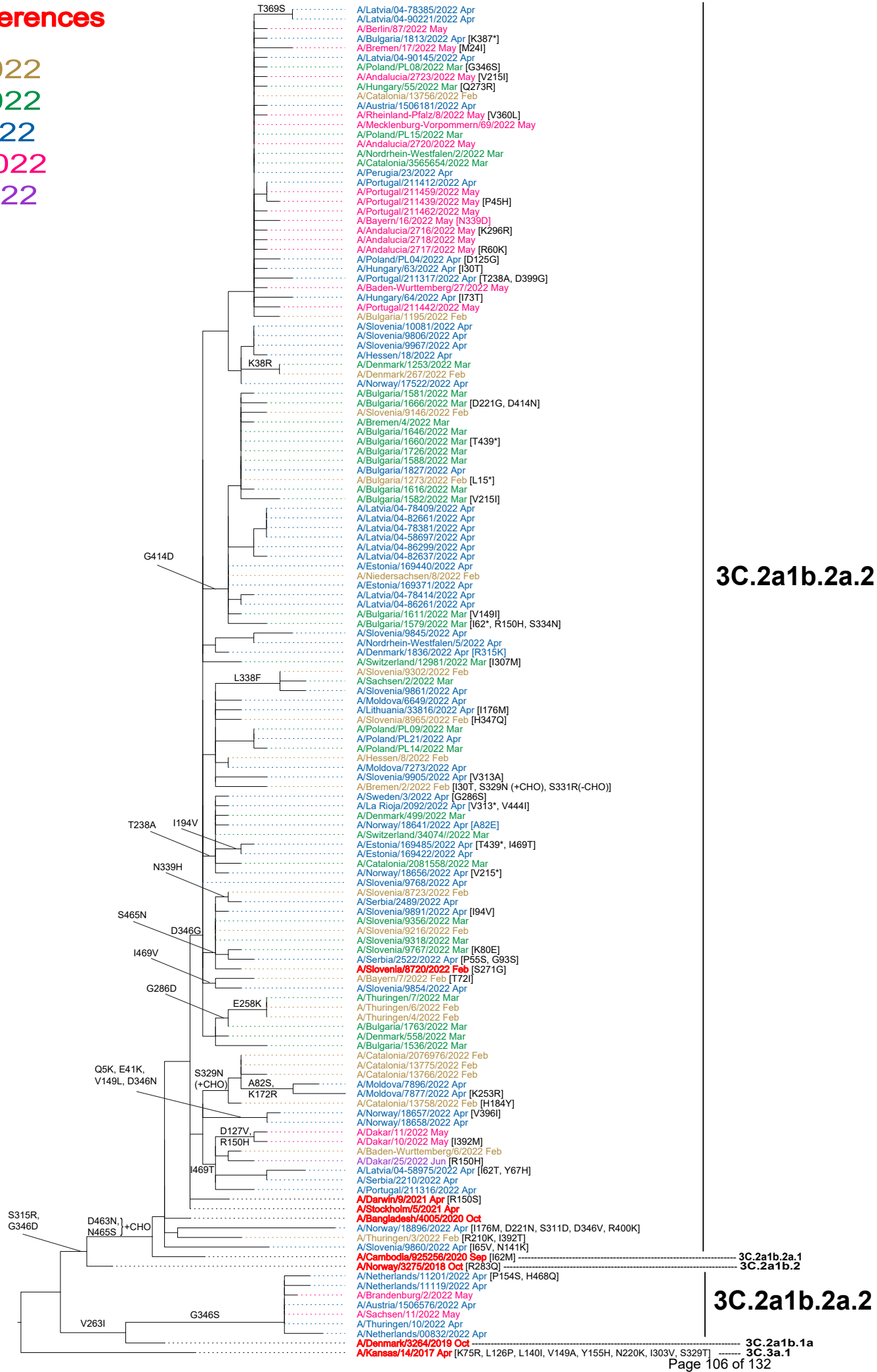


Figure 13. NA gene phylogeny of A(H3N2) viruses used in VN assays

VN References

Feb 2022
Mar 2022
Apr 2022
May 2022
Jun 2022



Influenza B virus analyses.

Influenza B viruses with collection dates after 2022-01-31 were received from 24 countries in four WHO Regions: African (n = 4; Ghana, Madagascar, South Africa and Togo), Eastern Mediterranean (n = 2; Egypt and Qatar), Western Pacific (n = 1; Hong Kong-SAR China) and European (n = 17) Regions. They represented 9% of the samples received with collection dates after 2022-01-31 (**Table 1**) and, of those ascribed to a lineage, 51% received as clinical specimens and 98% of the samples provided as virus isolates were successfully propagated (**Table 3**). A summary of the HI results obtained for these viruses is shown in **Table 4**. The summary is for antisera raised against the cell culture-propagated cultivar of the vaccine virus B/Austria/1359417/2021 (subclade V1A.3a.2 with additional **HA1** substitutions **A127T**, **P144L** and **K203R**) and an egg-propagated cultivar of B/Austria/1359417/2021 with no HA amino acid substitutions compared with the cell culture-propagated cultivar, an egg-propagated cultivar which acquired a substitution **G141R** in **HA1**, and against the former vaccine virus egg-propagated B/Washington/02/2019 (clade V1A.3).

B/Victoria-lineage.

HI analyses

The collection details and summary results for the 44 viruses with collection dates after 2022-01-31 that were assayed by HI, up to 2022-02-09, are shown in **Table 12**.

HI results for all propagated viruses analysed since the February 2022 report are shown in **Tables 14-1 to 14-7**. Test viruses in each table are sorted by date of collection and those with collection dates after 2022-01-31 are indicated in **Tables 14-2 to 14-7** (below the red horizontal lines). The HA genetic group is indicated. A summary of the results is shown in **Table 14-8**. **Table 14-2** includes the results for a test virus from CNIC.

For test viruses with collection dates after 2022-01-31, ferret antisera used were raised against the former vaccine viruses egg-propagated B/Colorado/06/2017 (clade V1A.1) and egg-propagated B/Washington/02/2019 (clade V1A.3), and two different cultivars of the egg-propagated vaccine virus B/Austria/1359417/2021 (subclade V1A.3a.2) – one with an HA sequence identical to the cell culture-propagated cultivar of B/Austria/1359417/2021 and one with a **HA1 G141R** substitution, and CNIC-2204B with its HA derived from B/Zhejiang-Shangcheng/11203/2021 which has acquired an egg-adaptive substitution **G141W** in **HA1** – this antiserum was used in **Table 14-2**. Additional antisera used were raised against cell culture-propagated cultivars of B/Austria/1359417/2021, B/Cote d'Ivoire/948/2020 (subclade V1A.3a.1) and three subclade V1A.3a.2, B/Paris/9878/2020, B/Gansu-Baiyin/1281/2021 and B/Stockholm/3/2022 (used in **Table 14-5**). An antiserum raised against cell culture-propagated B/Netherlands/11267/2022 (subclade V1A.3) was used in **Tables 14-6 and 14-7**.

Most test viruses analysed by HI fell in subclade V1A.3a.2 (n = 87) and a smaller number fell in clade V1A.3 (n = 11, all from the Netherlands).

In HI analyses of test viruses in subclade V1A.3a.2, 50 of 87 (57%) were recognised at titres within 4-fold of the homologous titre by the antiserum raised against egg-propagated B/Colorado/06/2017 but only 9 (10%) within 2-fold. The antiserum raised against egg-propagated B/Washington/02/2019 recognised only seven of the 87 test viruses in subclade V1A.3a.2 (8%) at titres within 4-fold of the homologous titre, none within 2-fold.

The antiserum raised against cell culture-propagated B/Cote d'Ivoire/948/2020, with **HA1** substitutions **N150K**, **G184E**, **N197D** (resulting in loss of a glycosylation site), **V220M**, **P241Q**, and **R279K** recognised 77 of 87 (89%) of the test viruses in subclade V1A.3a.2 at titres within 4-fold of the homologous titre and 19 (22%) within 2-fold.

The antiserum raised against the cell culture-propagated cultivar of B/Austria/1359417/2021 and the antiserum raised against the egg-propagated cultivar of B/Austria/1359417/2021 with an identical HA sequence recognised all subclade V1A.3a.2 viruses at titres within 4-fold of the homologous titre, 85 of 87 at titres within 2-fold. The antiserum raised against B/Stockholm/3/2022, used against 15 test viruses in subclade V1A.3a.2 (**Table 14-5**), recognised all at titres within 2-fold of the homologous titre. The antisera raised against cell culture-propagated B/Paris/9878/2020 and B/Gansu-Baiyin/1281/2021 each recognised 86 of the 87 test viruses at titres within 4-fold of the homologous titres, 99% and 95% within 2-fold.

The antiserum raised against egg-propagated B/Austria/1359417/2021 with the **HA1 G141R** substitution had a higher homologous HI titre and recognised none of the test viruses at titres within 2-fold of the homologous titre and only 12 of 87 (14%) within 4-fold. Likewise, the antiserum raised against CNIC-2204B, derived from B/Zhejiang-Shangcheng/11203/2021, with the egg-adaptive substitution **G141W** also recognised none of the five test viruses at titres within 2-fold of the homologous titre, although three were recognised at titres within 4-fold of the homologous titre. Nevertheless, the antisera raised against egg-propagated B/Austria/1359417/2021 with the **HA1 G141R** substitution recognised the test viruses in subclade V1A.3a.2 at absolute titres typically only 2- to 4-fold lower than the absolute titres seen with the other two antisera raised against B/Austria/1359417/2021.

The viruses falling in clade V1A.3 (n = 11) from the Netherlands were recognised by none of the antisera described above. The antiserum raised against cell culture-propagated B/Netherlands/11267/2022 (subclade V1A.3) was used in **Tables 14-6** and **14-7** recognised the three homologous test viruses at titres within 2-fold of the homologous titre. This antiserum recognised test viruses in subclade V1A.3a.2 poorly, with only 4/20 (20%) being recognised at titres within 4-fold of the homologous titre.

Genetic analysis

Figure 14a shows a phylogenetic tree for HA genes of B/Victoria lineage viruses analysed at the WIC, most with collection dates from 2022-02-01, with relevant reference virus HA gene sequences. **Figure 14b** shows a phylogenetic tree for HA genes of representative viruses as deposited in GISAID at the time of preparing this report. **Table 13** shows the geographic distribution of the 435 B/Victoria-lineage viruses

with collection dates after 2022-01-31 for which genetic information was deposited in GISAID at the time of preparing this report. An alignment of the HA glycoproteins of a selection of viruses is shown in **Figure 15**.

All recently circulating viruses analysed carried HA genes that fell into genetic clade **V1A.3** which encodes an HA with a three-amino acid deletion in **HA1 (Δ162-164)** and a **HA1 K136E** substitution. But for a number of recent viruses from the Netherlands, most viruses had HA genes encoding the substitutions **N150K**, **G184E**, **N197D** (resulting in the loss of a glycosylation site at residues 197-199, which formerly was usually lost in egg-propagated influenza B viruses) and **R279K** in **HA1**.

Within this group a major group encoding the **HA1** substitutions **A127T**, **P144L** and **K203R** (subclade V1A.3a.2, e.g. B/Austria/1359417/2021), with many in China encoding an **HA1** substitution **H122Q**. A large subgroup contains viruses with HA genes encoding the **HA1** substitution **D197E** (e.g. B/Stockholm/3/2022), sometimes with either **D129G** and/or **E183K**.

The viruses from the Netherlands in subclade V1A.3 have HA genes encoding **K75E**, **E128K**, **T155A**, **G184R** and **G230N** in **HA1**. A virus from East Africa collected in September 2021 (B/Kenya/186/2021) shares four of these five substitutions (no **G184R**).

Figure 16a shows the locations of amino acid substitutions that define viruses in subclades of group V1A.3a on a B/Brisbane/60/2008 HA structure, and **Figure 16b** shows the locations of amino acid substitutions that have been seen in the new genetic subgroup of viruses in the V1A.3 subclade that has circulated in the Netherlands on a B/Brisbane/60/2008 HA structure.

Figure 17a shows a phylogenetic tree for NA genes of B/Victoria lineage viruses analysed at the WIC most with collection dates from 2022-02-01, with relevant reference virus HA gene sequences. **Figure 17b** shows a phylogenetic tree for NA genes for representative viruses as deposited in GISAID at the time of preparing this report. An alignment of the NA glycoproteins of a selection of viruses is shown in **Figure 18**.

The NA gene phylogenies are largely congruent with those for HA, except for viruses from China where reassortment of HA and NA genes is evident. Viruses from China with HA genes encoding **HA1 H122Q** amino acid substitution have NA genes encoding **NA** substitutions **D53N**, **N59S** and **G233E**. Viruses falling in the HA subclade V1A.3a.2 have NA genes that fall into two groups. The largest of these groups contains viruses, including B/Paris/9878/2020 and B/Austria/1359417/2021, from locations around the world that have NA genes encoding the **NA** substitutions **P42Q**, **V71A**, **K343E**, **A395V** and **V401R**, some of which have NAs with an insertion of an amino acid in the stalk domain of the molecule (**ins 74L**), while others have **G233R** substitution (e.g. viruses detected in Brazil). The second of these groups, composed of recently circulating viruses from China, have NA genes encoding **NA V303I** and **K343E** substitutions. The V1A.3 viruses from the Netherlands fall in a separate group characterised by **NA T278S**, **G378E** and **K416R** substitutions.

B/Yamagata-lineage.

No B/Yamagata-lineage viruses with collection dates after 2022-01-31 have been received at the WIC.

Antiviral Analysis.

Susceptibility to the neuraminidase inhibitors (NAIs) oseltamivir and zanamivir was assessed phenotypically and by full NA gene sequencing, for 166 B/Victoria-lineage viruses detected and received within the course of the 2021-2022 influenza season (**Table 15**). All showed normal inhibition (NI: no reduced susceptibility) by both antivirals.

All of the B/Victoria-lineage full-length PA gene sequences generated at the Crick, from **151** individual viruses detected in the course of the 2021-2022 influenza season, encoded isoleucine at position 38 (**I38**) and showed no evidence of other substitutions associated with reduced susceptibility to baloxavir marboxil.

Conclusion.

Viruses of the B/Victoria-lineage circulated recently and viruses sequenced had HAs with a three amino acid deletion in **HA1** (**Δ162-164**) with the additional **HA1 K136E** amino acid substitution (clade **V1A.3**). The great majority of recent viruses had **HA1** amino acid substitutions **N150K**, **G184E**, **N197D/E** (resulting in the loss of a glycosylation site) and **R279K**, and most of the viruses with collection dates after 2022-01-31 received at the Francis Crick Institute (in shipments from National Influenza Centres) also encoded **HA1** substitutions **A127T**, **P144L** and **K203R**. These recently collected viruses were poorly recognised by antisera raised against the egg-propagated cultivar of B/Washington/02/2019 but were recognised very well by antisera raised against the cell culture-propagated cultivar of B/Austria/1359417/2021 and the egg-propagated cultivar of B/Austria/1359417/2021 with an identical HA sequence to the cell culture-propagated cultivar. An emergent group of clade **V1A.3** viruses from the Netherlands were antigenically and genetically distinct from both B/Washington/02/2019-like and B/Austria/1359417/2021-like viruses.

No viruses of the B/Yamagata-lineage with collection dates after 2022-01-31 have been received.

All influenza B/Victoria-lineage viruses showed normal inhibition by the NAIs oseltamivir and zanamivir and those viruses yielding full-length PA gene sequences all encoded proteins likely to be well inhibited by baloxavir marboxil.

Table 12. Summary of influenza B (Victoria-lineage) viruses collected after 2022-01-31, analysed by HI

MONTH	B Victoria lineage													
	Country	Number received	Number propagated ^{1,2}	Post-infection ferret antisera raised against										
				Clade V1A.3a.2 virus			Clade V1A.3a.2 virus			Clade V1A.3a.2 virus			Clade V1A.3 virus	
			≤2 fold ³	4 fold ³	≥8 fold ³	≤2 fold ⁴	4 fold ⁴	≥8 fold ⁴	≤2 fold ⁵	4 fold ⁵	≥8 fold ⁵	≤2 fold ⁶	4 fold ⁶	≥8 fold ⁶
2022														
FEBRUARY														
Estonia	1	1	1			1					1			1
Netherlands	1	1	1			1					1			1
Sweden	2	2	2				2				2			2
Togo	1	0												
MARCH														
Egypt	2	0												
Kyrgyzstan	6	6	6			6				3	3			6
Netherlands	4	4	4			4					4			4
Norway	2	2	2			2					2			2
Qatar	1	1	1			1					1			1
Russia	3	2	2			2					2			2
Sweden	3	3	3			3					3			3
Togo	13	0												
Ukraine	1	0												
APRIL														
France	1	1	1			1				1				1
Italy	1	1		1			1				1			1
Kyrgyzstan	6	6	6			6					6			6
Latvia	1	0												
Netherlands	20	16	5		11	5		11			16			16
Norway	3	1	1			1					1			1
Poland	11	5	4	1		4	1				5			5
Qatar	11	8	8			8				2	6		3	5
Russia	8	7	7			7				1	6			7
Serbia	2	2	2			2					2			2
Spain	1	1	1			1					1		1	
Sweden	1	1	1			1					1			1
Togo	4	0												
Ukraine	1	1	1			1				1				1
MAY														
Belgium	1	1	1			1					1			1
Denmark	1	1	1			1					1			1
France	1	1	1			1					1			1
Germany	2	2	2			2					2			2
Ghana	1	0												
Madagascar	4	0												
Netherlands	2	0												
Norway	3	3	3			3					3			3
Poland	1	1	1			1					1			1
Qatar	3	2	2			2					2		1	1
Romania	3	2	2			2					2		2	
Russia	4	4	4			4				2	2			4
Serbia	4	3	3			3					3			3
South Africa	1	1	1			1					1			1
Togo	1	0												
Ukraine	1	1	1			1				1				1
JUNE														
Denmark	1	1	1			1					1			1
Hong Kong	2	2	2			2				1	1			2
Madagascar	2	1	1			1					1			1
Russia	1	1	1			1					1			1
JULY														
Ghana	3	0												
Norway	1	0												
August														
Norway	2	0												
	156	99	86	2	11	86	2	11	0	12	87	0	7	92
24 Countries			86.9%	2.0%	11.1%	86.9%	2.0%	11.1%	0.0%	12.1%	87.9%	0.0%	7.1%	92.9%

1 Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2 Propagated to sufficient titre to perform HI assay

3 Compared to homologous titres for ferret antisera raised against tissue culture-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp + A127T, P144L, K203R

4 Compared to homologous titres for ferret antisera raised against egg-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp + A127T, P144L, K203R (G141)

5 Compared to homologous titres for ferret antisera raised against egg-propagated B/Austria/1359417/2021 - Clade V1A.3a.2 N150K gp + A127T, P144L, K203R (G141R)

6 Compared to homologous titres for ferret antisera raised against egg-propagated B/Washington/02/2019 - Clade V1A.3

Some samples not cultured because Ct value high (>30), failed sequence, identical sequence, mixed sequence or SARS-COV-2 positive

As of 2022-09-07

Figure 14a. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes (WIC sequences)

Vaccine viruses

**Reference viruses
(WIC)**

**Reference viruses
(CDC/VIDRL)**

Collection date

Feb 2022

Mar 2022

Apr 2022

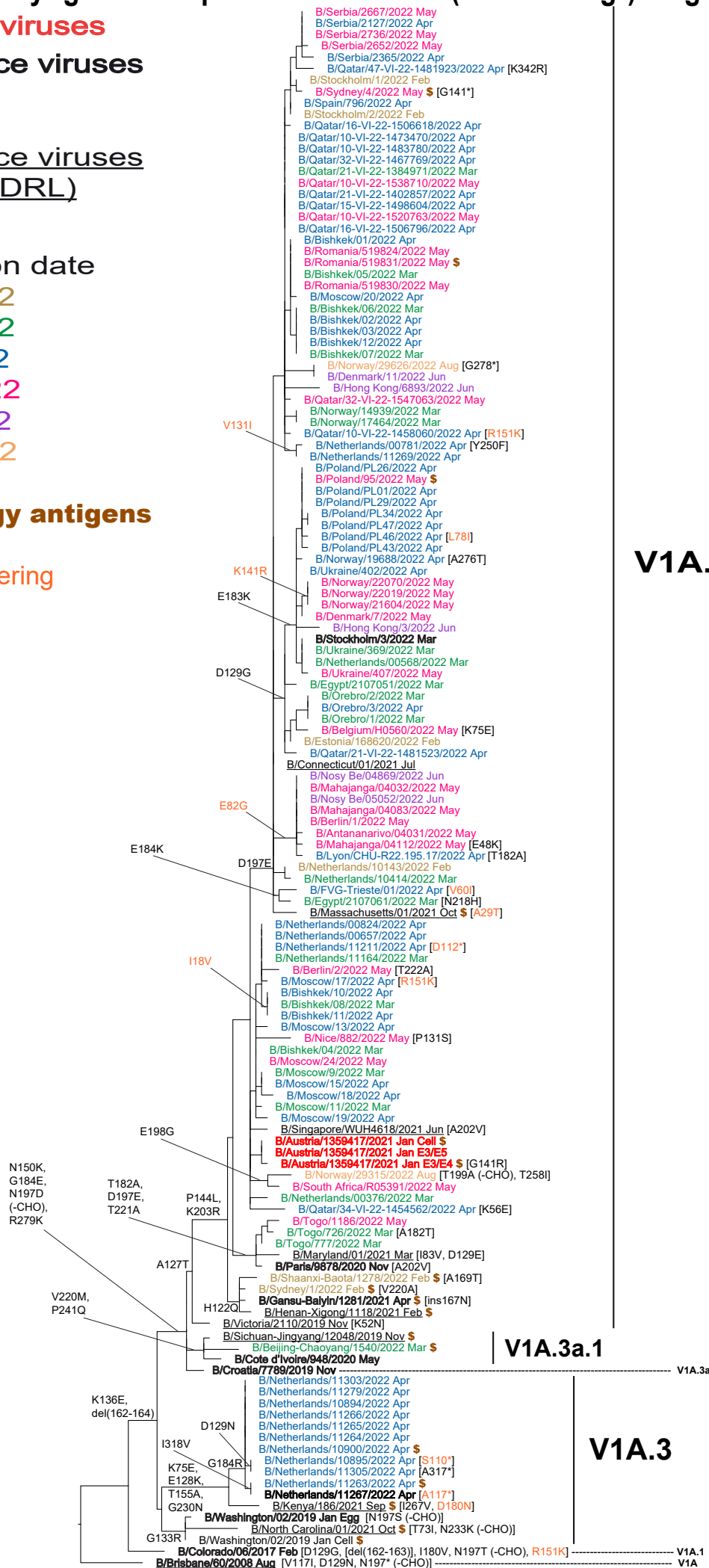
May 2022

Jun 2022

Aug 2022

\$ serology antigens

HA2 numbering



V1A.3a.2

V1A.3a.1

V1A.3

Table 13. Global count of B/Victoria-lineage HA sequences for viruses with collection dates after 2022-01-31

Clade	Group	Number	Country and (number of sequences)
V1A.3	K75E, E128K, T155A, G184R, G230N G133R	30 2	Netherlands (30) China (1), Guatemala (1)
V1A.3a	N150K, G184E, N197D(-cho), R279K	0	
V1A.3a.1	V220M, P241Q	24	China (24)
V1A.3a.2	A127T, P144L, K203R & H122Q & T182A, D197E, T221A & E198G & D197E & D197E, Q200P & D197E, E183K & D197E, E184K & D197E, D129G & D197E, D129G, E183K	40 172 3 2 102 14 3 2 18 23	France (1), Germany (1), Kyrgyzstan (4), Netherlands (8), Qatar (1), Russia (24), UAE (1) Australia (1), China (169), USA (2) Togo (3) Norway (1), South Africa (1) Australia (4), Bahrain (10), Denmark (1), France (1), Germany (1), Hong Kong (1), Kyrgyzstan (7), Madagascar (11), Netherlands (17), Norway (3), Qatar (13), Romania (3), Russia (3), Serbia (7), Singapore (4), Spain (1), Sweden (8), UAE (2), UK (4), USA (1) Brazil (14) Netherlands (1), UAE (2) Egypt (1), Italy (1) Belgium (1), Egypt (1), Estonia (1), Luxembourg (5), Netherlands (2), Qatar (1), Spain (1), Sweden (4), UAE (1), UK (1) Denmark (1), Hong Kong (1), Netherlands (2), Norway (7), Poland (8), Sweden (1), Ukraine (3)
	Total count	435	

WHO Region:

African (n = 15; 3 countries)

Americas (n = 18; 3 countries)

Eastern Mediterranean (n = 33; 4 countries)

European (n = 164; 18 countries)

South-East Asia (n = 0)

Western Pacific (n = 205; 4 countries)

Full-length HA sequences downloaded from GISAID and de-duplicated on 2022-09-05

Figure 15. HA protein alignment for B (Victoria-lineage) viruses

Vaccine virus:	Reference virus:	Genetic group:	HA1/HA2 boundary:	Potential N-linked glycosylation site	
					10 20 30 40 50 60 70 80 90 100
B/Brisbane/60/2008					DRICTGITSSNSPHVVKTATQGEVNVTVGVIPLTTPTKSHFANLKGTETRGKLCPRCLNCTDLDVALGRPKCTGKIP SARVSILHEVRPVTSGCFPIIMHD 1A
B/Colorado/06/2017					1A.1
B/Washington/02/2019 Cell					1A.3
B/Kenya/186/2021					1A.3
B/Netherlands/11267/2022					1A.3
B/Netherlands/11303/2022					1A.3
B/Croatia/7789/2019					1A.3a
B/Beijing-Chaoyang/1540/2022					1A.3a.1
B/Gansu-Baiyin/1281/2021					1A.3a.2
B/Sydney/1/2022					1A.3a.2
B/Paris/9878/2020					1A.3a.2
B/Maryland/01/2021					1A.3a.2
B/South Africa/R05391/2022					1A.3a.2
B/Austria/1359417/2021 Cell					1A.3a.2
B/Nice/882/2022					1A.3a.2
B/Bishkek/11/2022					1A.3a.2
B/Berlin/2/2022					1A.3a.2
B/FVG-Trieste/01/2022					1A.3a.2
B/Netherlands/10414/2022					1A.3a.2
B/Nosy Be/05052/2022					1A.3a.2
B/Qatar/21-VI-22-1481523/2022					1A.3a.2
B/Belgium/H0560/2022					1A.3a.2
B/Stockholm/3/2022					1A.3a.2
B/Norway/22070/2022					1A.3a.2
B/Poland/95/2022					1A.3a.2
B/Hong Kong/6893/2022					1A.3a.2
B/Romania/519830/2022					1A.3a.2
B/Qatar/10-VI-22-1538710/2022					1A.3a.2
B/Stockholm/2/2022					1A.3a.2
B/Serbia/2667/2022					1A.3a.2
					110 120 130 140 150 160 170 180 190 200
B/Brisbane/60/2008					RTKIRQLPNLLRGYEHIRLSTHNVINAEAPGGPYKIGTSGSCPNIITNGNGFFATMAWAVPKNDKNTATNPLTIEVPYICTEGEDQITVWGFHSDXETQ
B/Colorado/06/2017					1A.3a.2
B/Washington/02/2019 Cell					1A.3a.2
B/Kenya/186/2021					1A.3a.2
B/Netherlands/11267/2022					1A.3a.2
B/Netherlands/11303/2022					1A.3a.2
B/Croatia/7789/2019					1A.3a.2
B/Beijing-Chaoyang/1540/2022					1A.3a.2
B/Gansu-Baiyin/1281/2021					1A.3a.2
B/Sydney/1/2022					1A.3a.2
B/Paris/9878/2020					1A.3a.2
B/Maryland/01/2021					1A.3a.2
B/South Africa/R05391/2022					1A.3a.2
B/Austria/1359417/2021 Cell					1A.3a.2
B/Nice/882/2022					1A.3a.2
B/Bishkek/11/2022					1A.3a.2
B/Berlin/2/2022					1A.3a.2
B/FVG-Trieste/01/2022					1A.3a.2
B/Netherlands/10414/2022					1A.3a.2
B/Nosy Be/05052/2022					1A.3a.2
B/Qatar/21-VI-22-1481523/2022					1A.3a.2
B/Belgium/H0560/2022					1A.3a.2
B/Stockholm/3/2022					1A.3a.2
B/Norway/22070/2022					1A.3a.2
B/Poland/95/2022					1A.3a.2
B/Hong Kong/6893/2022					1A.3a.2
B/Romania/519830/2022					1A.3a.2
B/Qatar/10-VI-22-1538710/2022					1A.3a.2
B/Stockholm/2/2022					1A.3a.2
B/Serbia/2667/2022					1A.3a.2
					210 220 230 240 250 260 270 280 290 300
B/Brisbane/60/2008					MAKLYGDSKPKFTSSANGVTTHYVSQIGGFNPQTEDGGLPQSGRIIVDYMVKSGKTGTITYQRGILLPKQVWCASGRSKVIKGSLLPLIGEADCLHEKY
B/Colorado/06/2017					1A.3a.2
B/Washington/02/2019 Cell					1A.3a.2
B/Kenya/186/2021					1A.3a.2
B/Netherlands/11267/2022					1A.3a.2
B/Netherlands/11303/2022					1A.3a.2
B/Croatia/7789/2019					1A.3a.2
B/Beijing-Chaoyang/1540/2022					1A.3a.2
B/Gansu-Baiyin/1281/2021					1A.3a.2
B/Sydney/1/2022					1A.3a.2
B/Paris/9878/2020					1A.3a.2
B/Maryland/01/2021					1A.3a.2
B/South Africa/R05391/2022					1A.3a.2
B/Austria/1359417/2021 Cell					1A.3a.2
B/Nice/882/2022					1A.3a.2
B/Bishkek/11/2022					1A.3a.2
B/Berlin/2/2022					1A.3a.2
B/FVG-Trieste/01/2022					1A.3a.2
B/Netherlands/10414/2022					1A.3a.2
B/Nosy Be/05052/2022					1A.3a.2
B/Qatar/21-VI-22-1481523/2022					1A.3a.2
B/Belgium/H0560/2022					1A.3a.2
B/Stockholm/3/2022					1A.3a.2
B/Norway/22070/2022					1A.3a.2
B/Poland/95/2022					1A.3a.2
B/Hong Kong/6893/2022					1A.3a.2
B/Romania/519830/2022					1A.3a.2
B/Qatar/10-VI-22-1538710/2022					1A.3a.2
B/Stockholm/2/2022					1A.3a.2
B/Serbia/2667/2022					1A.3a.2

[illegible]

Figure 16a. Amino acid substitutions and deletions in the HA of recently circulating B/Victoria-lineage viruses mapped onto B/Brisbane/60/2008 HA structure

Substitutions defining subclade V1A.3 viruses shown in Black

Viruses displaying low reactivity with antisera raised against B/Washington/02/2019 (V1A.3):

The virus group (V1A.3a) with 4 substitutions, has evolved in two directions V220M [hidden] + P241Q (V1A.3a.1) and A127T + P144L + K203R + {T182A + D197E + T221A [hidden]} (V1A.3a.2) with both subgroups displaying low reactivity.

The V1A.3a.2 subgroup has dominated in the 2021-2022 season. Viruses with H122Q have largely been confined to China.

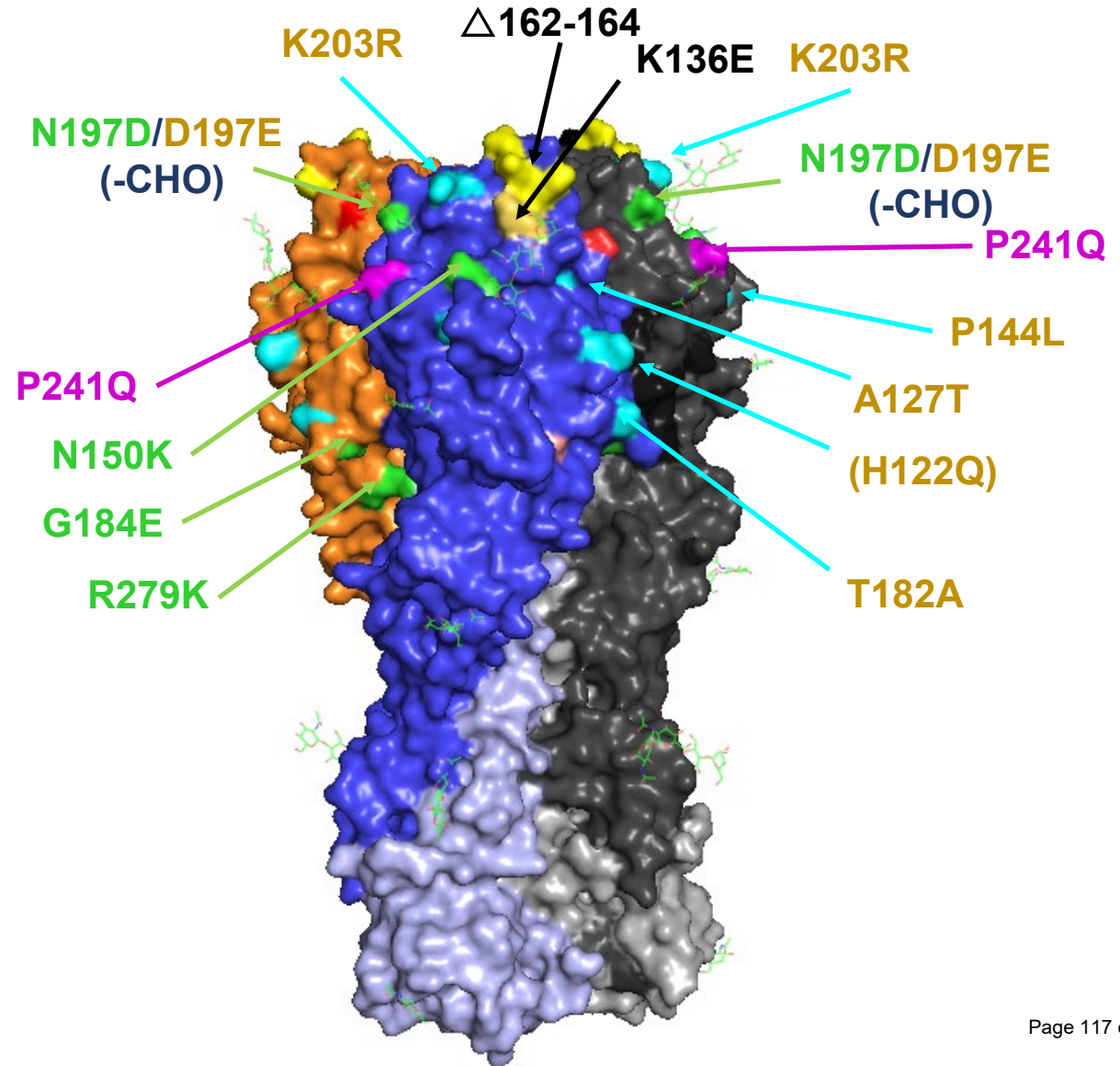


Figure 16b. Amino acid substitutions in the HA of recently emerged B/Victoria-lineage subclade V1A.3 viruses mapped onto B/Brisbane/60/2008 HA structure

Subclade V1A.3 viruses recently emerged and spread throughout the Netherlands contain the amino acid substitutions indicated compared to B/Washington/02/2019 HA.

These viruses were poorly recognised by the panel of post-infection ferret antisera used at the WIC, and an antiserum raised against B/Netherlands/11267/2022 poorly recognised not within the same cluster of viruses.

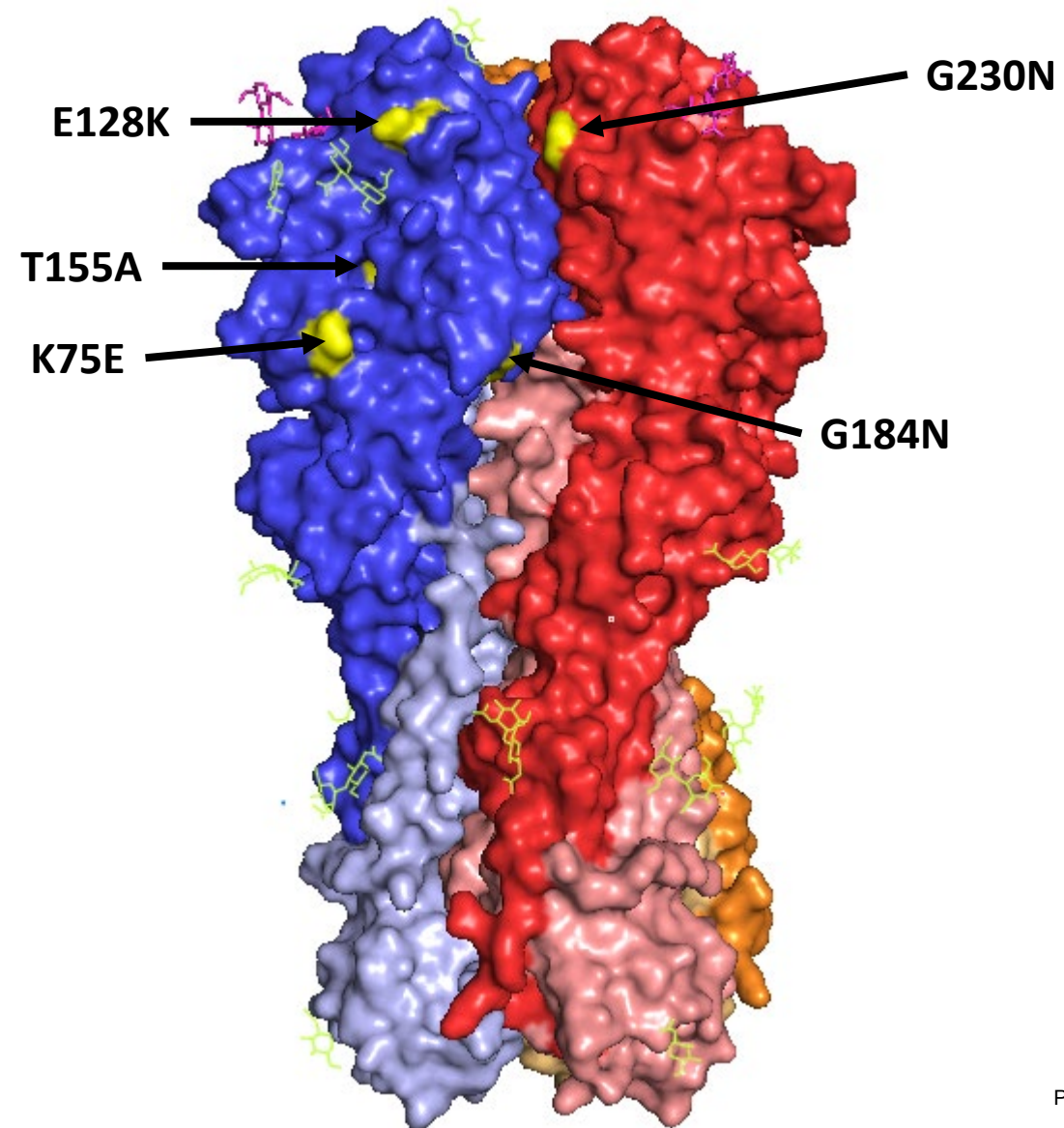


Figure 17a. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes (WIC Sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

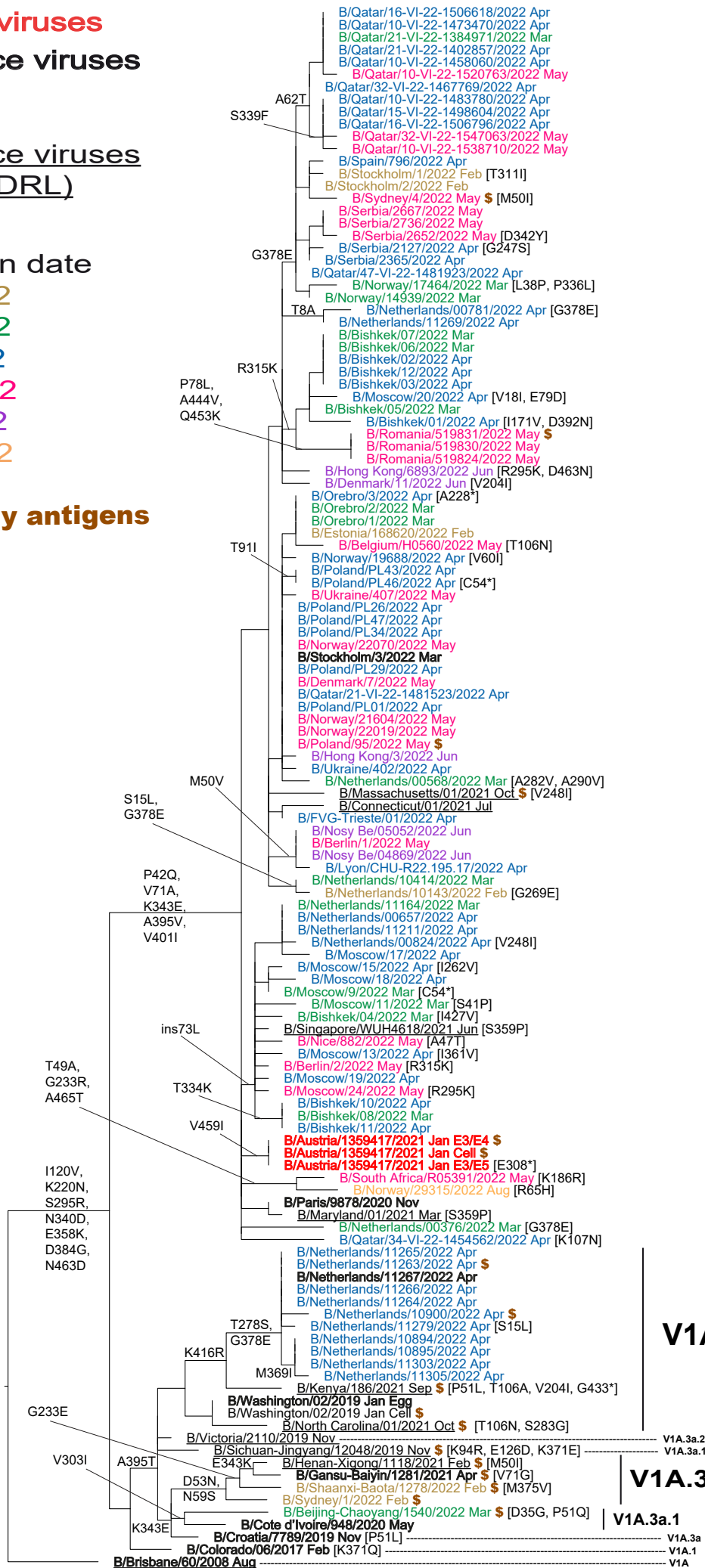
Apr 2022

May 2022

Jun 2022

Aug 2022

\$ serology antigens



V1A.3a.2

V1A.3

V1A.3a.2

V1A.3a.1

V1A.3a

V1A.1

V1A

Figure 17b. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes (non-WIC sequences)

Vaccine viruses

Reference viruses
(WIC)

Reference viruses
(CDC/VIDRL)

Collection date

Feb 2022

Mar 2022

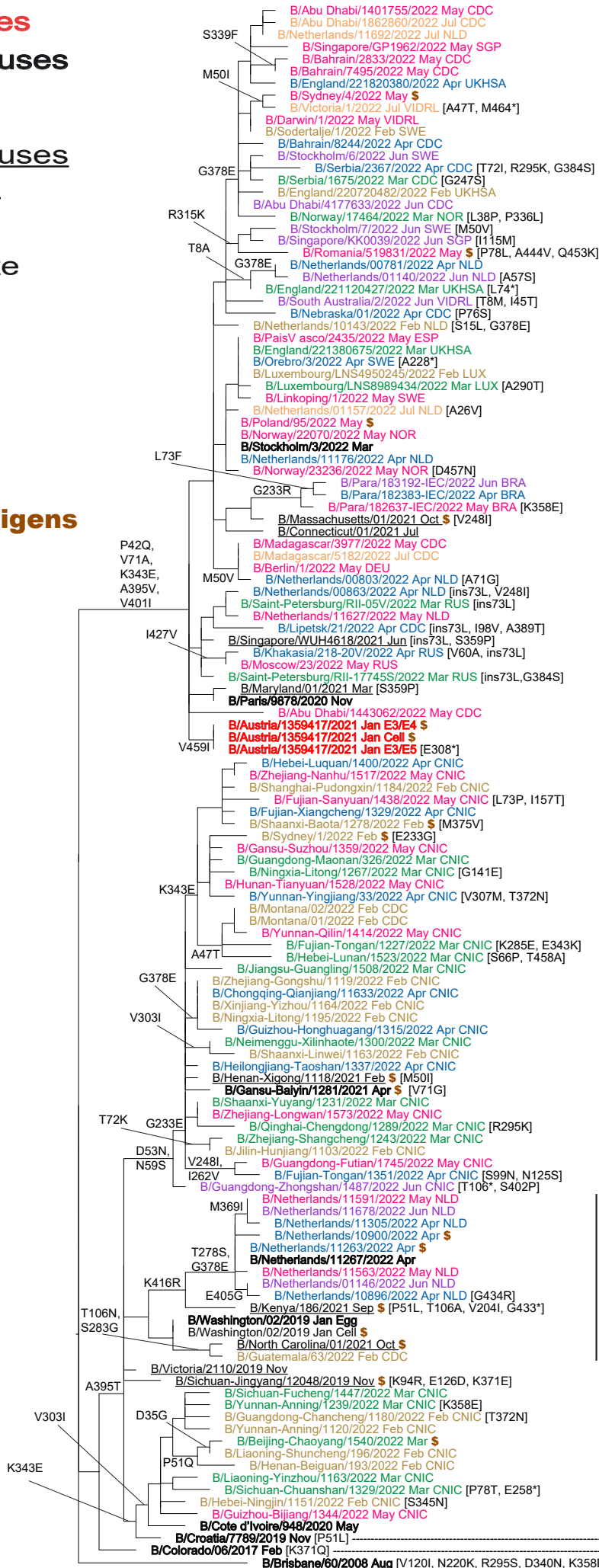
Apr 2022

May 2022

Jun 2022

Jul 2022

\$ serology antigens



V1A.3a.2

V1A.3a.1

V1A.3

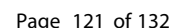
V1A.3a.2

V1A.3a

V1A.1

V1A

Vaccine virus: **Reference virus:** **Genetic group:** **Potential N-linked glycosylation site**



	310	320	330	340	350	360	370	380	390	400		
B/Brisbane/60/2008	PFVKLN	VETDTAEIRLMCTD	TYLDT	PRPNDGSI	TG	PCESNGDKGSGGI	KGGFVHQ	RMESKIGRWYSRT	MSKTERMG	GLYVKYDGD	FPWADSDALAFSGVM	1A
B/Colorado/06/2017				D			K		Q		G	1A.1
B/Washington/02/2019 Cell				D			K			G	T	1A.3
B/Kenya/186/2021			X							G		1A.3
B/Netherlands/11267/2022				D			K			E	G	1A.3
B/Netherlands/11303/2022				D			K		I	E	G	1A.3
B/Croatia/7789/2019				D	E		K				G	1A.3a
B/Beijing-Chaoyang/1540/2022		I		D	E		K				G	1A.3a.1
B/Gansu-Baiyin/1281/2021				D			K				G	1A.3a.2
B/Sydney/1/2022				D	E		K				G	1A.3a.2
B/Paris/9878/2020				D	E		K				G	1A.3a.2
B/Maryland/01/2021				D	E		KP				G	1A.3a.2
B/South Africa/R05391/2022				D	E		K				G	1A.3a.2
B/Austria/1359417/2021 Cell				D	E		K				G	1A.3a.2
B/Nice/882/2022				D	E		K				G	1A.3a.2
B/Bishkek/11/2022			K	D	E		K				G	1A.3a.2
B/Berlin/2/2022			K	D	E		K				G	1A.3a.2
B/FVG-Trieste/01/2022				D	E		K				G	1A.3a.2
B/Netherlands/10414/2022				D	E		K			E	G	1A.3a.2
B/Nosy Be/05052/2022				D	E		K				G	1A.3a.2
B/Qatar/21-VI-22-1481523/2022				D	E		K				G	1A.3a.2
B/Belgium/H0560/2022				D	E		K				G	1A.3a.2
B/Stockholm/3/2022				D	E		K				G	1A.3a.2
B/Norway/22070/2022				D	E		K				G	1A.3a.2
B/Poland/95/2022				D	E		K				G	1A.3a.2
B/Hong Kong/6893/2022				D	E		K				G	1A.3a.2
B/Romania/519830/2022			K	D	E		K				G	1A.3a.2
B/Qatar/10-VI-22-1538710/2022				FD	E		K			E	G	1A.3a.2
B/Stockholm/2/2022				D	E		K				G	1A.3a.2
B/Serbia/2667/2022				D	E		K			E	G	1A.3a.2

	410	420	430	440	450	460
B/Brisbane/60/2008	VSMKEPGWYSFGFEIKDKKCDVPCIGIEMVHDG	GGKETWHS	AAATAIYCLMGSGQLLWD	TVTGVN	MA	
B/Colorado/06/2017				D		
B/Washington/02/2019 Cell				D		
B/Kenya/186/2021		R	X			D
B/Netherlands/11267/2022		R				D
B/Netherlands/11303/2022		R				D
B/Croatia/7789/2019						D
B/Beijing-Chaoyang/1540/2022						D
B/Gansu-Baiyin/1281/2021						D
B/Sydney/1/2022						D
B/Paris/9878/2020	I					D
B/Maryland/01/2021	I					D
B/South Africa/R05391/2022	I					D.T
B/Austria/1359417/2021 Cell	I			I		D
B/Nice/882/2022	I					D
B/Bishkek/11/2022	I					D
B/Berlin/2/2022	I					D
B/FVG-Trieste/01/2022	I					D
B/Netherlands/10414/2022	I					D
B/Nosy Be/05052/2022	I					D
B/Qatar/21-VI-22-1481523/2022	I					D
B/Belgium/H0560/2022	I					D
B/Stockholm/3/2022	I					D
B/Norway/22070/2022	I					D
B/Poland/95/2022	I					D
B/Hong Kong/6893/2022	I					D
B/Romania/519830/2022	I			V	K	D
B/Qatar/10-VI-22-1538710/2022	I					D
B/Stockholm/2/2022	I			X		D
B/Serbia/2667/2022	I					D

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Table 14-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-03-11/2022-04-13

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				NEW									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Croatia 7789/19 MDCK	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R
	Passage history												
	Ferret number			Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F19/21 ^{*1}	F08/21 ^{*5}	F12/21 ^{*1}	F08/22 ^{*1}	NIB F01/21 ^{*1}	F15/21 ^{*1}	F44/21 ^{*1}
	Genetic group			V1A	V1A.1	V1A.3	V1A.3a	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	160	40	40	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	40	<	<	<	40	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	320	160	80	40	<	40	<	<
B/Croatia/7789/2019		V1A.3a	2019-11-11	MDCKx/MDCK3	640	160	80	640	320	<	40	160	80
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK4	320	40	40	320	640	<	40	160	<
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK5	640	160	<	320	160	640	320	1280	1280
B/Gansu-Baiyin/1281/2021		V1A.3a.2	2021-04-13	C1/C1/MDCK2	640	40	<	320	160	640	640	2560	1280
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	40	<	320	160	320	320	1280	1280
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3/E5	640	40	<	320	320	320	320	2560	1280
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E4	320	20	<	320	320	320	320	1280	640
TEST VIRUSES													
B/Slovenia/10026/2021		V1A.3a.2	2021-06-01	MDCK1/MDCK1	320	80	<	320	160	320	320	2560	1280
B/Toliary/11552/2021		V1A.3	2021-09-09	MDCK2	40	20	40	<	<	<	<	<	<
B/Antananarivo/11622/2021		V1A.3	2021-09-20	MDCK1	80	20	40	<	<	<	<	<	<
B/Toliary/11704/2021		V1A.3	2021-09-21	MDCK2	320	40	40	<	<	<	<	<	<
B/Antsirabe/11818/2021		V1A.3	2021-10-06	MDCK2	320	40	40	<	<	<	<	<	<
B/Toliary/11859/2021		V1A.3	2021-10-07	MDCK2	320	40	40	<	<	<	<	<	<
B/Toamasina/11911/2021		V1A.3	2021-10-13	MDCK3	20	20	40	<	<	<	<	<	<
B/Toliary/12085/2021		V1A.3	2021-10-19	MDCK2	80	20	40	<	<	<	<	<	<
B/Antsirabe/12055/2021		V1A.3	2021-10-28	MDCK2	80	20	40	<	<	<	<	<	<
B/Toliary/12167/2021		V1A.3	2021-11-02	MDCK2	80	20	40	<	<	<	<	<	<
B/Antsirabe/12267/2021		V1A.3	2021-11-09	MDCK2	80	20	40	<	<	<	<	<	<
B/Antananarivo/12258/2021		V1A.3	2021-11-10	MDCK1	40	10	40	<	<	<	<	<	<
B/Fianarantsoa/12422/2021		V1A.3	2021-11-12	MDCK1	80	20	40	<	<	<	<	<	<
B/Toliary/12566/2021		V1A.3	2021-11-16	MDCK2	40	20	40	<	<	<	<	<	<
B/Toamasina/12472/2021		V1A.3	2021-11-17	MDCK1	40	10	<	<	<	<	<	<	<
B/Mahajanga/12522/2021		V1A.3	2021-11-19	MDCK1	640	80	80	<	<	<	<	<	<
B/Toamasina/12585/2021		V1A.3	2021-11-20	MDCK1	160	10	40	<	<	<	<	<	<
B/Toliary/12683/2021		V1A.3	2021-11-22	MDCK2	80	10	<	<	<	<	<	<	<
B/Toamasina/12618/2021		V1A.3	2021-11-23	MDCK1	80	20	40	<	<	<	<	<	<

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not Done

Vaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 14-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-06-07

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	NEW CNIC-2204B Z-S/11203/21 Egg G141W
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F08/21 ⁵	F12/21 ¹	F08/22 ¹	NIB F01/21 ¹	F15/21 ¹	F44/21 ¹	F16/22 ¹
	Passage history												
	Ferret number												
	Genetic group			V1A	V1A.1	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008		2008-08-04	E4/E4	1280	160	40	<	<	<	<	<	<	<
B/Colorado/06/2017		2017-02-05	E5/E2	1280	320	40	<	<	<	40	<	<	<
B/Washington/02/2019		2019-01-19	E3/E2	1280	320	80	40	<	<	40	<	<	<
B/Cote d'Ivoire/948/2020		2020-05-28	MDCK4	320	40	40	320	<	40	160	<	80	80
B/Paris/9878/2020		2020-11-20	MDCK2	640	160	<	160	320	320	1280	1280	640	ND
B/Gansu-Baiyin/1281/2021		2021-04-13	C1/C1/MDCK2	640	40	<	160	640	640	2560	1280	320	320
B/Austria/1359417/2021		2021-01-09	SIAT1/MDCK4	640	40	<	160	320	320	1280	1280	320	320
B/Austria/1359417/2021 Isolate 2	G141	2021-01-09	E3/E3	640	40	<	320	320	320	2560	1280	640	320
B/Austria/1359417/2021 Isolate 2	G141R	2021-01-09	E3/E5	320	20	<	320	320	320	1280	640	5120	1280
CNIC-2204A (B/Zhejiang-Shangcheng/11203/2021)	G141W	2021-10-27	E2+2/E6/E1	80	10	<	40	40	40	320	160	160	320
TEST VIRUSES													
B/Zhejiang-Shangcheng/11203/2021		2021-10-27	E2/E2/E1	80	20	<	40	40	40	320	160	320	640
B/Poland/PL47/2022		2022-04-15	MDCK1	320	40	<	80	160	80	320	320	160	40
B/Poland/PL46/2022		2022-04-15	MDCK1	640	80	<	80	320	160	640	640	320	80
B/Poland/PL43/2022		2022-04-12	MDCK1	640	80	<	160	320	320	1280	1280	640	80
B/Poland/PL29/2022		2022-04-11	MDCK1	640	80	<	80	320	160	640	640	320	80
B/Poland/PL01/2022		2022-04-04	MDCK1	320	80	<	80	320	160	640	640	160	40

¹ Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not Done

Vaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
4-fold
8-fold
> 8-fold
< not recognised by the antiserum
≥ 160 (when homologous titre ≥ 2560)
≥ 160 (no homologous titre)

Table 14-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-06-29

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre								
				Post-infection ferret antiserum								
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F08/21 ^{*5}	F12/21 ^{*1}	F08/22 ^{*1}	NIB F01/21 ^{*1}	F15/21 ^{*1}	F44/21 ^{*1}
Genetic group				V1A	V1A.1	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES												
B/Brisbane/60/2008		2008-08-04	E4/E4	2560	80	20	<	<	<	<	<	<
B/Colorado/06/2017		2017-02-05	E5/E2	1280	320	20	<	<	<	<	<	<
B/Washington/02/2019		2019-01-19	E3/E2	1280	80	80	<	<	<	<	<	<
B/Cote d'Ivoire/948/2020		2020-05-28	MDCK4	160	20	<	320	<	<	80	40	80
B/Paris/9878/2020		2020-11-20	MDCK2	640	80	<	80	320	320	640	640	320
B/Gansu-Baiyin/1281/2021		2021-04-13	C1/C1/MDCK2	640	20	<	80	320	640	1280	1280	320
B/Austria/1359417/2021		2021-01-09	SIAT1/MDCK4	640	40	<	80	320	320	1280	1280	320
B/Austria/1359417/2021 Isolate 2	G141	2021-01-09	E3/E5	640	40	<	160	320	320	1280	1280	640
B/Austria/1359417/2021 Isolate 2	G141R	2021-01-09	E3/E5	320	40	<	160	160	320	1280	640	5120
TEST VIRUSES												
B/Stockholm/2/2022		2022-02-27	SIAT1/MDCK1	1280	80	<	160	320	320	1280	1280	320
B/Norway/14939/2022		2022-03-15	MDCK1	1280	80	<	160	320	320	1280	1280	640
B/Stockholm/3/2022		2022-03-22	SIAT1/MDCK1	1280	80	<	160	640	320	1280	1280	320
B/Norway/17464/2022		2022-03-30	MDCK1	1280	80	<	160	320	320	1280	1280	320
B/Orebro/1/2022		2022-03-30	SIAT1/MDCK1	640	80	<	160	320	320	1280	1280	320
B/Orebro/2/2022		2022-03-31	SIAT1/MDCK1	1280	80	<	160	320	320	1280	1280	320
B/Norway/19688/2022		2022-04-05	MDCK1	1280	80	<	80	320	320	1280	1280	320
B/Orebro/3/2022		2022-04-08	SIAT1/MDCK1	640	80	<	80	320	320	640	1280	320
B/Norway/21604/2022		2022-05-02	MDCK1	1280	80	<	80	320	320	1280	640	320
B/Berlin/1/2022		2022-05-03	P1/MDCK1	2560	160	<	160	640	320	1280	1280	640
B/Norway/22070/2022		2022-05-04	MDCK1	1280	80	<	80	320	320	1280	1280	320
B/Berlin/2/2022		2022-05-04	P1/MDCK1	1280	40	<	160	320	320	2560	1280	320
B/Norway/22019/2022		2022-05-09	MDCK1	640	80	<	80	320	320	640	640	320

*Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not Done

Vaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 14-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-07-15

Viruses	Other information	Passage history	Collection date	Passage history	Haemagglutination inhibition titre									
					Post-infection ferret antiserum									
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F08/21 ⁵	F12/21 ¹	F08/22 ¹	NIB F01/21 ¹	F15/21 ¹	F44/21 ¹	
		Genetic group			V1A	V1A.1	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES														
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	80	40	<	<	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	40	<	<	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E2	1280	80	80	<	<	<	<	<	<	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK4	320	20	<	640	<	<	80	40	160	
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	640	160	<	80	320	320	640	640	320	
B/Gansu-Baiyin/1281/2021		V1A.3a.2	2021-04-13	C1/C1/MDCK2	640	40	<	80	320	640	1280	1280	320	
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	40	<	80	320	320	1280	1280	320	
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3/E5	640	40	<	160	320	640	1280	1280	640	
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E5	320	20	<	160	160	320	1280	640	5120	
TEST VIRUSES														
B/Stockholm/1/2022		V1A.3a.2	2022-02-01	SIAT1/MDCK1	1280	160	<	160	640	640	1280	640	640	
B/Estonia/168620/2022		V1A.3a.2	2022-02-08	SIAT1/MDCK1	1280	160	<	160	640	320	1280	1280	320	
B/Netherlands/10143/2022		V1A.3a.2	2022-02-27	MDCK-MIX2/MDCK1	640	80	<	160	640	320	640	1280	320	
B/Netherlands/10414/2022		V1A.3a.2	2022-03-04	MDCK-MIX2/MDCK1	640	80	<	160	640	320	640	640	320	
B/Netherlands/00376/2022		V1A.3a.2	2022-03-09	hCK1/MDCK1	320	<	<	80	320	320	1280	640	320	
B/Moscow/9/2022		V1A.3a.2	2022-03-09	MDCK3/MDCK1	640	20	<	160	640	320	1280	1280	320	
B/Bishkek/08/2022		V1A.3a.2	2022-03-18	MDCK2/MDCK1	640	20	<	160	320	320	2560	640	320	
B/Bishkek/09/2022		no sequence	2022-03-24	MDCK1/MDCK1	640	20	<	160	320	320	2560	640	320	
B/Netherlands/00568/2022		V1A.3a.2	2022-03-28	hCK1/MDCK1	320	80	<	160	640	320	640	640	320	
B/Netherlands/11164/2022		V1A.3a.2	2022-03-31	MDCK-MIX2/MDCK1	1280	160	<	<	<	320	1280	1280	320	
B/Moscow/11/2022		V1A.3a.2	2022-03-31	MDCK1/MDCK1	640	20	<	80	640	320	1280	1280	320	
B/Serbia/2127/2022		V1A.3a.2	2022-04-04	P2/MDCK1	1280	80	<	160	320	320	1280	640	320	
B/Bishkek/10/2022		V1A.3a.2	2022-04-04	MDCK2/MDCK1	640	20	<	160	320	320	2560	640	320	
B/Moscow/13/2022		V1A.3a.2	2022-04-06	MDCK1/MDCK1	640	20	<	80	320	320	1280	640	320	
B/Moscow/19/2022		V1A.3a.2	2022-04-07	MDCK3/MDCK1	640	20	<	80	320	320	1280	640	320	
B/Netherlands/00657/2022		V1A.3a.2	2022-04-08	hCK1/MDCK1	320	<	<	80	320	320	640	640	320	
B/Bishkek/11/2022		V1A.3a.2	2022-04-10	MDCK1/MDCK1	640	20	<	160	320	320	2560	640	320	
B/Moscow/15/2022		V1A.3a.2	2022-04-12	MDCK1/MDCK1	320	<	<	80	640	320	1280	1280	320	
B/Bishkek/12/2022		V1A.3a.2	2022-04-16	MDCK1/MDCK1	640	80	<	160	320	320	1280	640	320	
B/Netherlands/11211/2022		V1A.3a.2	2022-04-17	MDCK-MIX2/MDCK1	40	80	<	160	640	320	1280	1280	320	
B/Netherlands/00781/2022		V1A.3a.2	2022-04-17	hCK1/MDCK1	640	160	<	160	640	320	1280	1280	320	
B/Serbia/2365/2022		V1A.3a.2	2022-04-18	MDCK3	1280	80	<	160	640	320	1280	640	320	
B/Netherlands/11269/2022		V1A.3a.2	2022-04-18	MDCK-MIX2/MDCK1	1280	160	<	160	640	320	1280	1280	160	
B/Moscow/20/2022		V1A.3a.2	2022-04-19	MDCK2/MDCK1	320	80	<	160	640	320	640	640	320	
B/Netherlands/00824/2022		V1A.3a.2	2022-04-21	hCK1/MDCK1	320	<	<	80	320	320	640	640	320	
B/Bishkek/01/2022		V1A.3a.2	2022-04-26	MDCK2/MDCK1	1280	80	<	160	320	320	1280	640	320	
B/Moscow/18/2022		V1A.3a.2	2022-04-27	MDCK1/MDCK1	320	40	<	160	640	320	1280	1280	320	
B/Bishkek/02/2022		V1A.3a.2	2022-04-28	MDCK1/MDCK1	1280	160	<	160	640	640	1280	1280	640	
B/Bishkek/03/2022		V1A.3a.2	2022-04-28	MDCK2/MDCK1	1280	80	<	160	320	320	1280	640	320	
B/Moscow/17/2022		V1A.3a.2	2022-04-29	MDCK1/MDCK1	320	40	<	160	640	320	1280	1280	320	
B/South Africa/R05391/2022		V1A.3a.2	2022-05-10	MDCK1/MDCK1	640	40	<	160	320	320	1280	640	320	
B/Serbia/2652/2022		V1A.3a.2	2022-05-11	P2/MDCK1	1280	80	<	160	640	320	1280	640	320	
B/Serbia/2667/2022		V1A.3a.2	2022-05-12	MDCK2	1280	160	<	160	320	320	1280	640	320	
B/Serbia/2736/2022		V1A.3a.2	2022-05-18	MDCK2	1280	160	<	160	640	320	1280	640	640	
B/Moscow/24/2022		V1A.3a.2	2022-05-20	MDCK3/MDCK1	320	40	<	160	640	320	1280	1280	320	
B/Netherlands/10895/2022	K75E, E128K, D129N, T155A, G164R, G230N	V1A.3	2022-04-02	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/10894/2022	K75E, E128K, T155A, G164R, G230N	V1A.3	2022-04-02	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/10900/2022	K75E, E128K, T155A, G164R, G230N	V1A.3	2022-04-04	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/11263/2022	K75E, E128K, T155A, G164R, G230N, I319V	V1A.3	2022-04-09	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/11264/2022	K75E, E128K, T155A, G164R, G230N	V1A.3	2022-04-13	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/11267/2022	K75E, E128K, T155A, G164R, G230N, I319V	V1A.3	2022-04-14	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/11265/2022	K75E, E128K, T155A, G164R, G230N	V1A.3	2022-04-14	MDCK-MIX2/MDCK1	40	<	<	<	<	<	<	<	<	
B/Netherlands/11305/2022	K75E, E128K, D129N, T155A, G164R, G230N	V1A.3	2022-04-22	MDCK-MIX2/MDCK1	80	<	<	<	<	<	<	<	<	

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not DoneVaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 14-5. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-07-26

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				NEW									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Stock 3/22 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R
Passage history	Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ¹²	F20/20 ¹²	F08/21 ^{*5}	F12/21 ^{*1}	F08/22 ^{*1}	F28/22 ^{*1}	NIB F01/21 ^{*1}	F15/21 ^{*1}	F44/21 ^{*1}			
Genetic group	V1A	V1A.1	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2		
REFERENCE VIRUSES													
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	1280	80	80	<	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	320	80	<	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	80	160	<	<	<	<	<	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK4	160	20	10	320	40	40	80	80	80
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	320	80	<	80	640	160	640	640	320
B/Gansu-Baiyin/1281/2021		V1A.3a.2	2021-04-13	C1/C1/MDCK2	320	20	<	80	320	320	640	1280	640
B/Stockholm/3/2022	D197E, D129G, E183K	V1A.3a.2	2022-03-22	SIAT1/MDCK2	320	80	10	80	640	320	1280	1280	640
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	320	20	<	80	320	320	640	1280	640
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3/E5	320	20	<	160	640	320	640	2560	1280
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E5	320	20	<	160	320	320	640	1280	2560
TEST VIRUSES													
B/Bishkek/04/2022		V1A.3a.2	2022-03-09	MDCK2	640	20	<	80	640	320	640	1280	640
B/Bishkek/05/2022		V1A.3a.2	2022-03-11	MDCK2/MDCK1	640	80	<	80	640	160	1280	640	320
B/Bishkek/06/2022		V1A.3a.2	2022-03-17	MDCK2/MDCK1	640	80	10	160	640	320	2560	1280	640
B/Bishkek/07/2022		V1A.3a.2	2022-03-17	MDCK2/MDCK1	640	80	10	160	640	320	1280	1280	640
B/Spain/796/2022		V1A.3a.2	2022-04-07	MDCK1	640	80	40	80	640	320	1280	640	320
B/FVG-Trieste/01/2022	E184K	V1A.3a.2	2022-04-15	MDCK3/MDCK1	640	40	<	40	320	160	640	320	160
B/Qatar/10-VI-22-1483780/2022		V1A.3a.2	2022-04-22	MDCK1	640	80	40	80	640	320	1280	640	320
B/Qatar/16-VI-22-1506618/2022		V1A.3a.2	2022-04-27	MDCK1	640	80	40	80	640	320	1280	640	320
B/Qatar/16-VI-22-1506796/2022		V1A.3a.2	2022-04-28	MDCK1	640	80	40	160	640	320	1280	1280	640
B/Qatar/10-VI-22-1520763/2022		V1A.3a.2	2022-05-02	MDCK1	640	80	40	80	640	320	1280	640	320
B/Poland/95/2022		V1A.3a.2	2022-05-07	MDCK1	640	40	<	80	640	160	640	640	320
B/Denmark/7/2022		V1A.3a.2	2022-05-16	MDCK2/MDCK1	640	40	<	<	320	160	640	640	320
B/Romania/519831/2022		V1A.3a.2	2022-05-26	MDCK2	640	80	40	80	640	320	1280	640	320
B/Romania/519824/2022		V1A.3a.2	2022-05-26	MDCK3	640	40	40	80	640	320	1280	640	320
B/Denmark/11/2022		V1A.3a.2	2022-06-13	MDCK2/MDCK1	640	40	<	80	320	160	640	640	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not Done

Vaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 14-6. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-08-23

					Haemagglutination inhibition titre										
Viruses	Other information	Passage history	Collection date	Passage history	Post-infection ferret antiserum										
					NEW										
					B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/CIV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R	B/Austria 1359417/21 Egg G141R
					Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F29/22 ¹¹	F08/21 ^{*5}	F12/21 ¹¹	F08/22 ¹¹	NIB F01/21 ¹¹	F15/21 ¹¹	F44/21 ¹¹	F30/22 ¹¹
					V1A	V1A.1	V1A.3	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES															
B/Brisbane/60/2008		V1A	2008-08-04	E4/E4	2560	160	40	<	<	<	<	<	<	<	<
B/Colorado/06/2017		V1A.1	2017-02-05	E5/E2	1280	640	80	<	<	<	<	<	<	<	<
B/Washington/02/2019		V1A.3	2019-01-19	E3/E3	640	160	160	<	<	<	<	<	<	<	<
B/Netherlands/11267/2022	G184R	V1A.3	2022-04-14	MDCK-MIX/MDCK1	40	10	<	320	<	<	<	<	<	<	<
B/Cote d'Ivoire/948/2020		V1A.3a.1	2020-05-28	MDCK4	320	40	40	40	640	40	40	160	80	80	40
B/Paris/9878/2020		V1A.3a.2	2020-11-20	MDCK2	640	80	10	<	160	640	160	1280	1280	320	160
B/Gansu-Baiyin/1281/2021		V1A.3a.2	2021-04-13	C1/C1/MDCK2	640	40	<	40	160	640	320	1280	1280	320	320
B/Austria/1359417/2021		V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	80	10	<	160	640	320	1280	1280	320	320
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3/E5	640	40	10	40	320	640	320	2560	1280	640	640
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E5	320	40	<	40	320	320	320	1280	1280	2560	>5120
TEST VIRUSES															
B/Saudi Arabia/22-13/2021		V1A.3a.2	2021-11-09	MDCK2	1280	80	10	40	160	640	320	1280	1280	640	320
B/Qatar/21-VI-22-1384971/2022		V1A.3a.2	2022-03-30	MDCK1	640	80	10	40	160	640	320	1280	1280	320	320
B/Qatar/21-VI-22-1402857/2022		V1A.3a.2	2022-04-02	MDCK2	1280	80	10	40	160	640	320	1280	1280	320	320
B/Lyon/CHU/R22.195.17/2022		V1A.3a.2	2022-04-09	MDCKx/MDCK1	1280	80	20	40	160	640	320	1280	1280	640	320
B/Qatar/10-VI-22-1458060/2022		V1A.3a.2	2022-04-16	MDCK1	1280	160	10	40	320	640	320	1280	2560	320	320
B/Ukraine/402/2022		V1A.3a.2	2022-04-18	MDCK1	1280	80	10	40	160	640	320	1280	1280	640	320
B/Qatar/10-VI-22-1473470/2022		V1A.3a.2	2022-04-20	MDCK1	1280	160	10	40	320	640	320	1280	1280	320	320
B/Qatar/21-VI-22-1481523/2022		V1A.3a.2	2022-04-21	MDCK1	640	80	10	40	160	640	320	1280	1280	320	320
B/Qatar/15-VI-22-1498604/2022		V1A.3a.2	2022-04-26	MDCK2	1280	160	20	40	320	1280	320	1280	2560	640	320
B/Belgium/H0560/2022		V1A.3a.2	2022-05-04	MDCK1/MDCK1	640	40	<	40	160	320	160	1280	1280	320	160
B/Nice/882/2022		V1A.3a.2	2022-05-05	MDCKx/MDCK1	640	80	10	40	160	320	320	1280	1280	320	320
B/Qatar/10-VI-22-1538710/2022		V1A.3a.2	2022-05-08	MDCK3	1280	80	10	40	160	640	320	1280	1280	320	320
B/Ukraine/407/2022		V1A.3a.2	2022-05-30	MDCK1	1280	160	10	40	160	640	320	1280	1280	640	320
B/Hong Kong/6893/2022		V1A.3a.2	2022-06-02	MDCK1	640	80	<	40	160	640	320	1280	1280	320	160
B/Hong Kong/3/2022		V1A.3a.2	2022-06-12	MDCK1/MDCK1	1280	160	20	40	160	640	320	1280	1280	640	320
B/Nosy Be/04869/2022		V1A.3a.2	2022-06-22	MDCK1	1280	80	20	40	160	640	320	1280	1280	320	320
B/Netherlands/11266/2022	G184R	V1A.3	2022-04-13	MDCK1	40	10	<	160	<	<	<	<	<	<	<
B/Netherlands/11303/2022	G184R	V1A.3	2022-04-19	MDCK1	80	10	<	320	<	<	<	<	<	<	<
B/Netherlands/11279/2022	G184R	V1A.3	2022-04-26	MDCK1	80	10	10	160	<	<	<	<	<	<	<

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not DoneVaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22Vaccine
SH 2022
NH 2022-23

< 4-fold

4-fold

8-fold

> 8-fold

< not recognised by the antiserum

≥ 160 (when homologous titre ≥ 2560)

≥ 160 (no homologous titre)

Table 14-7. Antigenic analyses of influenza B viruses (Victoria lineage) 2022-09-06

Viruses	Other information	Collection date	Passage history	Haemagglutination inhibition titre									
				Post-infection ferret antiserum									
				B/Bris 60/08 Egg	B/Colorado 06/17 Egg	B/Wash'ton 02/19 Egg	B/Neth 11267/22 MDCK	B/C/IV 948/20 MDCK	B/Paris 9878/20 MDCK	B/G-Baiyin 1281/21 MDCK	B/Austria 1359417/21 MDCK	B/Austria 1359417/21 Egg G141	B/Austria 1359417/21 Egg G141R
				Sh 539, 540, 543, 544, 570, 571, 574 ^{1,3}	F11/18 ²	F20/20 ²	F29/22 ¹	F08/21 ⁵	F12/21 ¹	F08/22 ¹	NIB F01/21 ¹	F15/21 ¹	F44/21 ¹
				Genetic group	V1A	V1A.1	V1A.3	V1A.3a.1	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2	V1A.3a.2
REFERENCE VIRUSES													
B/Brisbane/60/2008	V1A	2008-08-04	E4/E4	1280	160	40	<	<	<	<	<	<	<
B/Colorado/06/2017	V1A.1	2017-02-05	E5/E2	1280	640	80	<	<	<	<	<	<	<
B/Washington/02/2019	V1A.3	2019-01-19	E3/E3	640	160	160	<	<	<	<	<	<	<
B/Netherlands/11267/2022	V1A.3	2022-04-14	MDCK-MIX/MDCK1	40	10	<	320	<	<	<	<	<	<
B/Cote d'Ivoire/948/2020	V1A.3a.1	2020-05-28	MDCK4	320	40	40	40	640	40	40	160	80	80
B/Paris/9878/2020	V1A.3a.2	2020-11-20	MDCK2	640	80	10	<	160	640	160	1280	1280	320
B/Gansu-Baiyin/1281/2021	V1A.3a.2	2021-04-13	C1/C1/MDCK2	640	40	<	40	160	640	640	1280	1280	320
B/Austria/1359417/2021	V1A.3a.2	2021-01-09	SIAT1/MDCK4	640	80	10	<	160	640	320	2560	1280	320
B/Austria/1359417/2021 Isolate 2	G141	V1A.3a.2	2021-01-09	E3/E5	640	40	10	40	320	640	320	2560	640
B/Austria/1359417/2021 Isolate 2	G141R	V1A.3a.2	2021-01-09	E3/E5	320	40	<	40	320	320	320	1280	2560
TEST VIRUSES													
B/Moscow/RII-01/2022	V1A.3a.2	2022-04-28	MDCK3/MDCK1	1280	80	10	80	320	640	640	2560	2560	640
B/S. Petersburg/RII-09/2022	V1A.3a.2	2022-05-13	MDCK1/MDCK1	1280	80	20	80	320	640	640	2560	2560	640
B/S. Petersburg/RII-11/2022	V1A.3a.2	2022-05-25	MDCK1/MDCK1	1280	80	10	40	320	640	320	2560	2560	320
B/S. Petersburg/RII-15/2022	V1A.3a.2	2022-05-26	MDCK1/MDCK1	1280	160	20	80	320	640	640	1280	2560	640
B/S. Petersburg/RII-13/2022	V1A.3a.2	2022-06-06	MDCK1/MDCK1	1280	80	20	80	160	640	320	1280	1280	320

* Superscripts refer to antiserum properties (< relates to the lowest dilution of antiserum used):

¹ < = <40; ² < = <10; ³ hyperimmune sheep serum; ⁴ < = <20; ⁵ < = <80; ND = Not Done

Vaccine
SH 2020
NH 2020-21
SH 2021
NH 2021-22

Vaccine
SH 2022
NH 2022-23

< 4-fold
 4-fold
 8-fold
 > 8-fold
 < not recognised by the antiserum
 ≥ 160 (when homologous titre ≥ 2560)
 ≥ 160 (no homologous titre)

Table 14-8. Summary of HI results for B/Victoria-lineage viruses with collection dates after 2022-01-31 by genetic group

Genetic group	Antiserum raised against Fold reduction	B/Colorado 06/17 Egg F11/18* ⁴ V1A.1	B/Wash'ton 02/19 Egg F20/20* ⁴ V1A.3	B/Neth 11267/22 MDCK F29/22* ¹ V1A.3	B/CIV 948/20 MDCK F08/21* ⁵ V1A.3a.1	B/Paris 9878/20 MDCK F12/21* ¹ V1A.3a.2	B/G-Baiyin 1281/21 MDCK F08/22* ¹ V1A.3a.2	B/Stock 3/22 MDCK F28/22* ¹ V1A.3a.2	B/Austria 1359417/21 MDCK NIB F01/21* ¹ V1A.3a.2	B/Austria 1359417/21 Egg G141 F15/21* ¹ V1A.3a.2	B/Austria 1359417/21 Egg G141R F44/21* ¹ V1A.3a.2	CNIC-2204B Z-S/11203/21 Egg G141W F16/22* ¹ V1A.3a.2
V1A.3a.2	No tested	87	87	20	87	87	87	15	87	87	87 ^{\$}	5
	≤2	9	0	0	19	86	83	15	85	85	0	0
	%	10.4			21.8	98.9	95.4	100	97.7	97.7		
	4	41	7	4	58	0	3	0	2	2	12	3
	%	47.1	8.0	20.0	66.7		3.5		2.3	2.3	13.8	
	≥8	37	80	16	10	1	1	0	0	0	75	2
V1A.3	%	42.5	92.0	80.0	11.5	1.1	1.1				86.2	
	No tested	11	11	3	11	11	11		11	11	11	
	≤2	0	0	3	0	0	0		0	0	0	
	%											
	4	0	0	0	0	0	0		0	0	0	
	%											
	≥8	11	11	0	11	11	11		11	11	11	
	%	100	100		100	100	100		100	100	100	

All fold-reductions are based on homologous titres determined on the day of individual assays.

^{\$} All test viruses had HI titres of ≥ 160

Table 15. Antiviral (NAI) testing of isolates with collection dates after 2021-08-31*

Month of Collection	Number of viruses tested for phenotype ¹								Totals
	A(H1N1)pdm09		A(H3N2)		Influenza B-Victoria		Influenza B-Yamagata		
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2021									
September	31		61	1	23				116
October	41		59		12				112
November	43		133		26				202
December	20		147		4				171
2022									0
January	20		73		0				93
February	17		77		4				98
March	52		206		19				277
April	50		213		51				314
May	30		80		22				132
June	4		19		5				28
July	2		9						11
August	1								1
Totals	311	0	1077	1	166	0	0	0	1555

¹ Phenotype: NI (normal inhibition), RI (reduced inhibition), HRI (highly reduced inhibition) as defined in WHO Wkly. Epidemiol. Rec. (2012) 87, 369-374

* As of 2022-09-13

Virus name	Subtype/ Lineage	Collection date	Fold difference		NA substitution
			Oseltamivir	Zanamivir	
A/Tripoli/5646/2021	A(H3N2)	2021-09-07	NI	11.89 (RI)	A246V (NXT) - also in clinical specimen

Based on NA gene sequencing, no markers (amino acid substitutions) associated with reduced susceptibility to NAIs have been identified in any influenza-positive clinical specimens from which viruses have either not been recovered or are awaiting isolation attempts. Total number = 604 (75 N1, 515 N2, 14 BNA).

A(H3N2) viruses containing NA amino acid substitutions that have been associated with reduced sialidase activity based on MUNANA assay (Hussain S et al, 2021, J Gen Virol)

Virus name	Sample type	Collection date	Substitution	Status	Outcome (MUNANA)
A/Kosova/641/2021	cs	2021-12-23	S331R	To be isolated	Insufficient titre for IC ₅₀ determination
A/Bremen/2/2022	C1/SIAT1	2022-02-18	S331R	Propagated	
A/Belgium/G0096/2022	SIAT1	2022-03-07	S331R	To be propagated	
A/Belgium/S0822/2022	SIAT1	2022-03-18	S331R	To be propagated	
A/Belgium/S0826/2022	SIAT1	2022-03-19	S331R	To be propagated	Insufficient titre for IC ₅₀ determination
A/Belgium/G0153/2022	SIAT1	2022-03-23	S331R	To be propagated	
A/Belgium/S0894/2022	SIAT1/SIAT1	2022-03-26	S331R	Propagated	
A/Belgium/G0163/2022	SIAT1/SIAT1	2022-03-28	S331R	Propagated	
A/Switzerland/67859/2022	cs/SIAT1	2022-04-19	S331R	Propagated	Insufficient titre for IC ₅₀ determination
A/Argentina/2859/2022	cs	2022-05-15	S331R	To be isolated	
A/Argentina/2995/2022	cs	2022-06-21	S331R	To be isolated	
A/Baabda.LBN/7463/2021	cs/SIAT1	2021-11-24	S334R	Propagated	
A/Croatia/77367/2022	cs	2022-05-07	S334R	To be isolated	Insufficient titre for IC ₅₀ determination

Table 16. Influenza PA gene sequences generated at the WIC and screened for Markers of Baloxavir Marboxil reduced susceptibility

For viruses with collection dates after 2021-08-31

Virus with Marker	Collection Date	Subtype/lineage	Marker	Clinical specimen (CS) ¹	Virus isolate ²	Comments
A/Netherlands/10253/2022	2022-03-06	H3N2	E23G	N/A	Yes	CS sequenced by NIC: phenotypic reduced susceptibility confirmed at CDC/NIID
A/Estonia/169374/2022	2022-04-19	H3N2	L28P	N/A	Yes	
A/Madrid/1445/2022	2022-03-23	H3N2	L28P	N/A	Yes	CS sequenced by NIC
Total number of PA sequences generated at WIC				704	1388	Includes 136 CS/virus isolate pairs

Subtype/Lineage	Clinical specimen (CS) ¹	Virus isolate ²	CS/Virus pairs	Individual Viruses
H1N1pdm09	35	284	11	308
H3N2	627	976	106	1497
B/Victoria	42	128	19	151
Total number	704	1388	136	1956

¹ N/A = Not available (not provided by NIC); SF = available but sequence failed

² Yes = Marker present; No = Marker absent (only used if Marker in clinical specimen); NR = Virus not recovered (when sent as isolate)

As of 2022-09-14